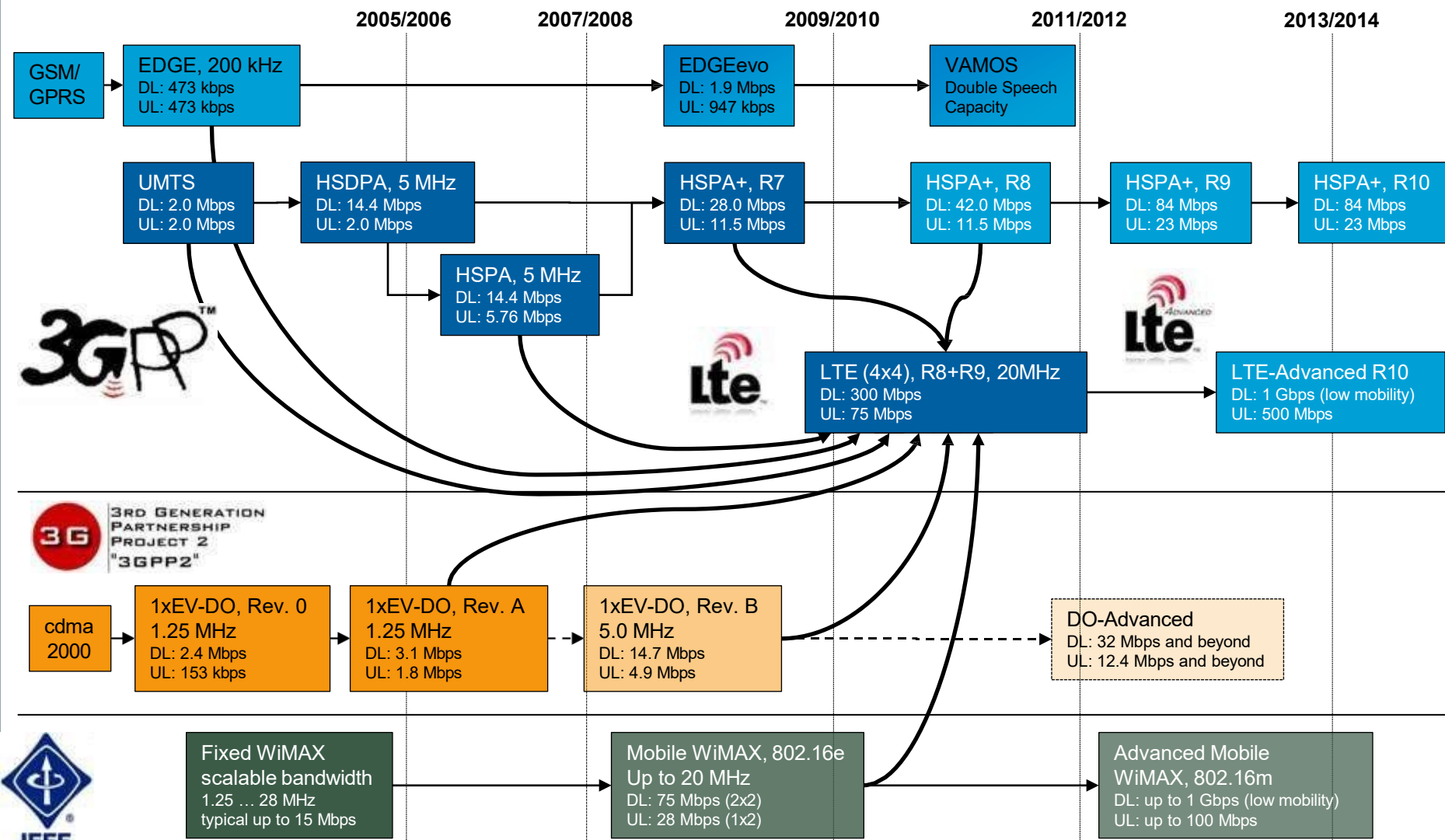


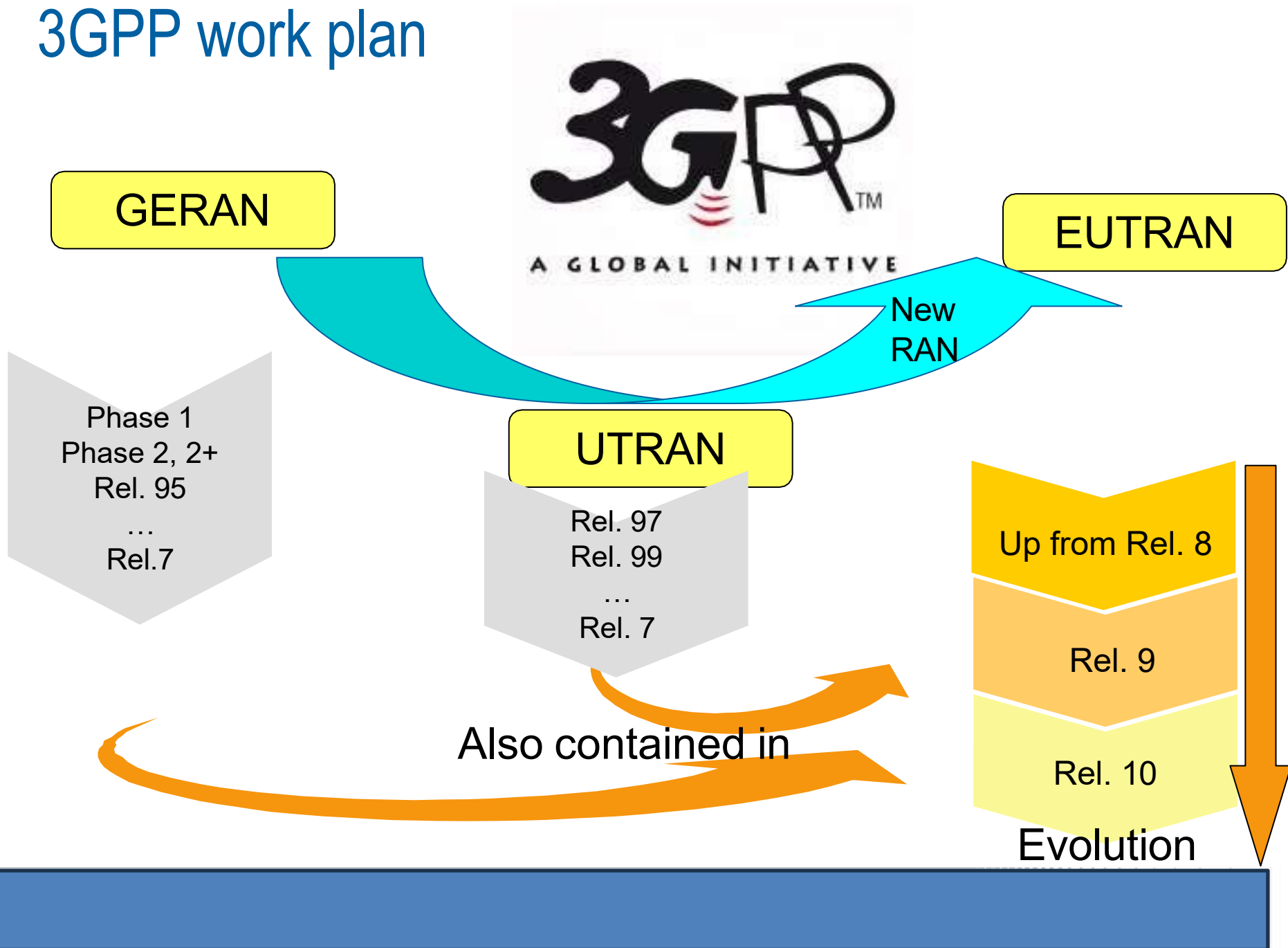
UMTS Long Term Evolution (LTE)

Dr Dhiraj Kumar Patel,
Assistant Professor,
Department of Electronics and Communication Engineering
IIIT Surat

Technology evolution path



3GPP work plan



Overview 3GPP UMTS evolution



3GPP release	3GPP Release 99/4	3GPP Release 5/6	3GPP Release 7	3GPP Release 8	3GPP Release 10
App. year of network rollout	2003/4	2005/6 (HSDPA) 2007/8 (HSUPA)	2008/2009	2010	
Downlink data rate	384 kbps (typ.)	14 Mbps (peak)	28 Mbps (peak)	LTE: 150 Mbps* (peak) HSPA+: 42 Mbps (peak)	100 Mbps high mobility 1 Gbps low mobility
Uplink data rate	128 kbps (typ.)	5.7 Mbps (peak)	11 Mbps (peak)	LTE: 75 Mbps (peak) HSPA+: 11 Mbps (peak)	
Round Trip Time	~ 150 ms	< 100 ms	< 50 ms	LTE: ~10 ms	

*based on 2x2 MIMO and 20 MHz operation

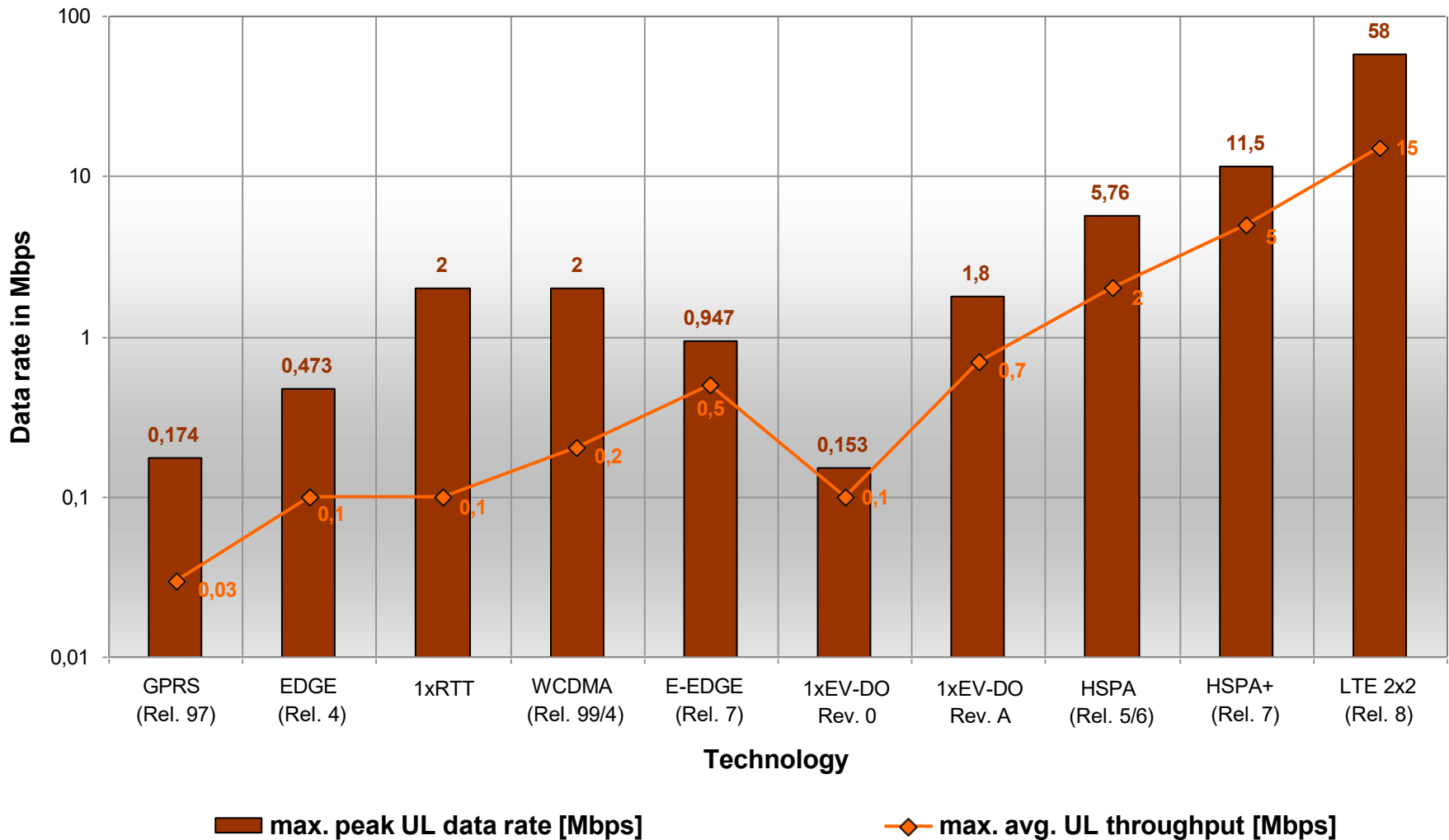
HSPA: High Speed D/U packet access
UMTS: Universal Mobile Telecommunication System

Why LTE?

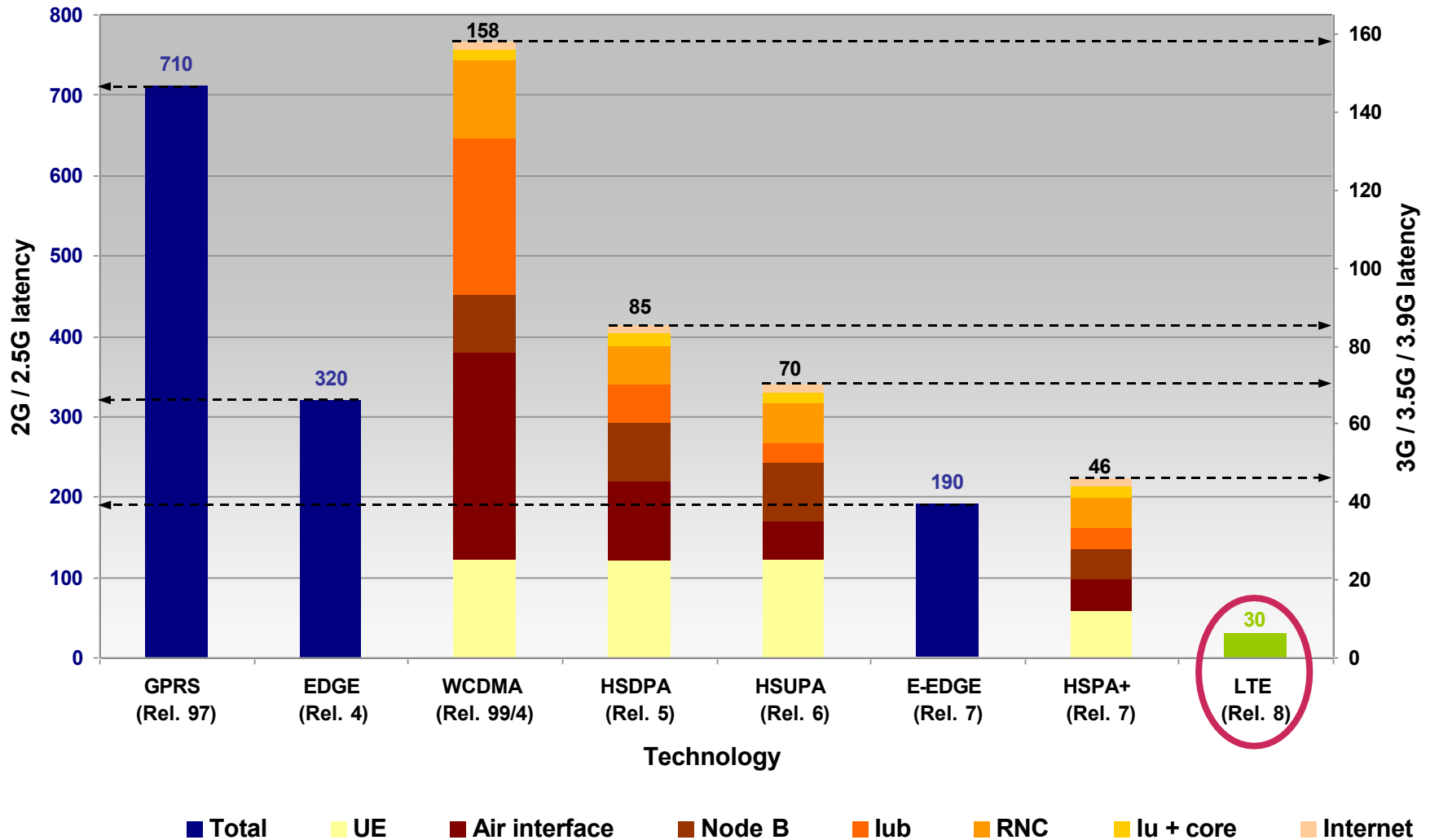
Ensuring Long Term Competitiveness of UMTS

- ❑ **LTE is the next UMTS evolution step after HSPA and HSPA+.**
- ❑ **LTE is also referred to as**
EUTRA(N) = Evolved UMTS Terrestrial Radio Access (Network).
- ❑ **Main targets of LTE:**
 - ❑ Peak data rates of 100 Mbps (downlink) and 50 Mbps (uplink)
 - ❑ Scaleable bandwidths up to 20 MHz
 - ❑ Reduced latency
 - ❑ Cost efficiency
 - ❑ Operation in paired (FDD) and unpaired (TDD) spectrum

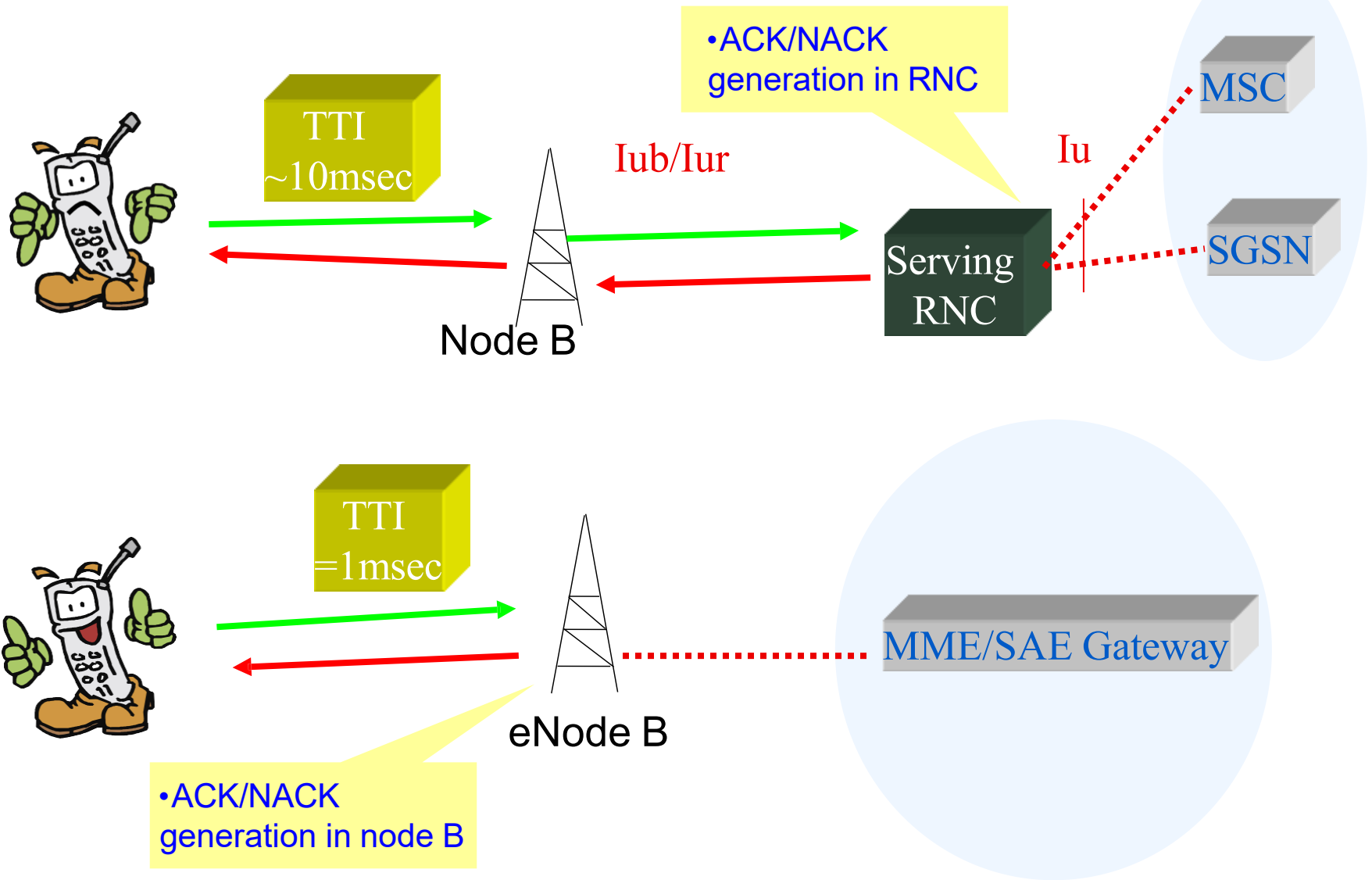
Peak data rates and real average throughput (UL)



Comparison of network latency by technology



Round Trip Time, RTT



RNC: Radio Network Controller
Ack/ Nack: Acknowledgement/ Negative Ack.

Major technical challenges in LTE

New radio transmission schemes (OFDMA / SC-FDMA)

FDD and TDD mode

MIMO multiple antenna schemes

Throughput / data rate requirements

Timing requirements (1 ms transm.time interval)

Multi-RAT requirements (GSM/EDGE, UMTS, CDMA)

Scheduling (shared channels, HARQ, adaptive modulation)

System Architecture Evolution (SAE)

Introduction to UMTS LTE: Key parameters

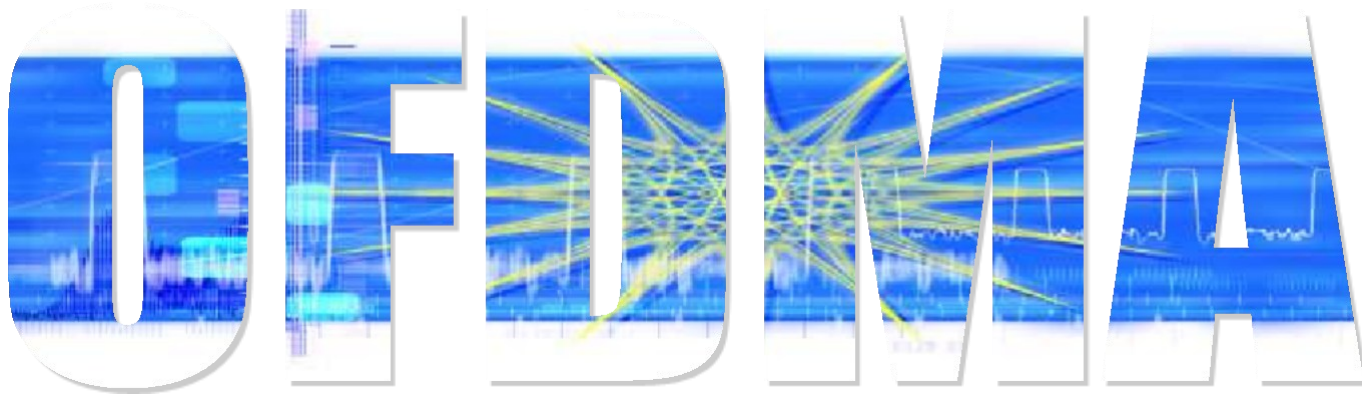
Frequency Range	UMTS FDD bands and UMTS TDD bands					
Channel bandwidth, 1 Resource Block=180 kHz	1.4 MHz	3 MHz	5 MHz	10 MHz	15 MHz	20 MHz
	6 Resource Blocks	15 Resource Blocks	25 Resource Blocks	50 Resource Blocks	75 Resource Blocks	100 Resource Blocks
Modulation Schemes	Downlink: QPSK, 16QAM, 64QAM Uplink: QPSK, 16QAM, 64QAM (optional for handset)					
Multiple Access	Downlink: OFDMA (Orthogonal Frequency Division Multiple Access) Uplink: SC-FDMA (Single Carrier Frequency Division Multiple Access)					
MIMO technology	Downlink: Wide choice of MIMO configuration options for transmit diversity, spatial multiplexing, and cyclic delay diversity (max. 4 antennas at base station and handset) Uplink: Multi user collaborative MIMO					
Peak Data Rate	Downlink: 150 Mbps (UE category 4, 2x2 MIMO, 20 MHz) 300 Mbps (UE category 5, 4x4 MIMO, 20 MHz) Uplink: 75 Mbps (20 MHz)					

LTE/LTE-A Frequency Bands (FDD)

E-UTRA Operating Band	Uplink (UL) operating band BS receive UE transmit			Downlink (DL) operating band BS transmit UE receive			Duplex Mode
	F _{UL_low} – F _{UL_high}			F _{DL_low} – F _{DL_high}			
1	1920 MHz	–	1980 MHz	2110 MHz	–	2170 MHz	FDD
2	1850 MHz	–	1910 MHz	1930 MHz	–	1990 MHz	FDD
3	1710 MHz	–	1785 MHz	1805 MHz	–	1880 MHz	FDD
4	1710 MHz	–	1755 MHz	2110 MHz	–	2155 MHz	FDD
5	824 MHz	–	849 MHz	869 MHz	–	894MHz	FDD
6	830 MHz	–	840 MHz	875 MHz	–	885 MHz	FDD
7	2500 MHz	–	2570 MHz	2620 MHz	–	2690 MHz	FDD
8	880 MHz	–	915 MHz	925 MHz	–	960 MHz	FDD
9	1749.9 MHz	–	1784.9 MHz	1844.9 MHz	–	1879.9 MHz	FDD
10	1710 MHz	–	1770 MHz	2110 MHz	–	2170 MHz	FDD
11	1427.9 MHz	–	1452.9 MHz	1475.9 MHz	–	1500.9 MHz	FDD
12	698 MHz	–	716 MHz	728 MHz	–	746 MHz	FDD
13	777 MHz	–	787 MHz	746 MHz	–	756 MHz	FDD
14	788 MHz	–	798 MHz	758 MHz	–	768 MHz	FDD
17	704 MHz	–	716 MHz	734 MHz	–	746 MHz	FDD
18	815 MHz	–	830 MHz	860 MHz	–	875 MHz	FDD
19	830 MHz	–	845 MHz	875 MHz	–	890 MHz	FDD
20	832 MHz	-	862 MHz	791 MHz	-	821 MHz	FDD
21	1447.9 MHz	-	1462.9 MHz	1495.9 MHz	-	1510.9 MHz	FDD
22	3410 MHz	-	3500 MHz	3510 MHz	-	3600 MHz	FDD

LTE/LTE-A Frequency Bands (TDD)

E-UTRA Operating Band	Uplink (UL) operating band BS receive UE transmit			Downlink (DL) operating band BS transmit UE receive			Duplex Mode
	F _{UL_low} – F _{UL_high}			F _{DL_low} – F _{DL_high}			
33	1900 MHz	–	1920 MHz	1900 MHz	–	1920 MHz	TDD
34	2010 MHz	–	2025 MHz	2010 MHz	–	2025 MHz	TDD
35	1850 MHz	–	1910 MHz	1850 MHz	–	1910 MHz	TDD
36	1930 MHz	–	1990 MHz	1930 MHz	–	1990 MHz	TDD
37	1910 MHz	–	1930 MHz	1910 MHz	–	1930 MHz	TDD
38	2570 MHz	–	2620 MHz	2570 MHz	–	2620 MHz	TDD
39	1880 MHz	–	1920 MHz	1880 MHz	–	1920 MHz	TDD
40	2300 MHz	–	2400 MHz	2300 MHz	–	2400 MHz	TDD
41	3400 MHz – 3600MHz			3400 MHz – 3600MHz			TDD

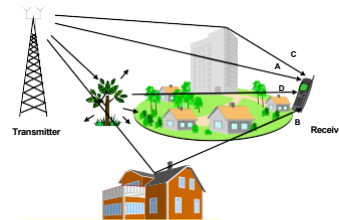
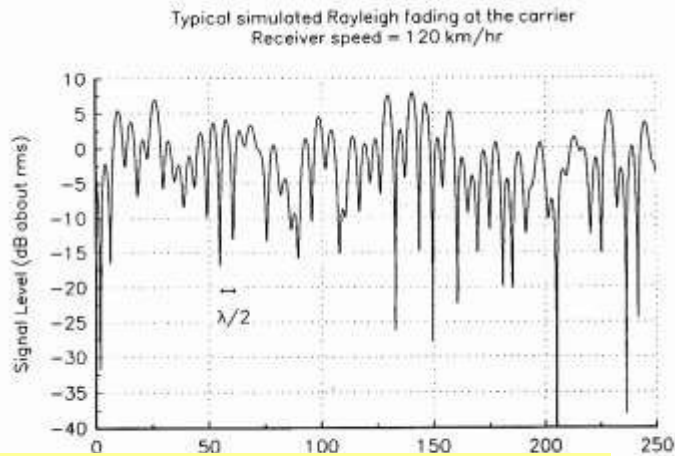


Orthogonal Frequency Division Multiple Access

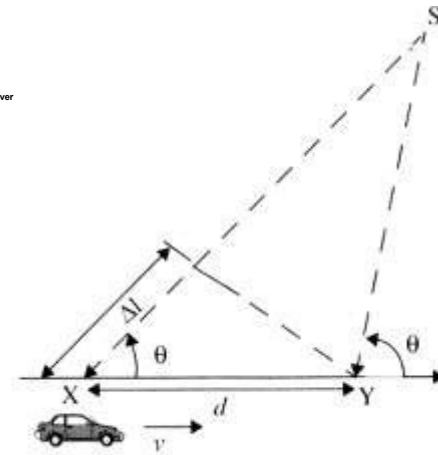
- I OFDM is **the** modulation scheme for LTE in downlink and uplink (as reference)
- I Some technical explanation about our physical base: radio link aspects

What does it mean to use the radio channel?

Using the radio channel means to deal with aspects like:

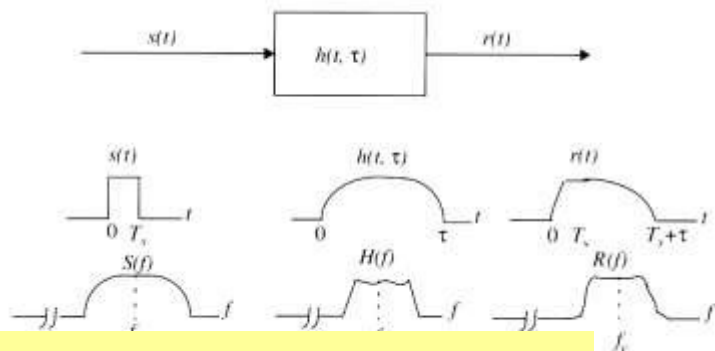


MPP

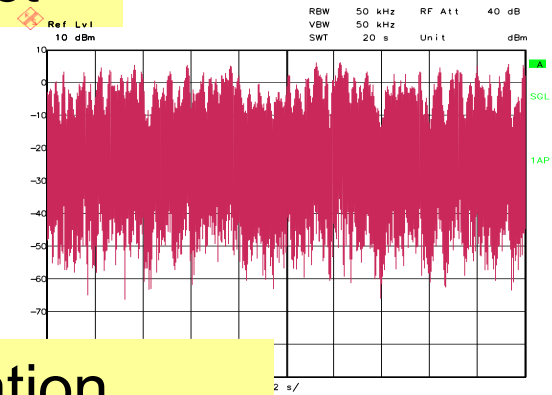


Time variant channel

Doppler effect

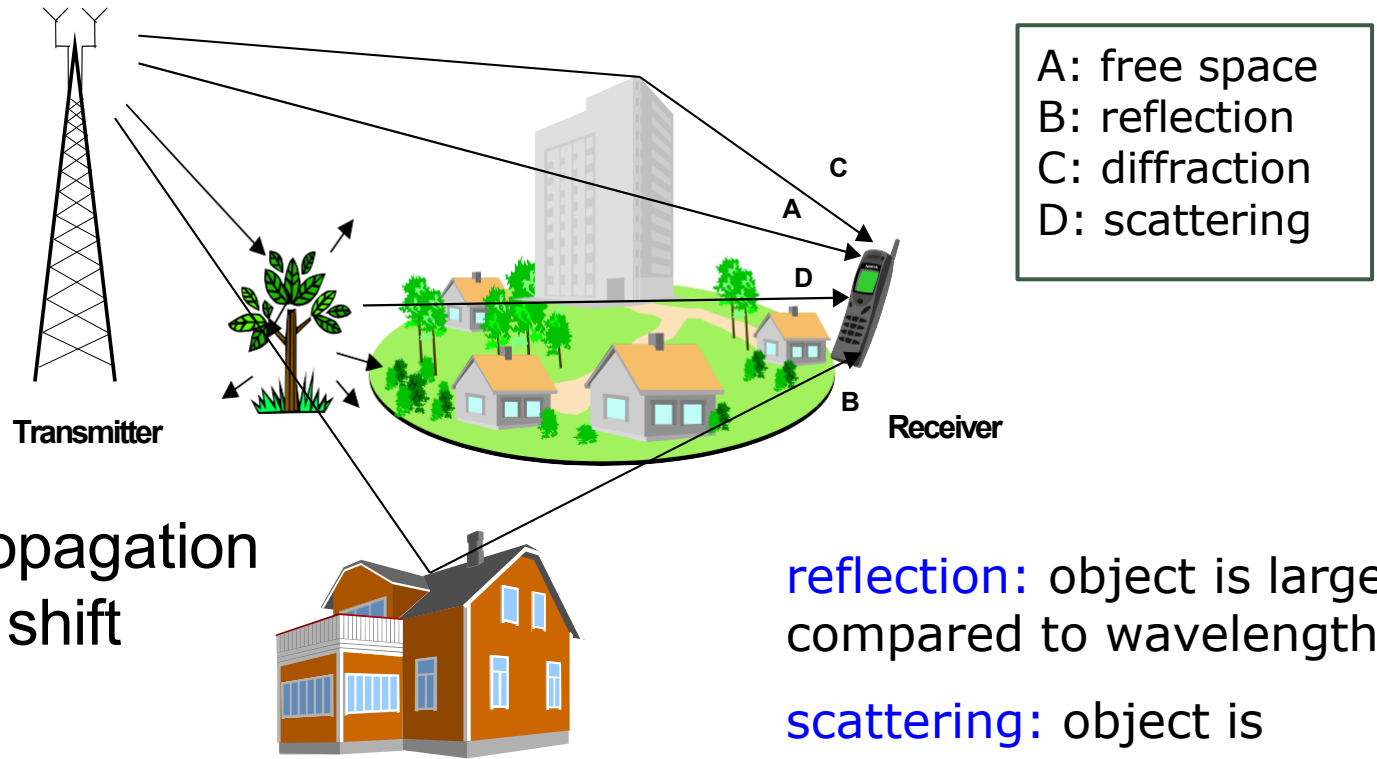


Frequency selectivity



attenuation

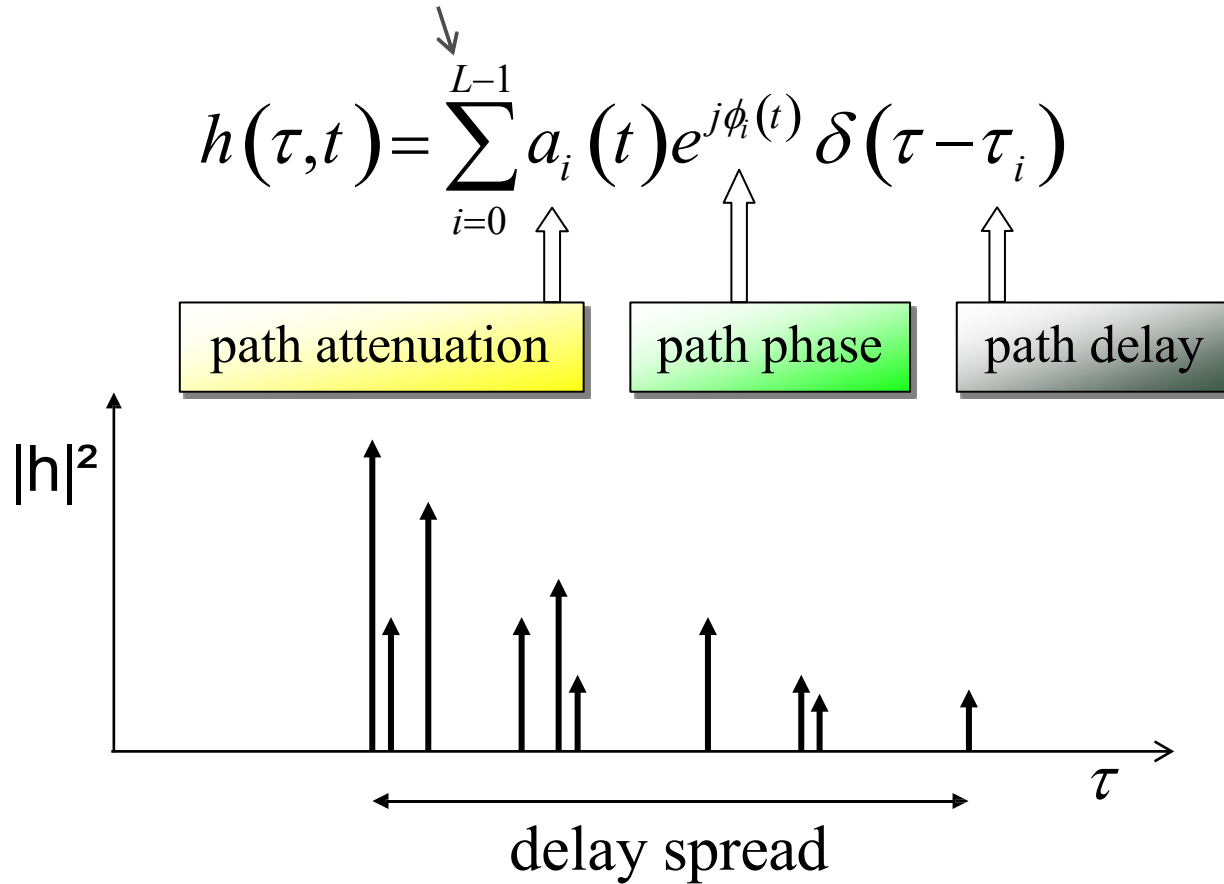
Still the same “mobile” radio problem: Time variant multipath propagation



Multipath Propagation
and Doppler shift

Multipath channel impulse response

The CIR consists of L resolvable propagation paths



Radio Channel – different aspects to discuss

Bandwidth



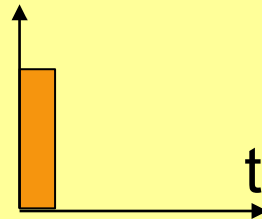
Wideband

or



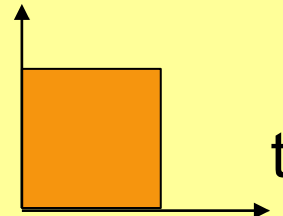
Narrowband

Symbol duration



Short symbol
duration

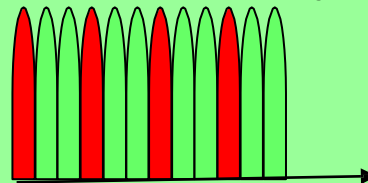
or



Long symbol
duration

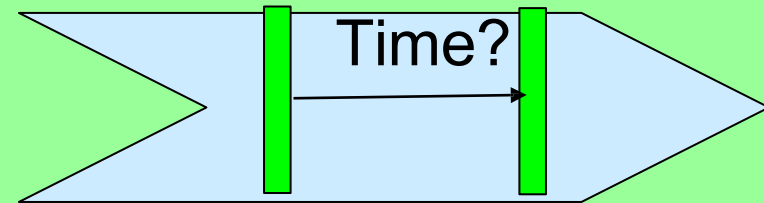
Channel estimation:
Pilot mapping

Frequency?



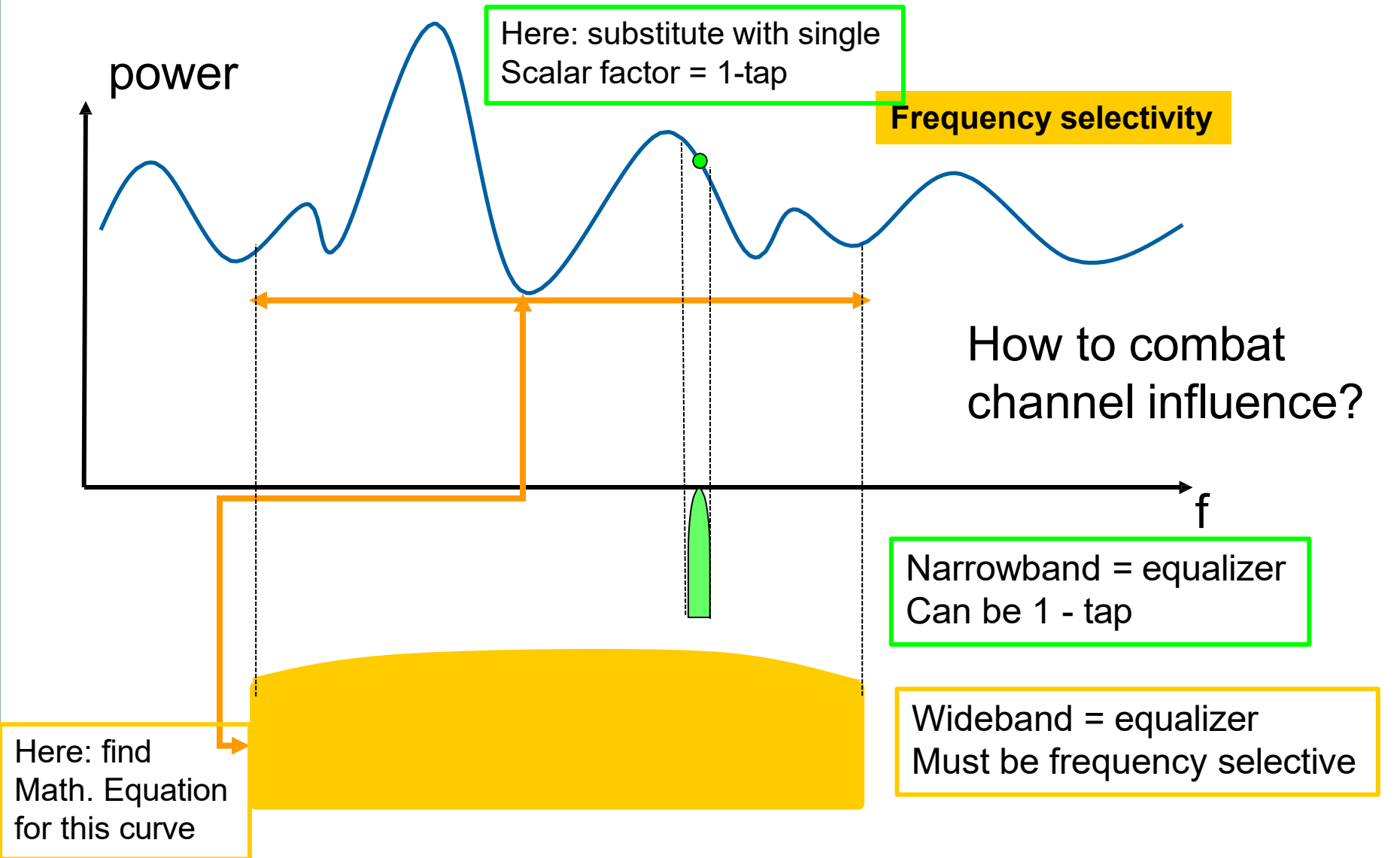
frequency distance of pilots?

Time?



Repetition rate of pilots?

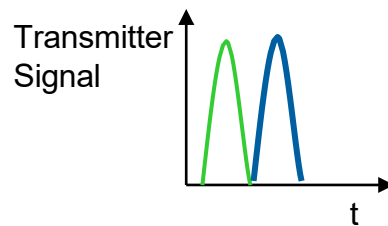
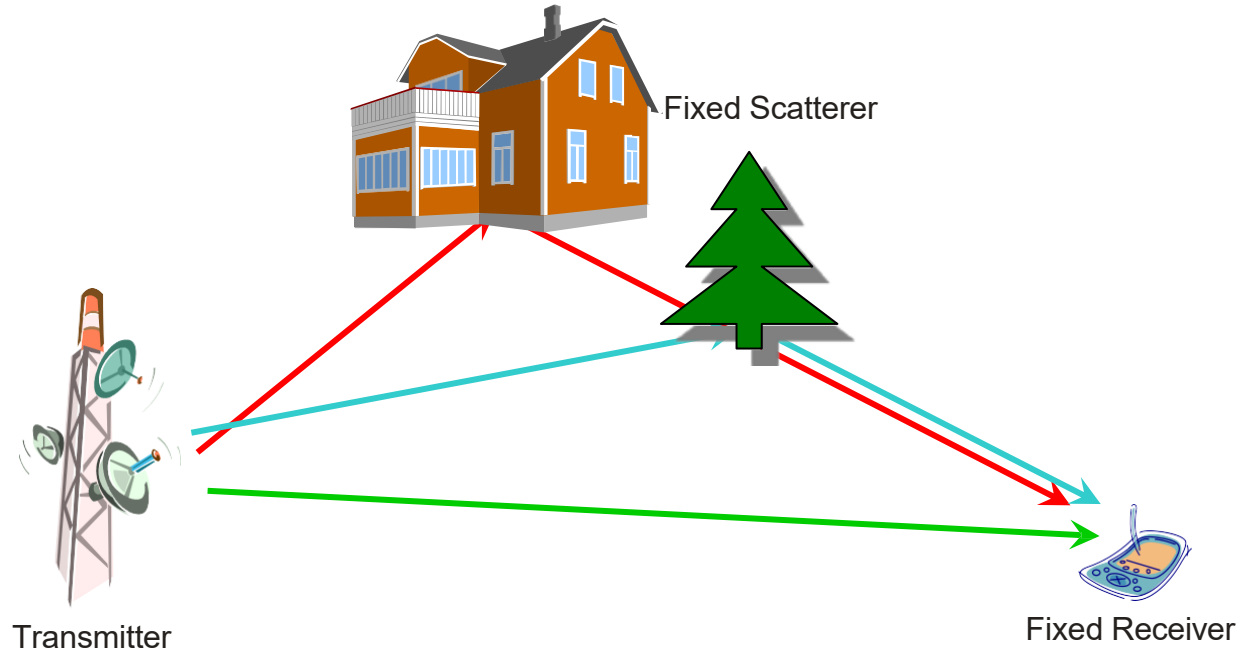
Frequency selectivity - Coherence Bandwidth



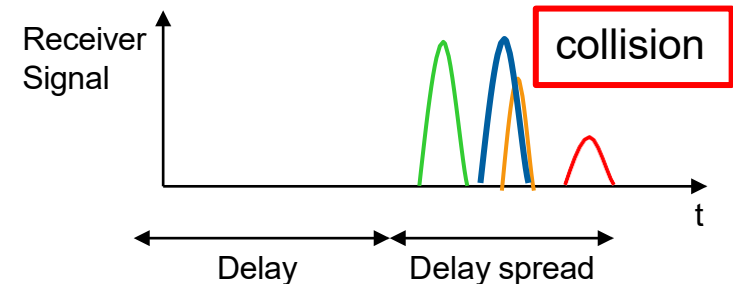
Time-Invariant Channel: Scenario

ISI: Inter Symbol Interference:
Happens, when
Delay spread >
Symbol time

Successive
symbols
will interfere

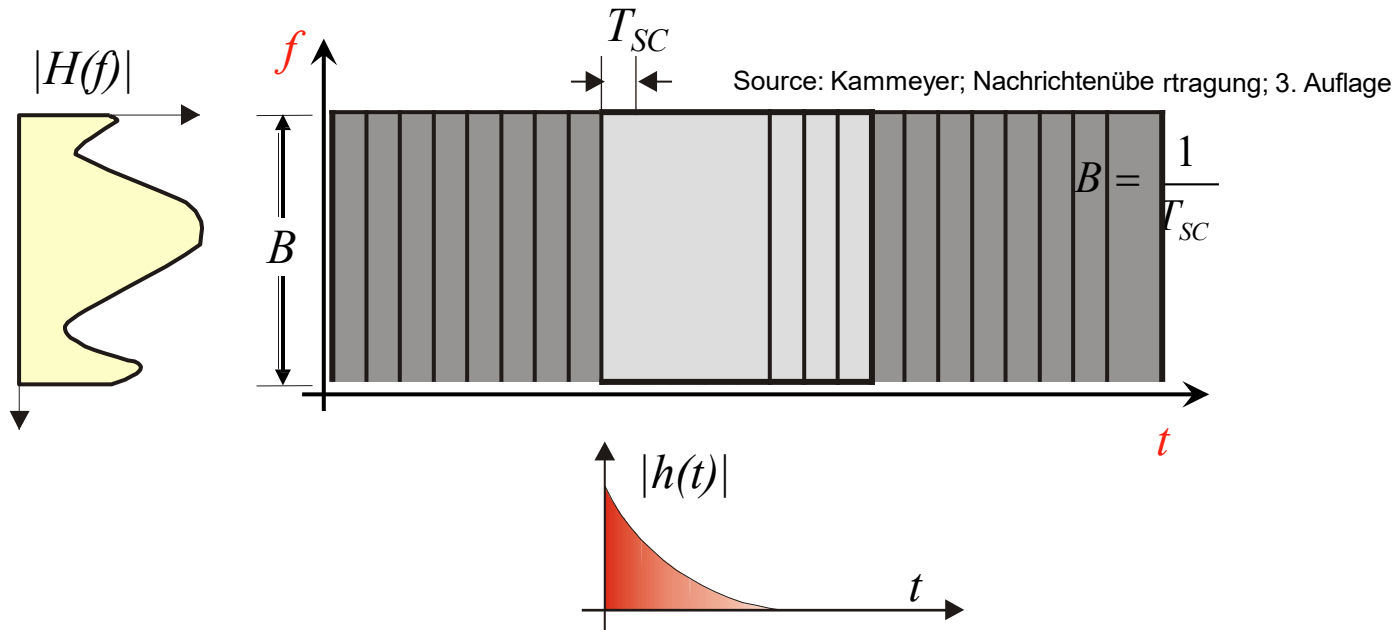


Channel Impulse Response, CIR



→time dispersive

Motivation: Single Carrier versus Multi Carrier



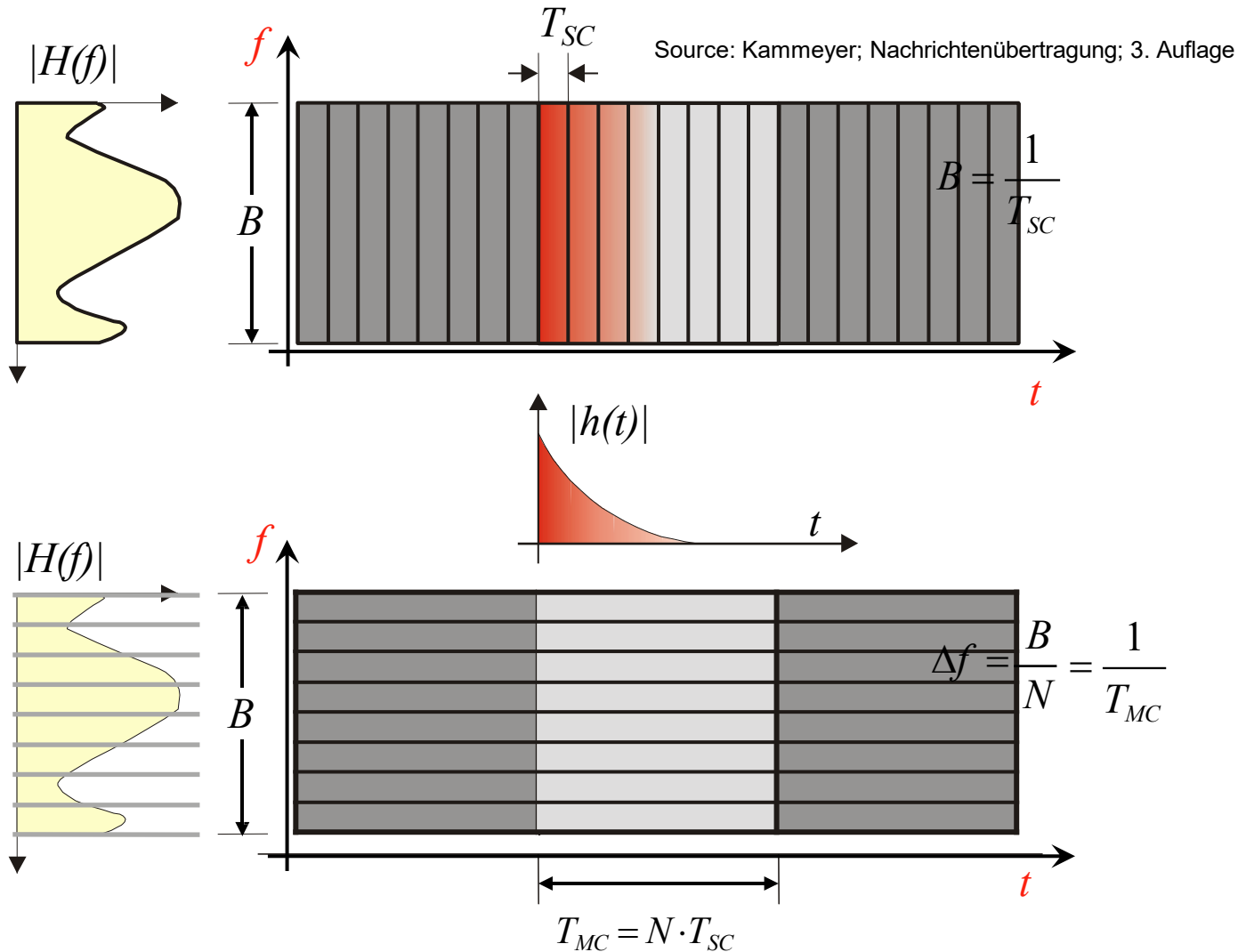
I Time Domain

- I Delay spread $>$ Symboltime T_{SC}
→ **Inter-Symbol-Interference (ISI)** → equalization effort

I Frequency Domain

- I Coherence Bandwidth $B_c <$ Systembandwidth B
→ **Frequency Selective Fading** → equalization effort

Motivation: Single Carrier versus Multi Carrier

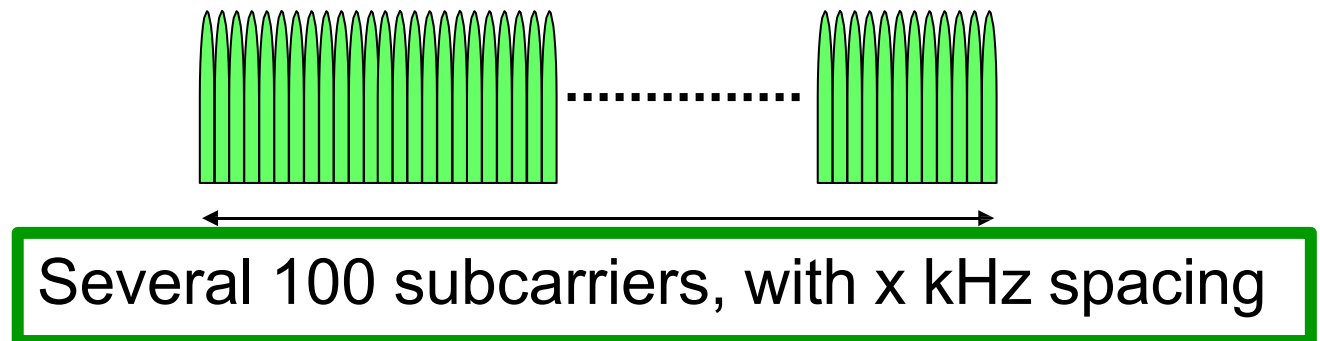


What is OFDM?

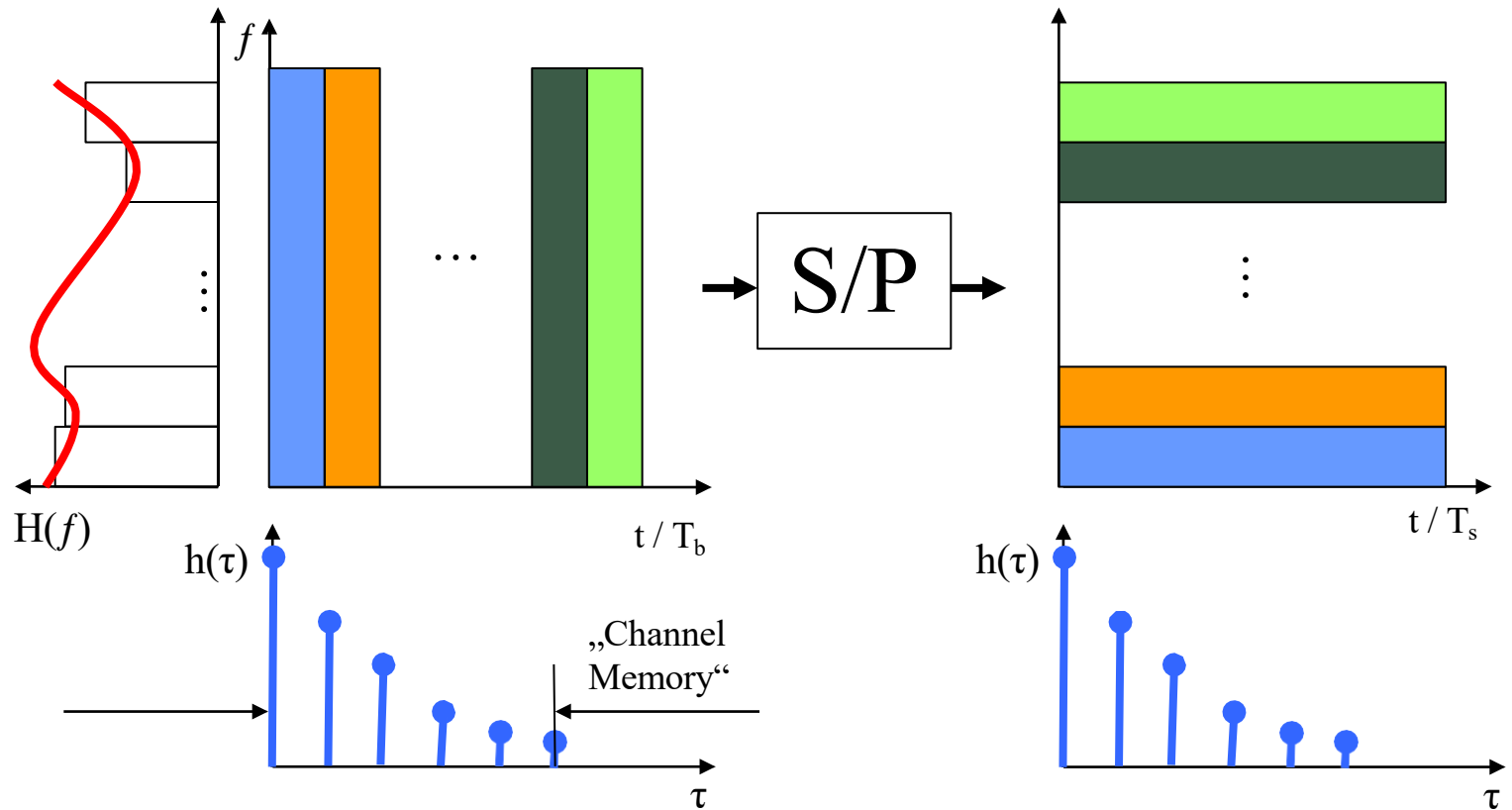
Single carrier
transmission,
e.g. WCDMA



Orthogonal
Frequency
Division
Multiplex



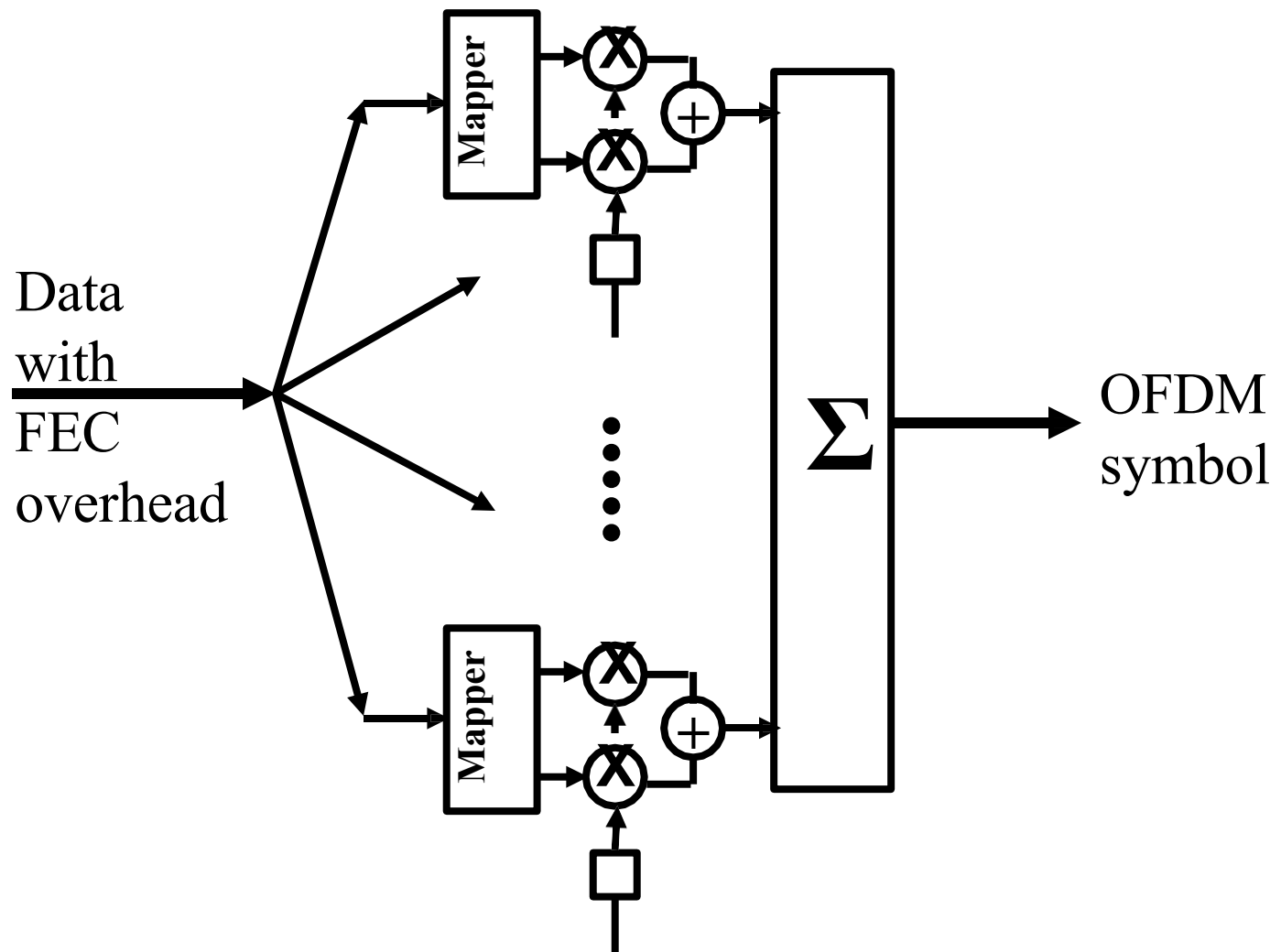
Idea: Wide/Narrow band conversion



**One high rate signal:
Frequency selective fading**

**N low rate signals:
Frequency flat fading**

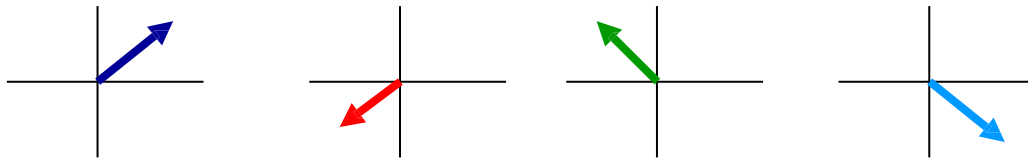
COFDM



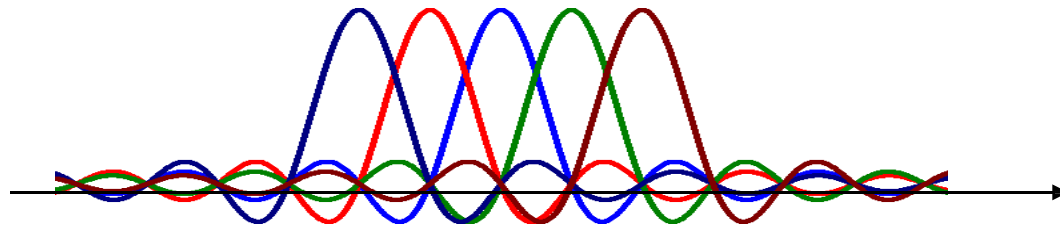
OFDM signal generation

00 11 10 10 01 01 11 01

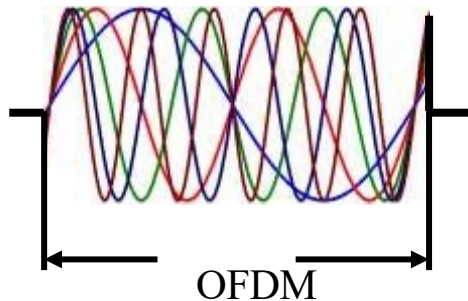
e.g. QPSK



$$h^*(\sin_{j\omega t} + \cos_{j\omega t}) \quad h^*(\sin_{j\omega t} + \cos_{j\omega t}) \Rightarrow \sum h^* (\sin.. + \cos...)$$



Frequency



time

Fourier Transform, Discrete FT

Fourier Transform

$$H(f) = \int_{-\infty}^{+\infty} h(t) e^{2\pi j f t} dt;$$

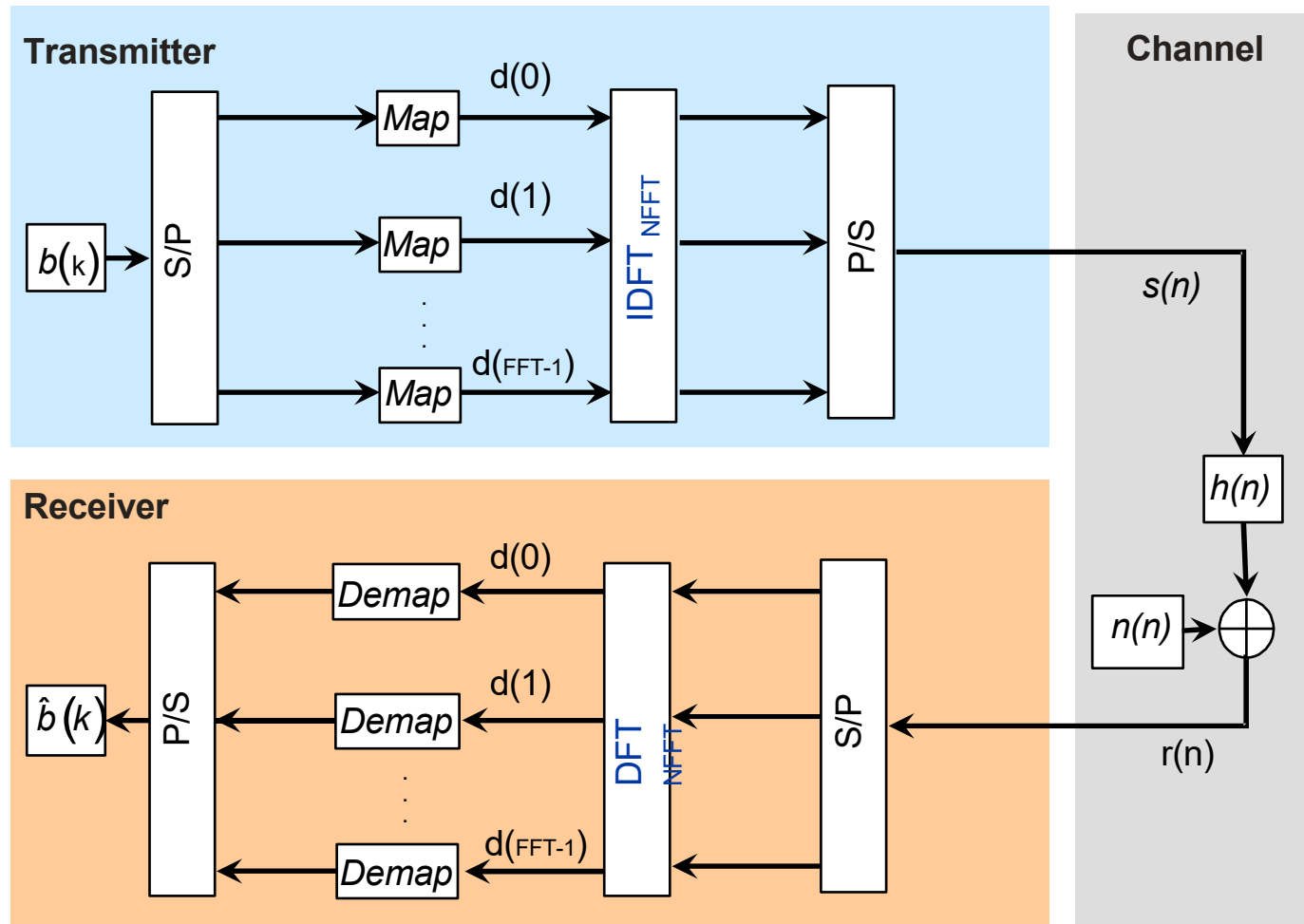
$$h(t) = \int_{-\infty}^{+\infty} H(f) e^{-2\pi j f t} df;$$

Discrete Fourier Transform (DFT)

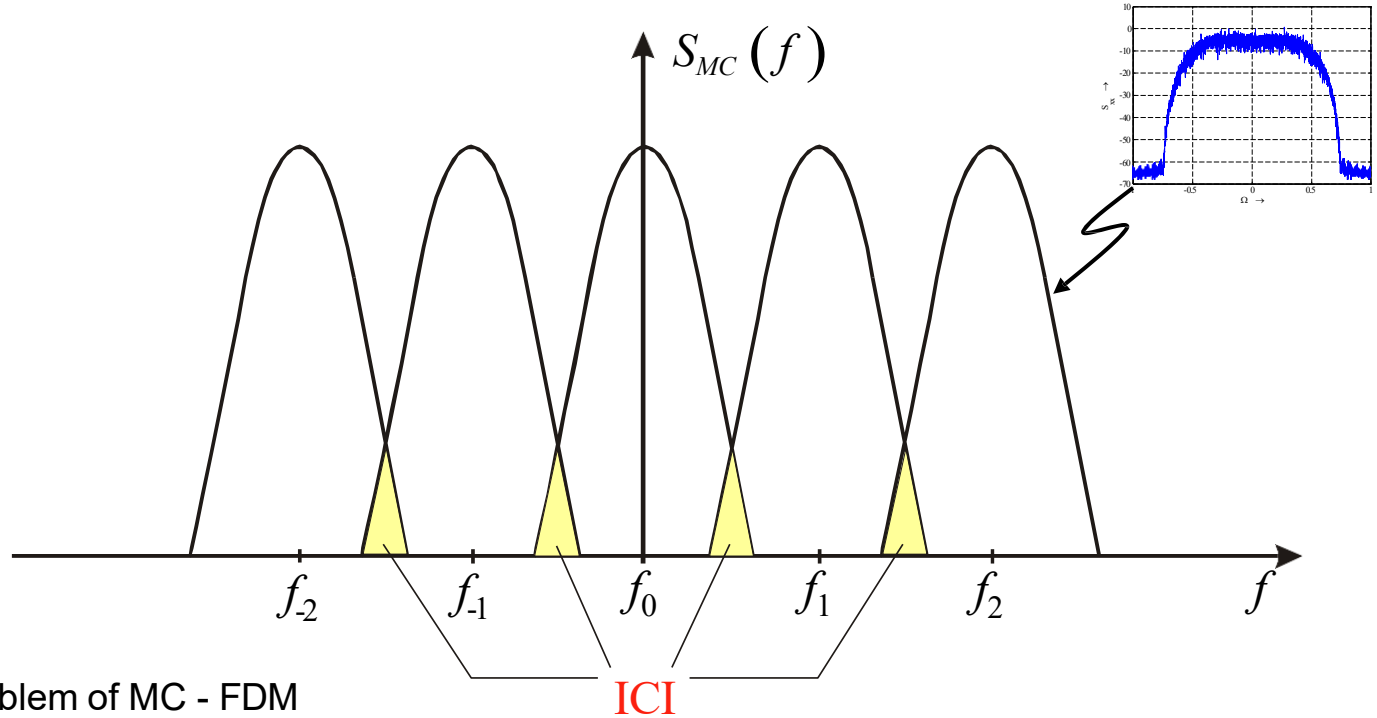
$$H_n = \sum_{k=0}^{N-1} h_k e^{2\pi j k n / N} = \sum_{k=0}^{N-1} h_k \cos(2\pi k \frac{n}{N}) + j \sum_{k=0}^{N-1} h_k \sin(2\pi k \frac{n}{N});$$

$$h_k = \frac{1}{N} \sum_{n=0}^{N-1} H_n e^{-2\pi j k \frac{n}{N}};$$

OFDM Implementation with FFT (Fast Fourier Transformation)



Inter-Carrier-Interference (ICI)



Problem of MC - FDM

Overlapp of neighbouring subcarriers

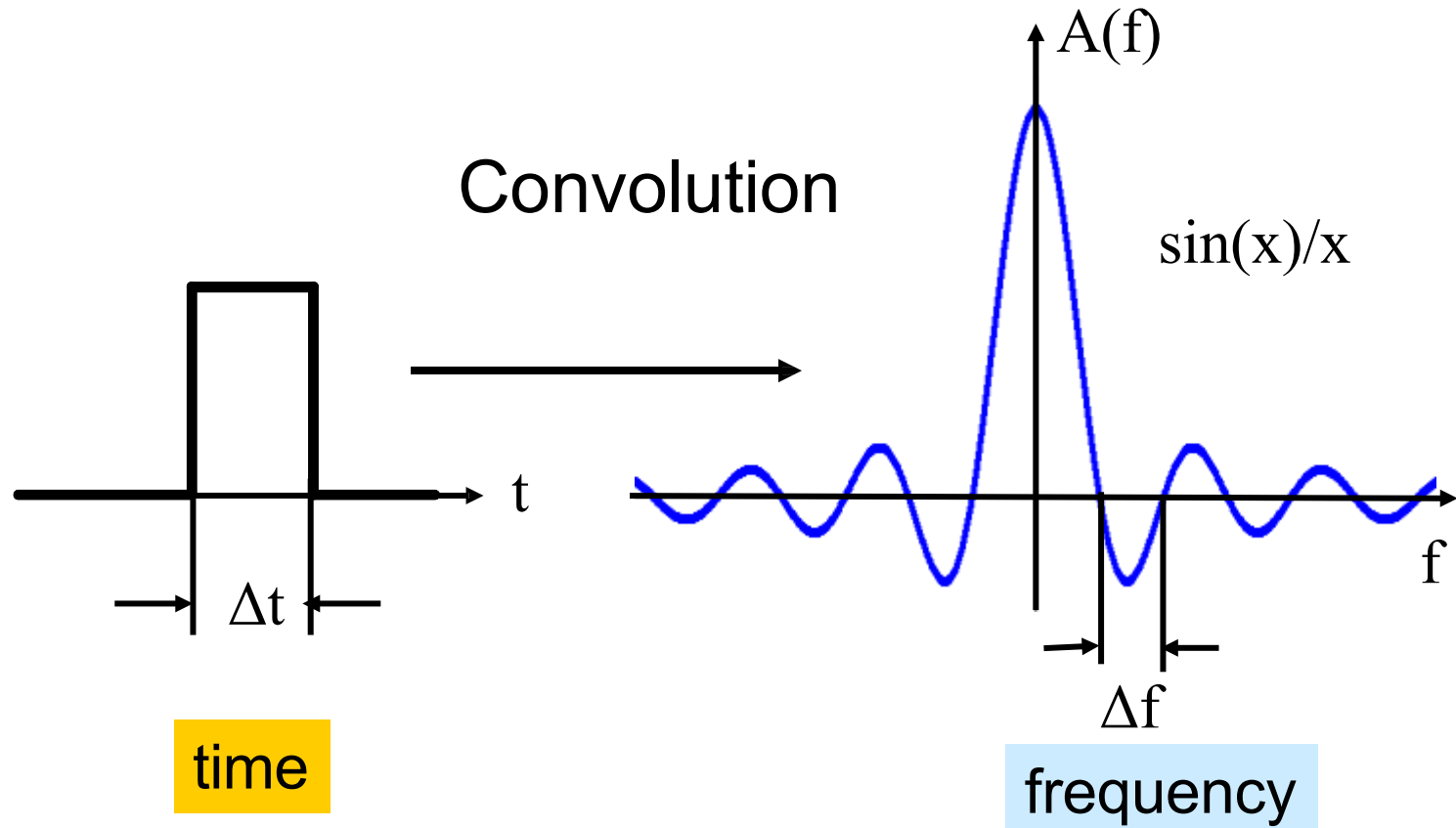
→ **Inter Carrier Interference (ICI).**

Solution

“Special” transmit $g_s(t)$ and receive filter $g_r(t)$ and frequencies f_k allows orthogonal subcarrier

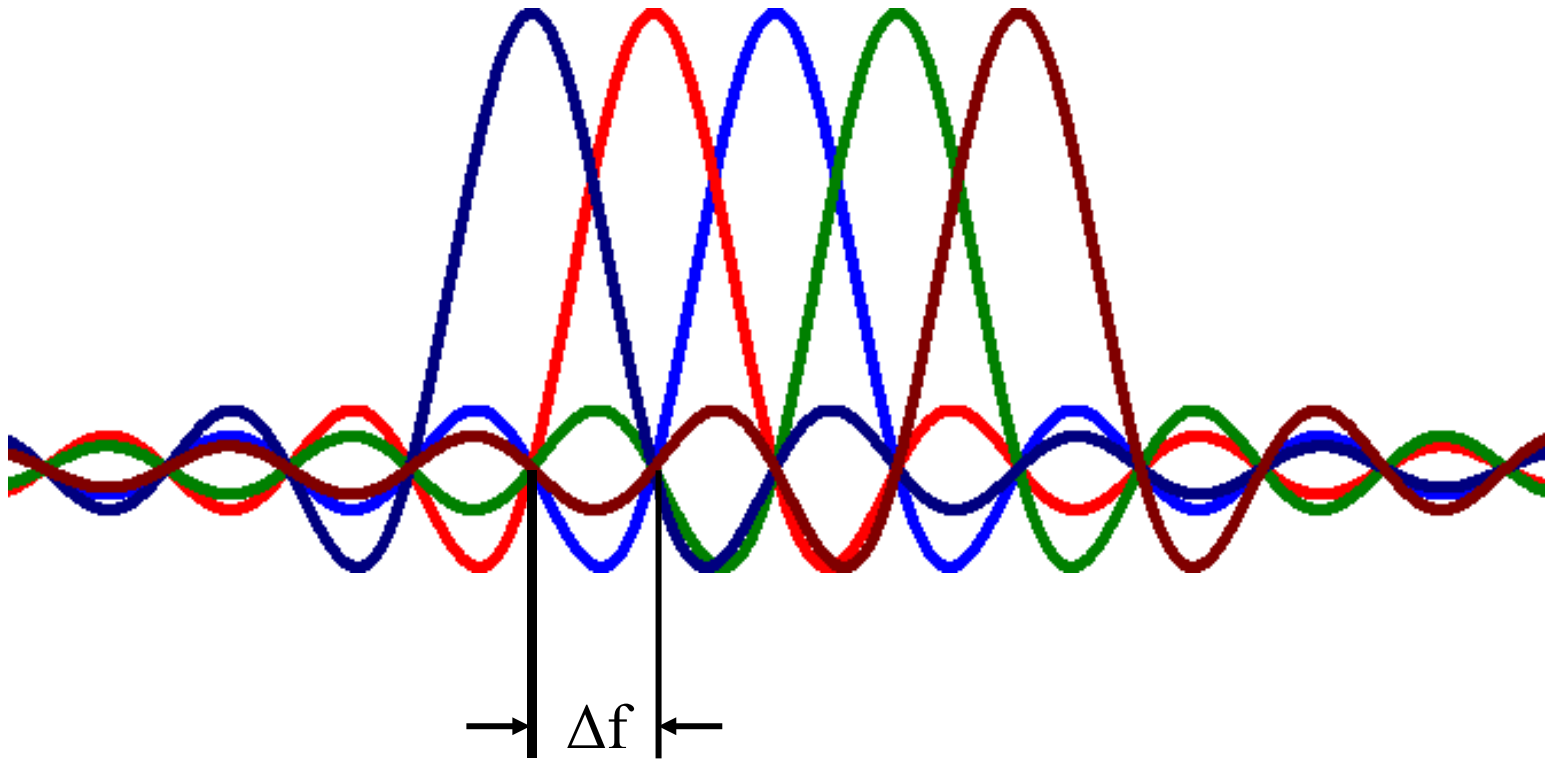
→ **Orthogonal Frequency Division Multiplex (OFDM)**

Rectangular Pulse

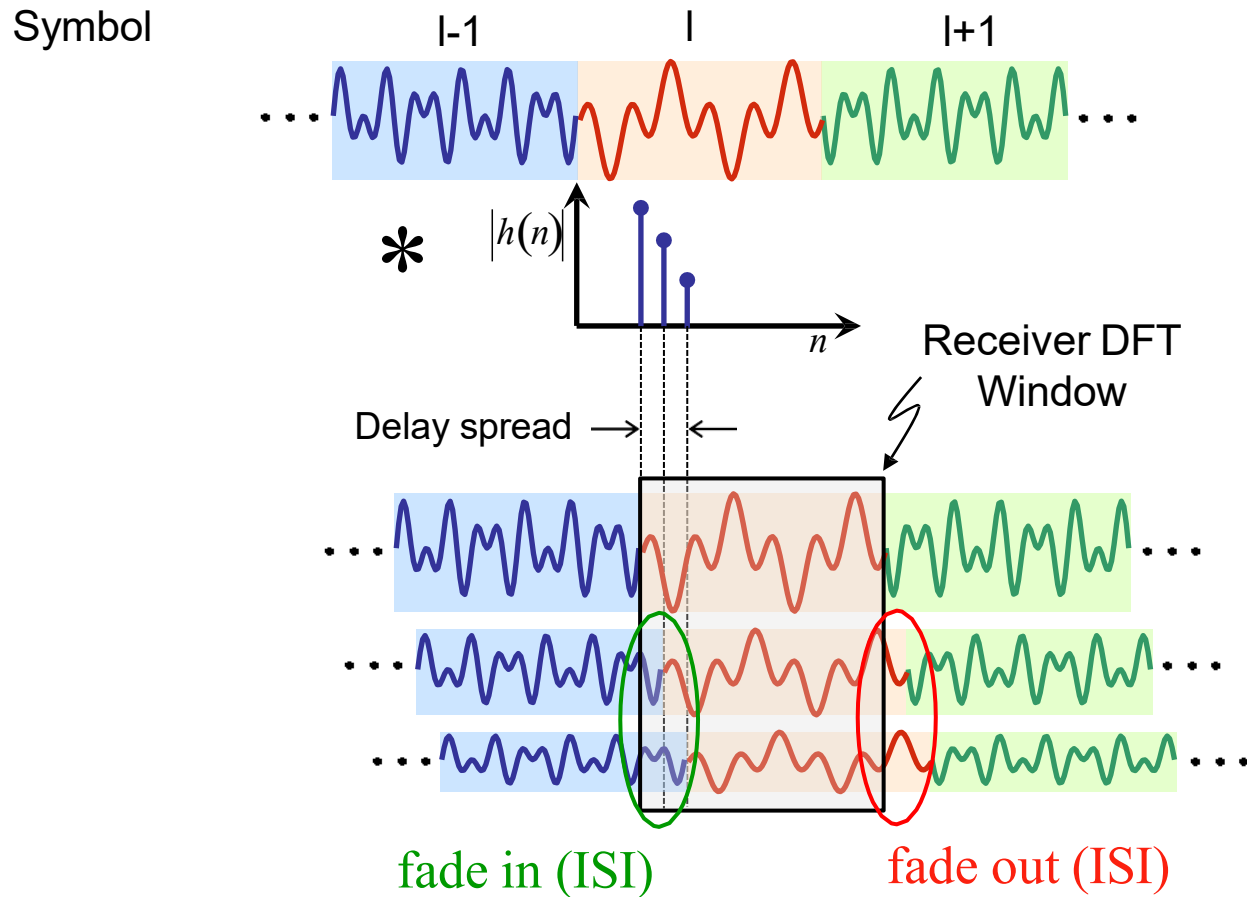


Orthogonality

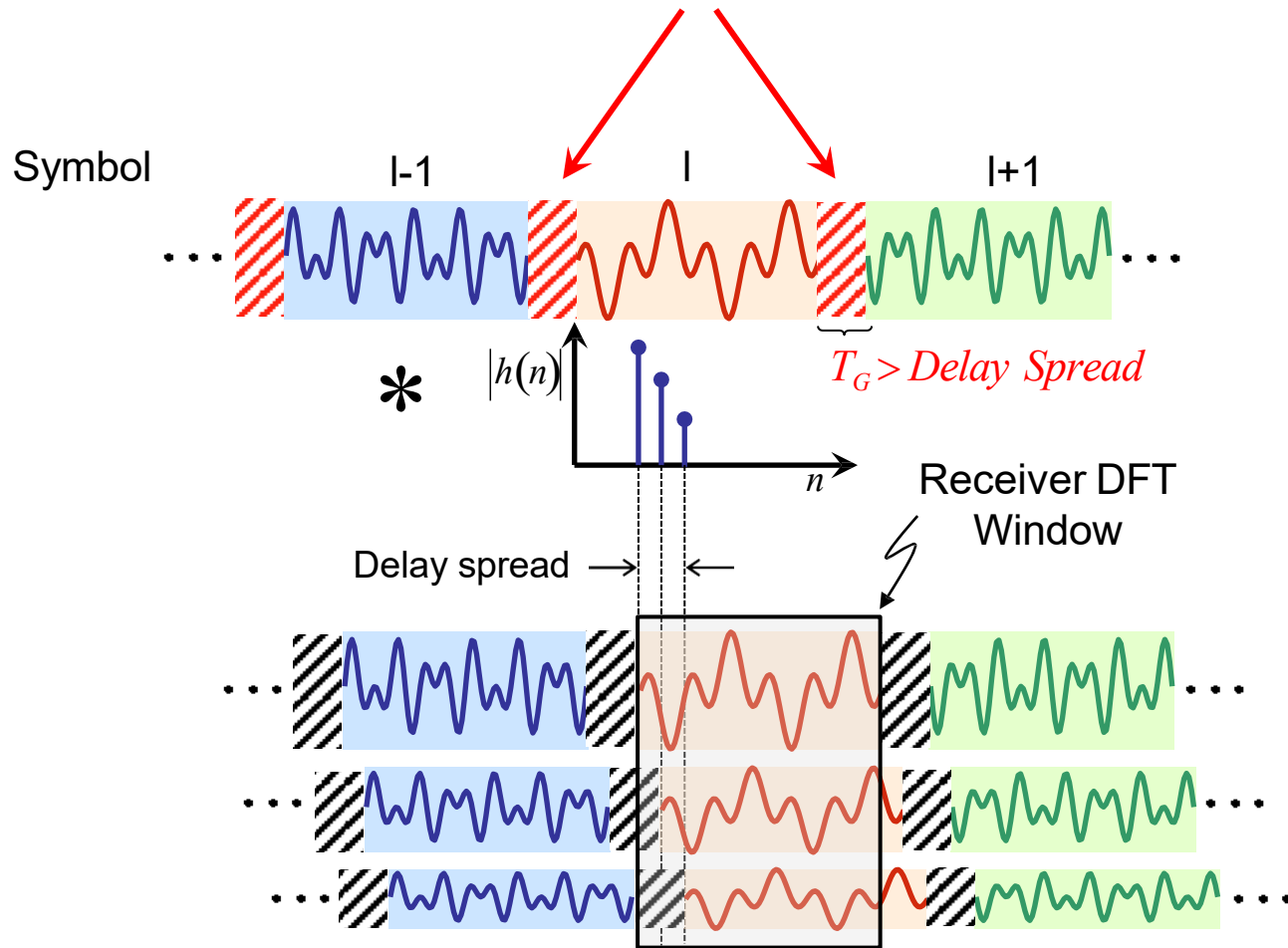
Orthogonality condition: $\Delta f = 1/\Delta t$



ISI and ICI due to channel

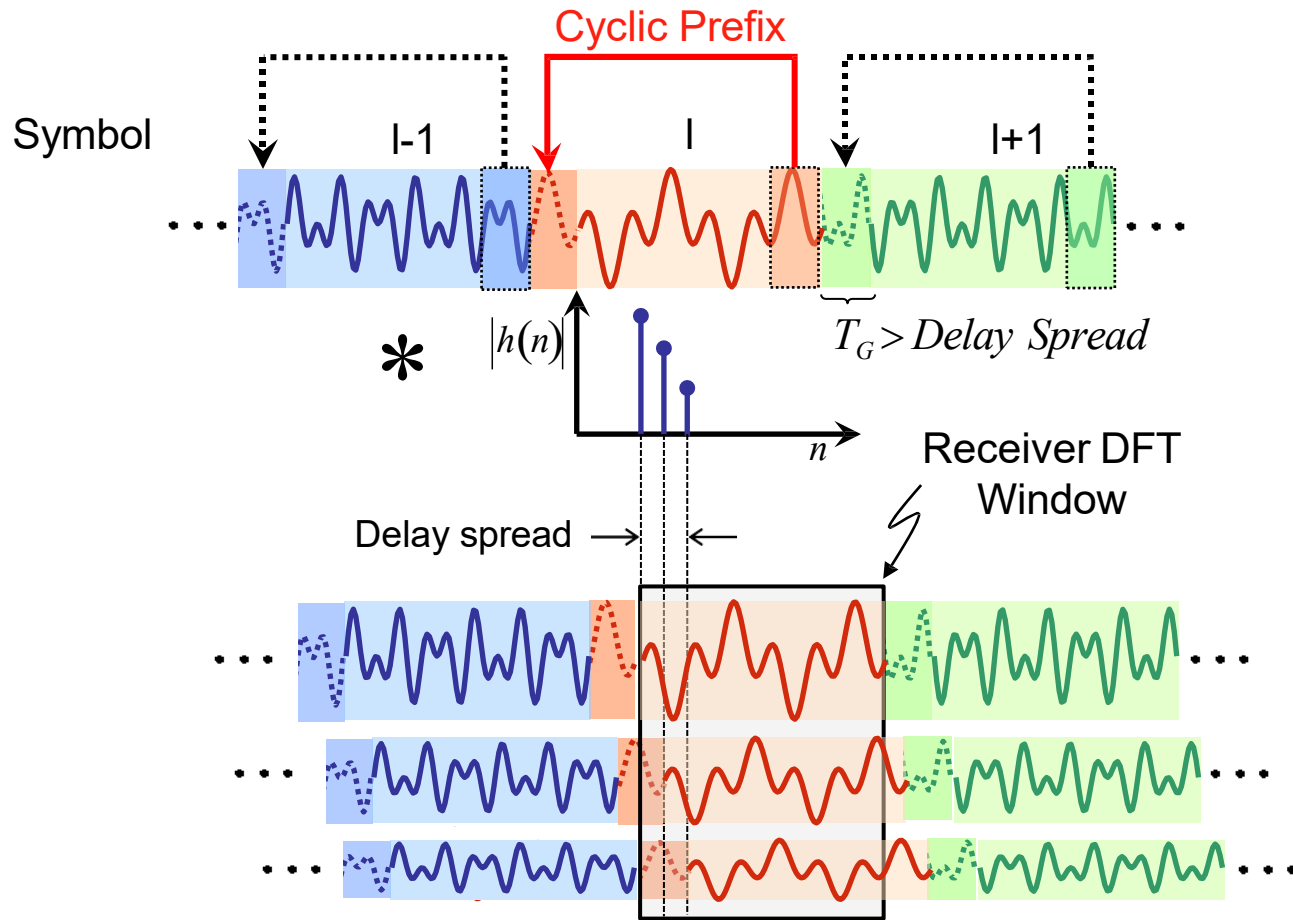


ISI and ICI: Guard Interval



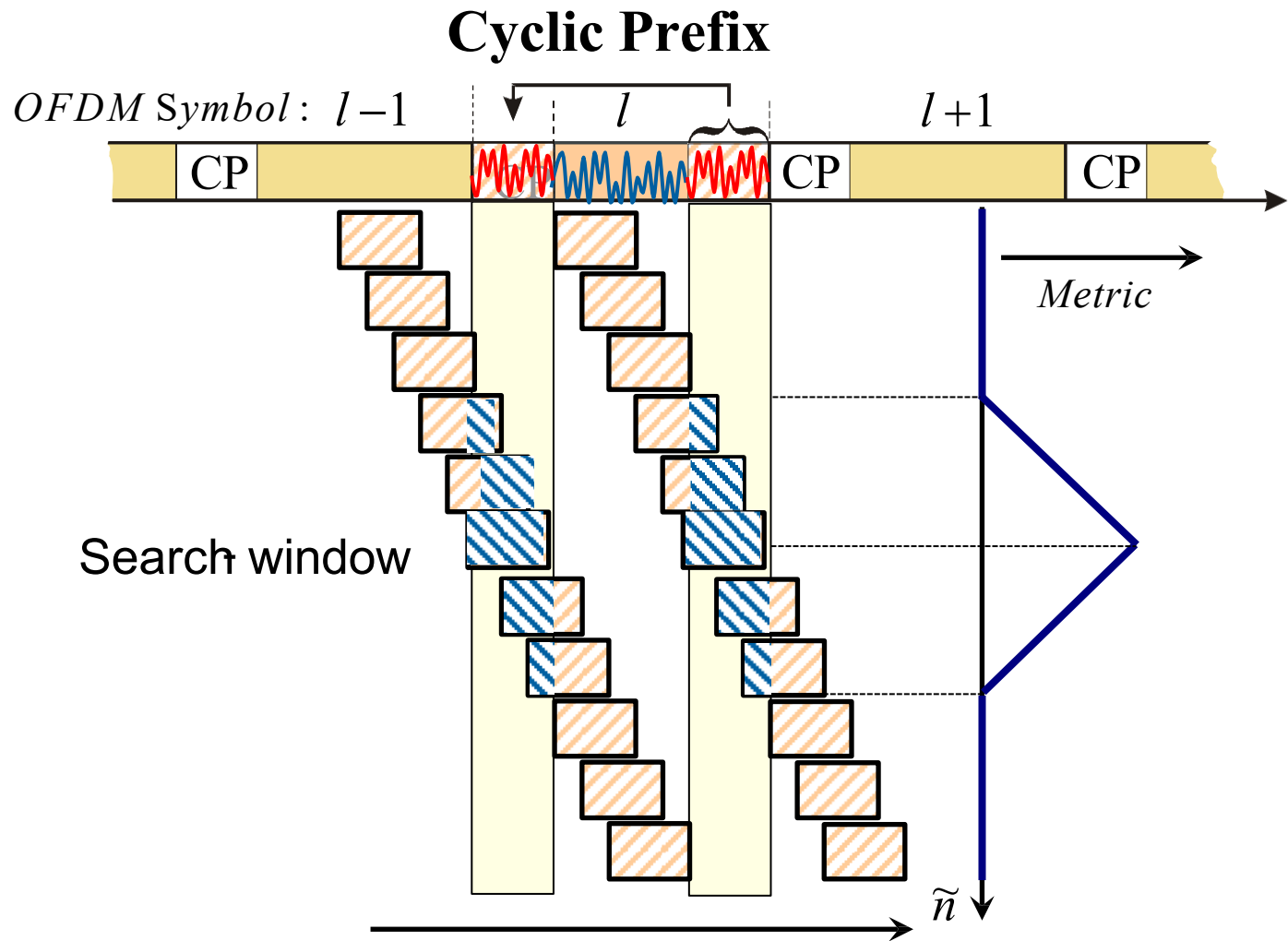
Guard Interval guarantees the suppression of ISI!

Guard Interval as Cyclic Prefix



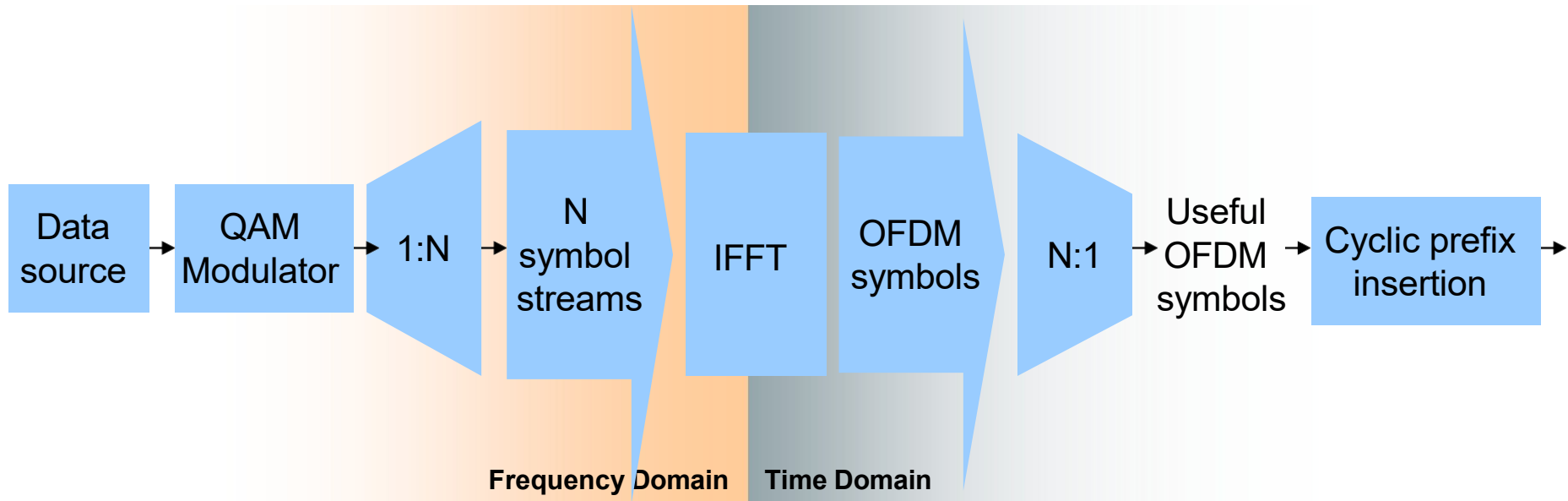
Cyclic Prefix guarantees the suppression of ISI and ICI!

Synchronisation



DL CP-OFDM signal generation chain

- OFDM signal generation is based on Inverse Fast Fourier Transform (IFFT) operation on transmitter side:



- On receiver side, an FFT operation will be used.

OFDM: Pros and Cons

Pros:

- 😊 scalable data rate
- 😊 efficient use of the available bandwidth
- 😊 robust against fading
- 😊 1-tap equalization in frequency domain

Cons:

- 😞 high crest factor or PAPR. Peak to average power ratio
- 😞 very sensitive to phase noise, frequency- and clock-offset
- 😞 guard intervals necessary (ISI, ICI) → reduced data rate

MIMO =

Multiple Input Multiple Output Antennas

MIMO is defined by the number of Rx / Tx Antennas and not by the Mode which is supported

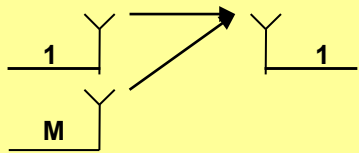
Mode



SISO

Single Input Single Output

Typical today's wireless Communication System

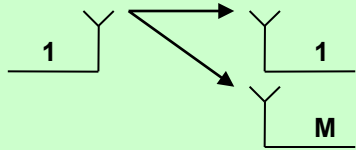


MISO

Multiple Input Single Output

Transmit Diversity

- Maximum Ratio Combining (MRC)
- Matrix A also known as STC
- Space Time / Frequency Coding (STC / SFC)

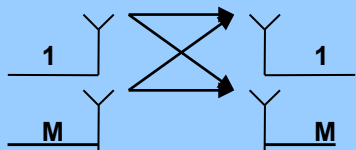


SIMO

Single Input Multiple Output

Receive Diversity

- Maximum Ratio Combining (MRC)



MIMO

Multiple Input Multiple Output

Definition is seen from Channel
Multiple In = Multiple Transmit Antennas

Receive / Transmit Diversity

Spatial Multiplexing (SM) also known as:

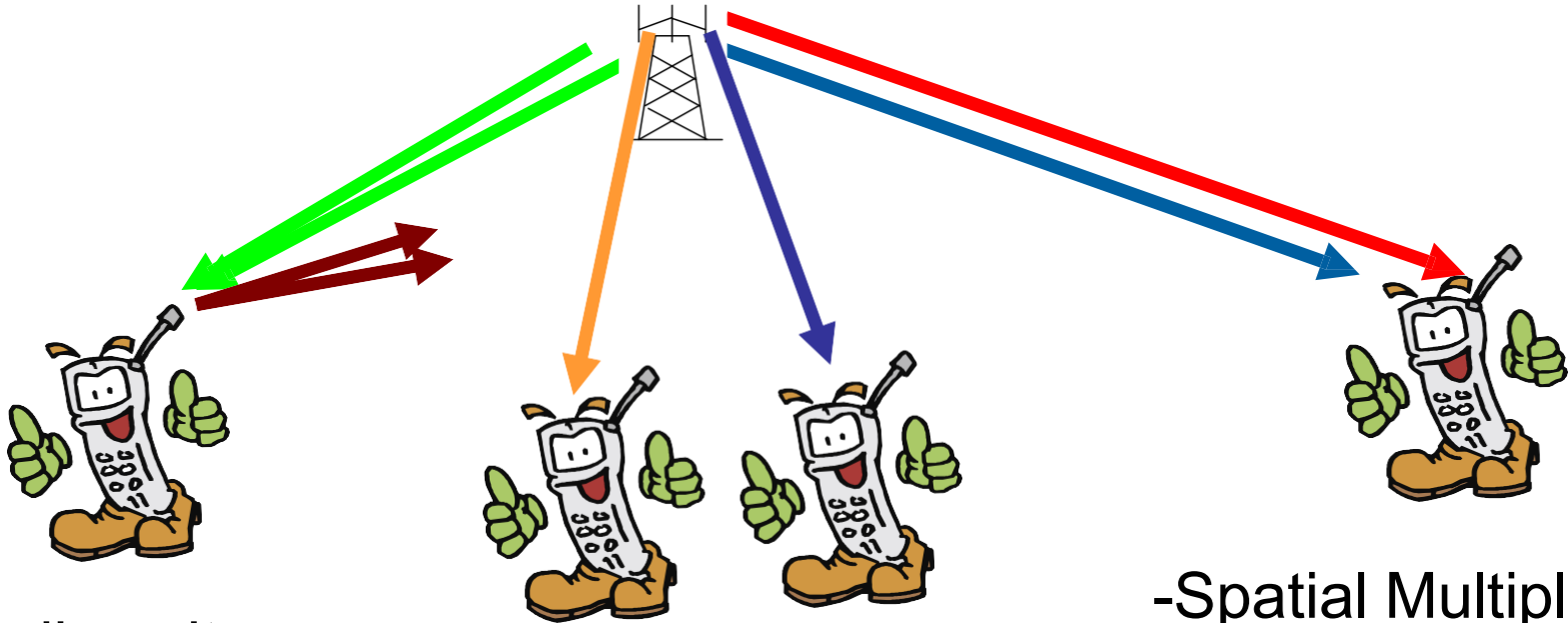
- Space Division Multiplex (SDM)
- True MIMO
- Single User MIMO (SU-MIMO)
- Matrix B

Space Division Multiple Access (SDMA) also known as:

- Multi User MIMO (MU MIMO)
- Virtual MIMO
- Collaborative MIMO

Beamforming

MIMO modes in LTE



- Tx diversity
- Beamforming
- Rx diversity

Better S/N

-Multi-User MIMO

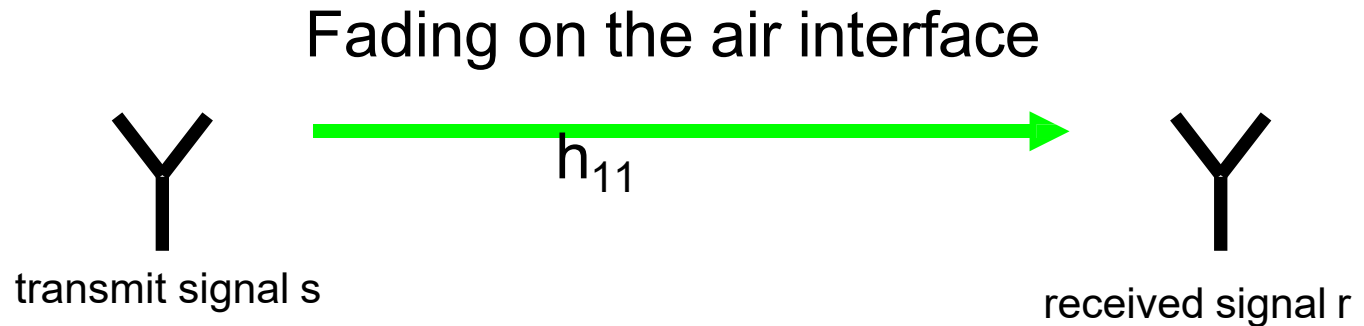
Increased
Throughput at
Node B

-Spatial Multiplexing

Increased
Throughput per
UE

Diversity – some thoughts

The SISO channel:



$$r = h_{11} * s + n = |h_{11}| e^{j\varphi} s + n$$

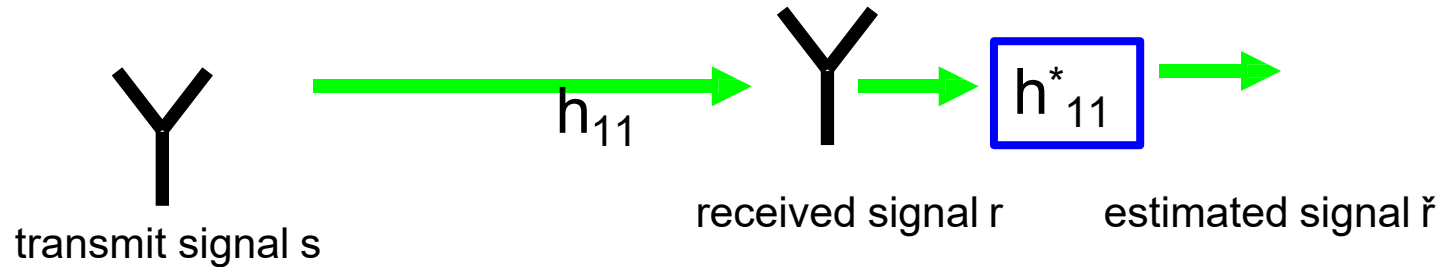
Amplitude
scaling

phase
rotation

The transmit signal is modified in amplitude and phase plus additional noise

Diversity – some thoughts: matched filter

The SISO channel and matched filter (maximum ratio combining):



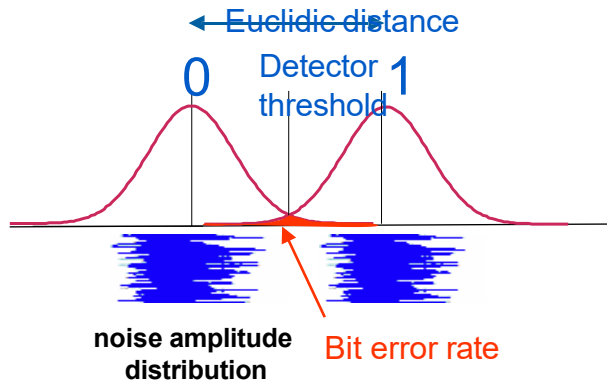
$$\tilde{r} = h_{11}^* r = h_{11}^* h_{11} s + h_{11}^* n = |h_{11}| e^{-j\varphi} |h_{11}| e^{j\varphi} s + h_{11}^* n = |h_{11}|^2 s + h_{11}^* n$$

Idea: matched filter multiplies received signal with conjugate of channel $\rightarrow$ maximizes SNR

Transmitted signal s is estimated as:

$$\hat{s} = \frac{\tilde{r}}{|h_{11}|^2}$$

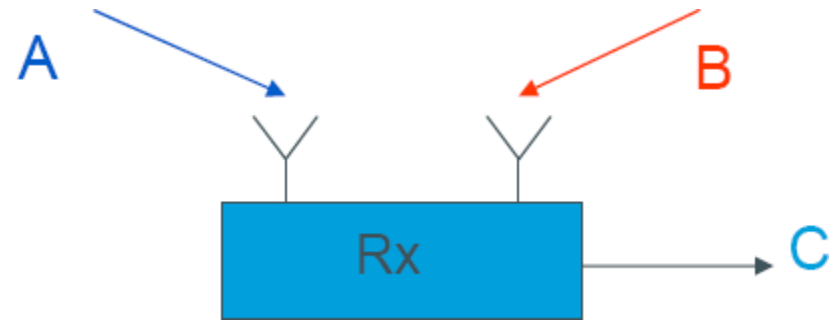
Diversity – some thoughts: performance of SISO



Decay with SNR, only one channel available -
> fading will deteriorate

Modulation scheme	Bit error probability	Data rate = bits per symbol
BPSK	$1/(4 \cdot \text{SNR})$	1
QPSK	$1/(2 \cdot \text{SNR})$	2
16 QAM	$5/(2 \cdot \text{SNR})$	4

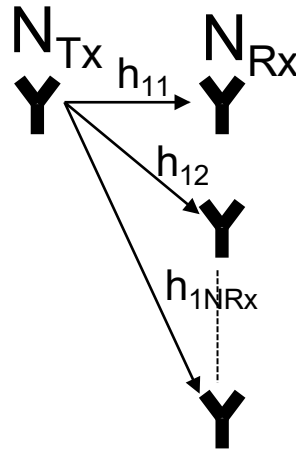
RX Diversity



Maximum Ratio Combining depends on different fading of the two received signals. In other words decorrelated fading channels

Receive diversity gain – SIMO: $1 \cdot N_{Rx}$

	time
Transmit antenna	s



Receive antenna 1

$$r_1 = h_{11}s + n_1$$

Receive antenna 2

$$r_2 = h_{12}s + n_2$$

Receive antenna N_{Rx}

$$r_{nrx} = h_{1nrx}s + n_{nrx}$$

$$\begin{aligned} \tilde{r} &= h_{11}^*s_1 + h_{12}^*s_2 + \dots + h_{1n_{Rx}}^*s_{n_{Rx}} \\ &= \left(|h_{11}|^2 + |h_{12}|^2 + \dots + |h_{1n_{Rx}}|^2 \right) s + h_{11}^*n_1 + h_{12}^*n_2 + \dots + h_{1n_{Rx}}^*n_{n_{Rx}} \end{aligned}$$

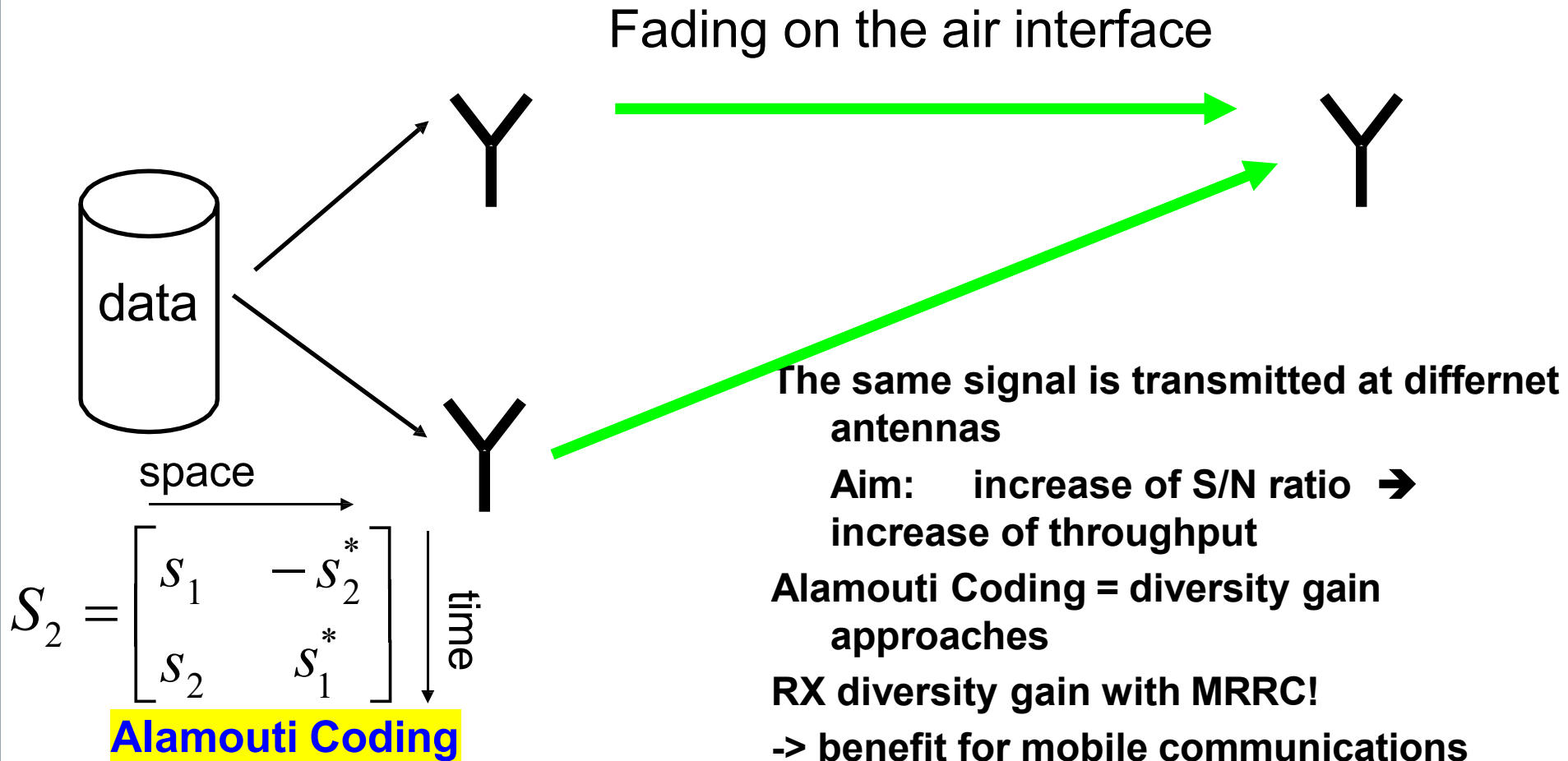
signal after maximum ratio combining

Probability for errors

$$P_e \sim \frac{1}{SNR^{n_{Rx}}}$$

Said:
diversity
order n_{Rx}

TX Diversity: Space Time Coding



Space Time Block Coding according to Alamouti

(when no channel information at all available at transmitter)

From slide before: $d^e_1 = h_1^* \cdot r_1 + h_2 \cdot r_2^* = (|h_1|^2 + |h_2|^2) \cdot d_1 + n'_1$
 $d^e_2 = h_2^* \cdot r_1 + h_1 \cdot r_2^* = (|h_1|^2 + |h_2|^2) \cdot d_2 + n'_2$

Probability for errors (Alamouti)

$$P_e \sim \frac{1}{SNR^2}$$

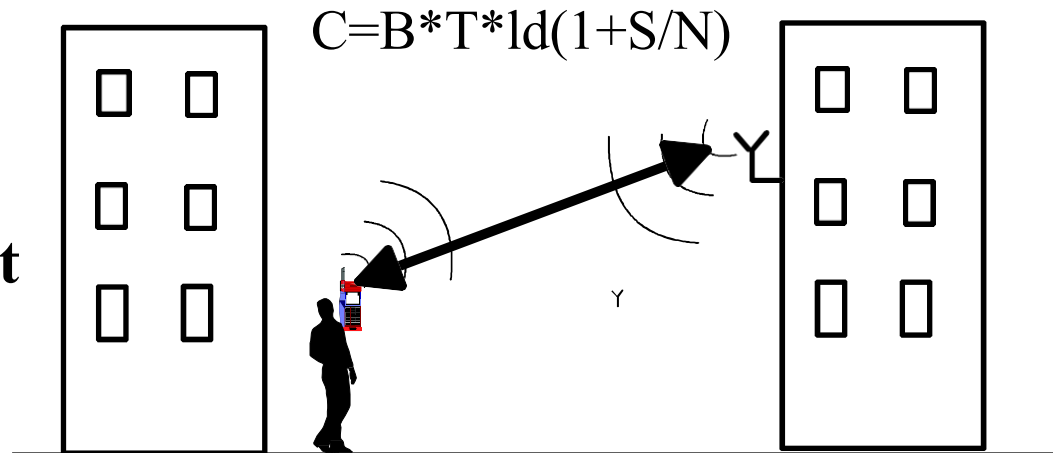
Compare
with Rx diversity

$$P_e \sim \frac{1}{SNR^{n_{Rx}}}$$

Alamouti coding is full diversity gain
and full rate, but it only works for
2 antennas
(due to Alamouti matrix is orthogonal)

MIMO Spatial Multiplexing

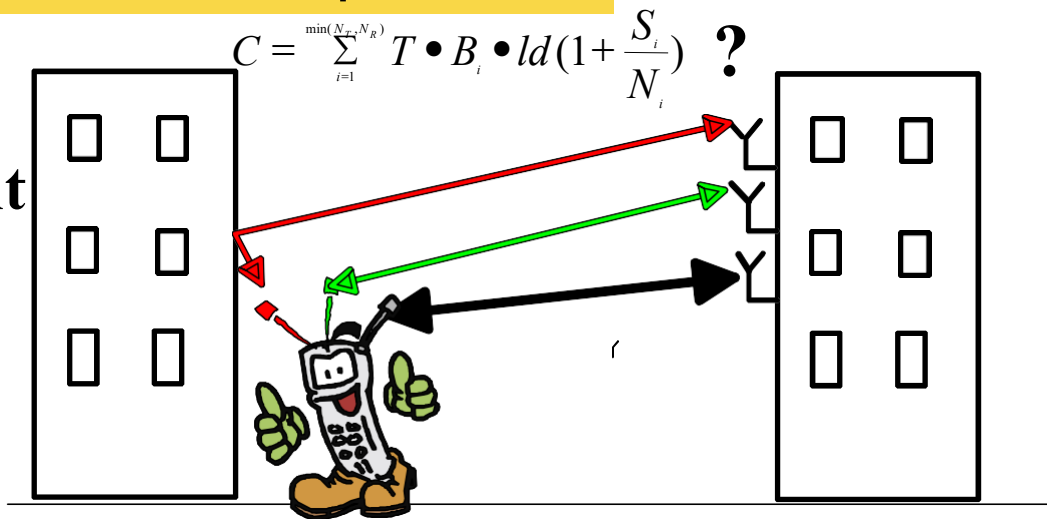
SISO:
Single Input
Single Output



Higher capacity without additional spectrum!

MIMO:
Multiple Input
Multiple Output

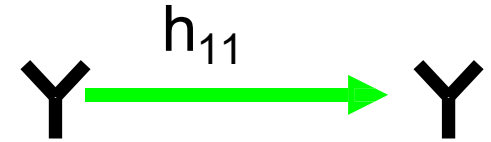
Increasing
capacity per cell



Spatial multiplexing – capacity aspects

Ergodic mean capacity of a SISO channel calculated as:

$$C = E_H \{ \log_2 (1 + \rho |h_{11}|^2) \}$$



$$r = s * h_{11} + n$$

Received signal r with
sent signal s , channel h_{11} and
AWGN with $\sigma=n$

$$\rho = \frac{P_s}{\sigma^2}$$

represents the signal to noise ratio SNR
at the receiver branch

Or simplified:

$$C = B * \log_2 \left(1 + \frac{S}{N} \right)$$

With B = bandwidth
and S/N = signal to
noise ratio

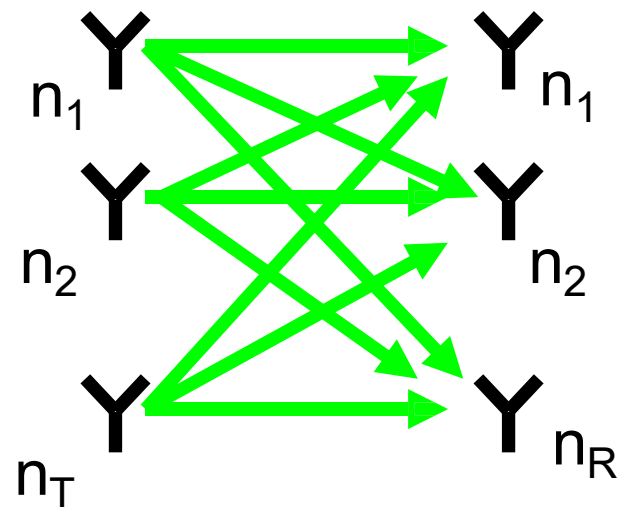
Spatial multiplexing – capacity aspects

Ergodic mean capacity of a MIMO channel is even worse 😞

$$C = E_H \left\{ \log_2 \left[\det \left(I_{n_R} + \frac{\rho}{n_T} H H^H \right) \right] \right\}$$

I_{n_R} is an Identity matrix with size $n_T \times n_R$

H^H is the Hermetian complex

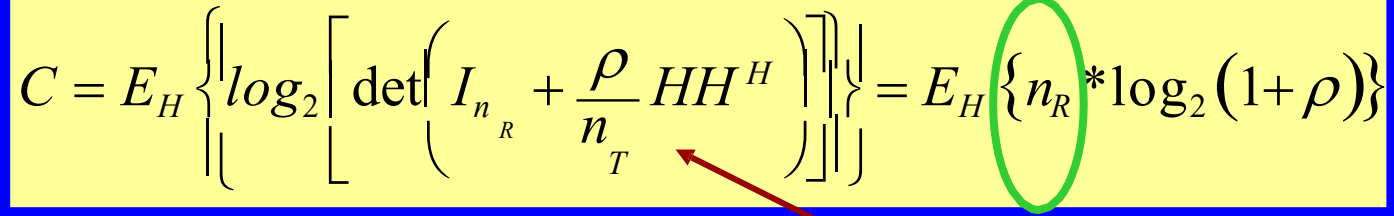


$$r = s * H + n$$

Received signal r with
sent signal s , channel H and
AWGN with $\sigma=n$

Spatial multiplexing – capacity aspects

Some theoretical ideas:

$$C = E_H \left\{ \log_2 \left[\det \left(I_{n_R} + \frac{\rho}{n_T} HH^H \right) \right] \right\} = E_H \{ n_R * \log_2 (1 + \rho) \}$$


⇒ We increase the number of transmit antennas to ∞ , and see:

$$\lim_{n_T \rightarrow \infty} \frac{HH^H}{n_T} = I_{n_R}$$

⇒ So the result is, if the number of Tx antennas is infinity, the capacity depends on the number of Rx antennas:

After this heavy mathematics the result: If we increase the number of Tx and Rx antennas, we can increase the capacity!

The MIMO promise

- I Channel capacity grows linearly with antennas ☺

$$\text{Max Capacity} \sim \min(N_{TX}, N_{RX})$$

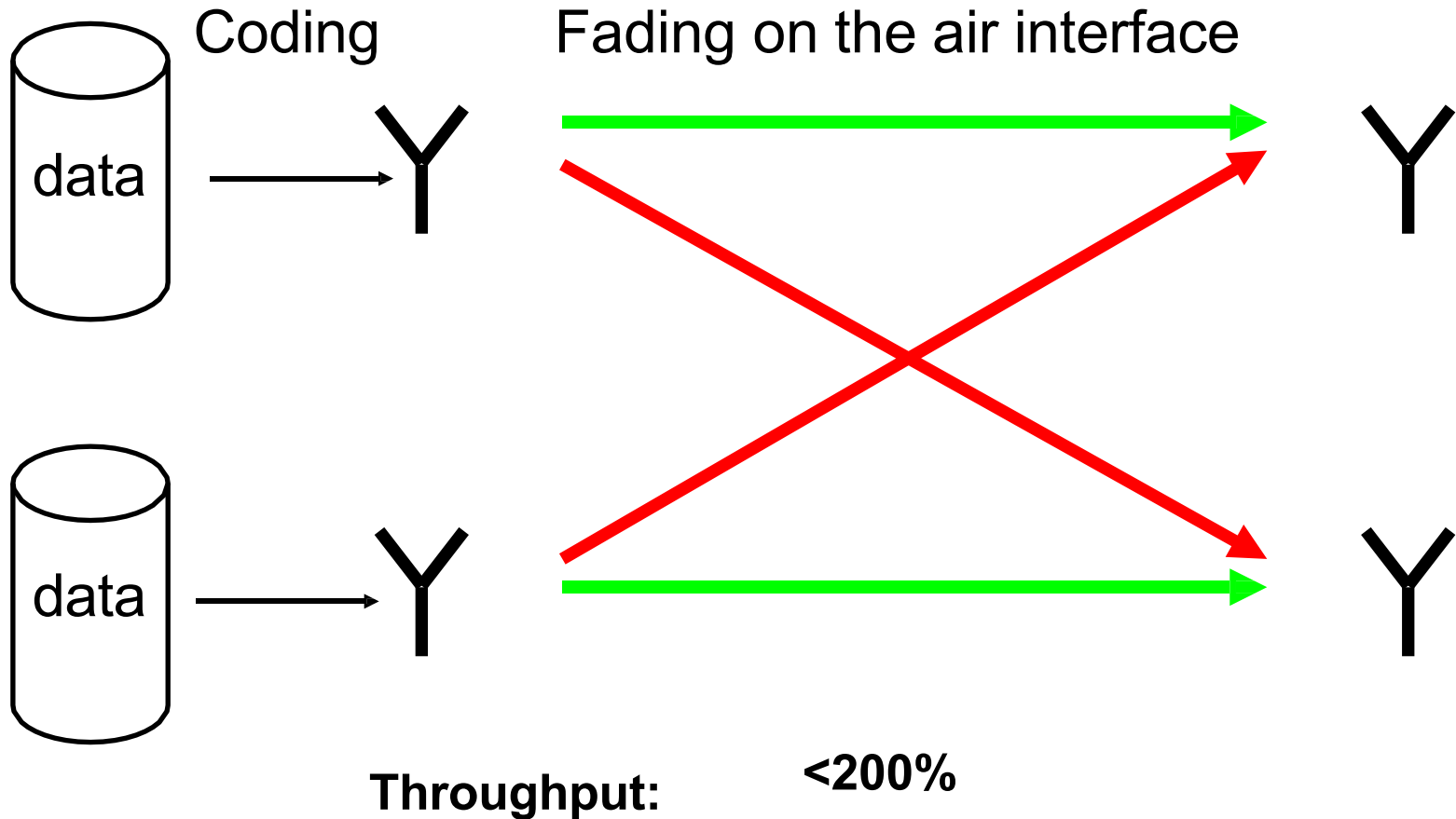
- I **Assumptions** ☹

- I Perfect channel knowledge
- I Spatially uncorrelated fading

- I **Reality** ☹

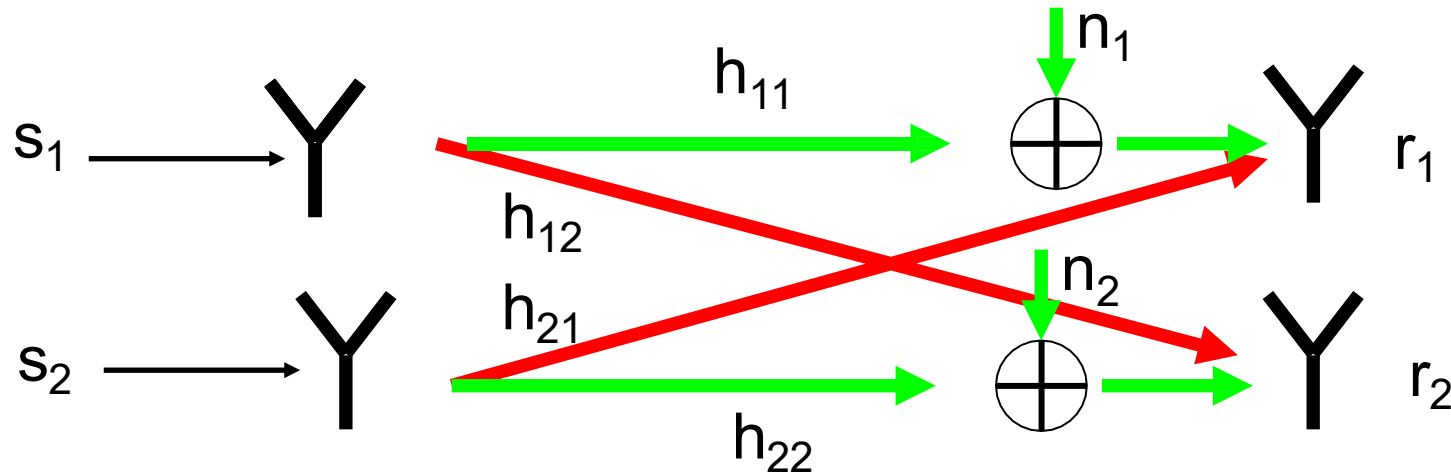
- I Imperfect channel knowledge
- I Correlation $\neq 0$ and rather unknown

Spatial Multiplexing



Spatial Multiplexing: We increase the throughput but we also increase the interference!

MIMO – capacity calculations, e.g. 2x2 MIMO



This results in the equations:

$$r_1 = s_1 * h_{11} + s_2 * h_{21} + n_1$$

$$r_2 = s_2 * h_{22} + s_1 * h_{12} + n_2$$

Or as matrix:

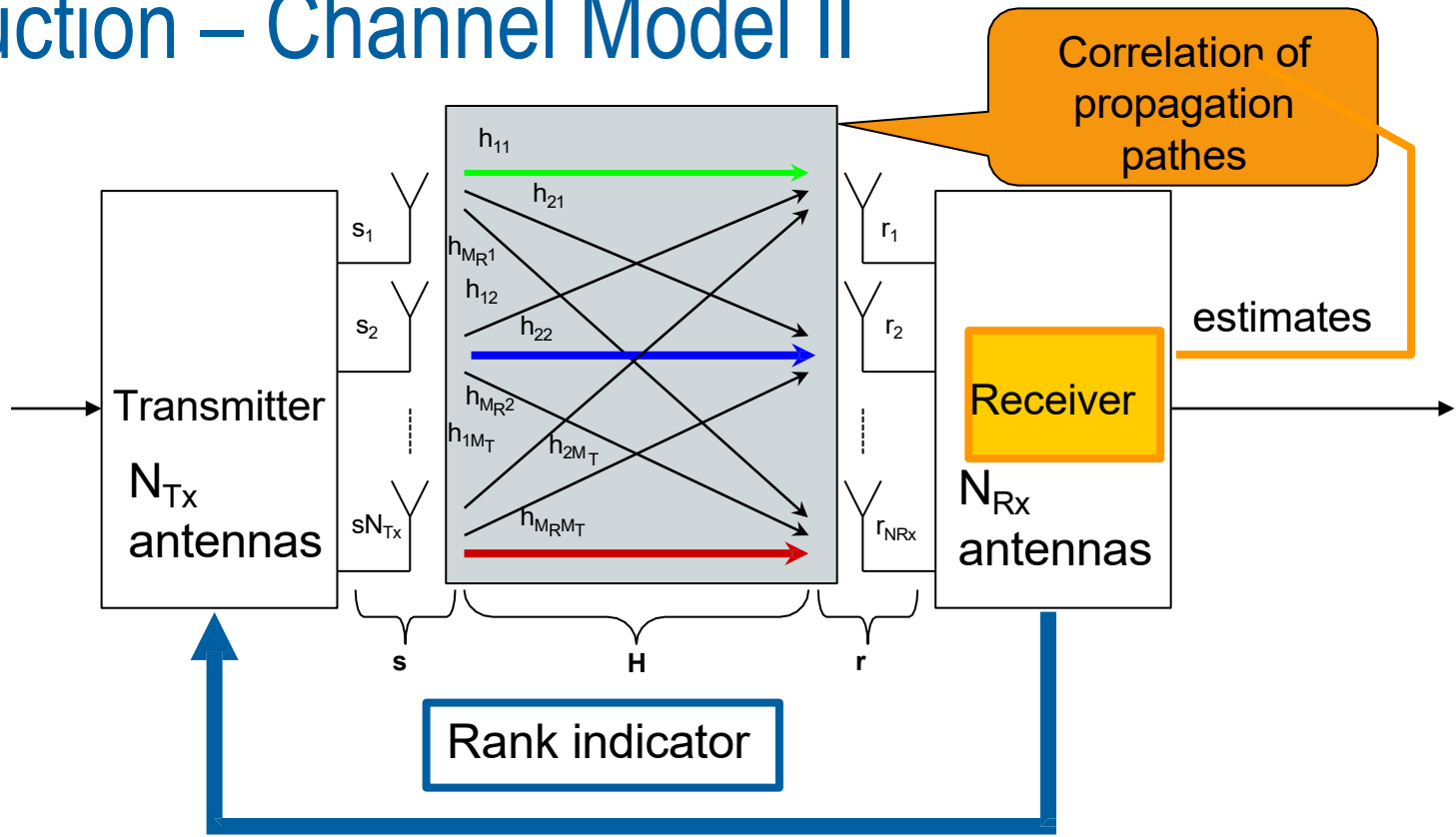
$$\begin{bmatrix} r_1 \\ r_2 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} * \begin{bmatrix} s_1 \\ s_2 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \end{bmatrix}$$

General written as:

$$r = s * H + n$$

To solve this equation, we have to know H

Introduction – Channel Model II



Capacity $\sim \min(N_{Tx}, N_{Rx}) \rightarrow \text{max. possible rank!}$

But effective rank depends on channel, i.e. the correlation situation of $\mathbf{H}$

Spatial Multiplexing prerequisites

Decorrelation is achieved by:

I Decorrelated data content on each spatial stream

difficult

I Large antenna spacing

Channel
condition

I Environment with a lot of scatters near the antenna
(e.g. MS or indoor operation, but not BS)

I Precoding

Technical
assist

I Cyclic Delay Diversity

But, also possible
that decorrelation
is not given

MIMO: channel interference + precoding

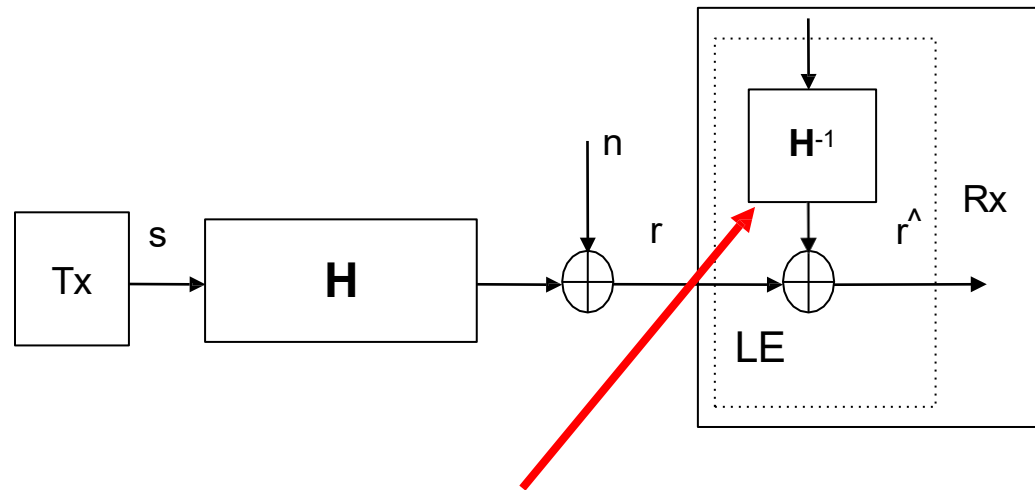
MIMO channel models: different ways to combat against channel impact:

- I.: Receiver cancels impact of channel
- II.: Precoding by using codebook. Transmitter assists receiver in cancellation of channel impact
- III.: Precoding at transmitter side to cancel channel impact

MIMO: Principle of linear equalizing

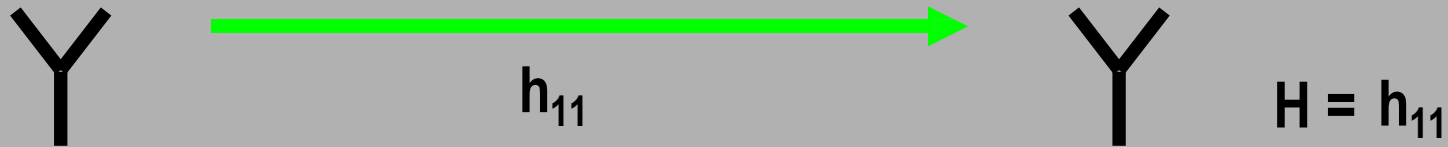
$$R = S^*H + n$$

Transmitter will send reference signals or pilot sequence to enable receiver to estimate H .

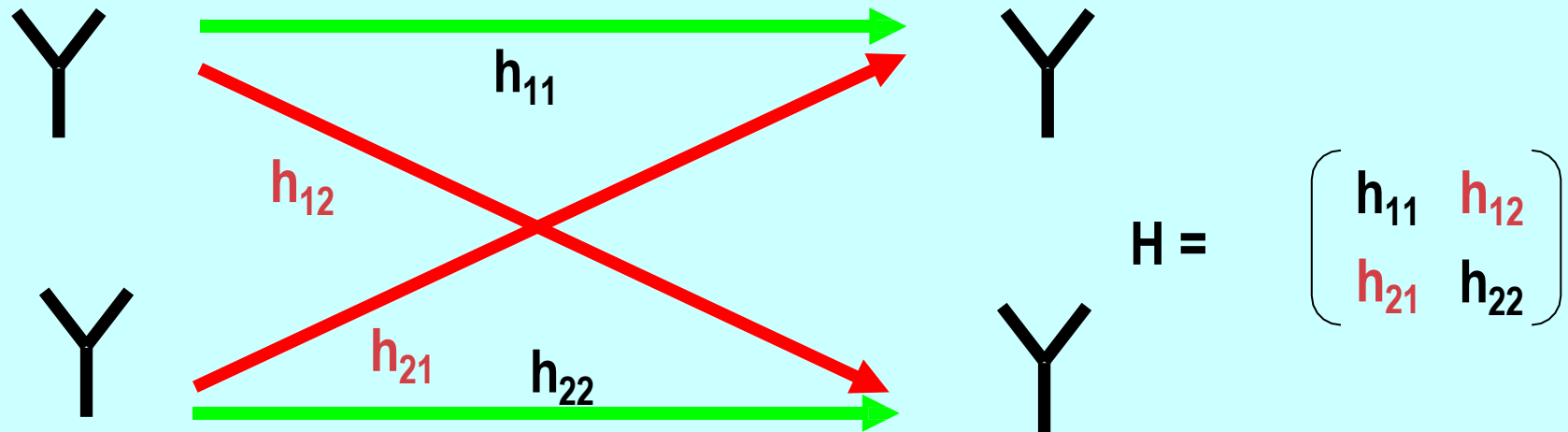


The receiver multiplies the signal r with the Hermetian conjugate complex of the transmitting function to eliminate the channel influence.

Linear equalization – compute power increase

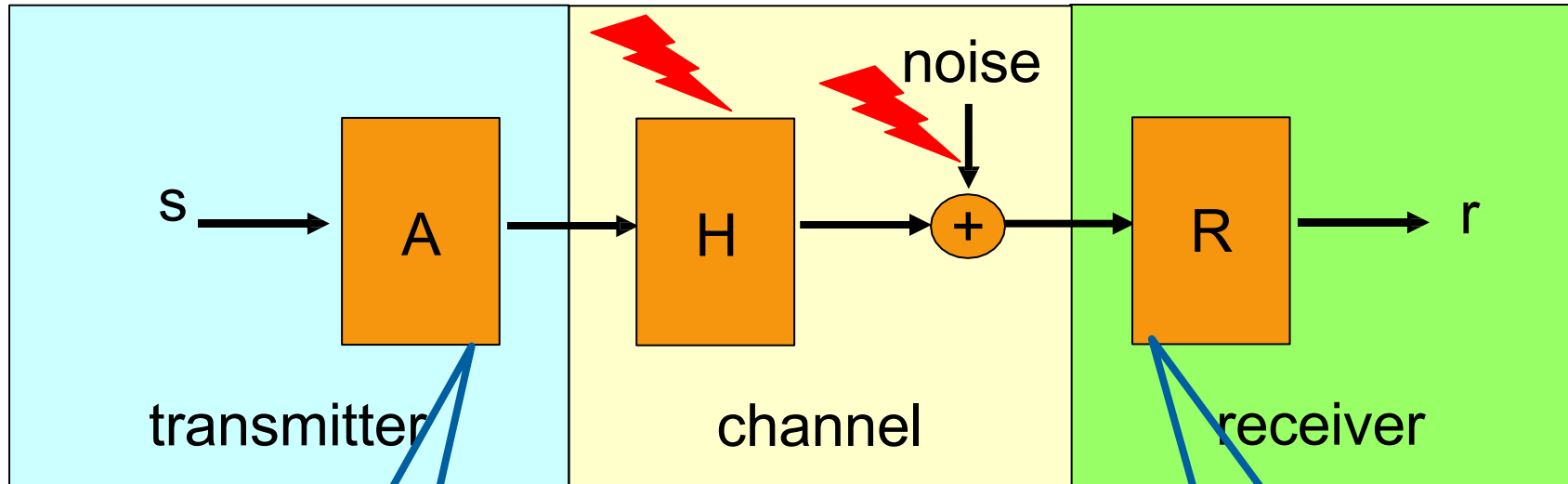


SISO: Equalizer has to estimate 1 channel



2x2 MIMO: Equalizer has to estimate 4 channels

transmission – reception model

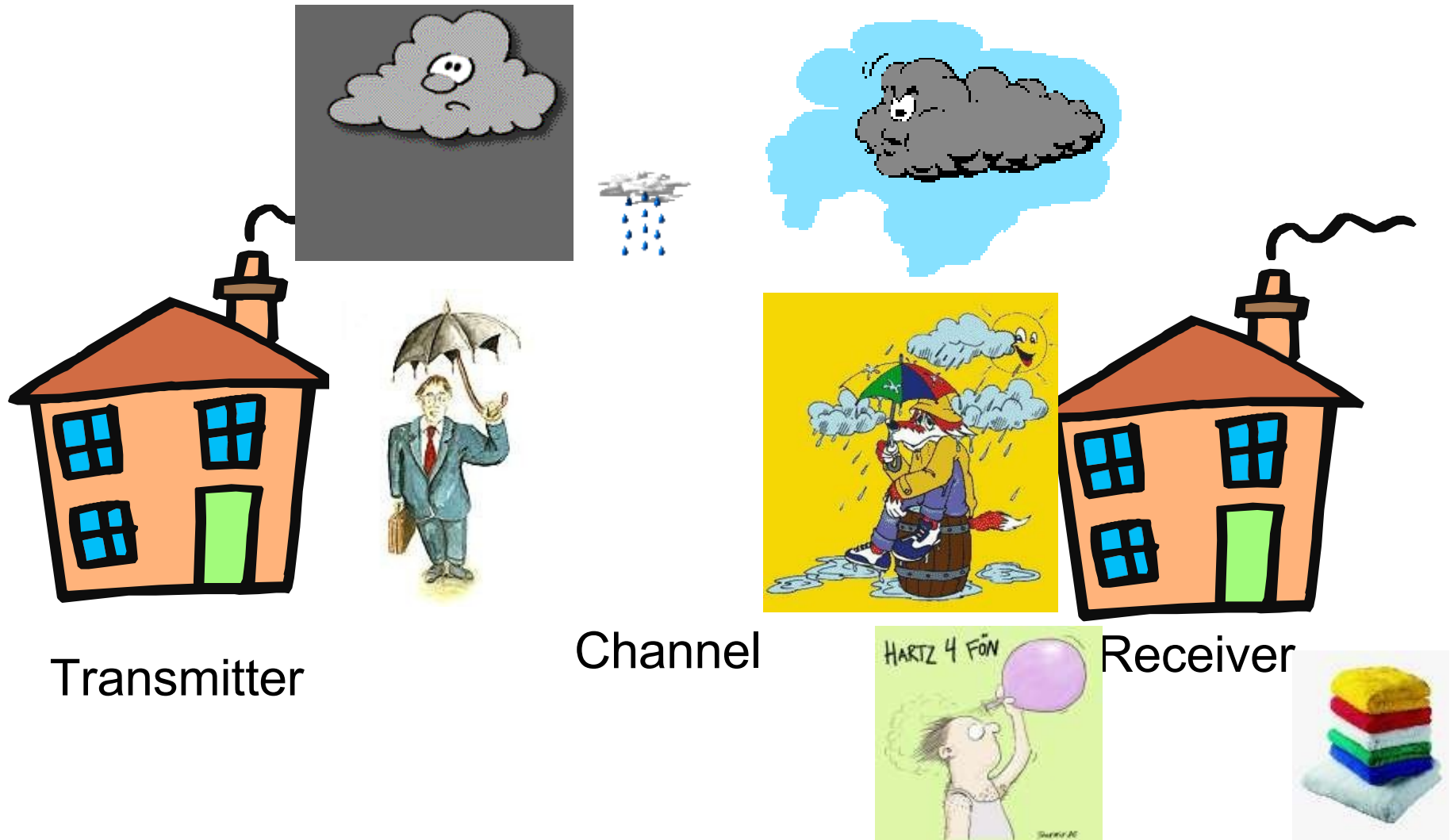


- Modulation,
- Power
- „precoding“,
- etc.

Linear equalization
at receiver is not
very efficient, i.e.
noise can not be cancelled

- detection,
- estimation
- Eliminating channel impact
- etc.

MIMO – work shift to transmitter



MIMO Precoding in LTE (DL)

Spatial multiplexing – Code book for precoding

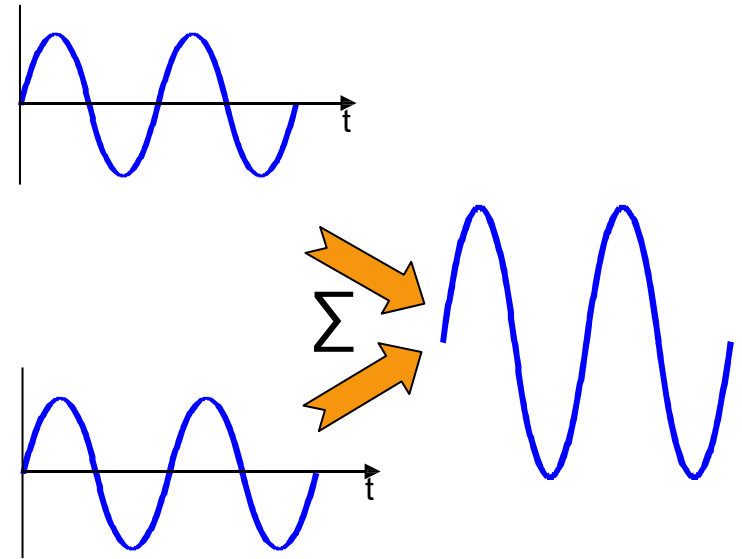
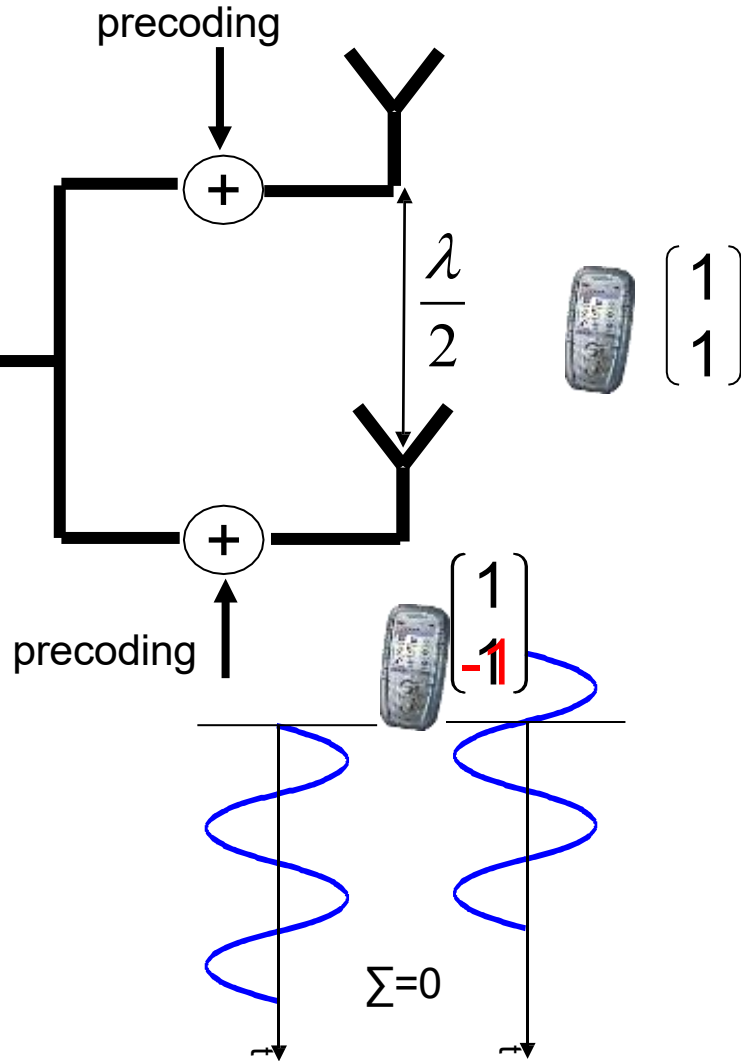
Code book for 2 Tx:

Codebook index	Number of layers ν	
	1	2
0	$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$
1	$\begin{bmatrix} 0 \\ 1 \end{bmatrix}$	$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$
2	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$	$\frac{1}{2} \begin{bmatrix} 1 & 1 \\ j & -j \end{bmatrix}$
3	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$	-
4	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ j \end{bmatrix}$	-
5	$\frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -j \end{bmatrix}$	-

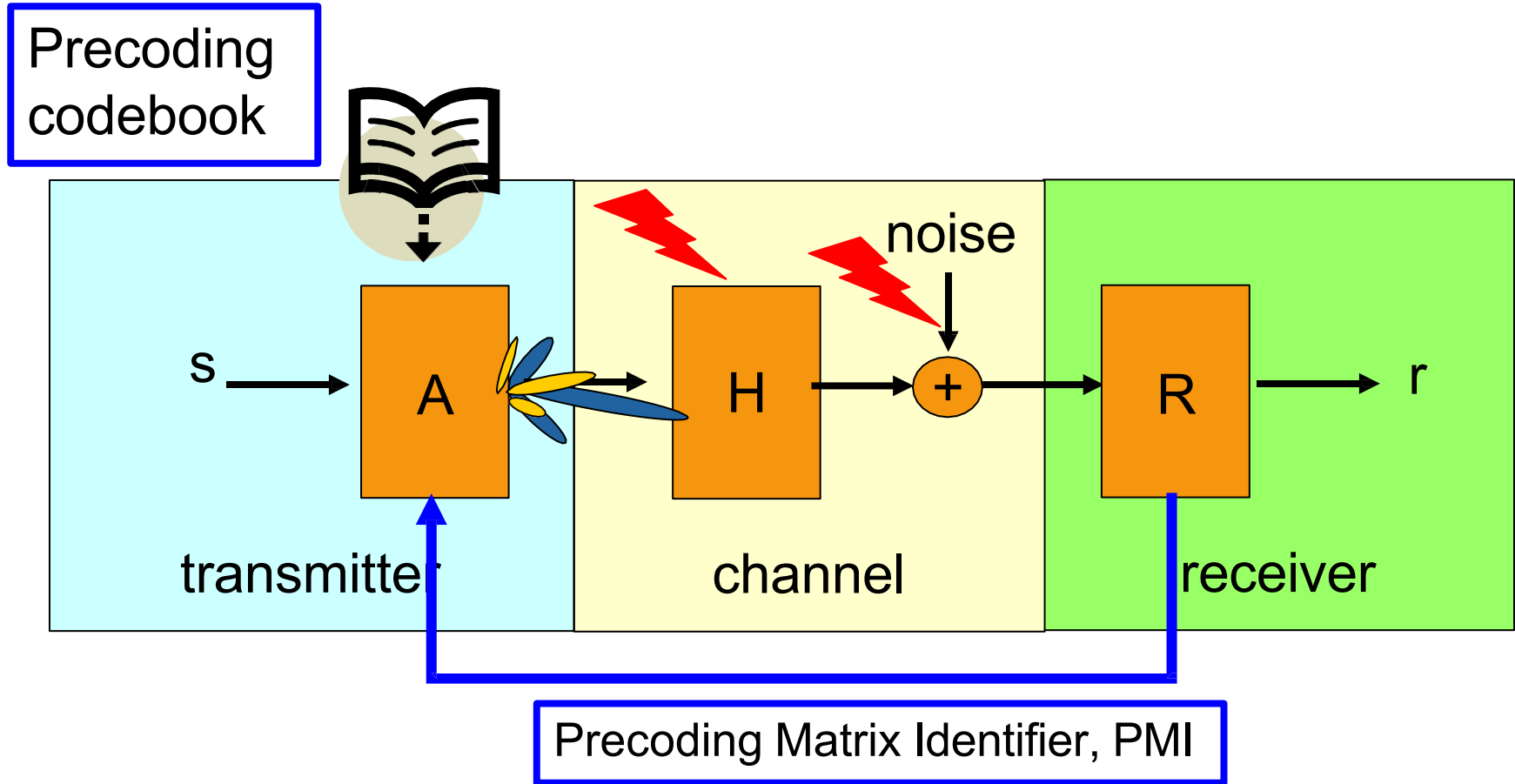
Additional multiplication of the layer symbols with codebook entry

MIMO precoding

$\begin{pmatrix} \text{Ant1} \\ \text{Ant2} \end{pmatrix}$



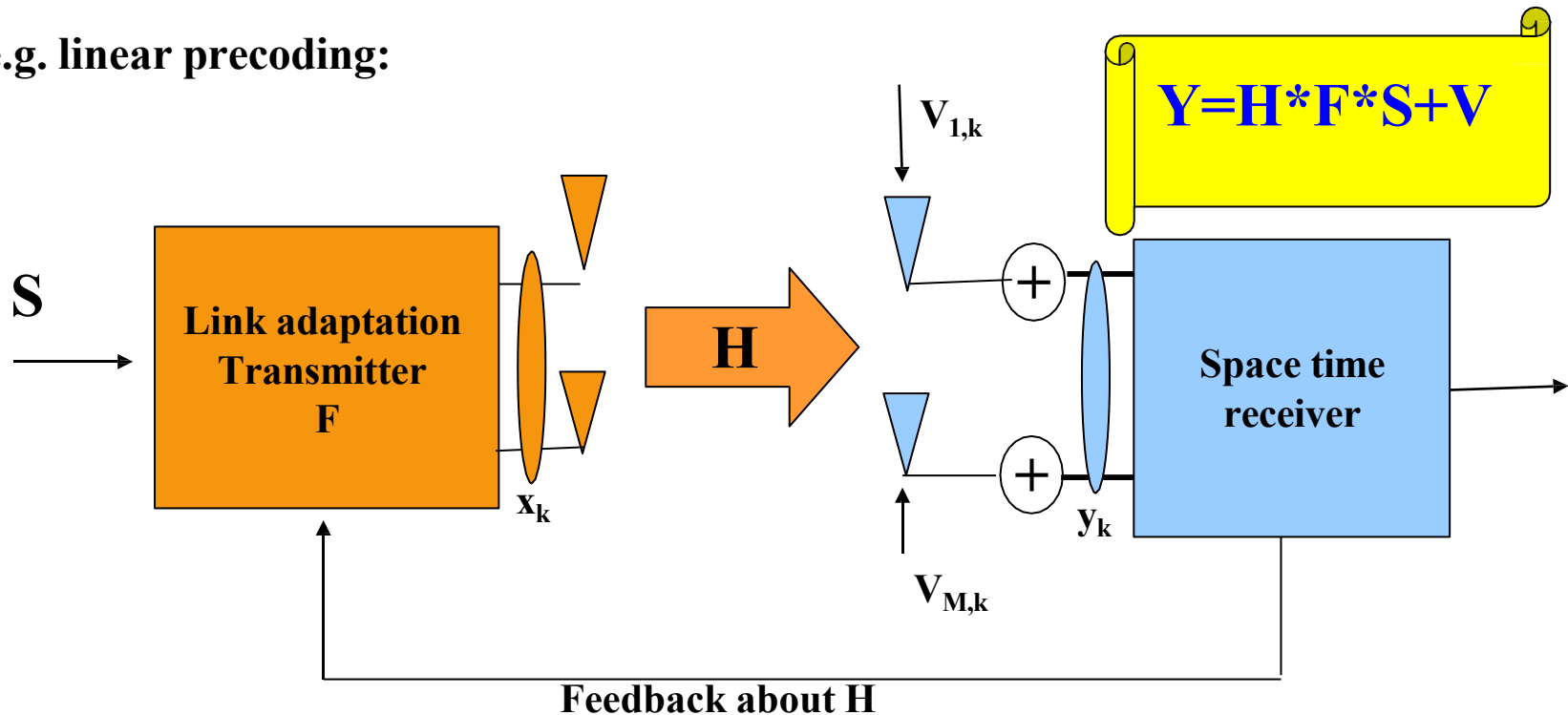
MIMO – codebook based precoding



Codebook based precoding creates some kind of „beamforming light“

MIMO: avoid inter-channel interference – future outlook

e.g. linear precoding:



Idea: F adapts transmitted signal to current channel conditions

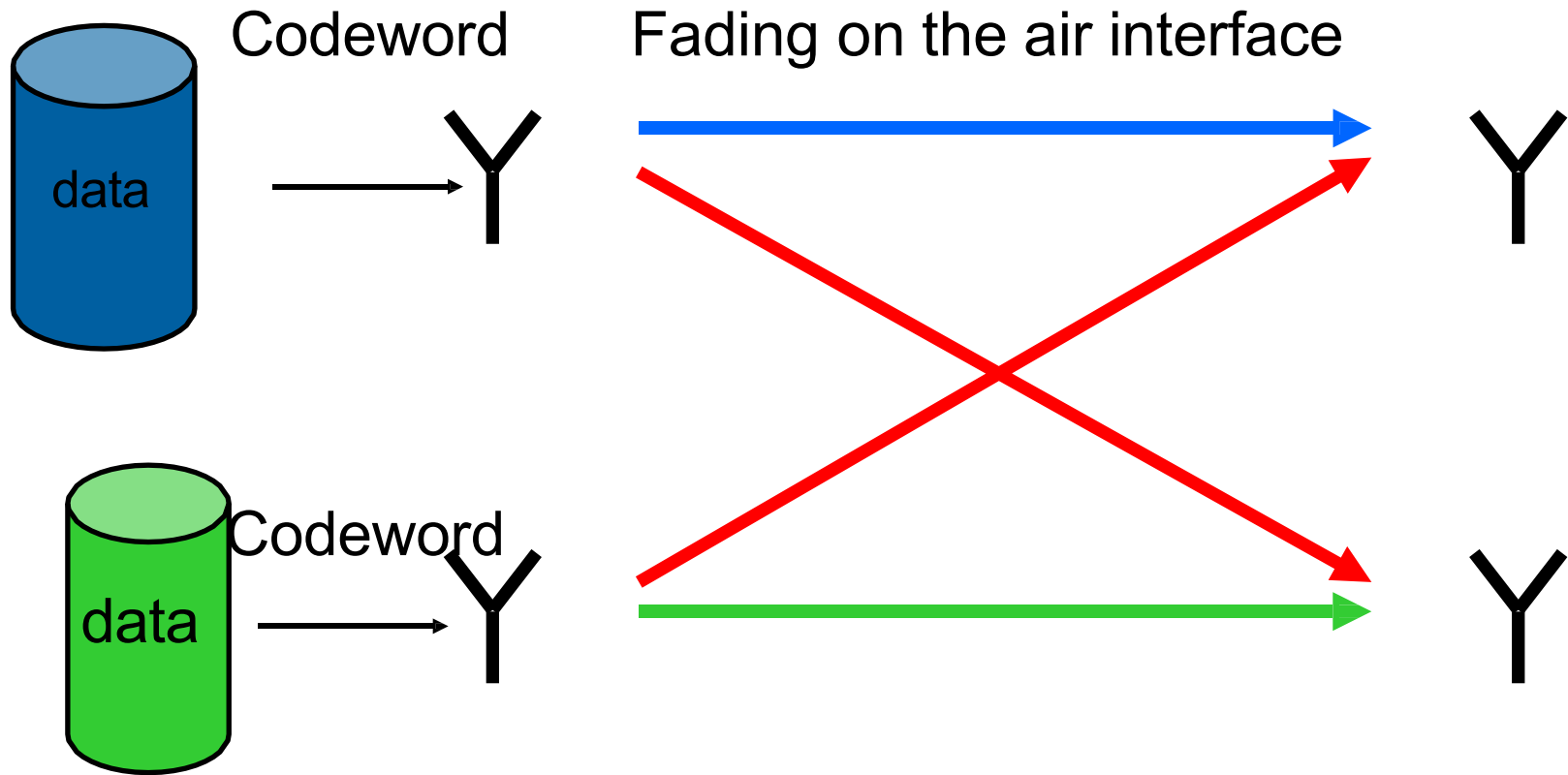
MAS: „Dirty Paper“ Coding – future outlook

I Multiple Antenna Signal Processing: „Known Interference“

- I Is like NO interference
- I Analogy to writing on „dirty paper“ by changing ink color accordingly



Spatial Multiplexing



Spatial Multiplexing: We like to distinguish the 2 useful
Propagation passes: 
How to do that? => one idea is SVD

Idea of Singular Value Decomposition

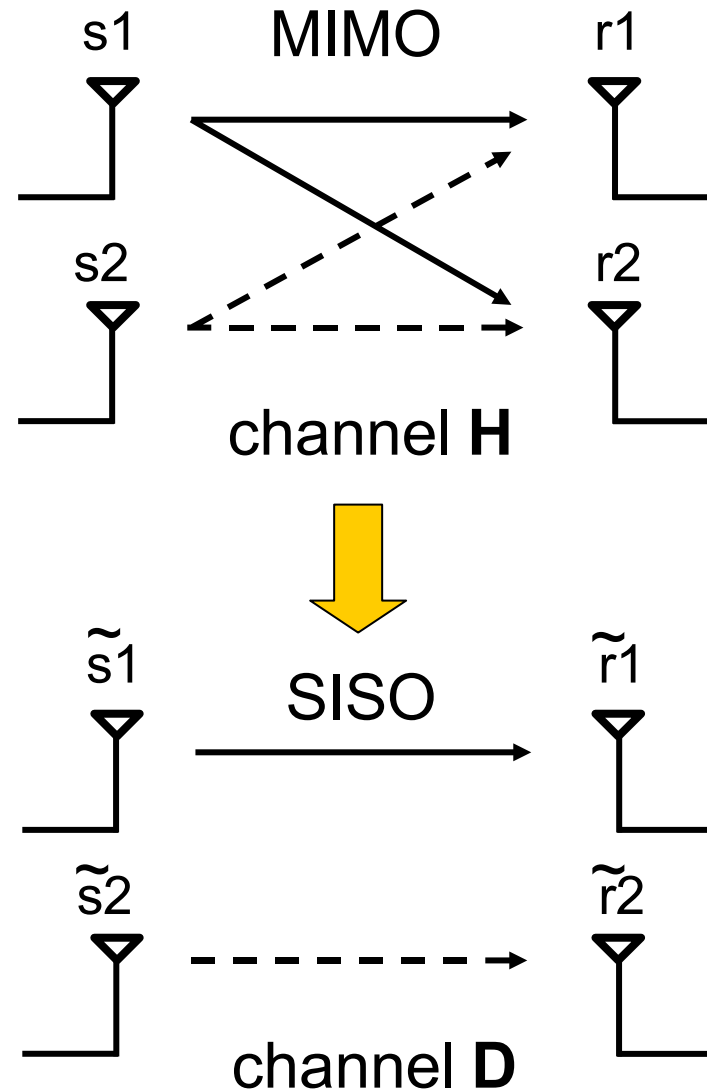
know

$$\underline{r} = \mathbf{H} \underline{s} + \underline{n}$$

**Singular Value
Decomposition**

wanted

$$\tilde{\underline{r}} = \mathbf{D} \tilde{\underline{s}} + \tilde{\underline{n}}$$



Singular Value Decomposition (SVD)

$$\underline{r} = \begin{bmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{bmatrix} \underline{s} + \underline{n}$$

$$\underline{r} = \mathbf{U} \begin{bmatrix} d_1 & h_{12} \\ h_{21} & d_2 \end{bmatrix} (\mathbf{V}^*)^T \underline{s} + \underline{n}$$

$$\begin{aligned} (\mathbf{U}^*)^T \underline{r} &= (\mathbf{U}^*)^T \mathbf{U} \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} (\mathbf{V}^*)^T \underline{s} + (\mathbf{U}^*)^T \underline{n} \\ \tilde{\underline{r}} &= \begin{bmatrix} d_1 & 0 \\ 0 & d_2 \end{bmatrix} \tilde{\underline{s}} + \tilde{\underline{n}} \end{aligned}$$

Singular Value Decomposition (SVD)

$$\underline{r} = \mathbf{H} \underline{s} + \underline{n}$$

$$\mathbf{H} = \mathbf{U} \mathbf{\Sigma} (\mathbf{V}^*)^T$$

$\mathbf{U} = [u_1, \dots, u_n]$ eigenvectors of $(\mathbf{H}^*)^T \mathbf{H}$

$\mathbf{V} = [v_1, \dots, v_m]$ eigenvectors of $\mathbf{H} (\mathbf{H}^*)^T$

$$\mathbf{\Sigma} = \begin{bmatrix} \sigma_1 & 0 & 0 & \vdots & \\ 0 & \sigma_2 & 0 & \vdots & 0 \\ 0 & 0 & \sigma_3 & \vdots & \\ \vdots & 0 & \vdots & \ddots & \\ 0 & 0 & \vdots & \vdots & 0 \end{bmatrix}$$

λ_i eigenvalues of $(\mathbf{H}^*)^T \mathbf{H}$

singular values $\sigma_i = \sqrt{\lambda_i}$

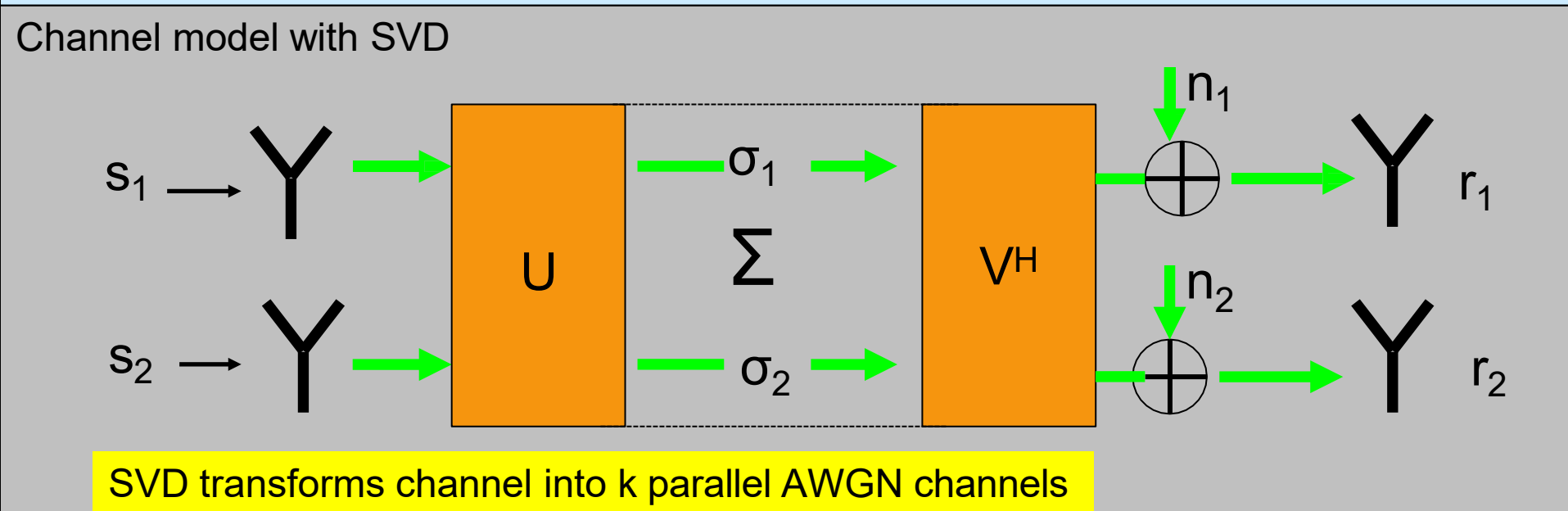
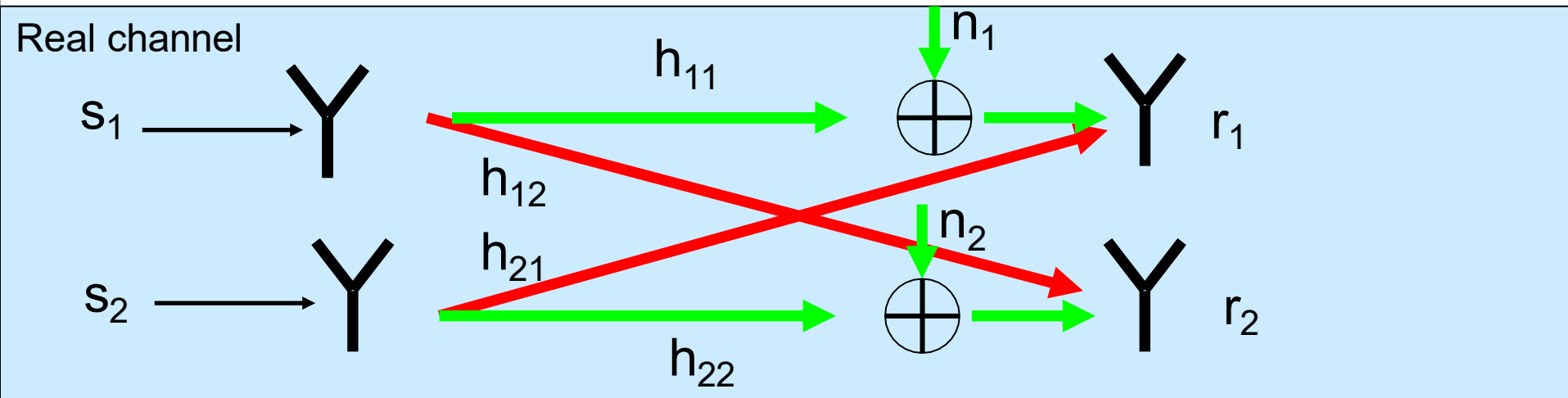
$$\tilde{\underline{r}} = \mathbf{\Sigma} \tilde{\underline{s}} + \tilde{\underline{n}}$$

$$\tilde{\underline{r}} = (\mathbf{U}^*)^T \underline{r}$$

$$\tilde{\underline{s}} = (\mathbf{V}^*)^T \underline{s}$$

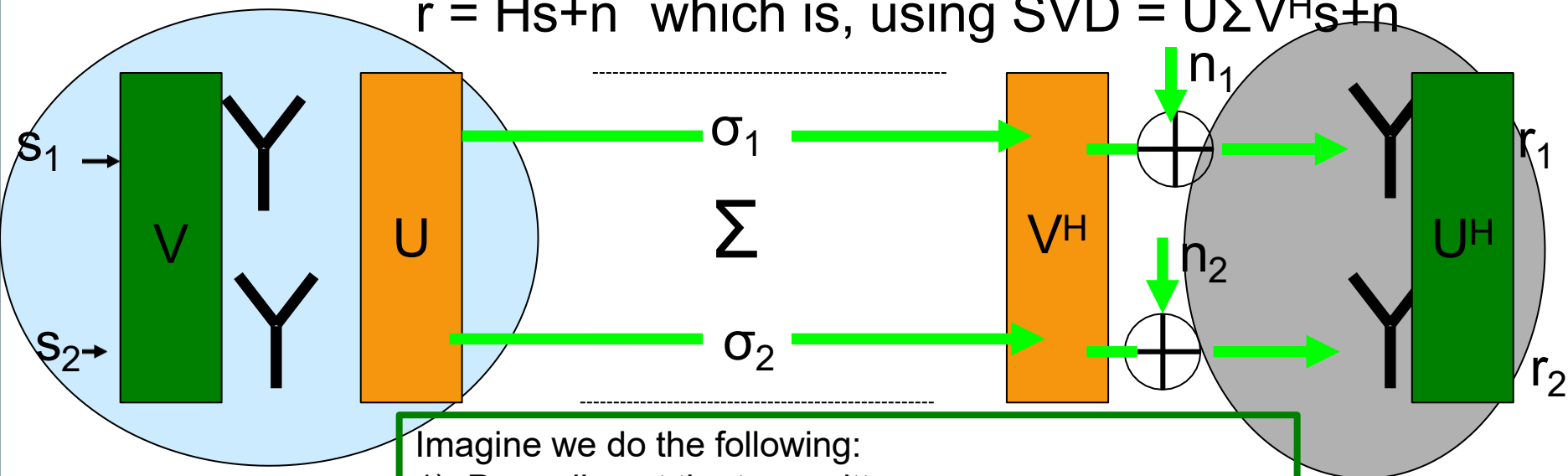
$$\tilde{\underline{n}} = (\mathbf{U}^*)^T \underline{n}$$

MIMO and singular value decomposition SVD



MIMO: Signal processing considerations

MIMO transmission can be expressed as $r = Hs + n$ which is, using SVD $= U \Sigma V^H s + n$



Imagine we do the following:

1) Precoding at the transmitter:

Instead of transmitting s , the transmitter sends $s = V^* s$

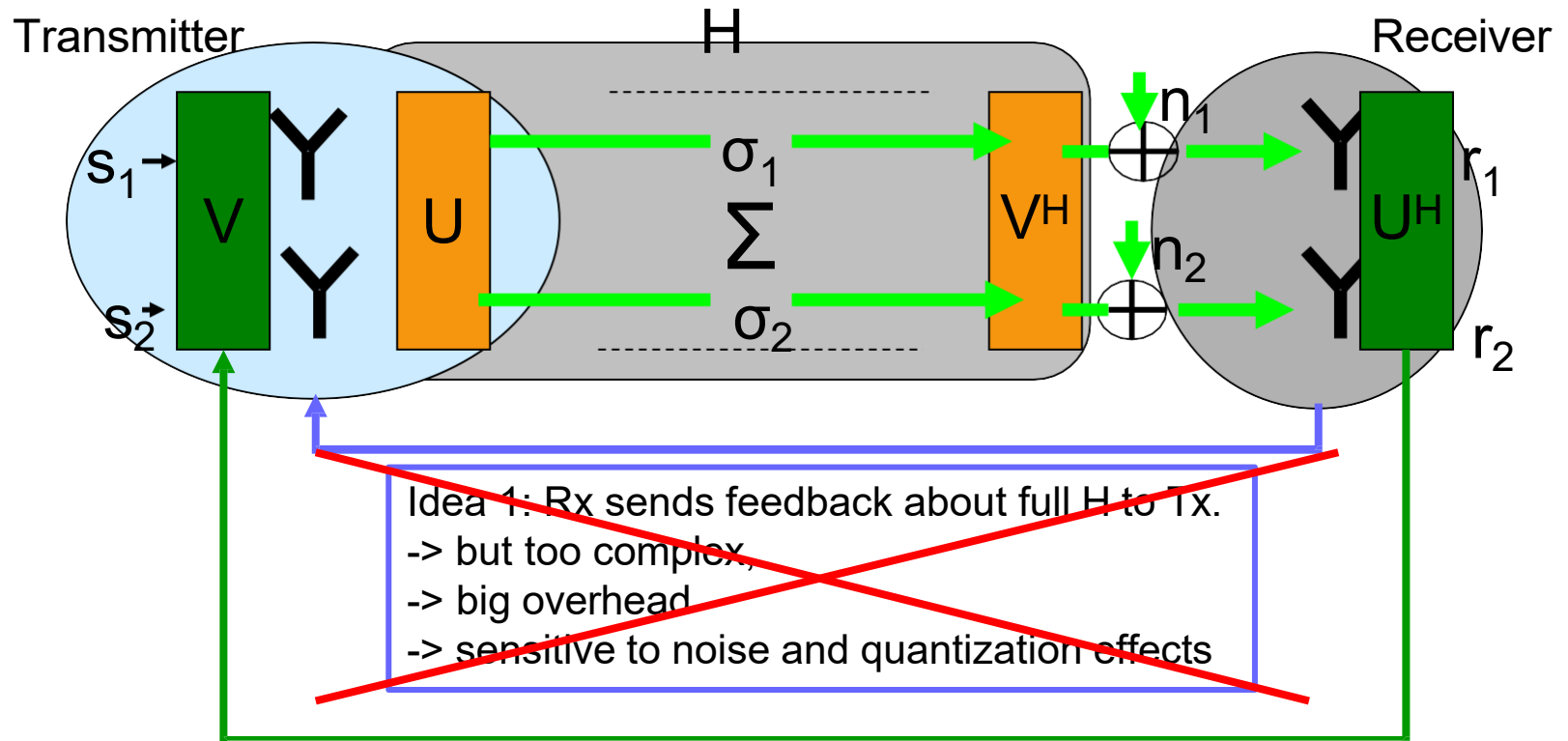
2) Signal processing at the receiver

Multiply the received signal with U^H , $r = r^* U^H$

So after signal processing the whole signal can be expressed as:

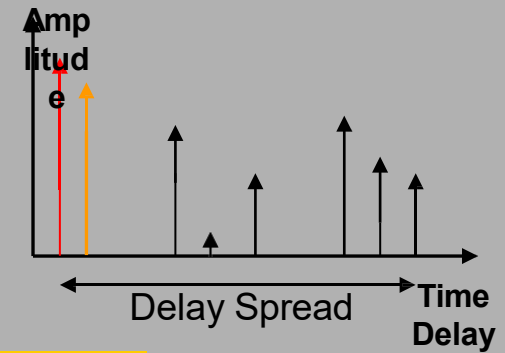
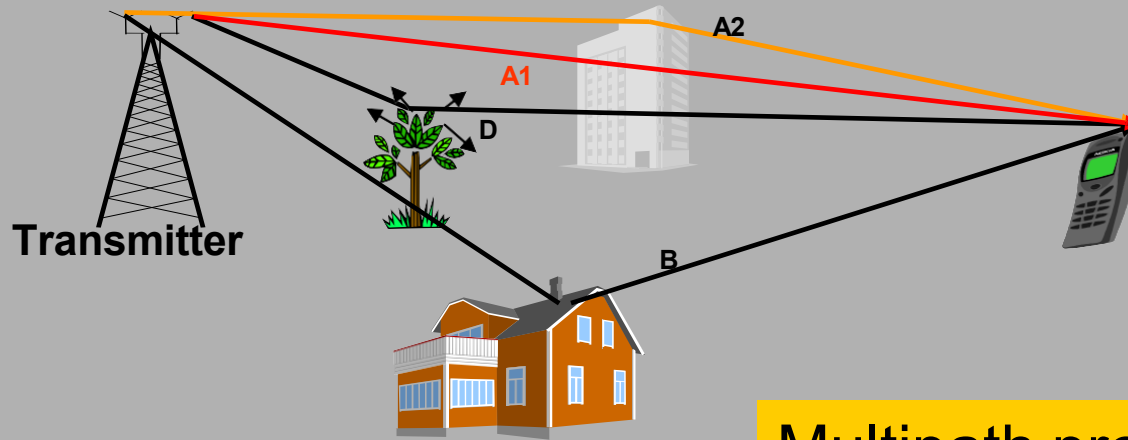
$$r = U^H (U \Sigma V^H V s + n) = \underbrace{U^H U}_{=I_{nTnT}} \underbrace{\Sigma V^H V}_{=I_{nTnT}} s + U^H n = \Sigma s + U^H n$$

MIMO: limited channel feedback

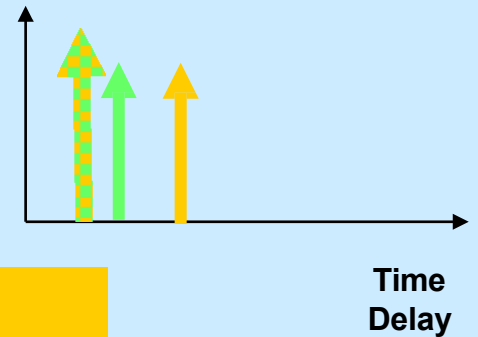
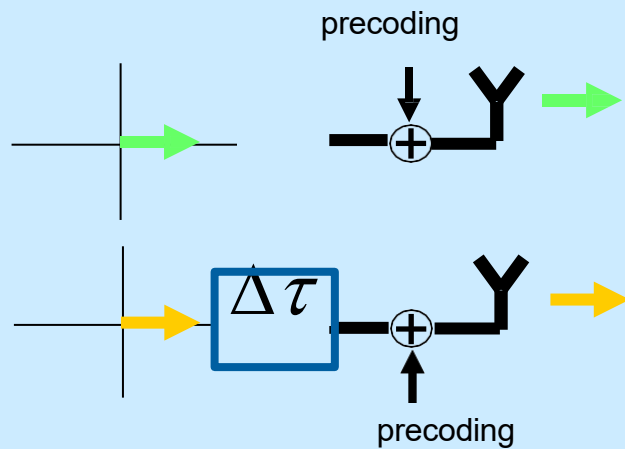


Idea 2: Tx does not need to know full H , only unitary matrix V
-> define a set of unitary matrices (codebook) and find one matrix in the codebook that maximizes the capacity for the current channel H
-> these unitary matrices from the codebook approximate the singular vector structure of the channel
=> Limited feedback is almost as good as ideal channel knowledge feedback

Cyclic Delay Diversity, CDD



Multipath propagation

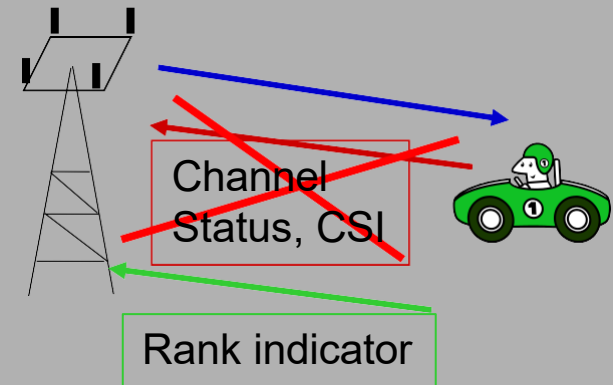


No multipath propagation

„Open loop“ und „closed loop“ MIMO

Open loop (**No** channel knowledge at transmitter)

$$r = Hs + n$$



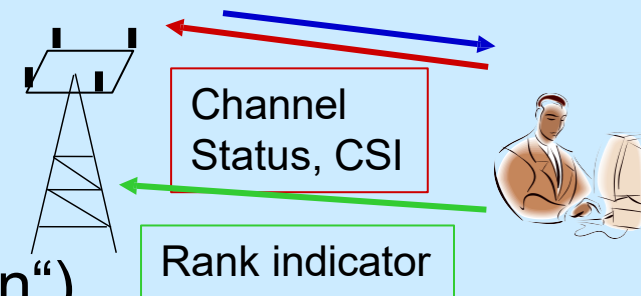
Closed loop (With channel knowledge at transmitter)

$$r = HWs + n$$

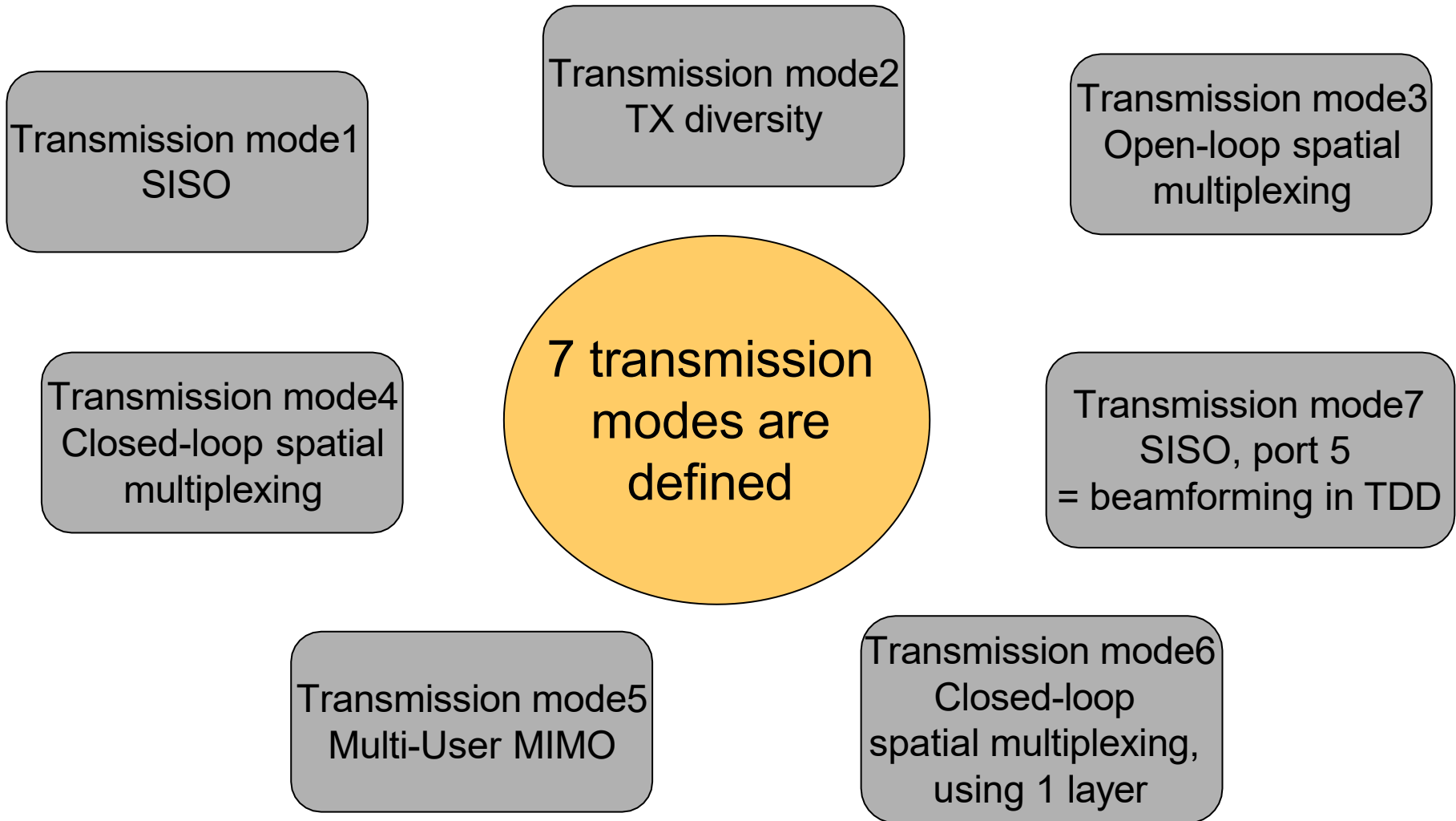


Adaptive **Precoding matrix** („Pre-equalisation“)

Feedback from receiver needed (closed loop)



MIMO transmission modes

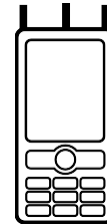
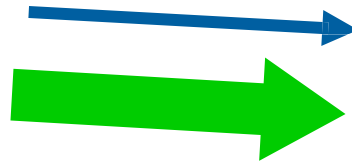
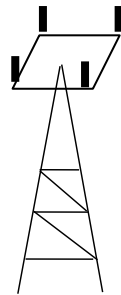


Transmission mode is given by higher layer IE: *AntennaInfo*

MIMO transmission modes

Transmission mode 1
SISO

the classic:
1Tx + 1RX
antenna



PDCCH indication via
DCI format 1 or 1A

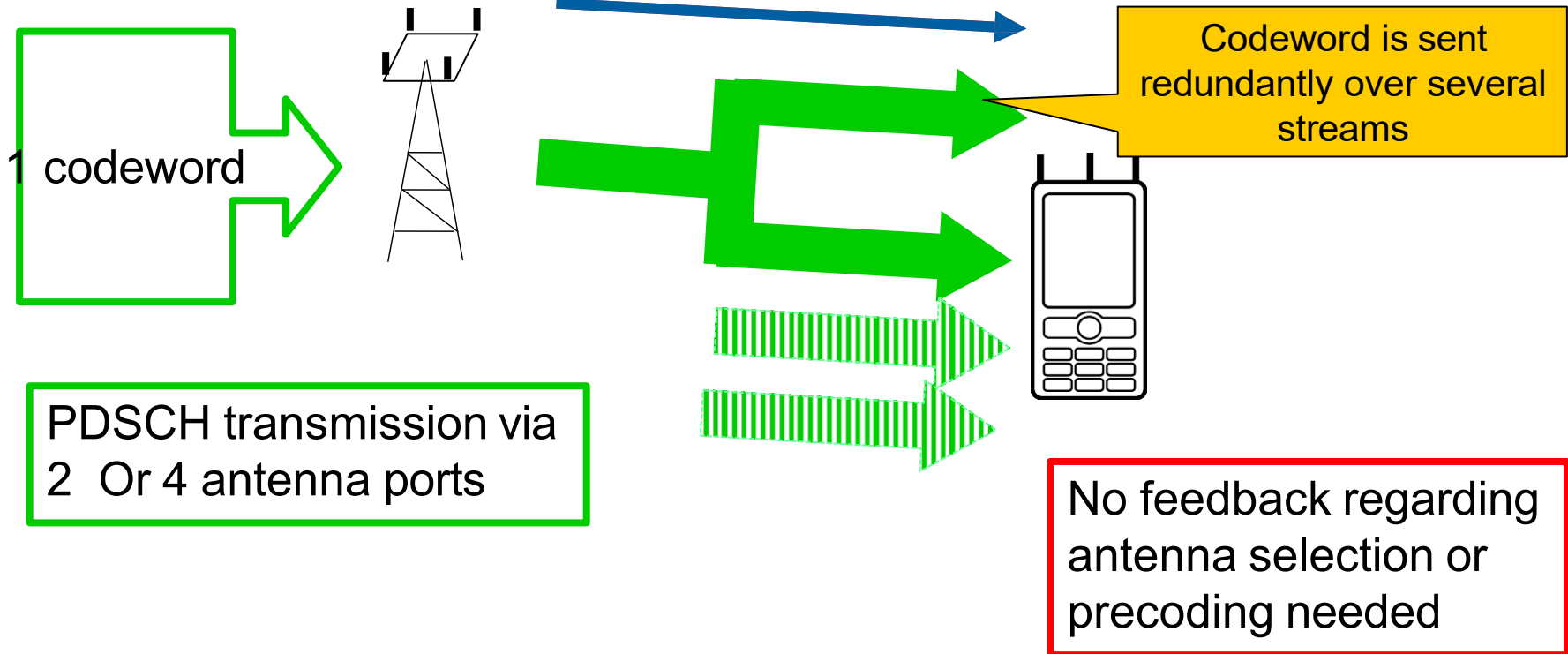
PDSCH transmission via
single antenna port 0

No feedback regarding
antenna selection or
precoding needed

MIMO transmission modes

Transmission mode 2

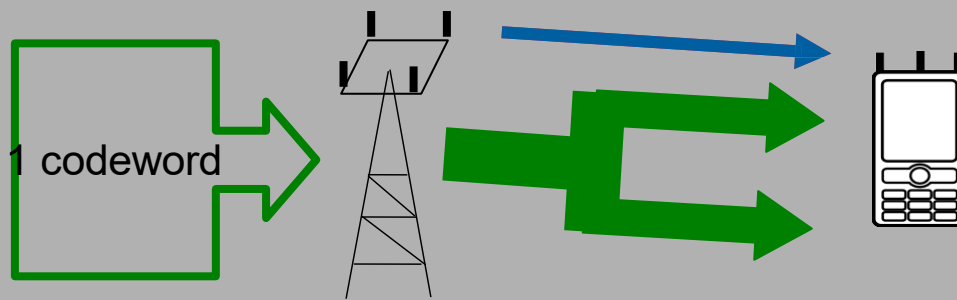
Transmit
diversity



MIMO transmission modes

Transmission mode 3

Transmit diversity or Open loop spatial multiplexing

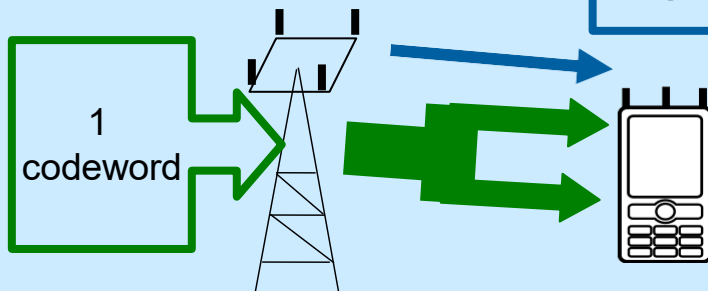


PDCCH indication via
DCI format 1A

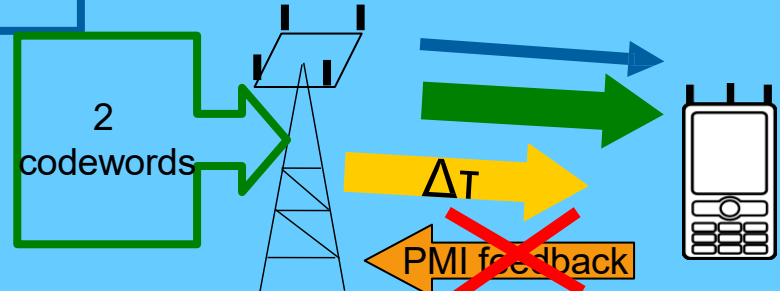
PDSCH transmission
Via 2 or 4 antenna ports

No feedback regarding
antenna selection or
precoding needed

PDCCH indication via
DCI format 2A



PDSCH spatial multiplexing
with 1 layer

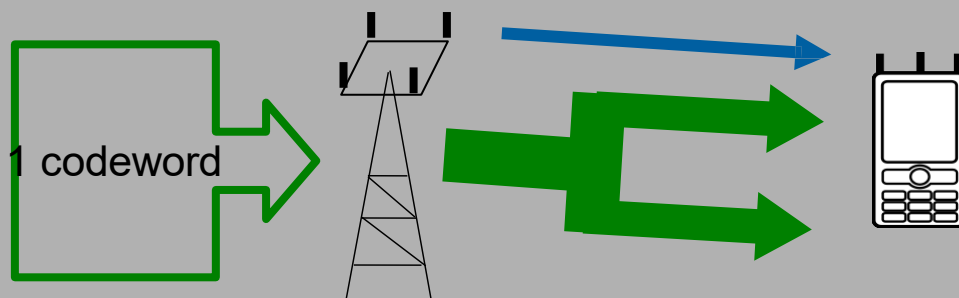


PDSCH spatial multiplexing, using CDD

MIMO transmission modes

Transmission mode 4

Transmit diversity or Closed loop spatial multiplexing

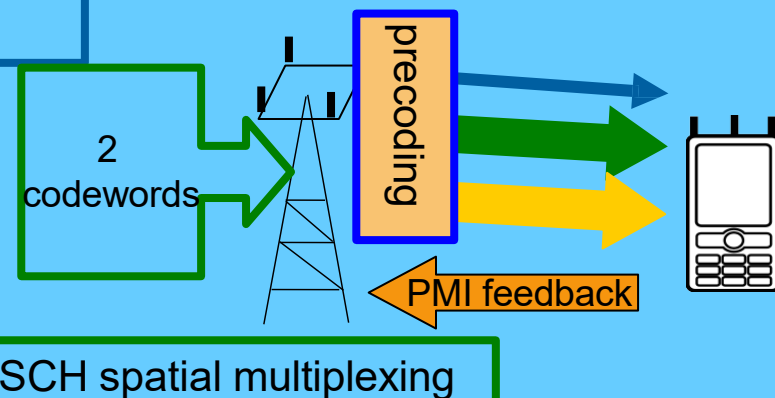
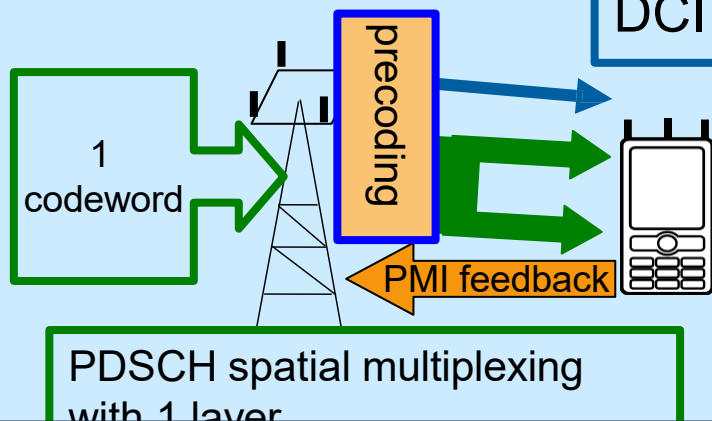


Closed loop MIMO =
UE feedback needed regarding
precoding and antenna
selection

PDCCH indication via
DCI format 1A

PDSCH transmission
Via 2 or 4 antenna ports

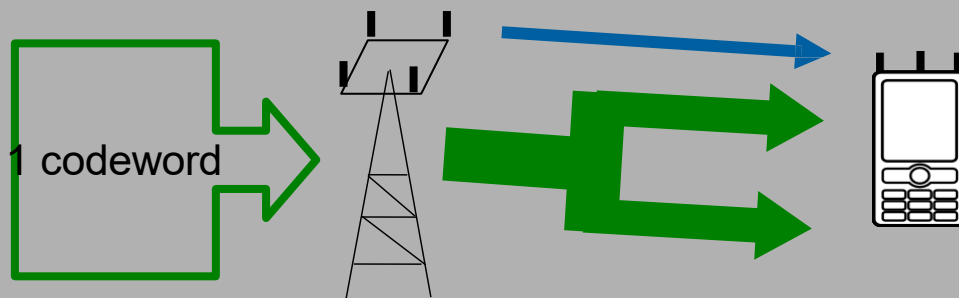
PDCCH indication via
DCI format 2



MIMO transmission modes

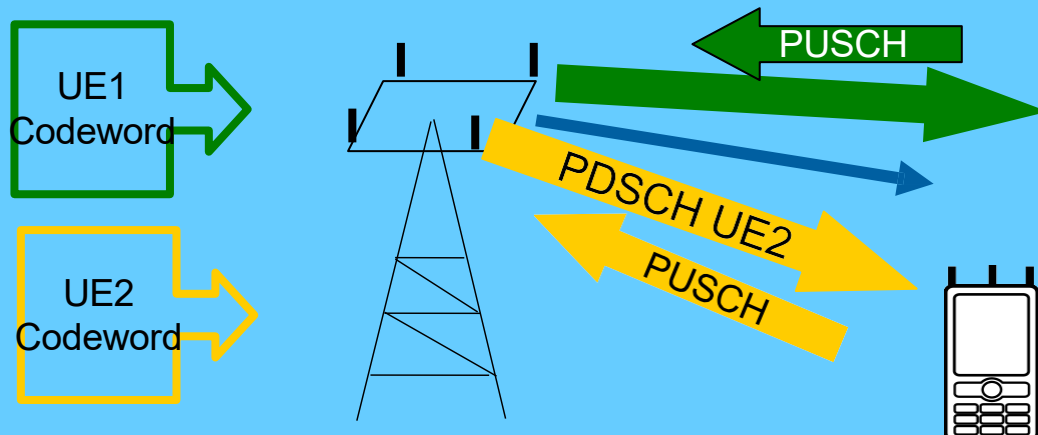
Transmission mode 5

Transmit diversity or
Multi User MIMO



PDCCH indication via
DCI format 1A

PDSCH transmission
Via 2 or 4 antenna ports



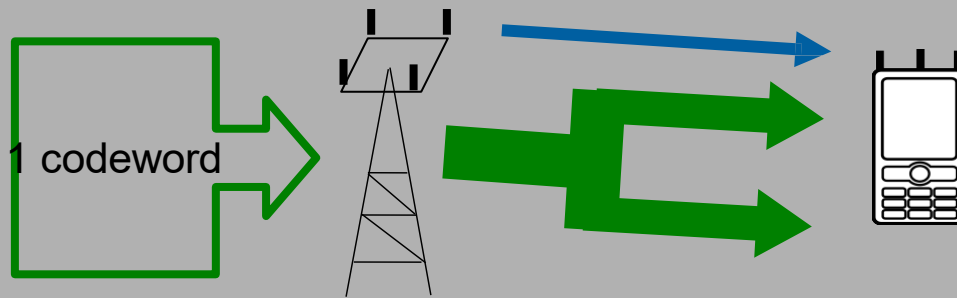
PDCCH indication
via DCI format 1D

PDSCH multiplexing to several UEs.
PUSCH multiplexing in Uplink

MIMO transmission modes

Transmission mode 6

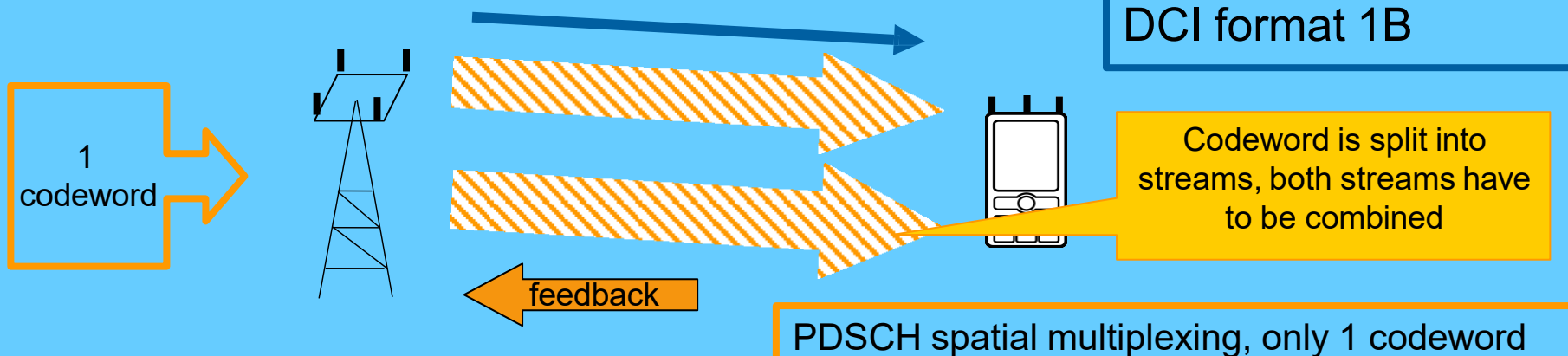
Transmit diversity or Closed loop spatial multiplexing **with 1 layer**



Closed loop MIMO =
UE feedback needed regarding
precoding and antenna
selection

PDCCH indication via
DCI format 1A

PDSCH transmission
via 2 or 4 antenna ports



PDCCH indication via
DCI format 1B

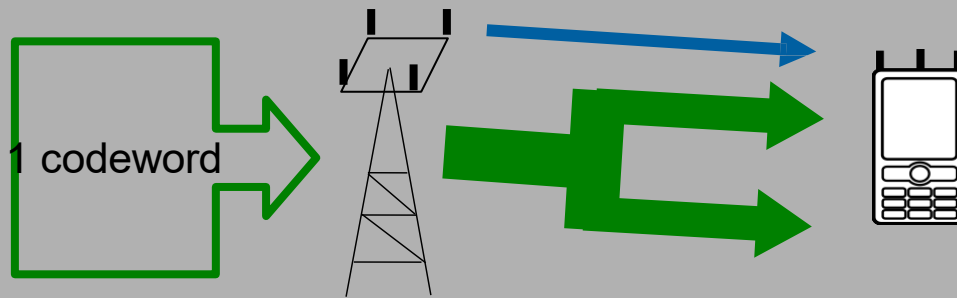
Codeword is split into
streams, both streams have
to be combined

PDSCH spatial multiplexing, only 1 codeword

MIMO transmission modes

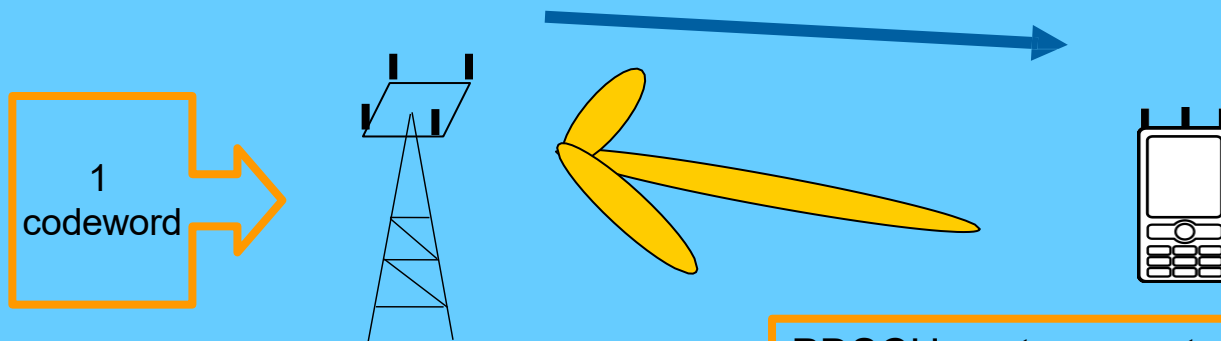
Transmission mode 7

Transmit diversity or beamforming



PDCCH indication via
DCI format 1A

PDSCH transmission
via 1, 2 or 4 antenna ports



PDCCH indication via
DCI format 1

PDSCH sent over antenna port 5 = beamforming

Beamforming

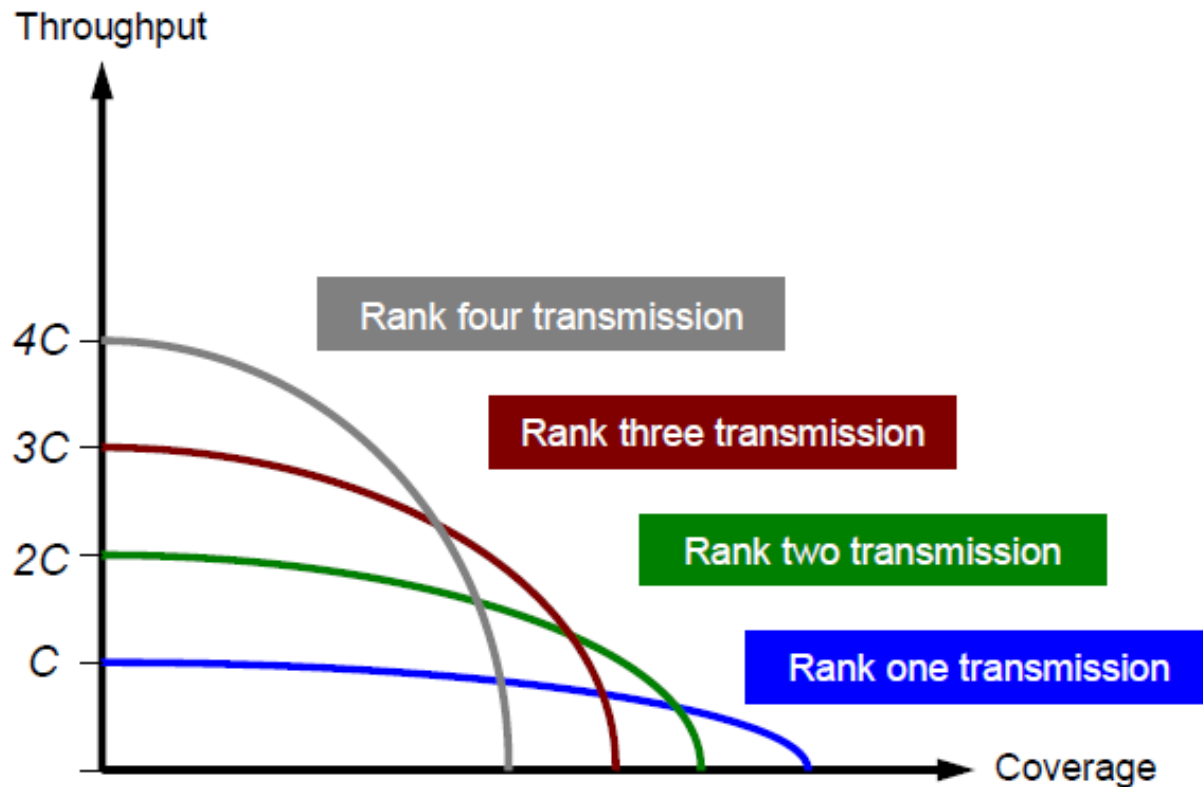
Adaptive Beamforming

- Classic way
- Antenna weights to adjust beam
- Directional characteristics
- Specific antenna array geometrie
- Dedicated pilots required

Closed loop precoded beamforming

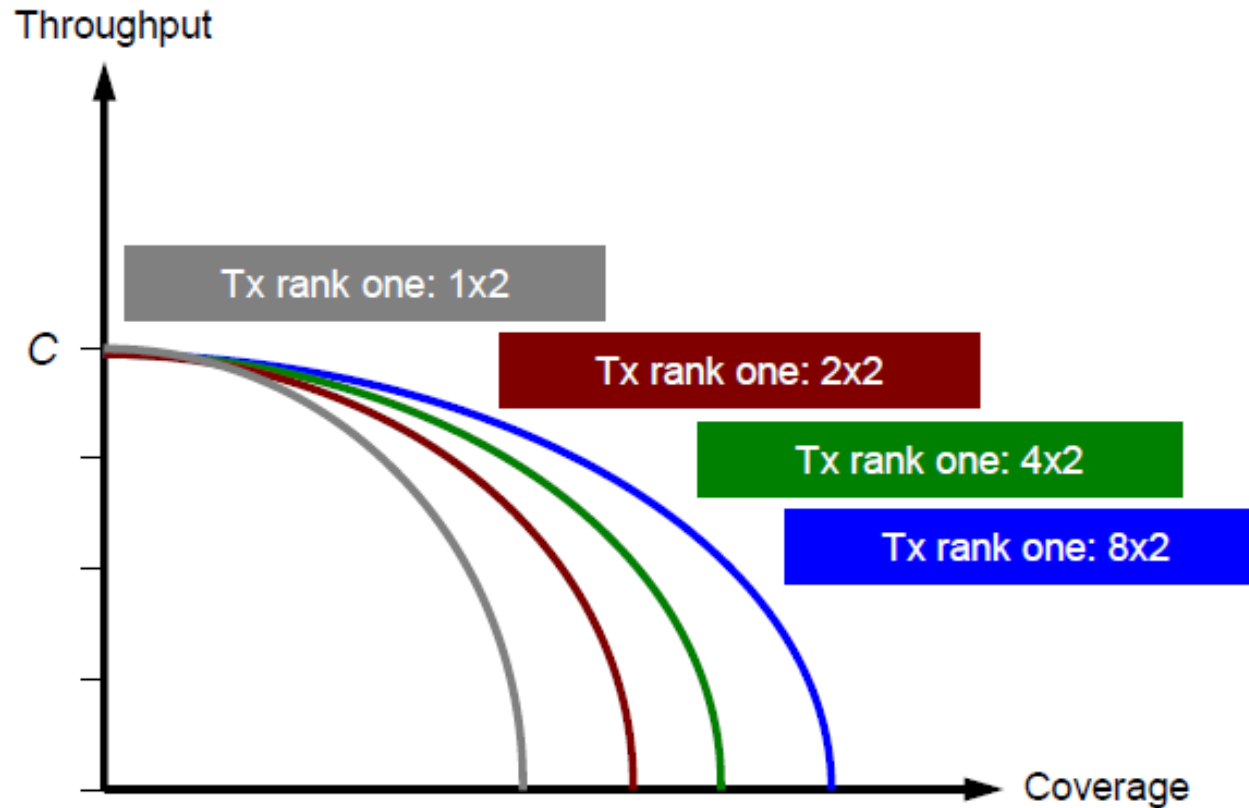
- Kind of MISO with channel knowledge at transmitter
- Precoding based on feedback
- No specific antenna array geometrie
- Common pilots are sufficient

Spatial multiplexing vs beamforming



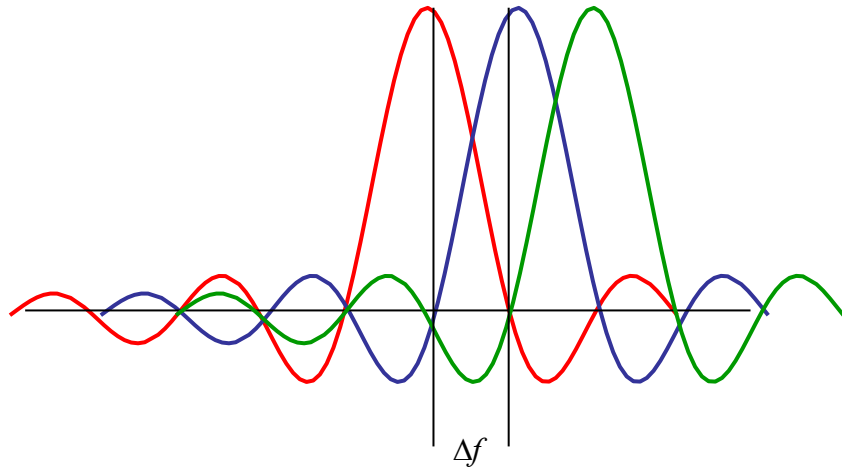
Spatial multiplexing increases throughput, but loses coverage

Spatial multiplexing vs beamforming



Beamforming increases coverage

Basic OFDM parameter



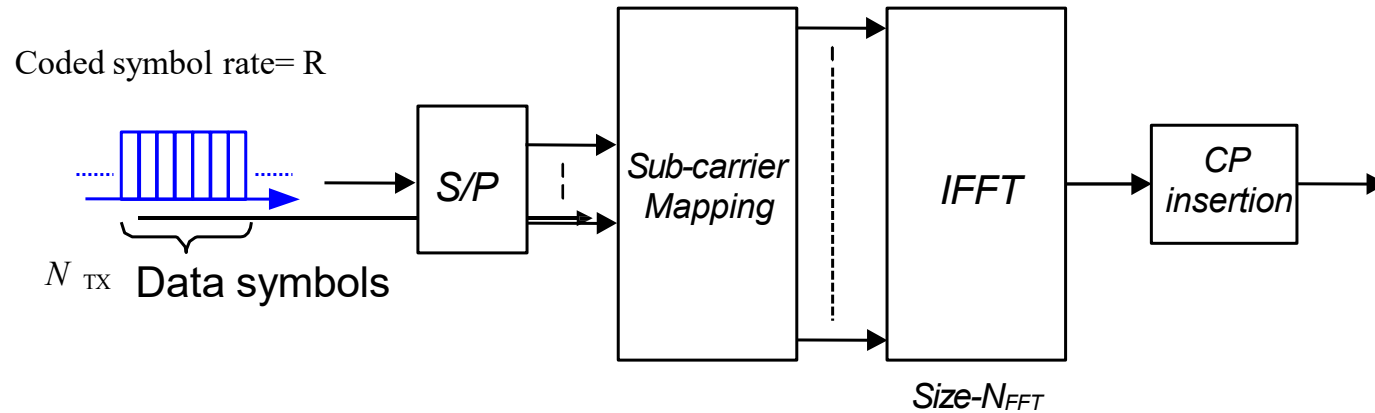
LTE

$$\Delta f = 15 \text{ kHz} = \frac{1}{T}$$

$$F_s = N_{FFT} \cdot \Delta f$$

$$F_s = \frac{N_{FFT}}{256} \cdot 3.84 \text{ Mcps}$$

$$N_{FFT} = 2048$$

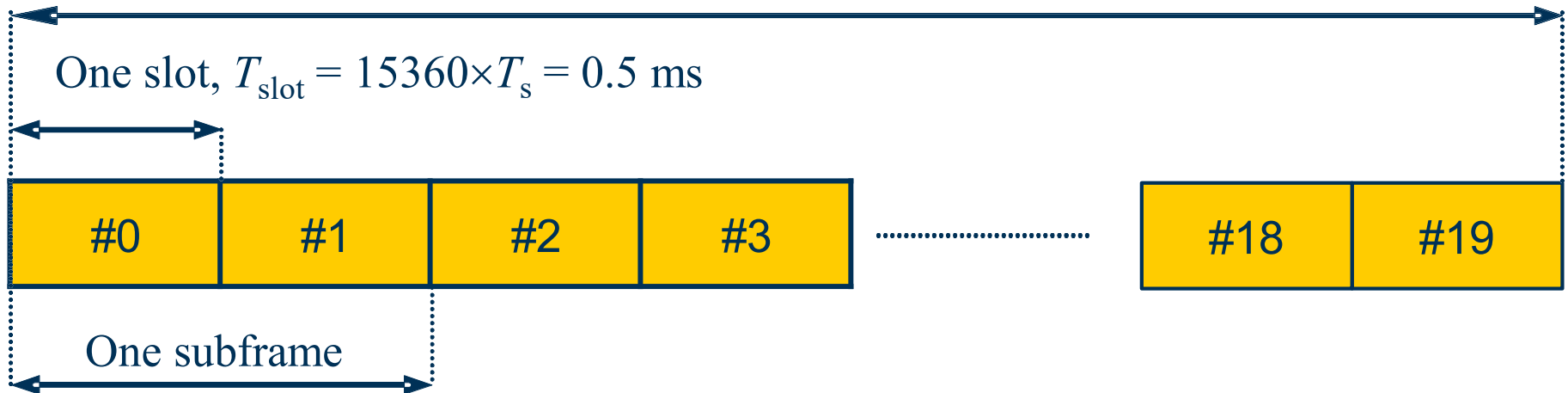


LTE Downlink:

Downlink slot and (sub)frame structure

Symbol time, or number of symbols per time slot is not fixed

One radio frame, $T_f = 307200 \times T_s = 10$ ms

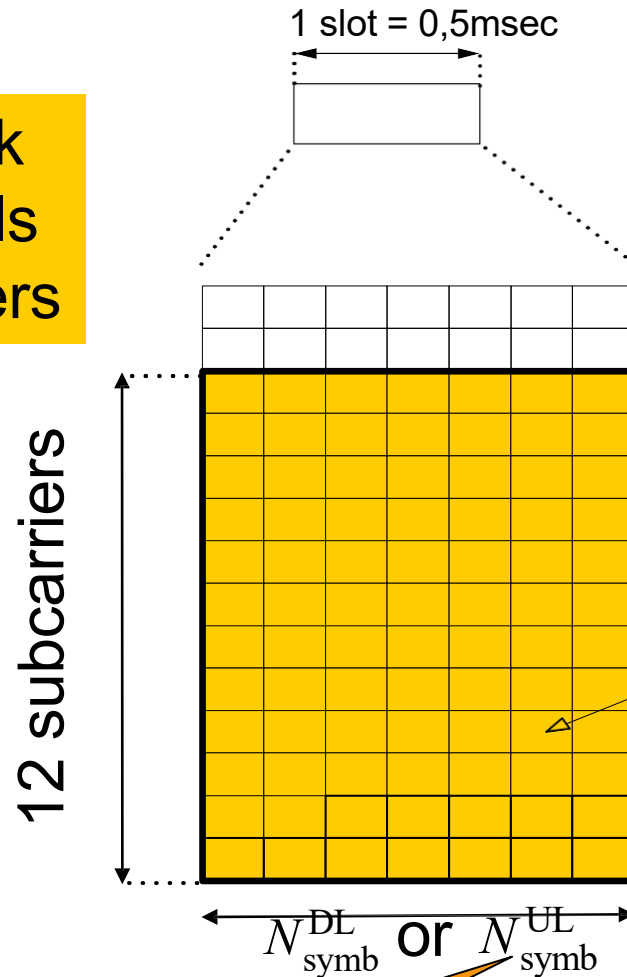


We talk about 1 slot, but the minimum resource is 1 subframe = 2 slots !!!!!

$$T_s = 1 / (15000 \times 2048) \quad T_s = 32.522 \text{ ns}$$

Resource block definition

Resource block
= 6 or 7 symbols
In 12 subcarriers



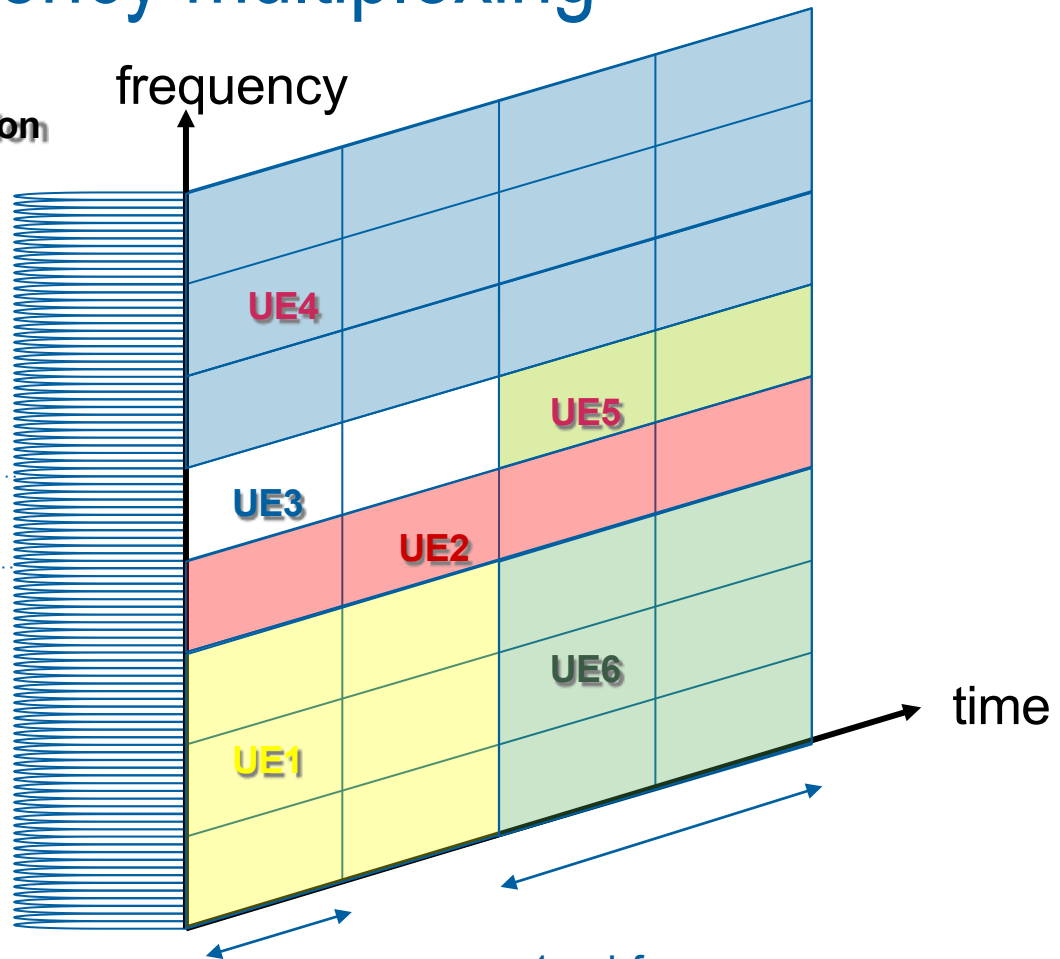
6 or 7,
Depending on
cyclic prefix

LTE Downlink OFDMA time-frequency multiplexing

QPSK, 16QAM or 64QAM modulation

1 resource block =
180 kHz = 12 subcarriers

Subcarrier spacing = 15 kHz



*TTI = transmission time interval

** For normal cyclic prefix duration

1 slot = 0.5 ms =
7 OFDM symbols**

1 subframe =
1 ms = 1 TTI* =
1 resource block pair

LTE: new physical channels for data and control

Physical Control Format Indicator Channel PCFICH:
Indicates Format of PDCCH



Physical Downlink Control Channel PDCCH:
Downlink and uplink scheduling decisions



Physical Downlink Shared Channel PDSCH: Downlink data



Physical Hybrid ARQ Indicator Channel PHICH:
ACK/NACK for uplink packets



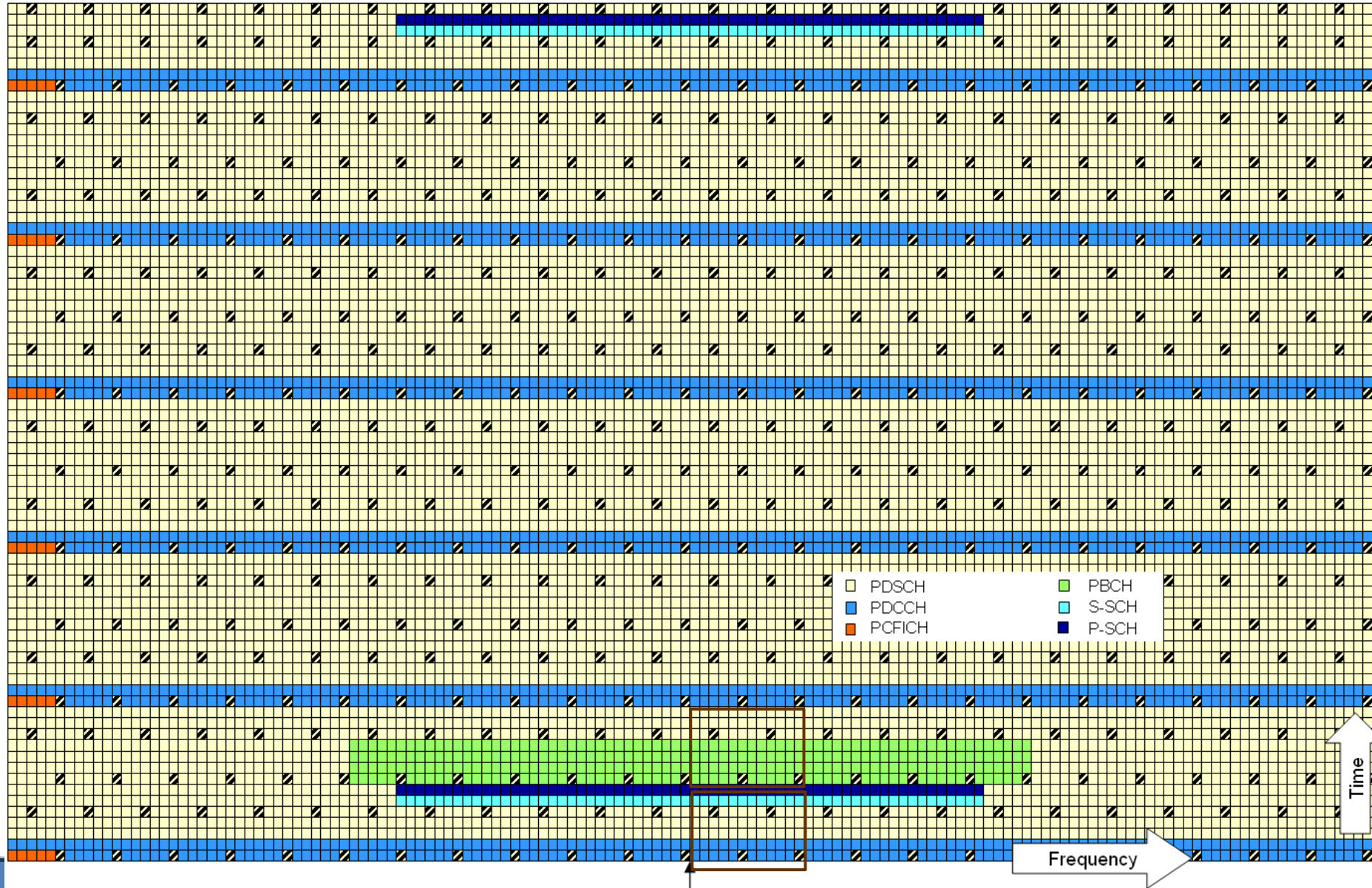
Physical Uplink Shared Channel PUSCH: Uplink data



Physical Uplink Control Channel PUCCH:
ACK/NACK for downlink packets, scheduling requests, channel quality info



LTE Downlink: FDD channel mapping example

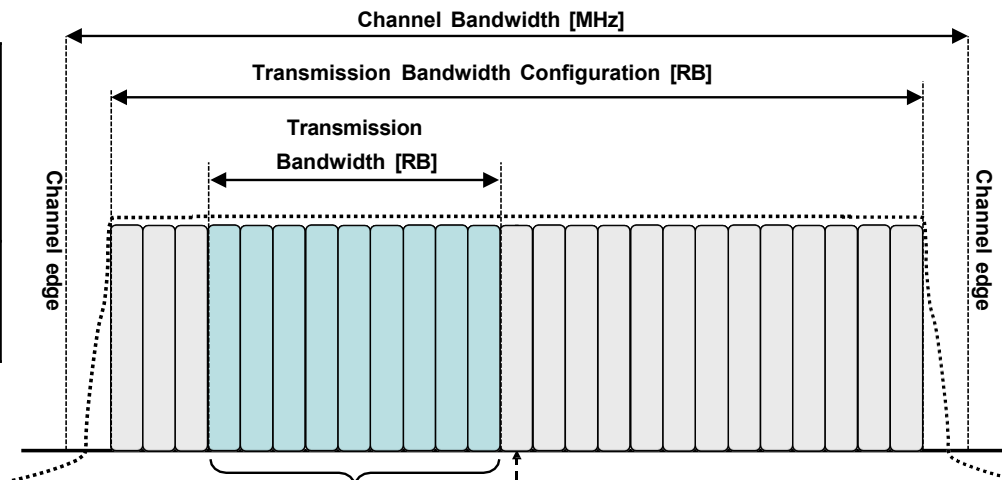


LTE – spectrum flexibility

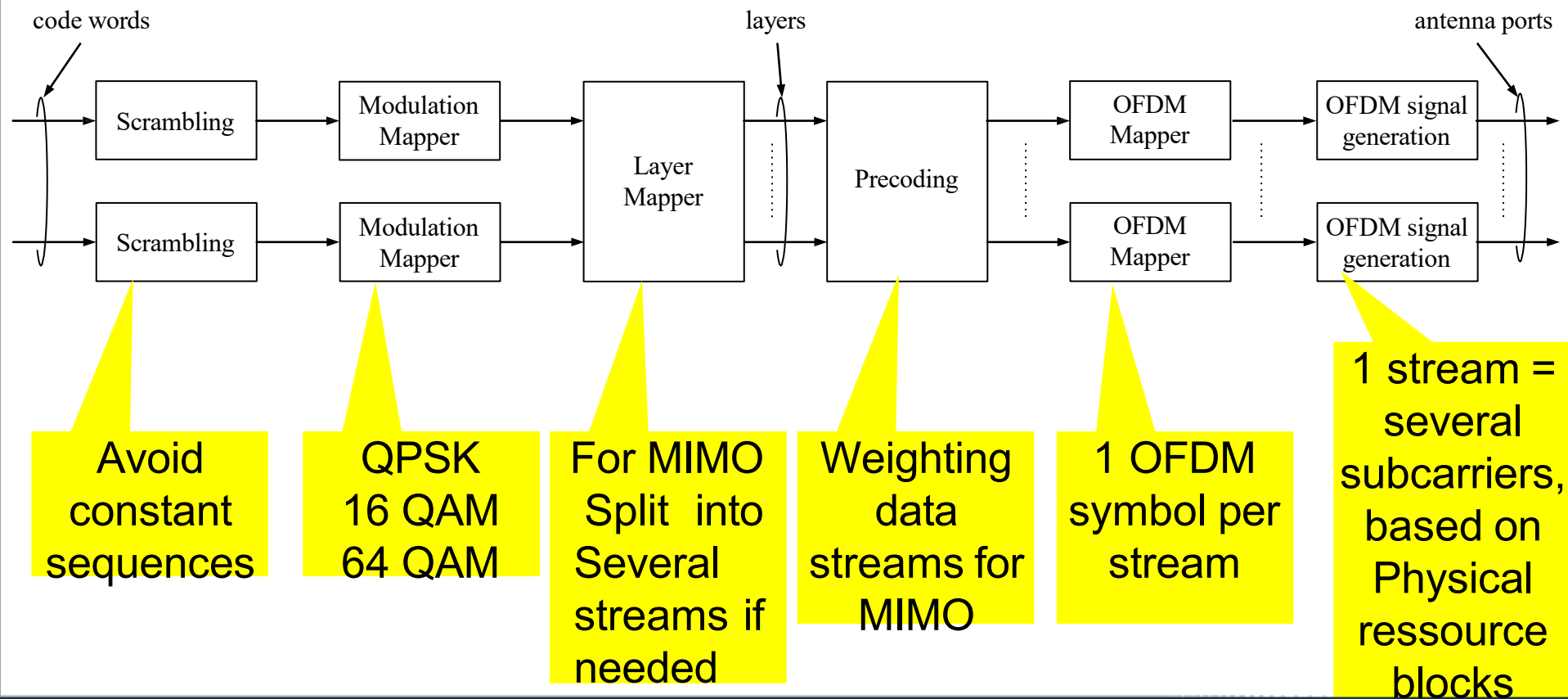
- ❑ LTE physical layer supports any bandwidth from 1.4 MHz to 20 MHz in steps of 180 kHz (resource block)
- ❑ Current LTE specification supports only a subset of 6 different system bandwidths
- ❑ All UEs must support the maximum bandwidth of 20 MHz

Channel bandwidth BW_{Channel} [MHz]	1.4	3	5	10	15	20
FDD and TDD mode	6	15	25	50	75	100

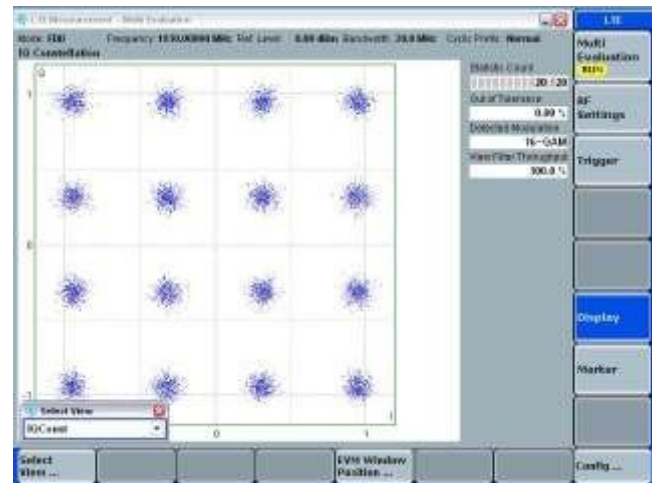
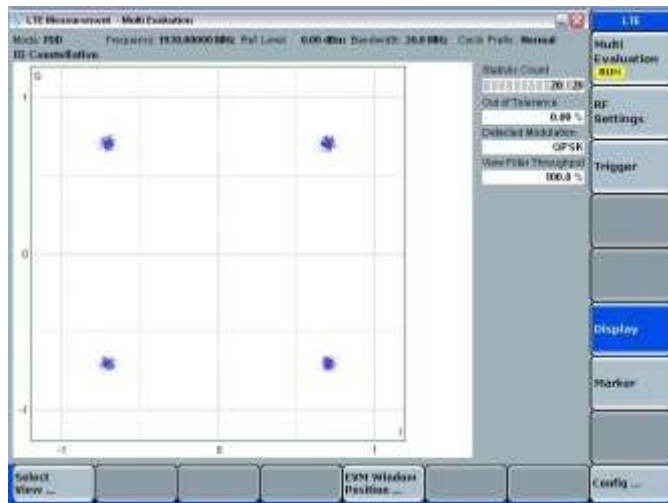
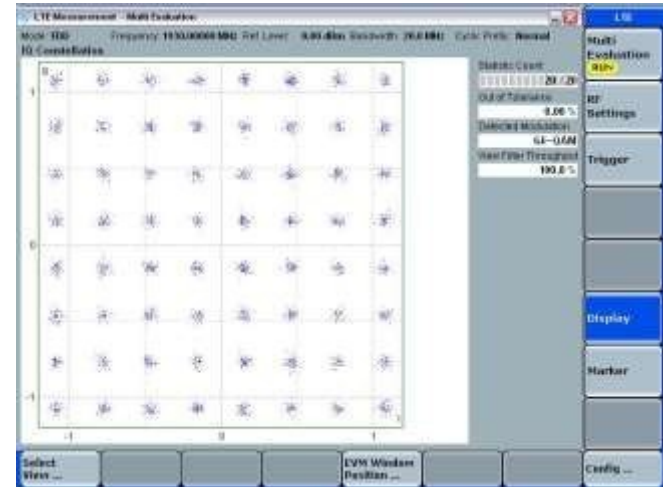
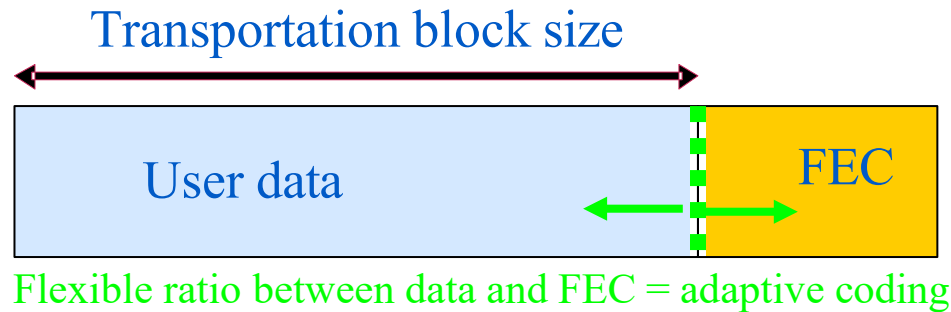
number of resource blocks



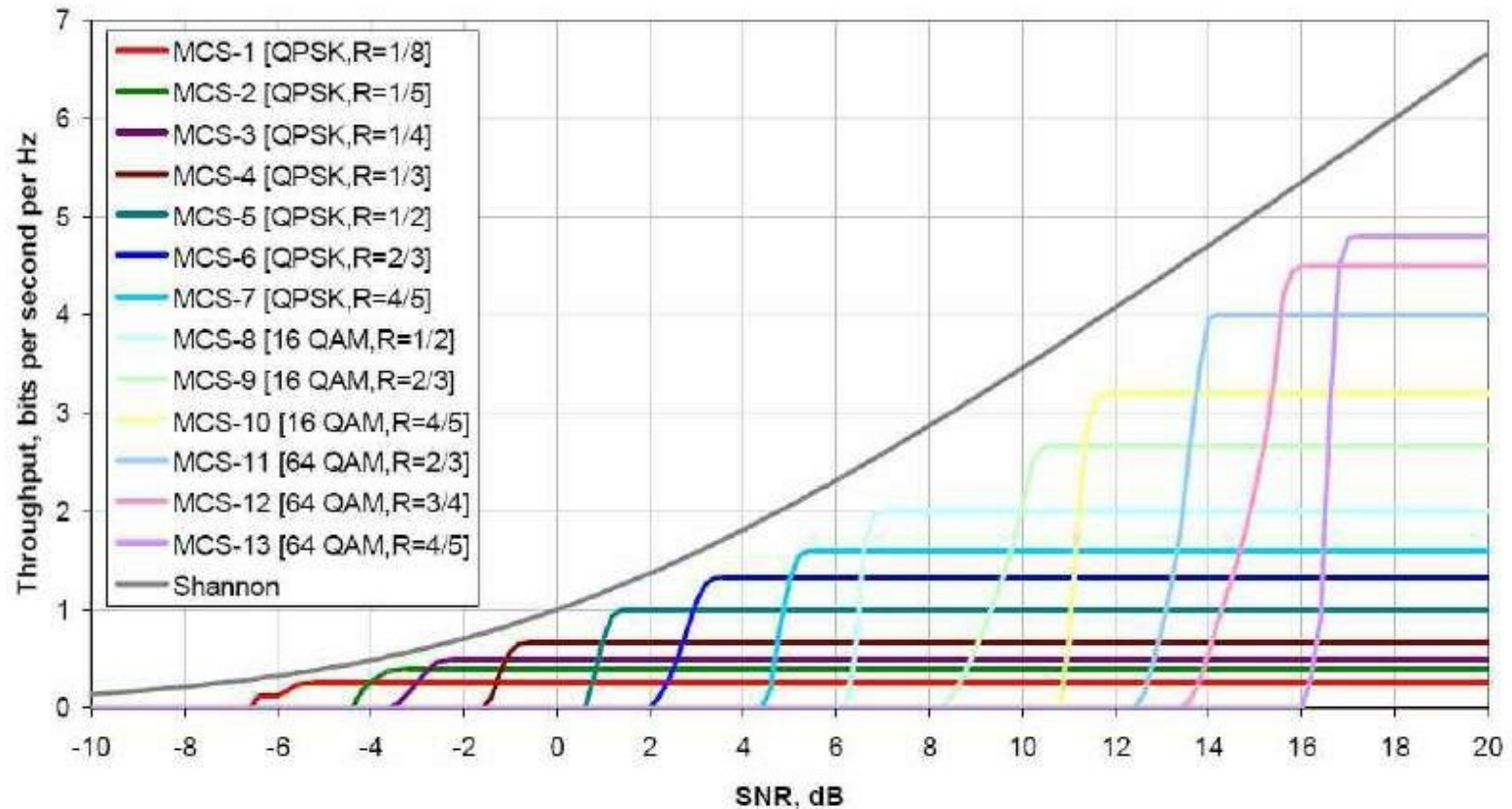
LTE Downlink: baseband signal generation



Adaptive modulation and coding

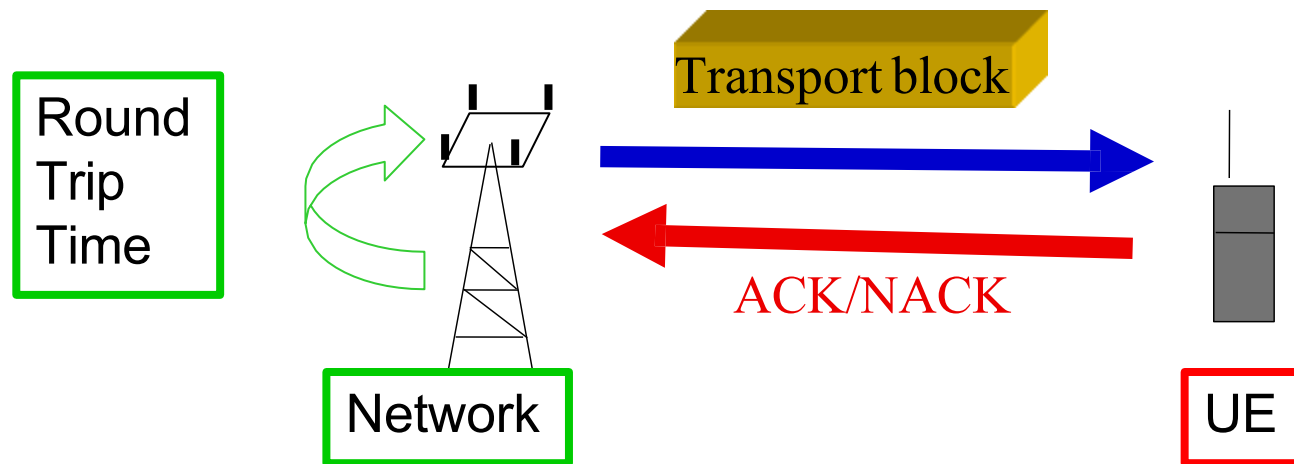


Channel Coding Performance



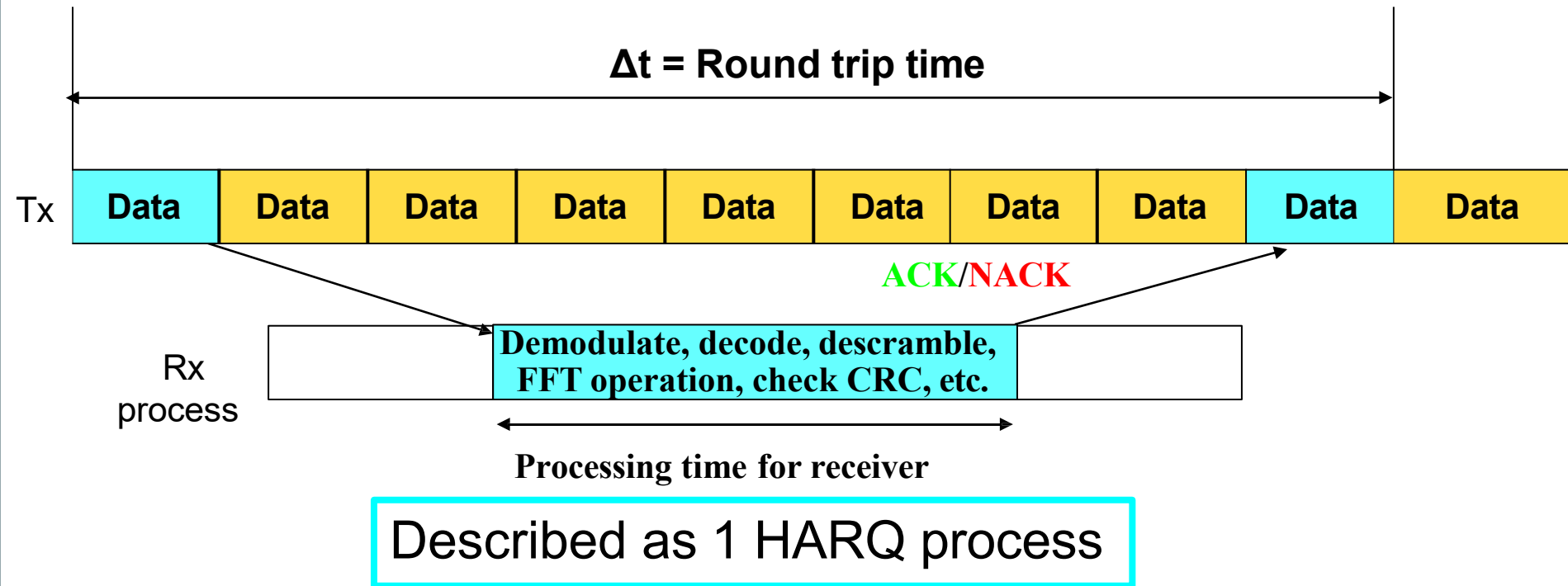
Automatic repeat request, latency aspects

- Transport block size = amount of data bits (excluding redundancy!)
- TTI, Transmit Time Interval = time duration for transmitting 1 transport block

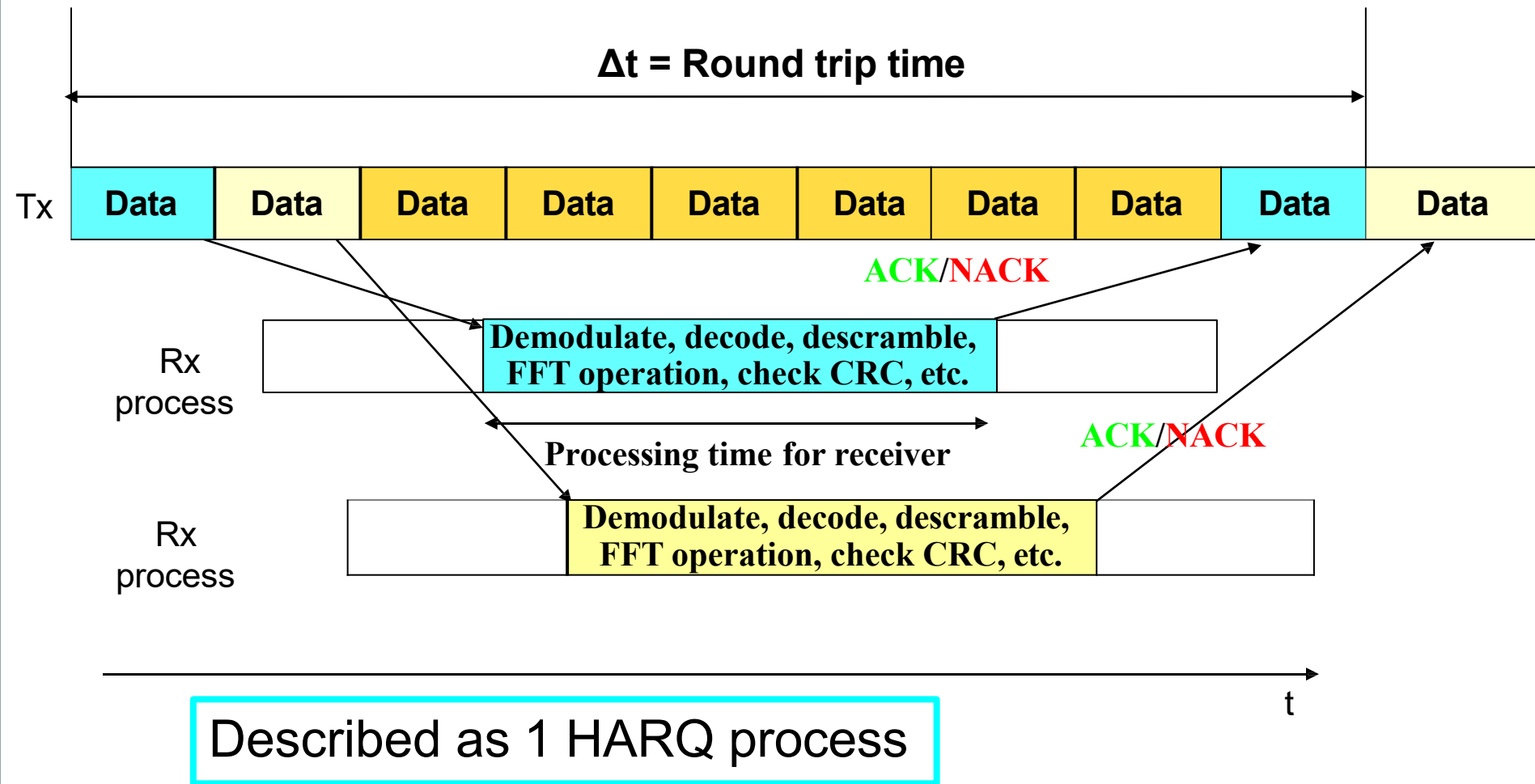


Immediate acknowledged or non-acknowledged feedback of data transmission

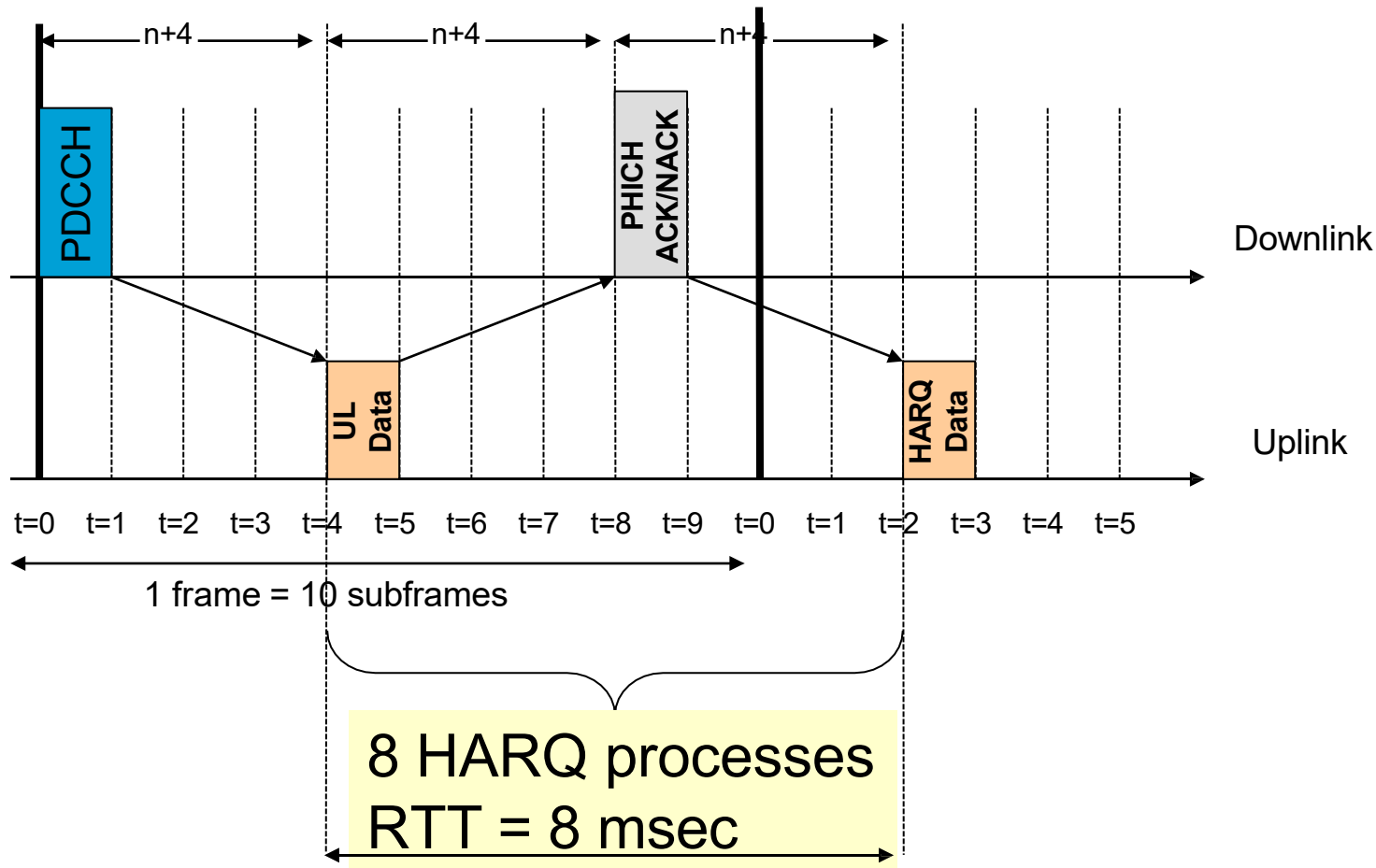
HARQ principle: Stop and Wait



HARQ principle: Multitasking



LTE Round Trip Time RTT



HARQ principle: Soft combining

IT i is a e am l o h n e co i g

Reception of first transportation block.
Unfortunately containing transmission errors

HARQ principle: Soft combining

I h i n x m l e f c h a n l c i n g

Reception of **retransmitted**
transportation block.
Still containing transmission errors

HARQ principle: Soft combining

1st transmission with puncturing scheme P1

I T i is a e a m l o h n e c o i g

2nd transmission with puncturing scheme P2

I h i i n x m l e f c h a n l c i n g

Soft Combining = Σ of transmission 1 and 2

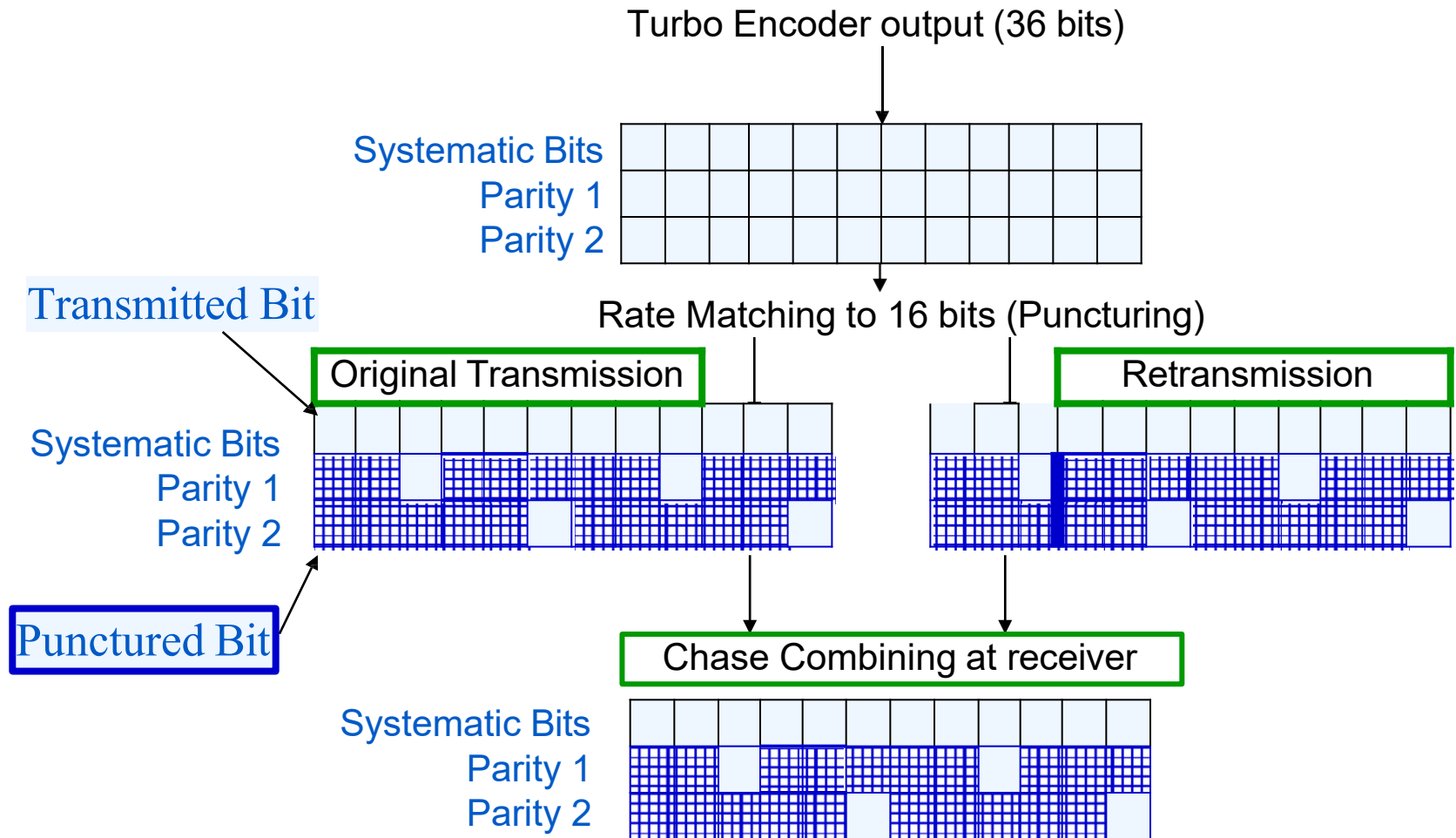
I Thi is an exam le of channel co ing

Final decoding

I This is an example of channel coding

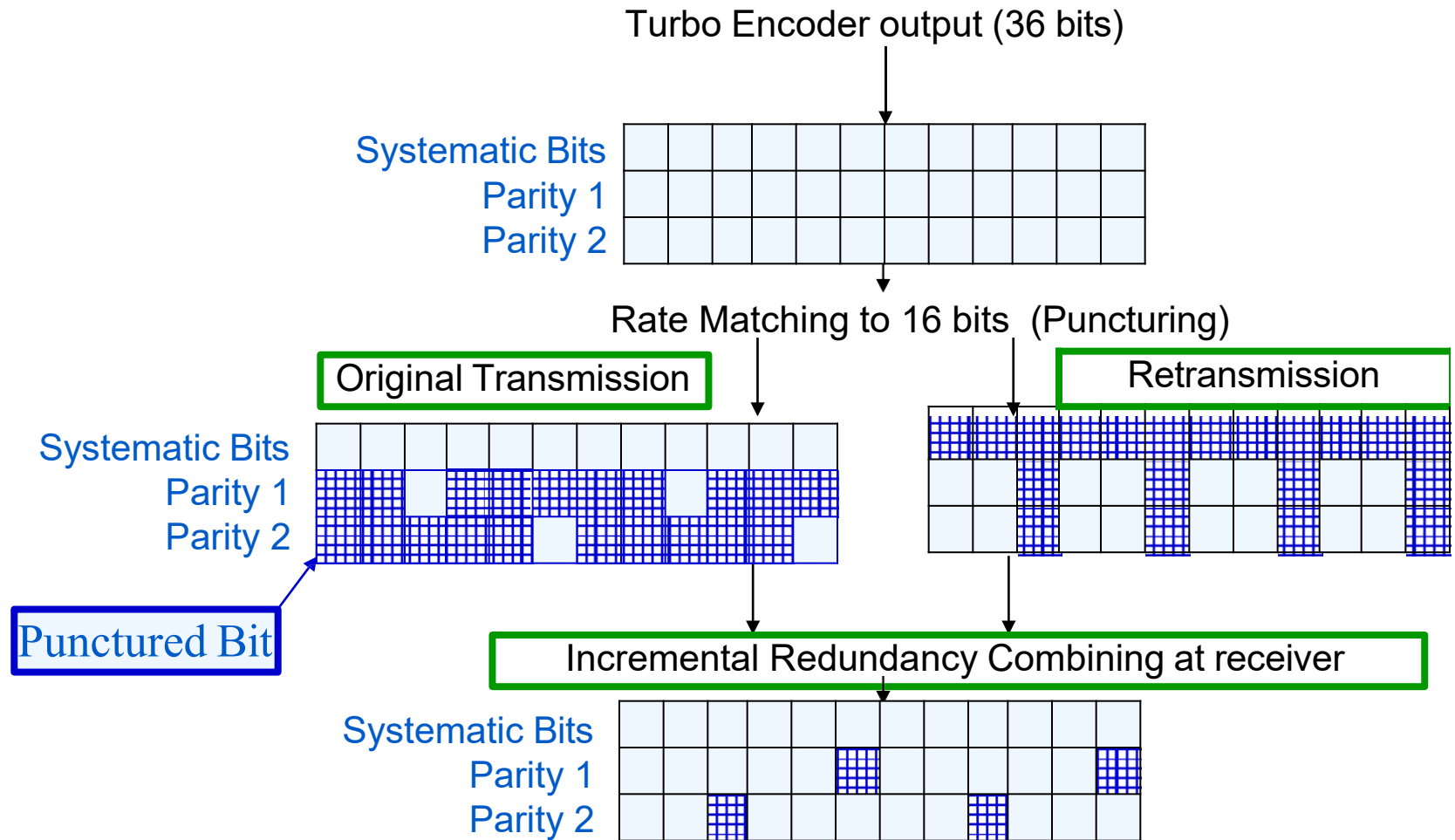
Hybrid ARQ

Chase Combining = identical retransmission



Hybrid ARQ

Incremental Redundancy

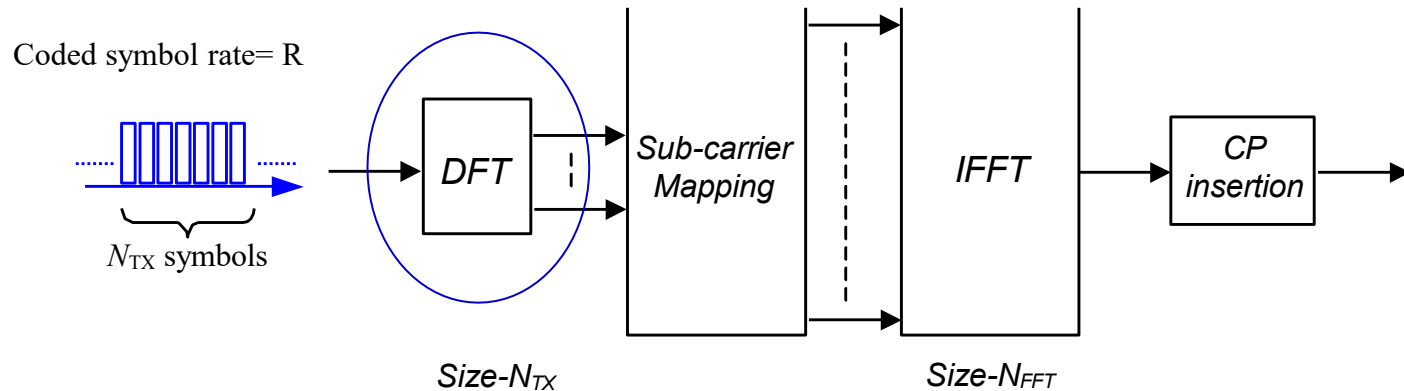


LTE Physical Layer: SC-FDMA in uplink

Single Carrier Frequency Division
Multiple Access

LTE Uplink:

How to generate an SC-FDMA signal in theory?

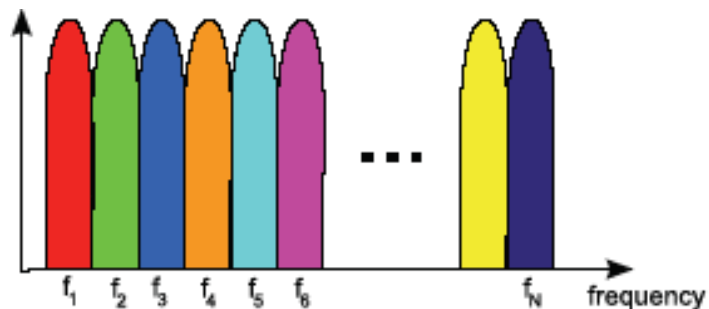


- ❖ LTE provides QPSK, 16QAM, and 64QAM as uplink modulation schemes
- ❖ DFT is first applied to block of N_{TX} modulated data symbols to transform them into frequency domain
- ❖ Sub-carrier mapping allows flexible allocation of signal to available sub-carriers
- ❖ IFFT and cyclic prefix (CP) insertion as in OFDM
- ❖ Each subcarrier carries a portion of superposed DFT spread data symbols
- ❖ Can also be seen as “**pre-coded OFDM**” or “**DFT-spread OFDM**”

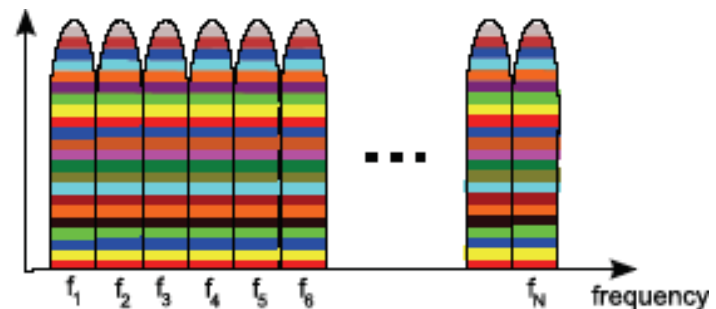
LTE Uplink:

How does the SC-FDMA signal look like?

- ❖ In principle similar to OFDMA, **BUT**:
 - ❖ In OFDMA, each sub-carrier only carries information related to **one specific symbol**
 - ❖ In SC-FDMA, each sub-carrier contains information of **ALL** transmitted symbols



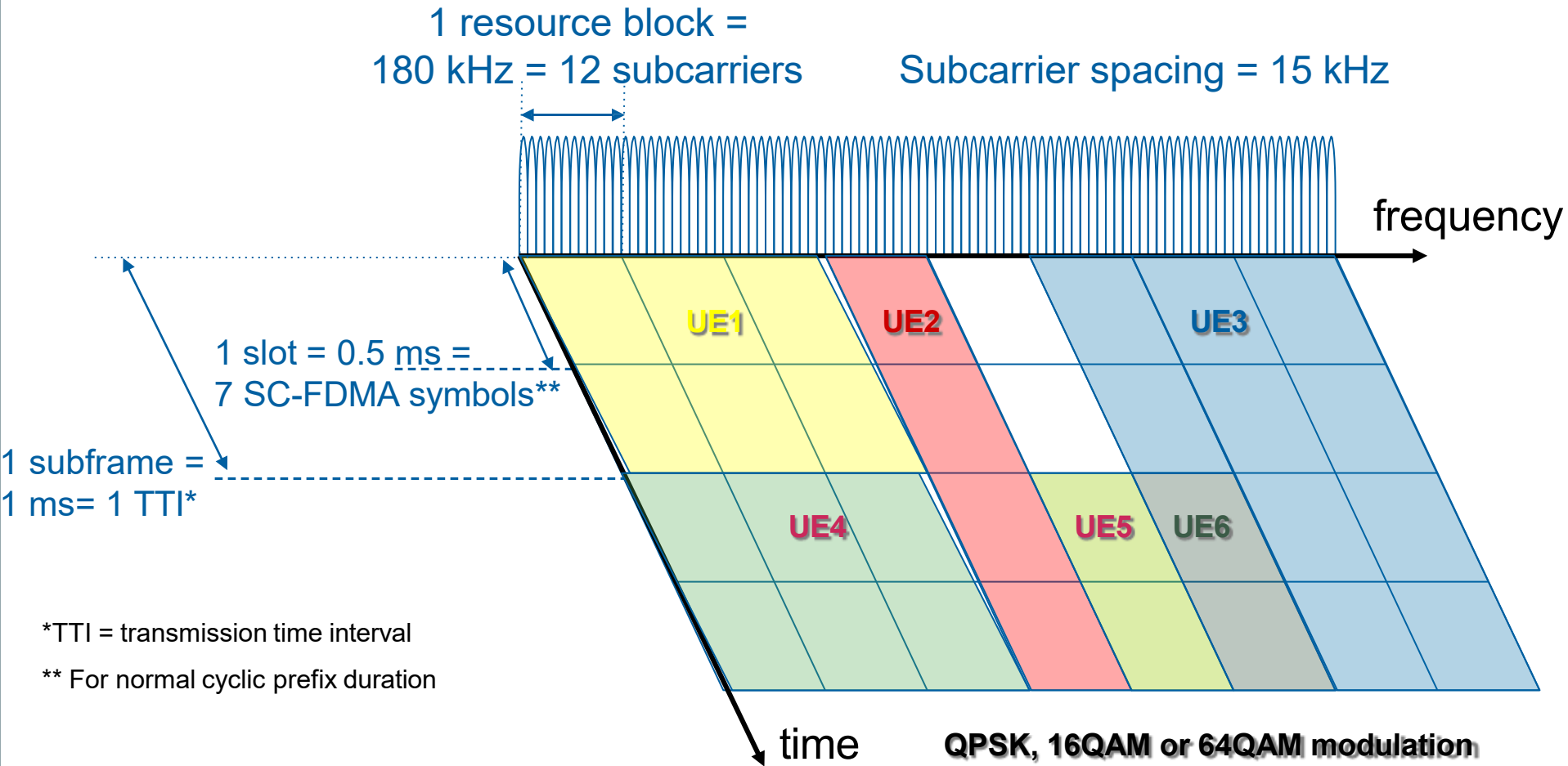
(a) OFDM subcarriers



(b) DFT-s-OFDM subcarriers

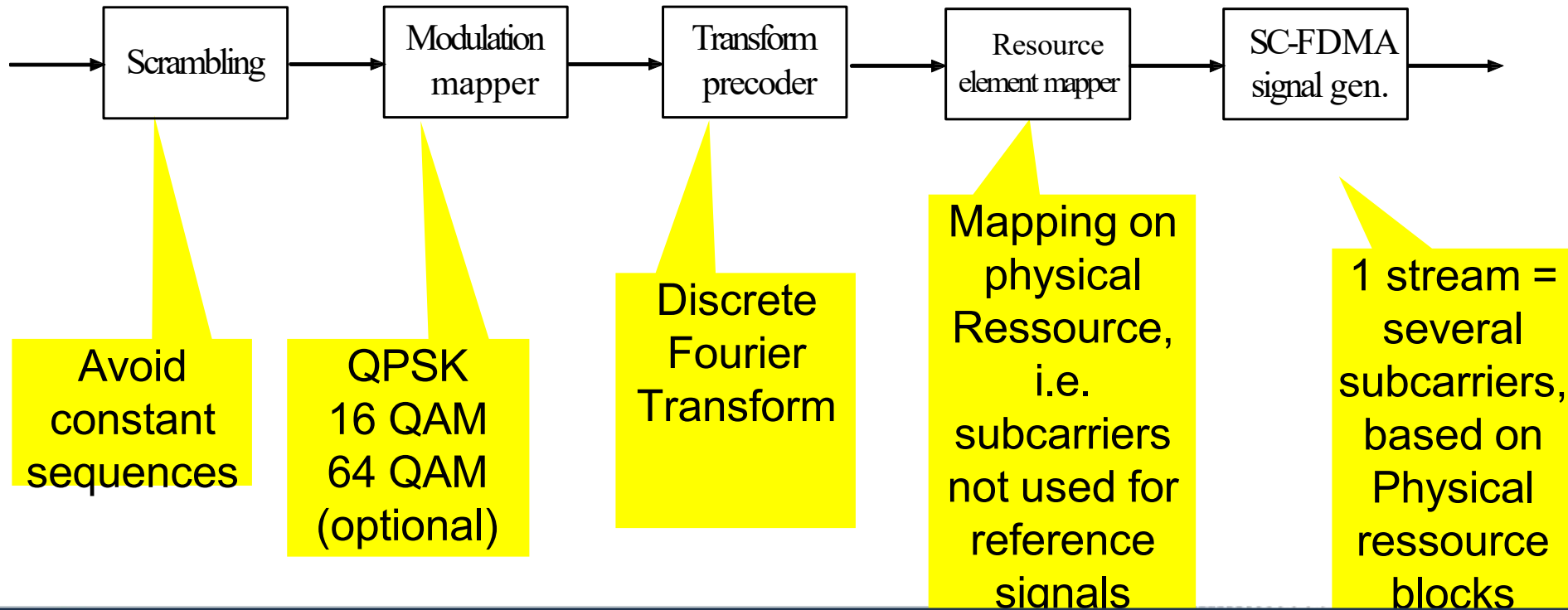
LTE uplink

SC-FDMA time-frequency multiplexing



LTE Uplink: baseband signal generation

UE specific
Scrambling code



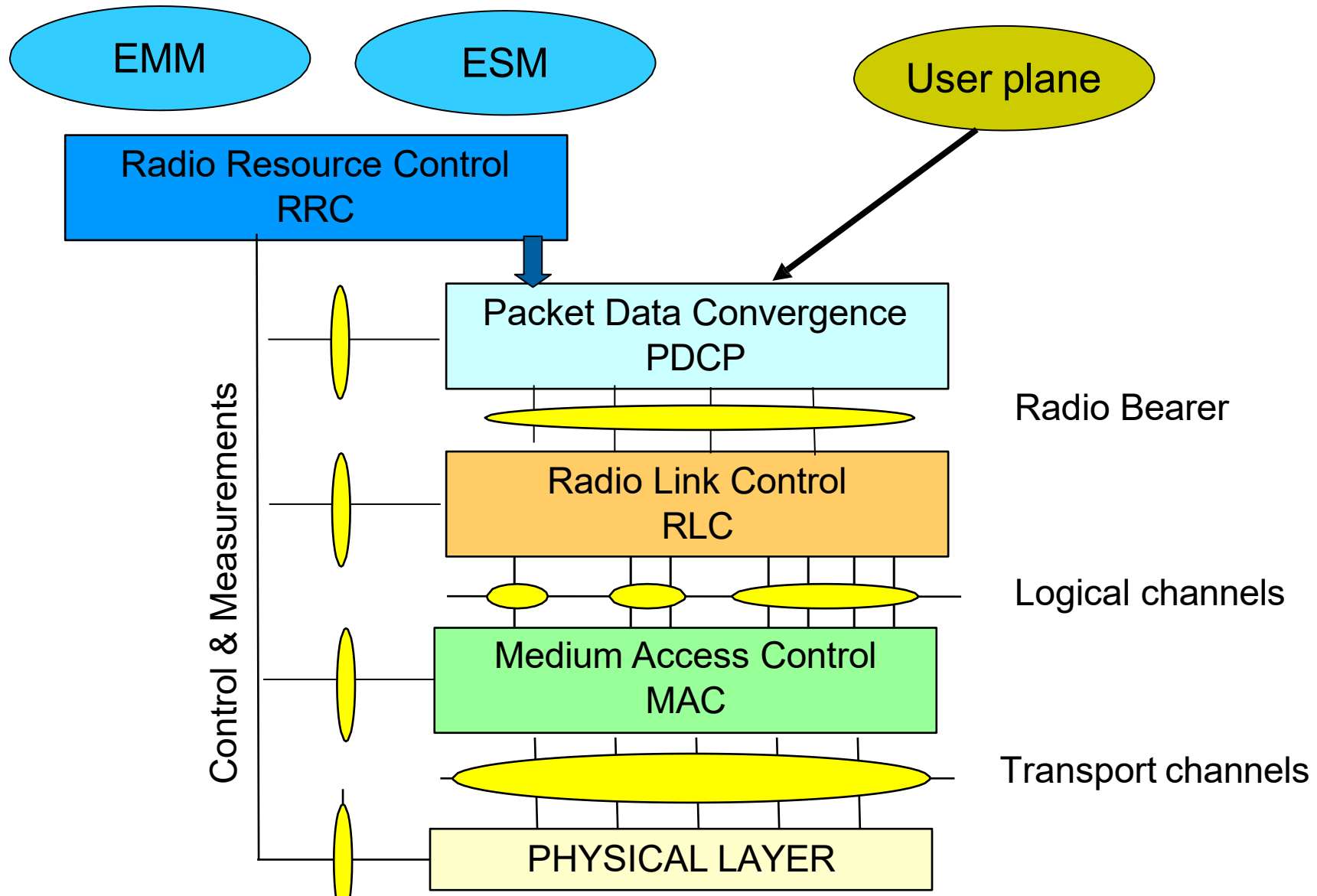
LTE Protocol Architecture

LTE Protocol Architecture

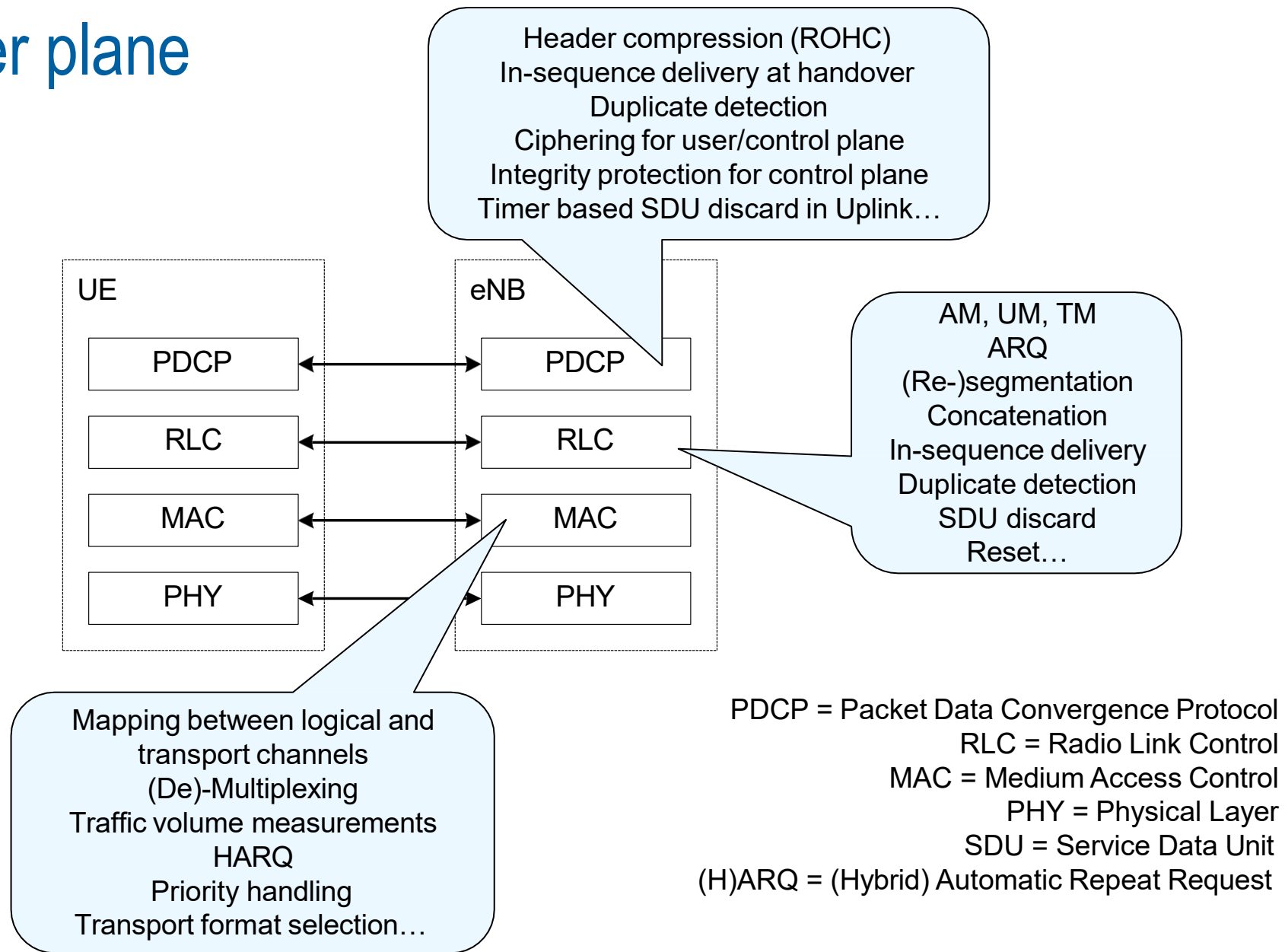
Reduced complexity

- Reduced number of transport channels
- Shared channels instead of dedicated channels
- Reduction of Medium Access Control (MAC) entities
- Streamlined concepts for broadcast / multicast (MBMS)
- No inter eNodeB soft handover in downlink/uplink
- No compressed mode
- Reduction of RRC states

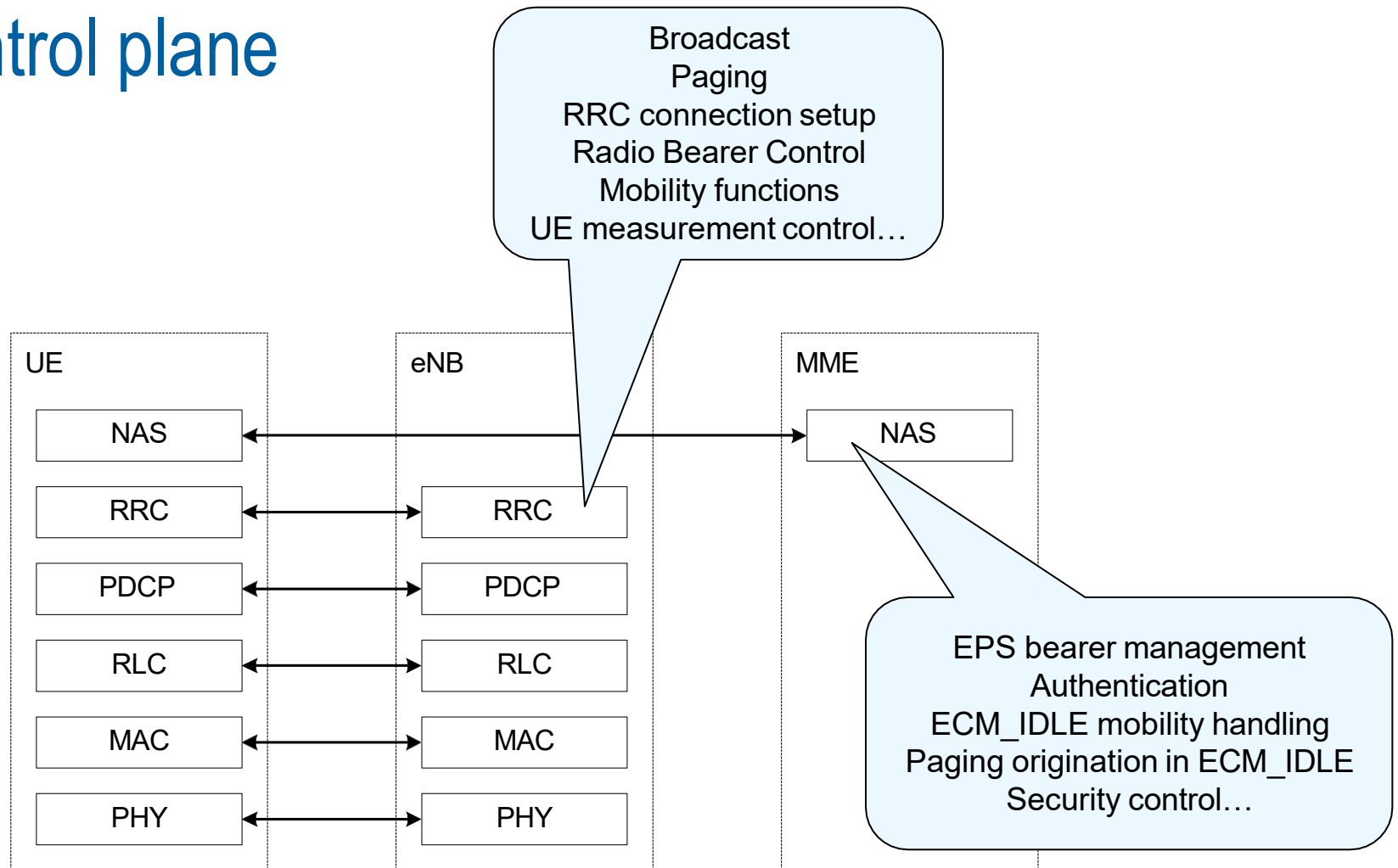
EUTRAN stack: protocol layers overview



User plane

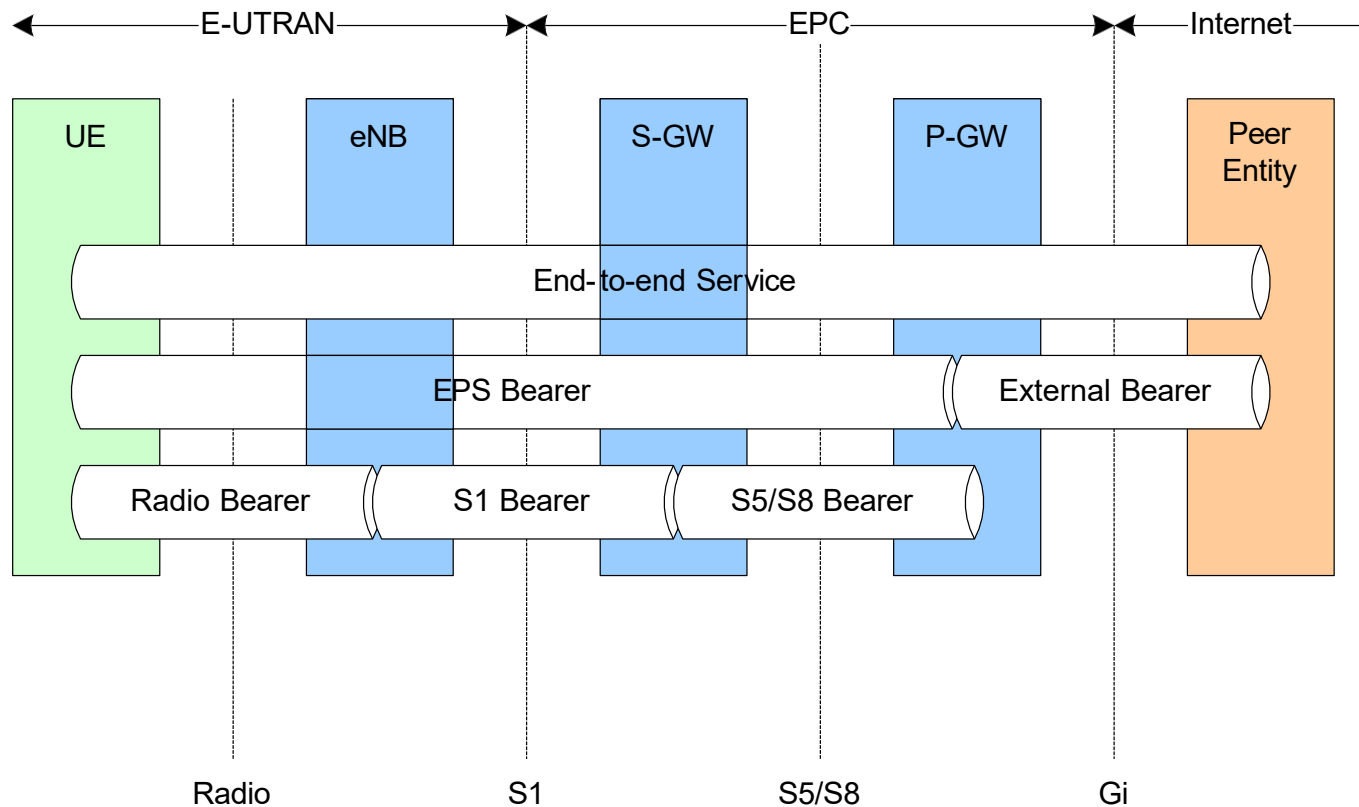


Control plane

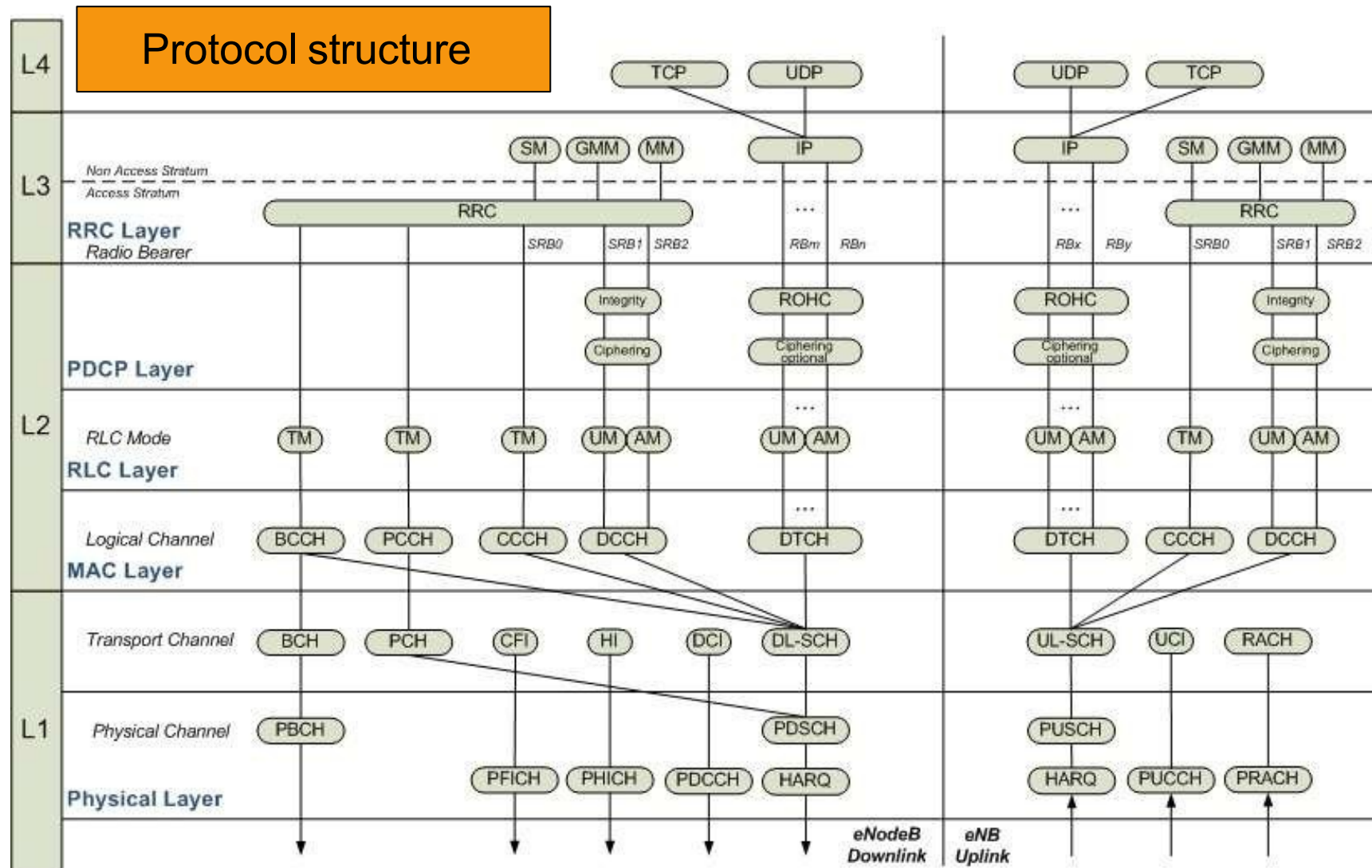


EPS = Evolved packet system
RRC = Radio Resource Control
NAS = Non Access Stratum
ECM = EPS Connection Management

EPS Bearer Service Architecture

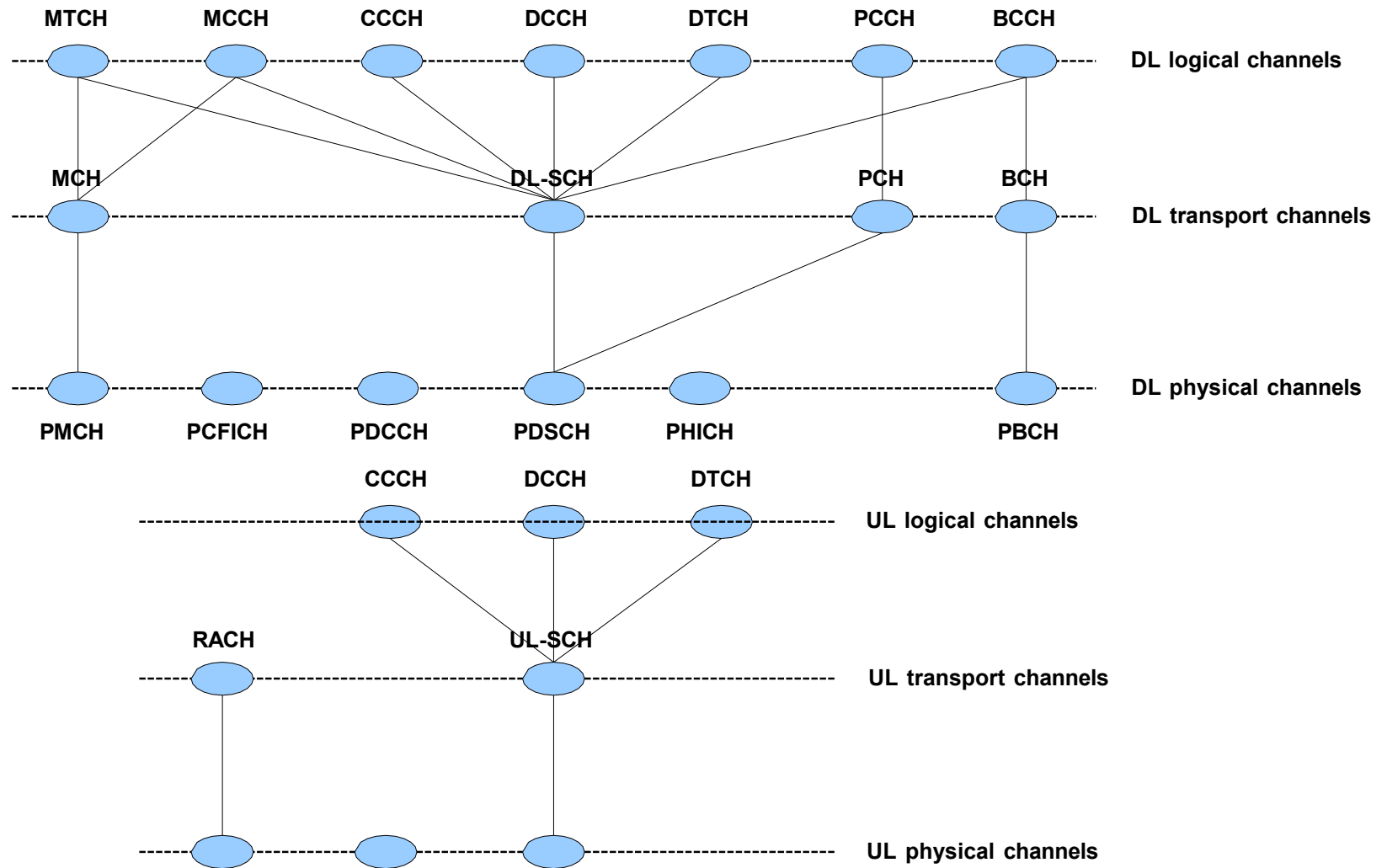


Channel structure: User + Control plane



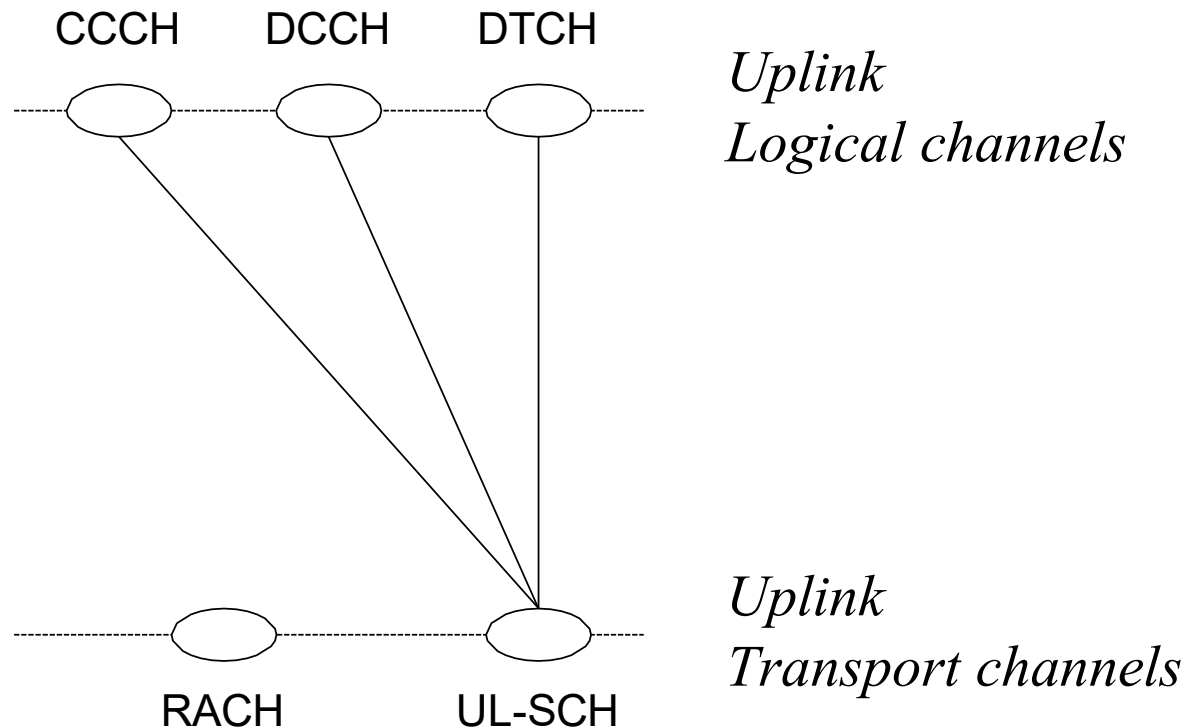
LTE channel mapping

LTE – channels



LTE – uplink channels

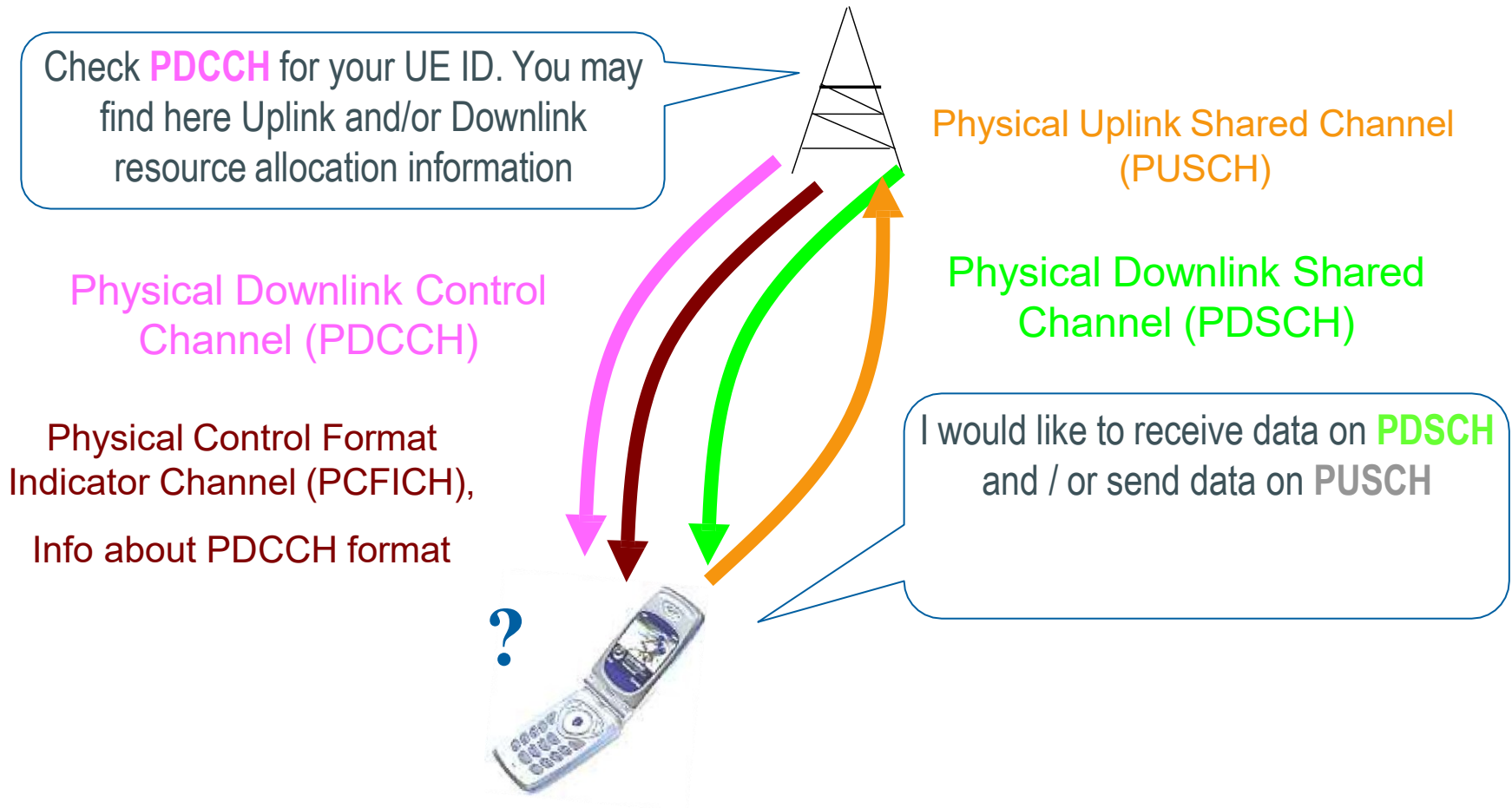
Mapping between logical and transport channels



LTE resource allocation principles

LTE resource allocation

Scheduling of **downlink** and uplink data

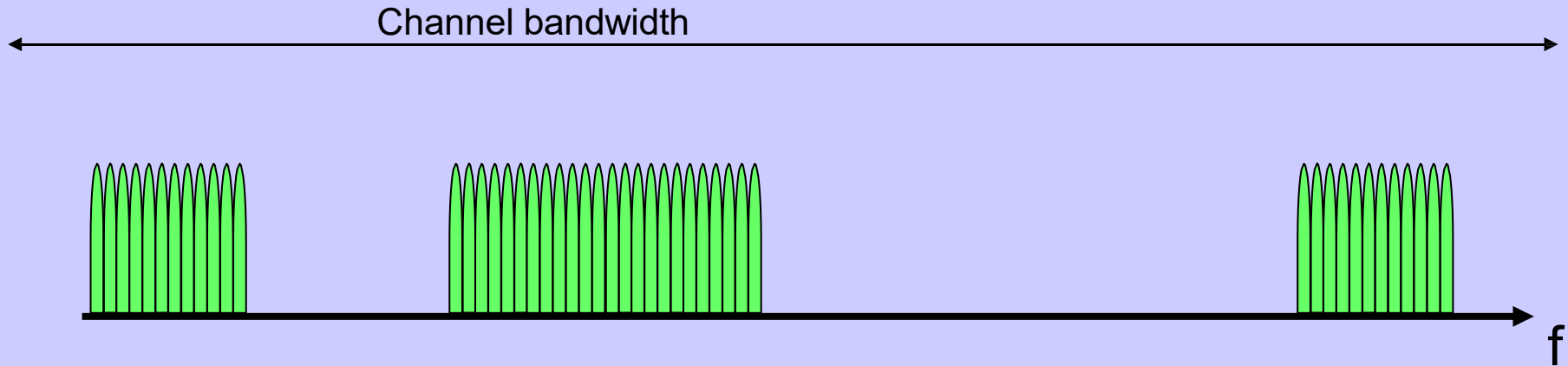


Resource allocation types in LTE

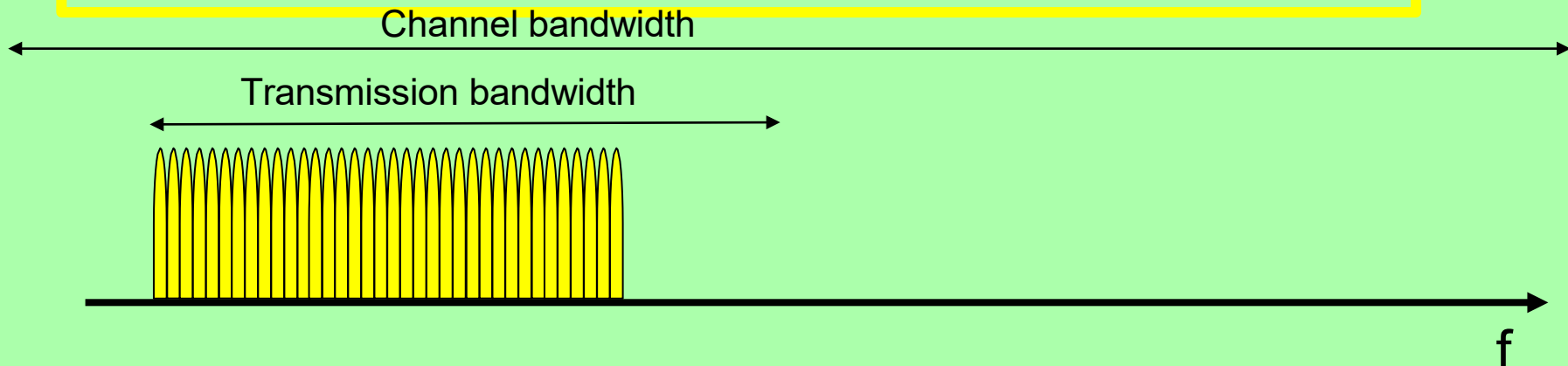
Allocation type	DCI Format	Scheduling Type	Antenna configuration
Type 0 / 1	DCI 1	PDSCH, one codeword	SISO, TxDiversity
	DCI 2A	PDSCH, two codewords	MIMO, open loop
	DCI 2	PDSCH, two codewords	MIMO, closed loop
Type 2	DCI 0	PUSCH	SISO
	DCI 1A	PDSCH, one codeword	SISO, TxDiversity
	DCI 1C	PDSCH, very compact codeword	SISO

Resource allocation types in LTE

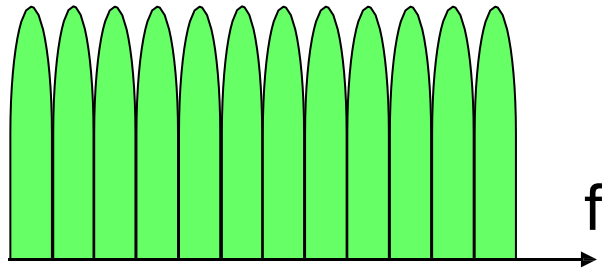
Type 0 and 1 for distributed allocation in frequency domain



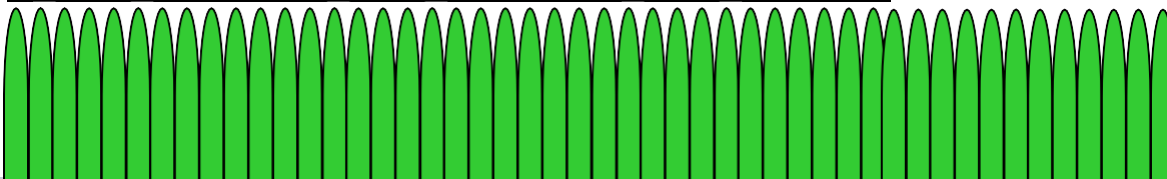
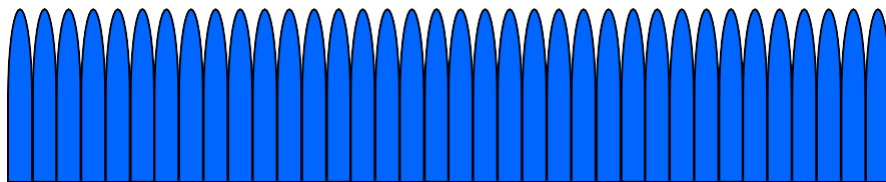
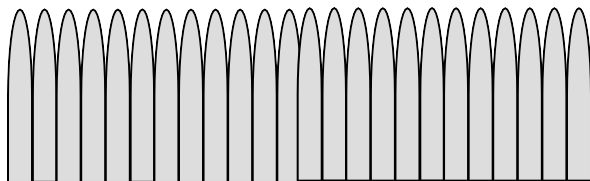
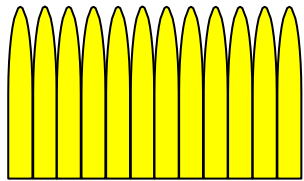
Type 2 for contiguous allocation in frequency domain



Resource Block Group



Reminder: 1 resource block =
12 subcarriers in frequency domain



Resource allocation is performed
based on **resource block groups**.
1 resource block group may consist of
1, 2, 3 or 4 resource blocks

Resource block groups,
RBG sizes

Resource allocation type 0

Type 0 (for distributed frequency allocation of Downlink resource, SISO and MIMO possible)

- I Bitmap to indicate which resource block groups, RBG are allocated**
- I One RBG consists of 1-4 resource blocks**

Channel bandwidth	RBG size P
≤10	1
11-26	2
27-63	3
64-110	4

- I Granularity is RBG size**

- I Number of resource block groups N_{RBG} is given as:**

$$N_{RBG} = \left\lceil N_{RB}^{DL} / P \right\rceil$$

- I Allocation bitmap has same length than N_{RBG}**

Resource allocation type 0 example

Calculation example for type 0:

- Channel bandwidth = 10MHz
- > 50 resource blocks
- > Resource block group RBG size = 3
- > bitmap size = 17

if $N_{RB}^{DL} \bmod P > 0$ then one of the RBGs is of size $N_{RB}^{DL} - P \cdot \left\lfloor N_{RB}^{DL} / P \right\rfloor$

i.e. here $50 \bmod 3 = 16$, so the last resource block group has the size 2.

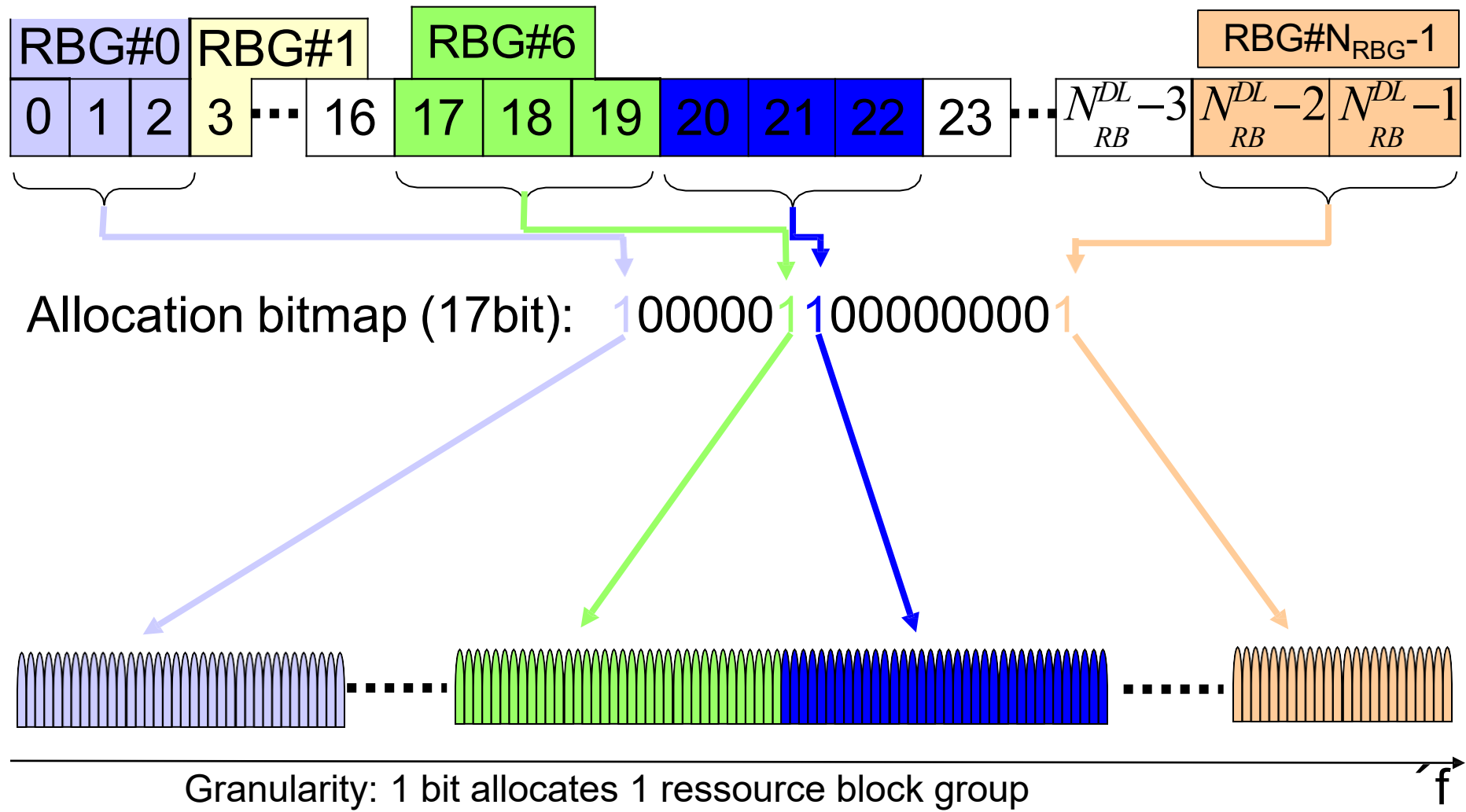
-> some allocations are not possible, e.g. here you can allocate 48 or 50 resource blocks, but not 49!

$$N_{RBG} = \left\lceil N_{RB}^{DL} / P \right\rceil = \text{round up, i.e. } \left\lceil 3.5 \right\rceil = 4 \quad \text{remainder}$$

$$\left\lfloor N_{RB}^{DL} / P \right\rfloor = \text{round down, i.e. } \left\lfloor 3.49 \right\rfloor = 3$$

Resource allocation type 0: example

Channel bandwidth = 10MHz -> 50 RBs -> RBG size = 3 -> number of RBGs = 17



Resource allocation type 1

Type 1 (for distributed frequency allocation of Downlink resource, SISO and MIMO possible)

One Resource Block Group consists of 1-4 resource blocks:

I RBs are divided into $\lceil \log_2(P) \rceil$ RBG subsets

I Granularity is resource block

Channel bandwidth	RBG size P
≤10	1
11-26	2
27-63	3
64-110	4

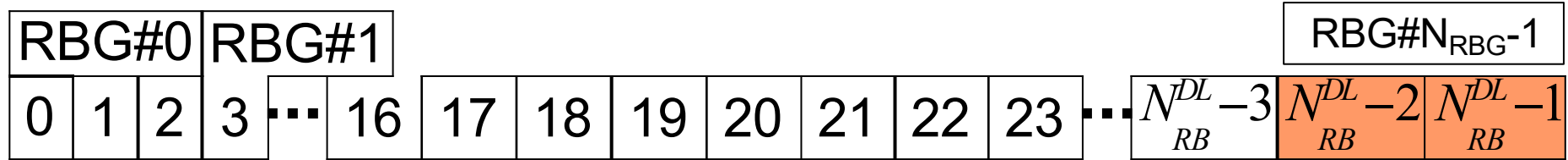
I Bitmap indicates RBs inside a RBG subset allocated to the UE

I Resource block assignment consists of 3 fields:

- I** Field to indicate the selected RBG
- I** Field to indicate a shift of the resource allocation
- I** Field to indicate the specific RB within a RBG subset

Resource allocation type 1

Channel bandwidth = 10MHz -> 50 RBs -> RBG size = 3 -> number of RBGs = 17



Size 18 resource blocks



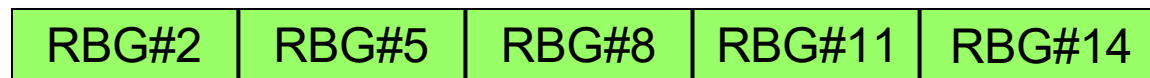
RBG subset #0

Size 17 resource blocks



RBG subset #1

Size 15 resource blocks



RBG subset #2

P= Number of RBG subsets with length:

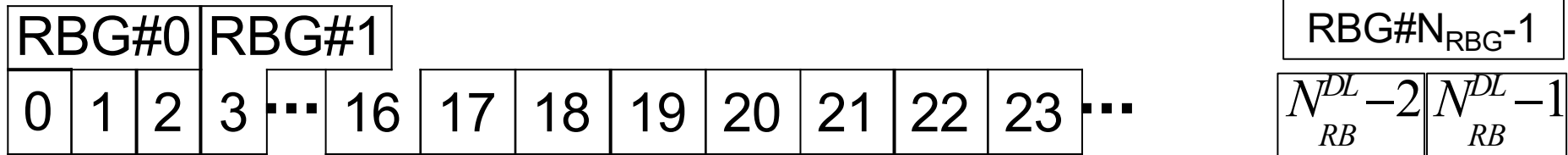
$$\lceil \log_2(P) \rceil$$

Number of bits to indicate subset

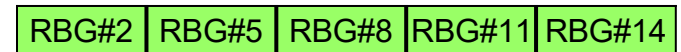
$$N_{RB}^{RBG \text{ subset}}(p) = \begin{cases} \left\lfloor \frac{N_{RB}^{DL} - 1}{P^2} \right\rfloor \cdot P + P & , p < \left\lfloor \frac{N_{RB}^{DL} - 1}{P} \right\rfloor \bmod P \\ \left\lfloor \frac{N_{RB}^{DL} - 1}{P^2} \right\rfloor \cdot P + (N_{RB}^{DL} - 1) \bmod P + 1 & , p = \left\lfloor \frac{N_{RB}^{DL} - 1}{P} \right\rfloor \bmod P \\ \left\lfloor \frac{N_{RB}^{DL} - 1}{P^2} \right\rfloor \cdot P & , p > \left\lfloor \frac{N_{RB}^{DL} - 1}{P} \right\rfloor \bmod P \end{cases}$$

Resource allocation type 1

Channel bandwidth = 10MHz -> 50 RBs -> RBG size = 3 -> number of RBGs = 17



RBG subset #0



RBG subset #2

Size 17 resource blocks



RBG subset #1



Resource blocks assignment

Field 1: RBG subset selection

Field 2: offset shift indication

Field 3: resource block allocation

Allocation bitmap (17bit): 010 000110000000001

RBG subset#1 is selected

Size 14 bits

Bit = 0, no shift

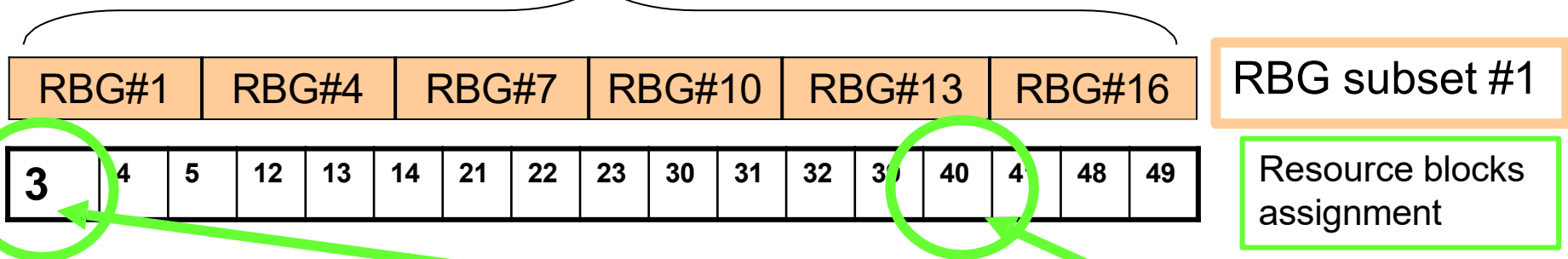
Resource allocation type 1

Channel bandwidth = 10MHz -> 50 RBs -> RBG size = 3 -> number of RBGs = 17

The meaning of the shift offset bit:

Number of resource blocks in one RBG subset is bigger than the allocation bitmap
-> you can not allocate all the available resource blocks
-> offset shift to indicate which RBs are assigned

17 resource blocks belonging to the RBG subset#1



Allocation bitmap (17bit):

010100000000000001

RBG subset#1 is selected

Bit = 0, no shift

14 bits in allocation bitmap

Resource allocation type 1

Channel bandwidth = 10MHz -> 50 RBs -> RBG size = 3 -> number of RBGs = 17

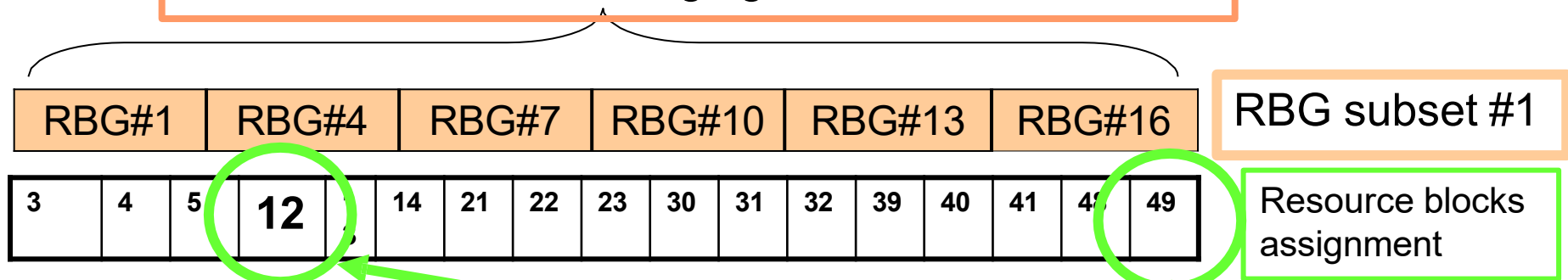
The meaning of the shift offset bit:

Number of resource blocks in one RBG subset is bigger than the allocation bitmap

-> you can not allocate all the available resource blocks

-> offset shift to indicate which RBs are assigned

17 resource blocks belonging to the RBG subset #1



Allocation bitmap (17bit):

011100000000000001

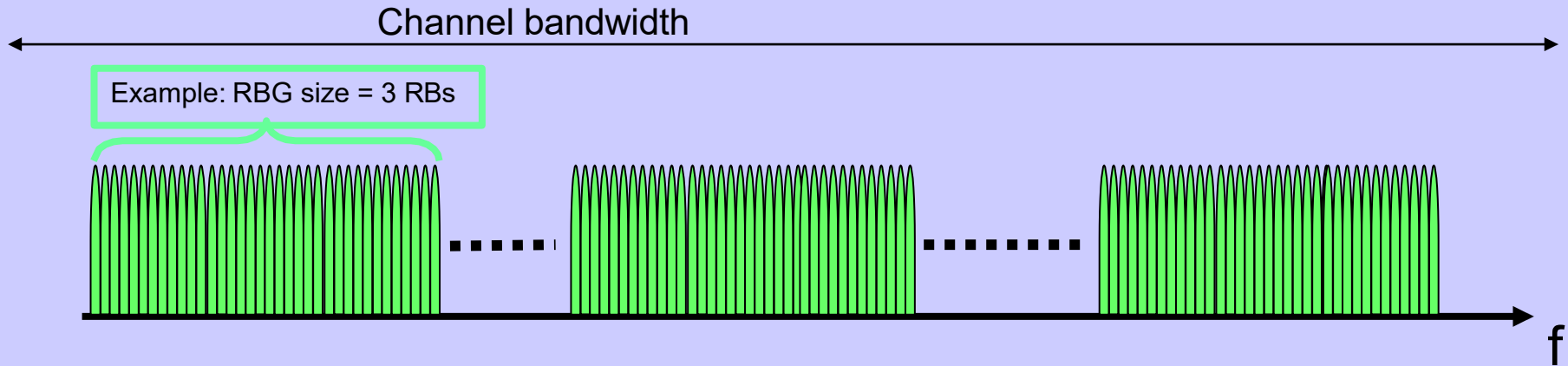
RBG subset#1 is selected

Bit = 1, offset shift

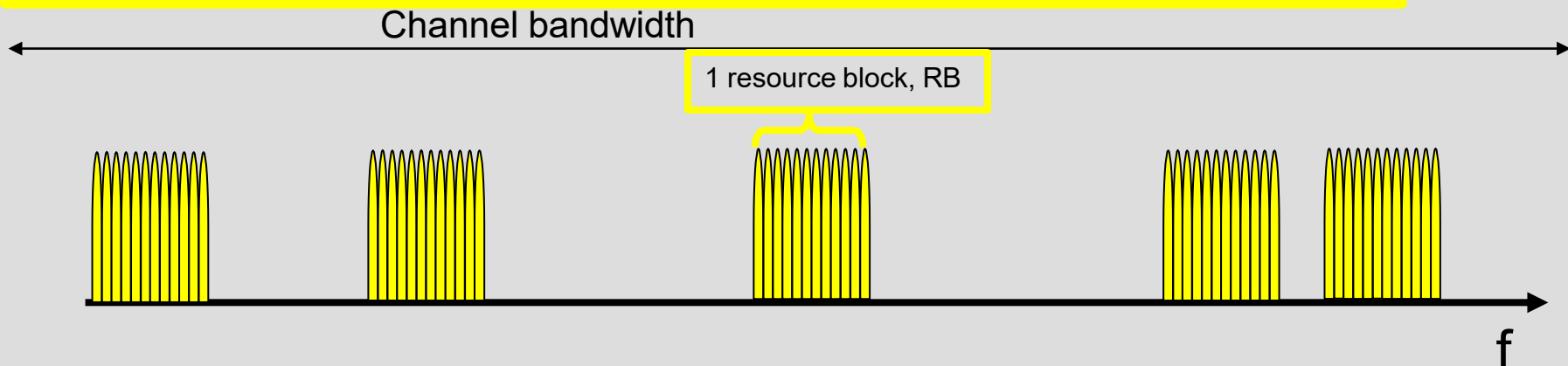
14 bits in allocation bitmap

Resource allocation types in LTE

Type 0 allows the allocation based on **resource block groups** granularity

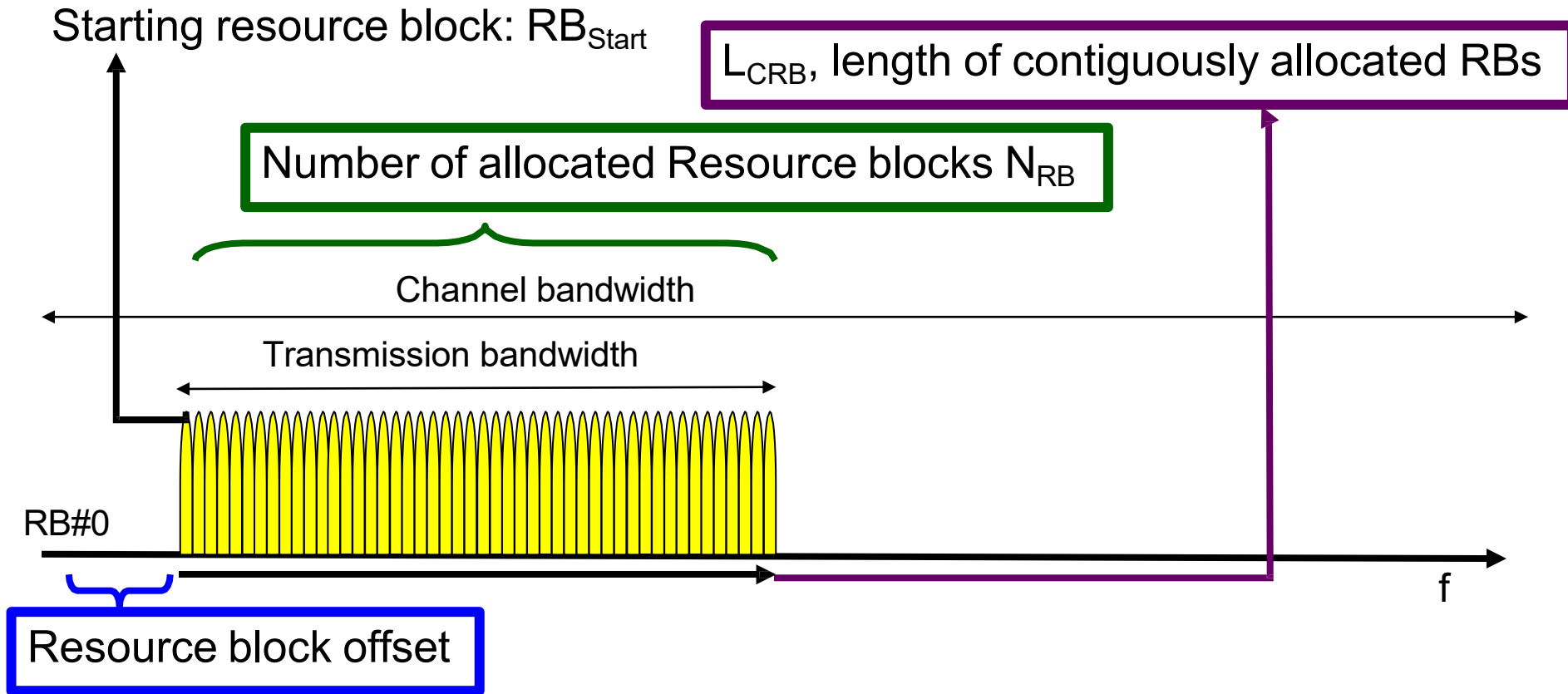


Type 1 allows the allocation based on **resource block** granularity



Resource allocation type 2

Localized mode



Resource indication value, RIV

Type 0: bitmap used to indicate the resource allocation:

$\lceil N_{RB}^{DL} / P \rceil$ = Length of allocation bitmap (17bit): 10000011000000001

RIV = bin to dec conversion

Type 1: bitmap used to indicate the resource allocation, with 3 fields:
Resource block group subset, shift indicator + resource allocation

Allocation bitmap (17bit): 01110000000000000

Type 2: TS 36.213 section 7.1.6.3. gives formula to calculate RIV:

if $(L_{CRBs} - 1) \leq \lfloor N_{RB}^{DL} / 2 \rfloor$

then

$$RIV = N_{RB}^{DL} (L_{CRBs} - 1) + RB_{start}$$

else

$$RIV = N_{RB}^{DL} (N_{RB}^{DL} - L_{CRBs} + 1) + (N_{RB}^{DL} - 1 - RB_{start})$$

Resource indication value, RIV in type 2 allocation

How to calculate the RIV value in allocation type 2, according to TS 36.213

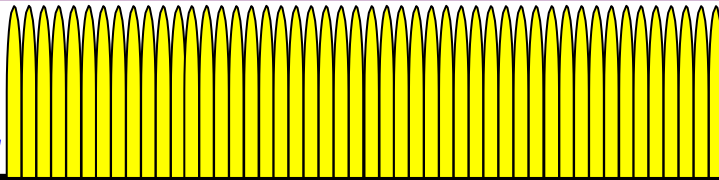
Starting resource block: $RB_{start}=5$

L_{CRB} , length of contiguously allocated RBs=20

Assumptions and given:

Localized mode, $RB_{start} = 5$, $L_{CRB} = 20$

$N_{RB}^{DL} = 50$



$$RIV = N_{RB}^{DL} (L_{CRBs} - 1) + RB_{start}$$

Formula from TS
36.213

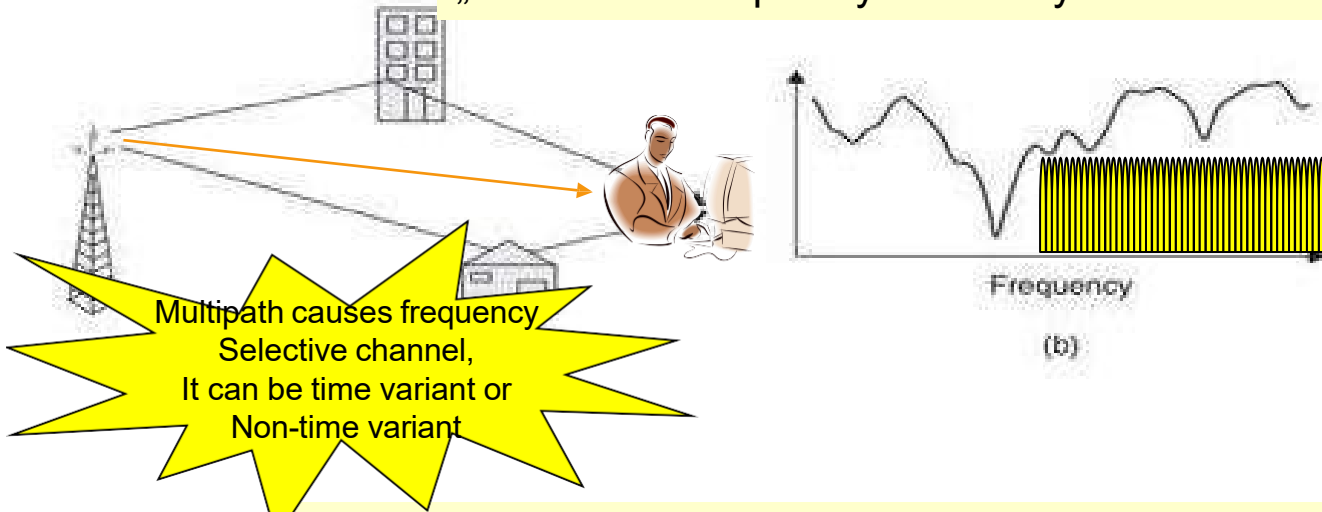
Here:

$$RIV = 50 * (20-1) + 5$$

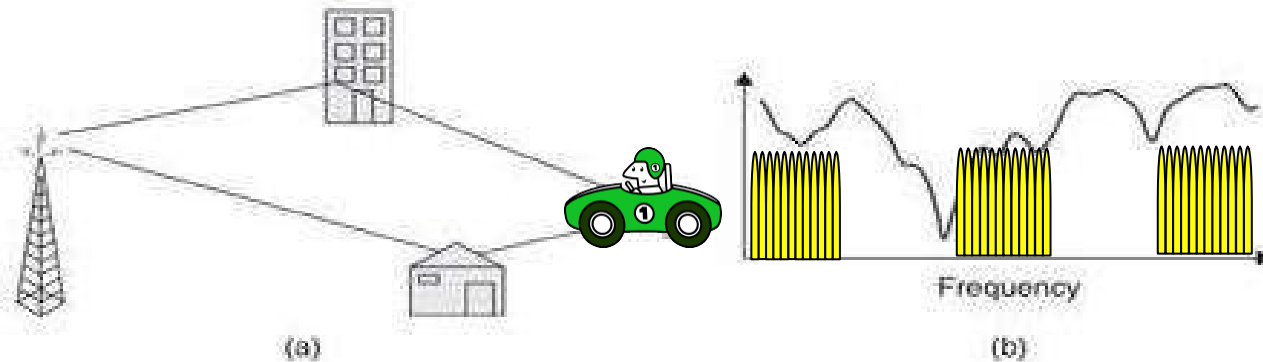
$$RIV = 955$$

Benefit of localized or distributed mode

„static UE“: frequency selectivity is not time variant -> localized allocation



„high velocity UE“: frequency selectivity is time variant -> distributed allocation



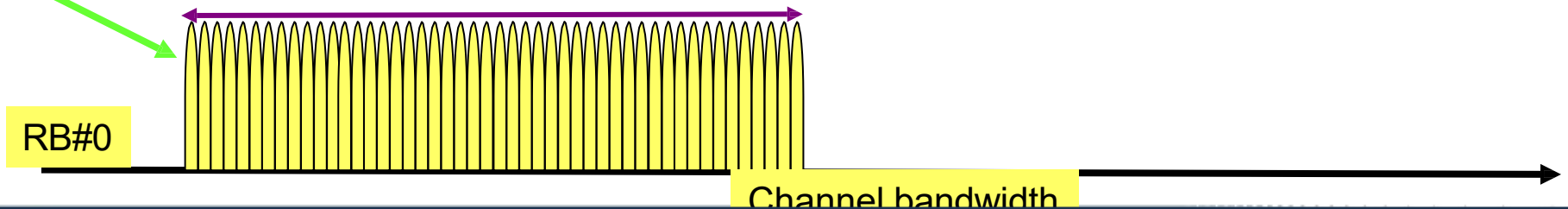
Resource allocation Uplink

Allocation type	DCI Format	Scheduling Type	Antenna configuration
Type 0 / 1	DCI 1	PDSCH, one codeword	SISO, TxDiversity
	DCI 2A	PDSCH, two codewords	MIMO , open loop
	DCI 2	PDSCH, two codewords	MIMO , closed loop
Type 2	DCI 0	PUSCH	SISO
	DCI 1A	PDSCH, one codeword	SISO, TxDiversity
	DCI 1C	PDSCH, very compact codeword	SISO

LTE Uplink uses Type 2 allocation

Starting resource block: RB_{Start}

L_{CRB} , length of contiguously allocated RBs



LTE Uplink: allocation of UL ressource

Scheduled UL
bandwidth

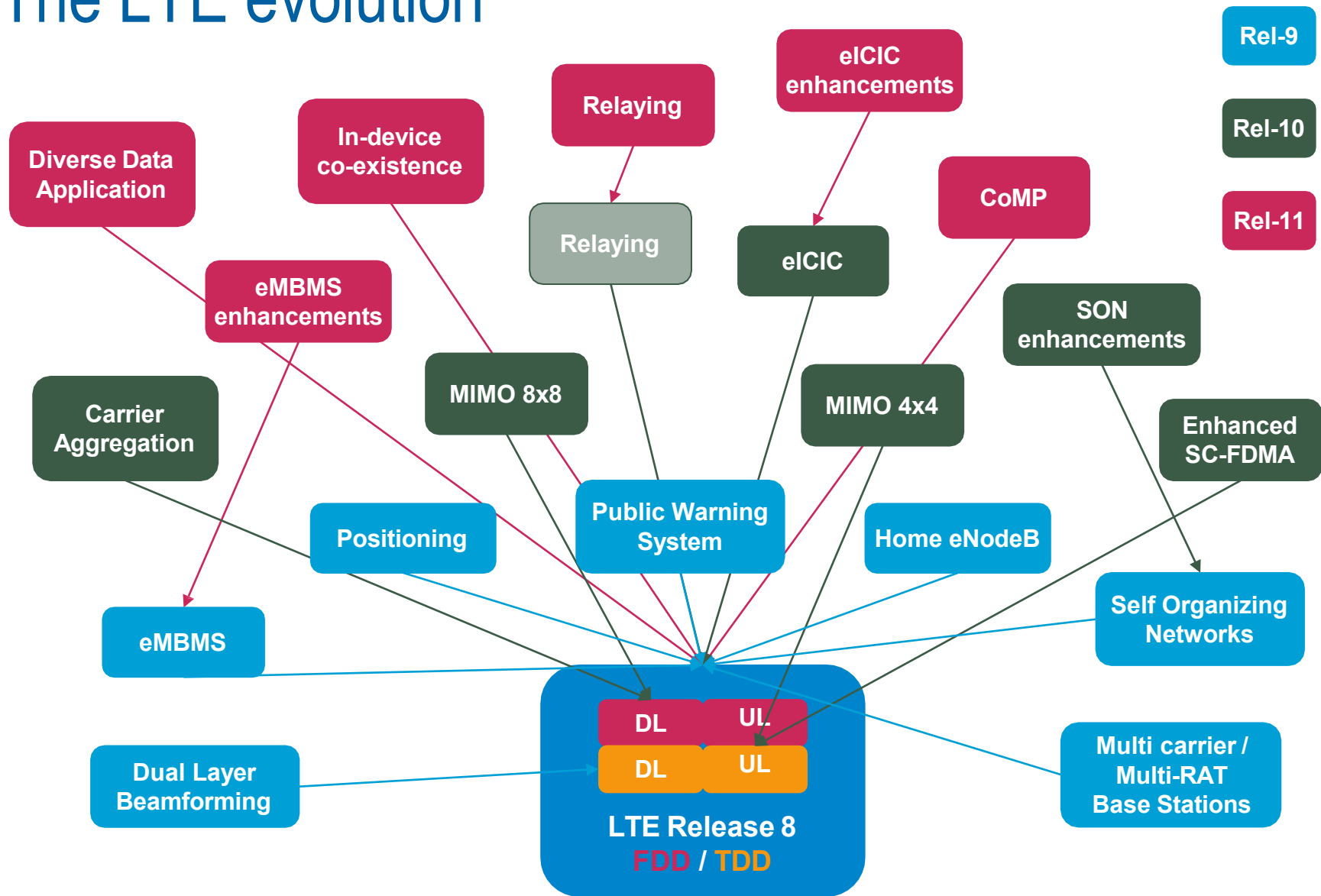
UL bandwidth
configuration

$$M_{\text{RB}}^{\text{PUSCH}} = 2^{\alpha_2} \cdot 3^{\alpha_3} \cdot 5^{\alpha_5} \leq N_{\text{RB}}^{\text{UL}}$$

Scheduled number of ressource blocks in UL must fullfill formula above(α_x are integer). Possible values are:

1	2	3	4	5	6	8	9	10	12
15	16	18	20	24	25	27	30	32	36
40	45	48	50	54	60	64	72	75	80
81	90	96	100						

The LTE evolution



What are antenna ports?

I 3GPP TS 36.211(Downlink)

“An antenna port is defined such that the channel over which a symbol on the antenna port is conveyed can be inferred from the channel over which another symbol on the same antenna port is conveyed.”



I What does that mean?

- I The UE shall demodulate a received signal – which is transmitted over a certain antenna port – based on the channel estimation performed on the reference signals belonging to this (same) antenna port.

What are antenna ports?

I **Consequences of the definition**

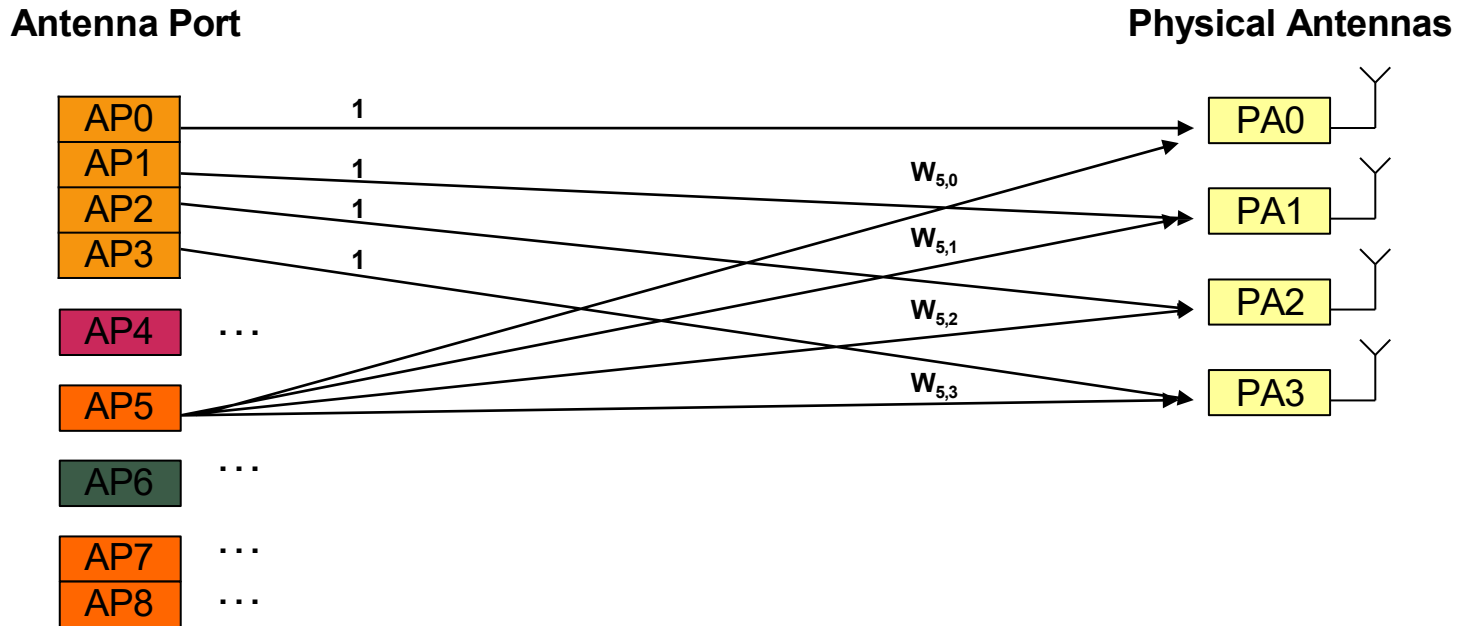
- I There is one sort of reference signal per antenna port
- I Whenever a new sort of reference signal is introduced by 3GPP (e.g. PRS), a new antenna port needs to be defined (e.g. Antenna Port 6)

I **3GPP defines the following antenna port / reference signal combinations for downlink transmission:**

- I Port 0-3: Cell-specific Reference Signals (CS-RS)
- I Port 4: MBSFN-RS
- I Port 5: UE-specific Reference Signals (DM-RS): single layer (TX mode 7)
- I Port 6: Positioning Reference Signals (PRS)
- I Port 7-8: UE-specific Reference Signals (DM-RS): dual layer (TX mode 8)
- I Port 7-14: UE specific Reference Signals for Rel. 10
- I Port 15 – 22: CSI specific reference signals, channel status info in Rel. 10

What are antenna ports?

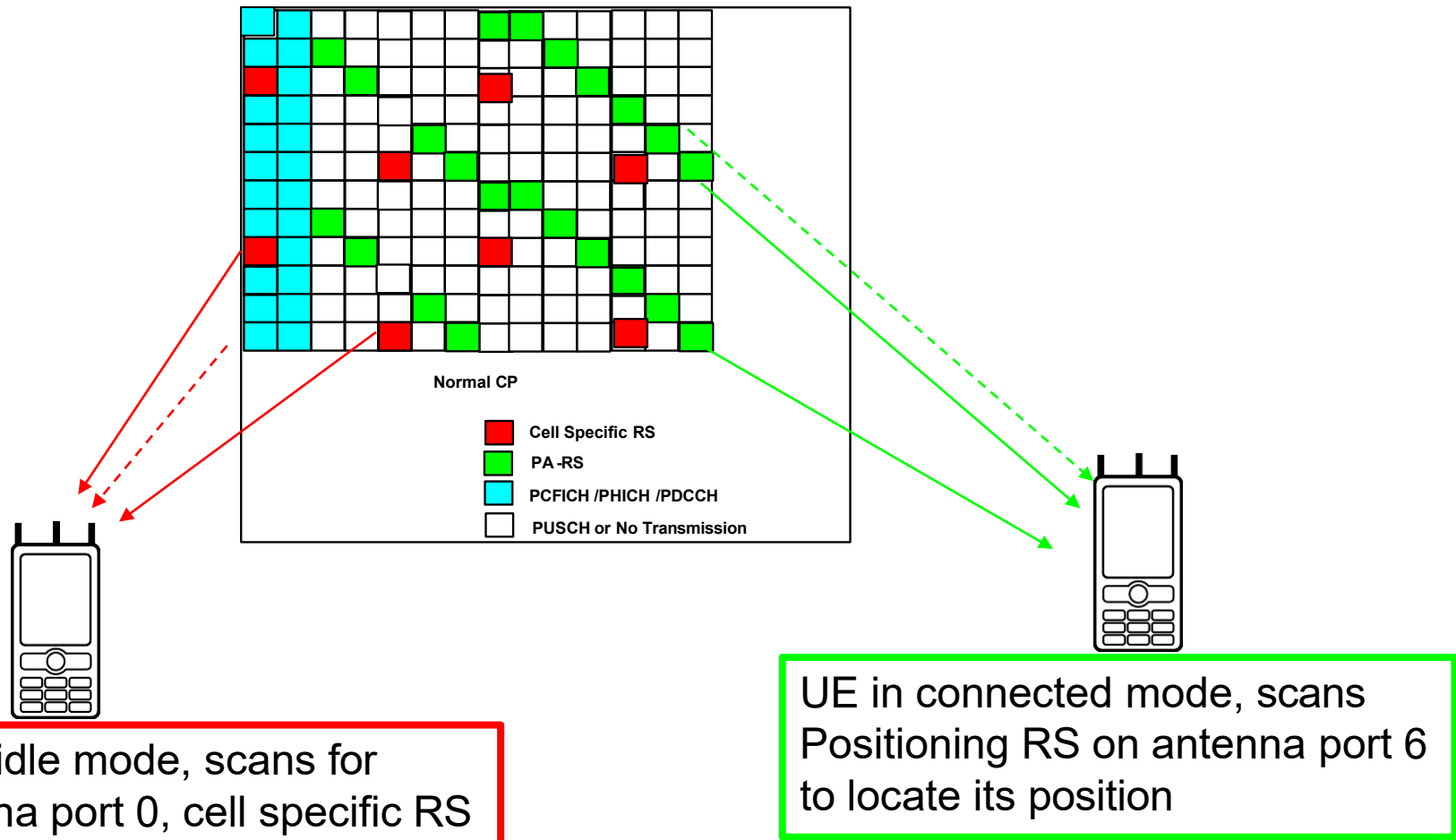
I Mapping „Antenna Port“ to „Physical Antennas“



The way the "logical" antenna ports are mapped to the "physical" TX antennas lies completely in the responsibility of the base station. There's no need for the base station to tell the UE.

LTE antenna port definition

Antenna ports are linked to the reference signals
-> one example:



eMBMS

Physical Layer Scenarios

I Dedicated and mixed mode.

- I Dedicated: carrier is only for MBMS = Single-cell MBMS.
- I MBMS/Unicast mixed mode: MBMS and user data are transmitted using time division duplex. Certain subframes carry MBMS data.

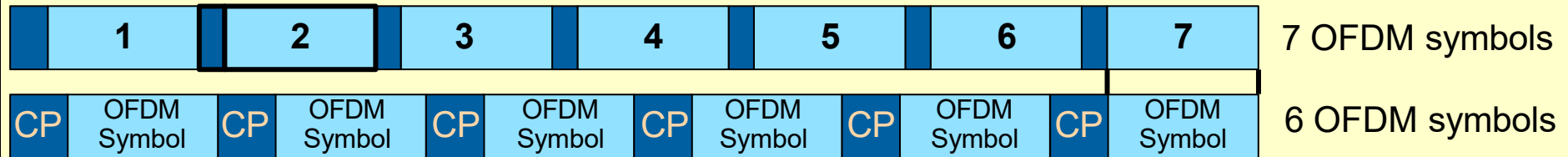
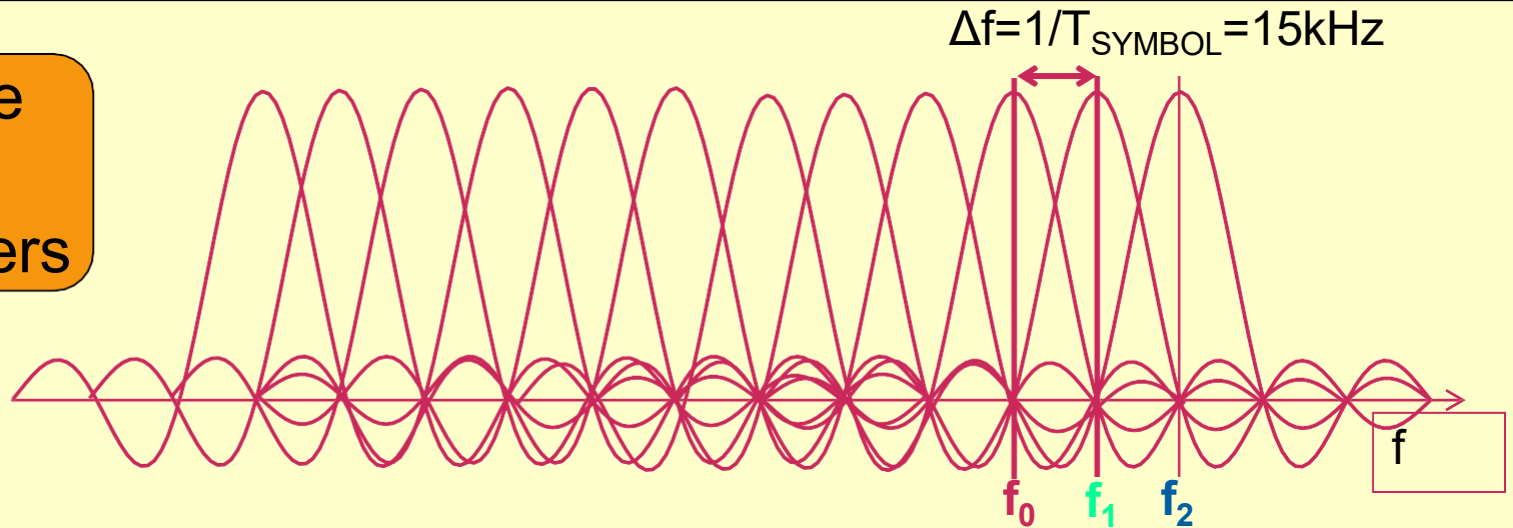
I Dedicated mode (single-cell scenario) offers use of new subcarrier spacing, longer cyclic prefix (CP), 3 OFDM symbols.

Configuration	OFDM Symbols	Sub-carrier	Cyclic Prefix Length in Samples	Cyclic Prefix Length in μs
Normal CP $\Delta f = 15 \text{ kHz}$	7	12	160 for 1 st symbol 144 for other symbols	5.2 for 1 st symbol 4.7 for other symbols
Extended CP $\Delta f = 15 \text{ kHz}$	6		512	16.7
Extended CP $\Delta f = 7.5 \text{ kHz}$	3	24	1024	33.3

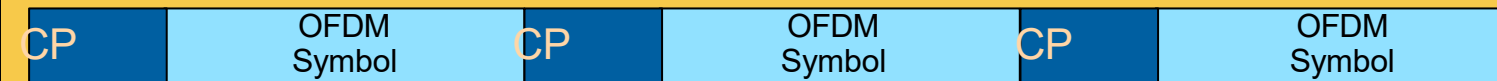
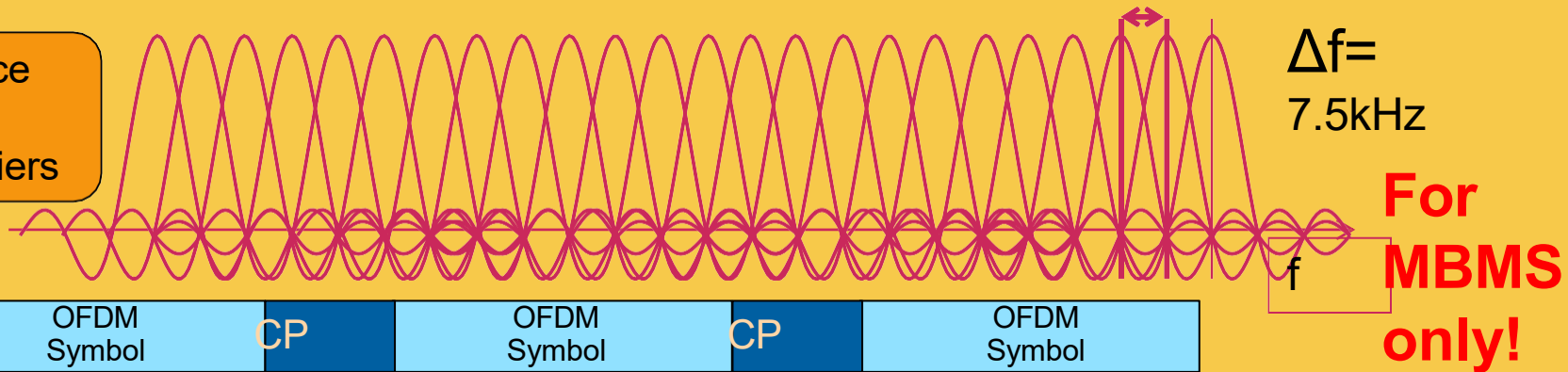
eMBMS
(Single cell scenario)

Resource block definition for MBMS

1 Resource Block = 12 subcarriers



1 Resource Block = 24 subcarriers



Multimedia Broadcast Messaging Services, MBMS



Broadcast:
Public info for
everybody

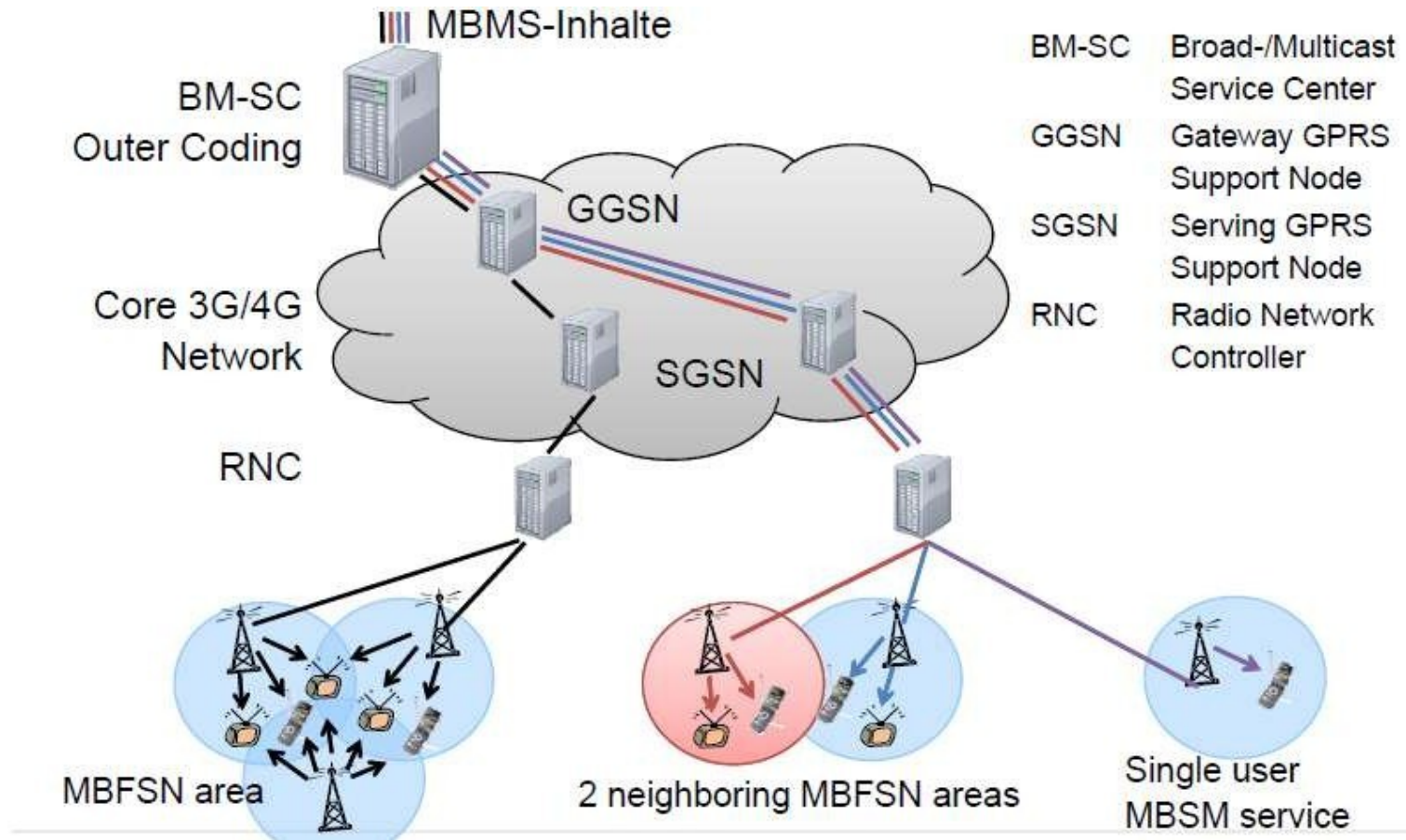


Unicast:
Private info for dedicated user

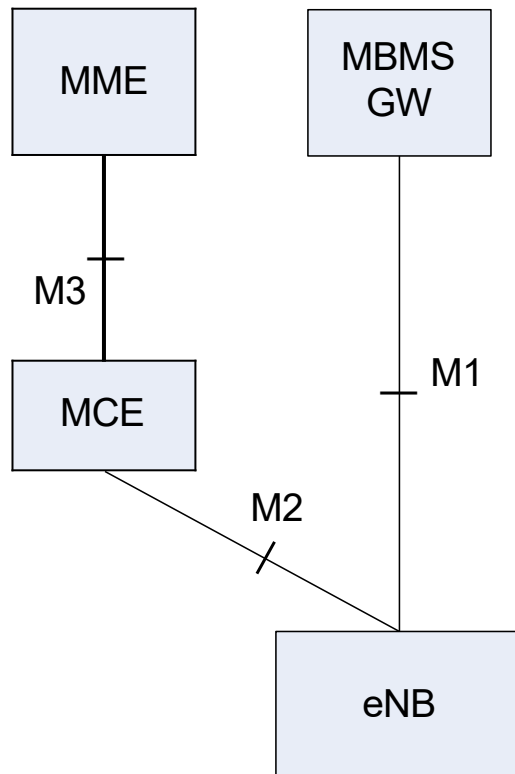
Multicast:
Common info for
User after authentication



LTE MBMS architecture



MBMS in LTE



MBMS GW: MBMS Gateway

MCE: Multi-Cell/Multicast Coordination Entity

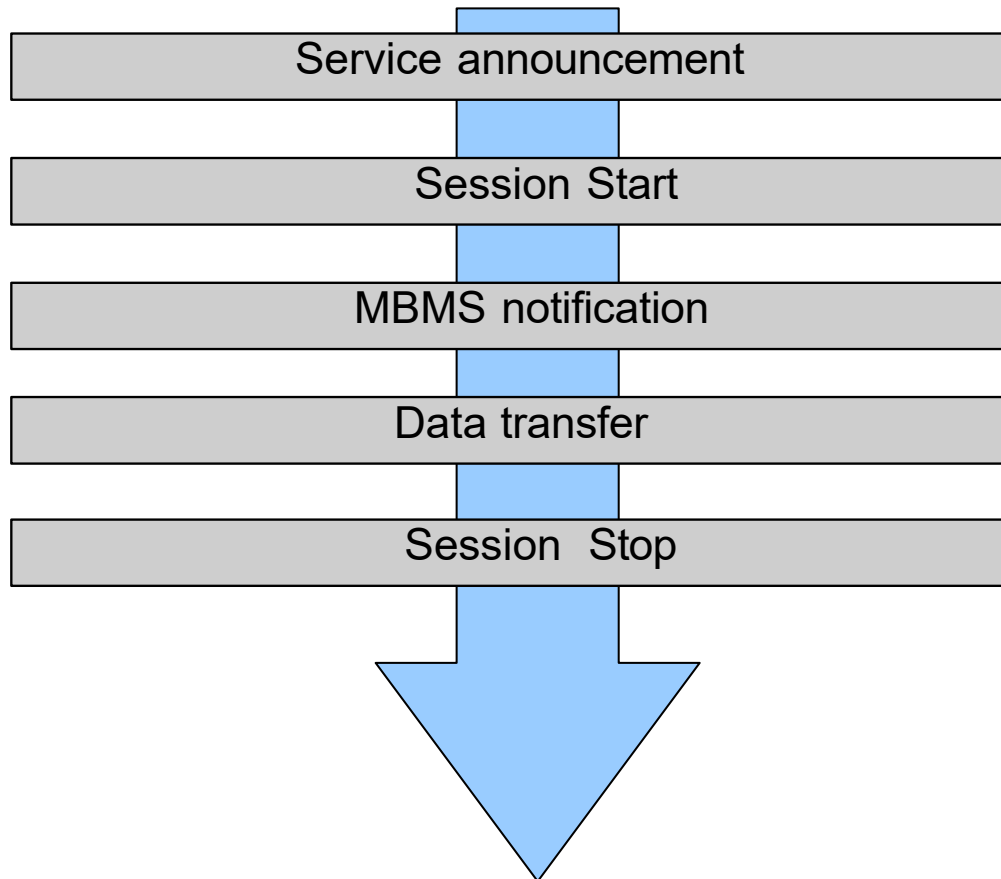
M1: user plane interface

M2: E-UTRAN internal control plane interface

M3: control plane interface between E-UTRAN and EPC

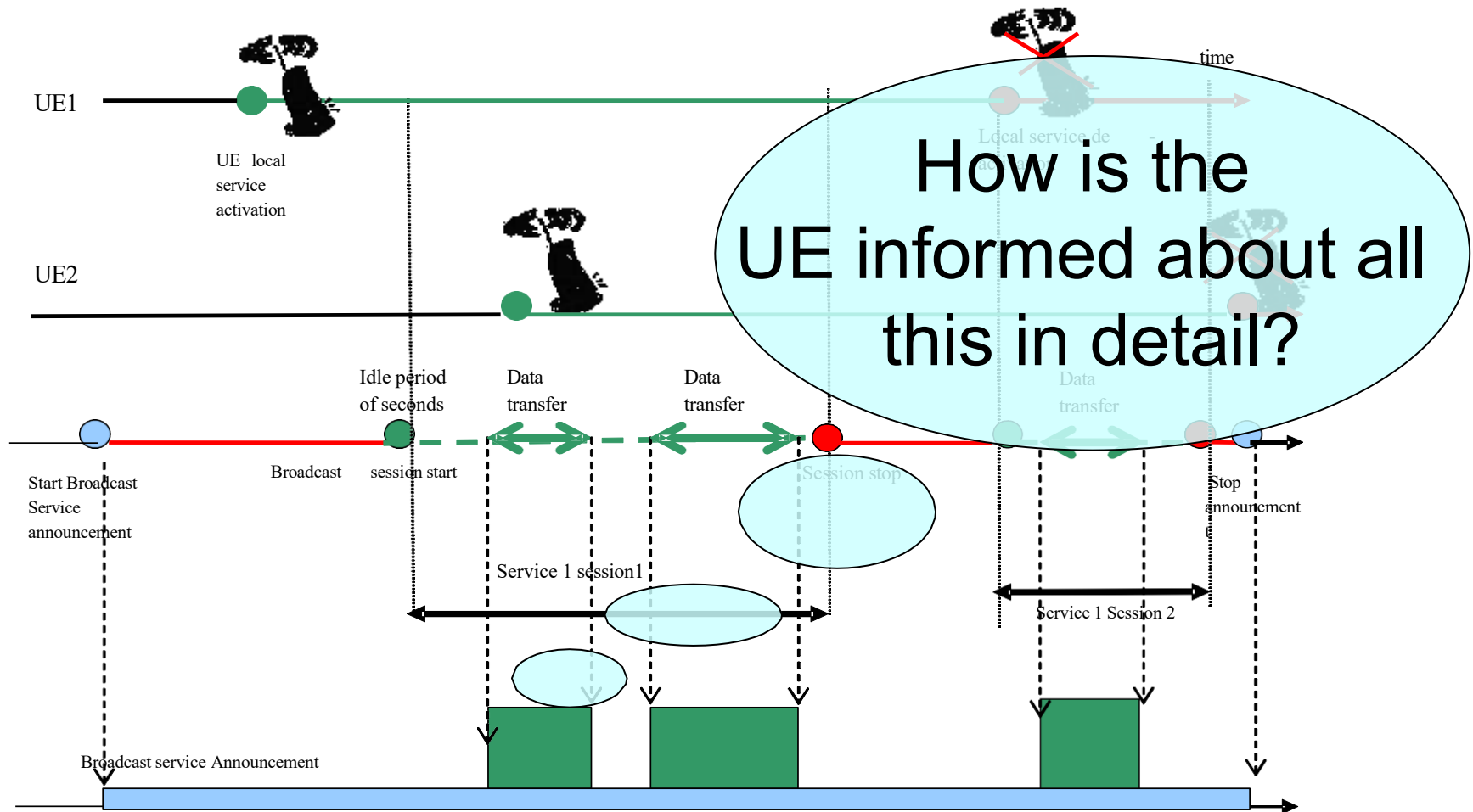
Logical architecture for MBMS

MBMS broadcast service provision



See 3GPP TS23.246, Section 4.4.3

MBMS broadcast service provision

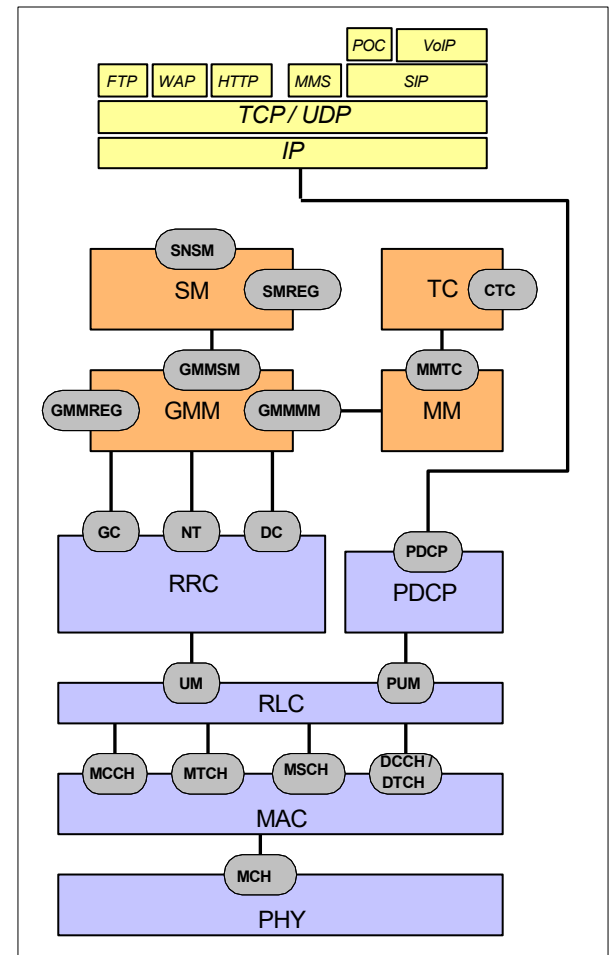


MBMS Signaling

I New logical Channels:

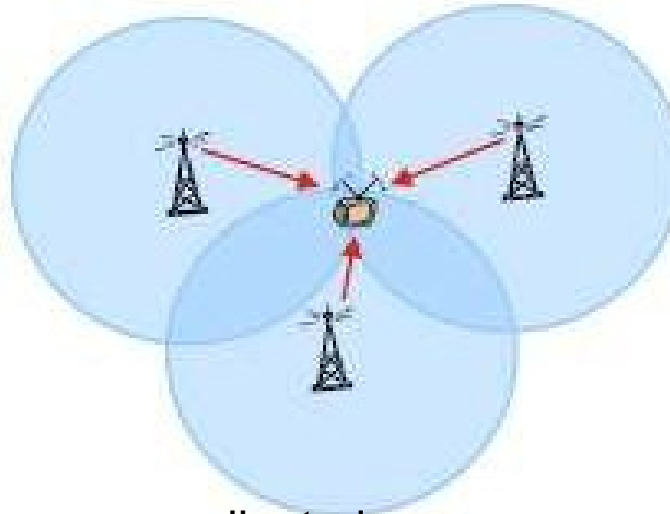
- I MBMS point-to-multipoint Control Channel (MCCH)
- I MBMS point-to-multipoint Traffic Channel (MTCH)
- I MBMS point-to-multipoint Scheduling Channel (MSCH)
- I Reception of MSCH is optional
- I MxCH channels are mapped on Multicast Channel MCH

New SIB 13 informs about MBMS configuration



MBSFN

MBSFN, Multicast/Broadcast Single Frequency Network

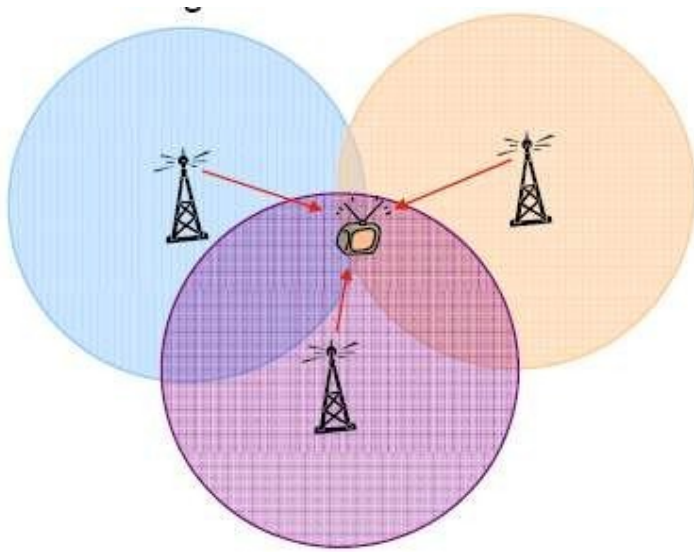


Cells belonging to MBSFN area are co-ordinated and transmit a time-synchronized common waveform
eNodeBs within an MBSFN area are synchronized.

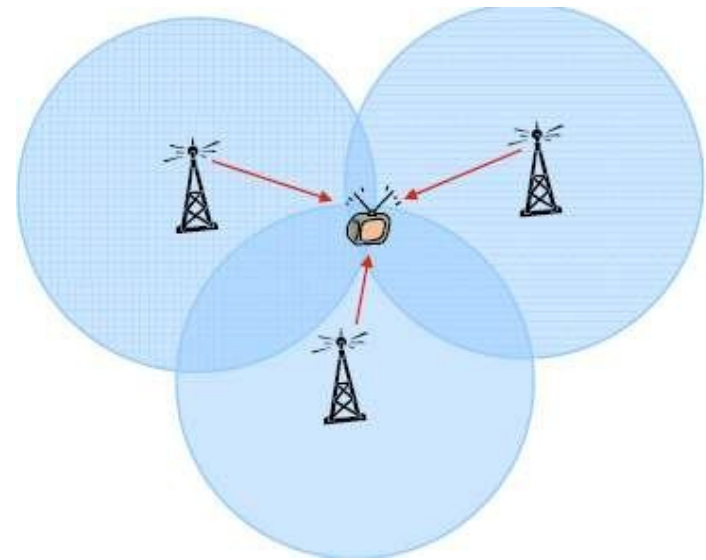
From point of view of the terminal, this appears to be a single transmission as if originating from one large cell (with correspondingly large delay spread).

Cyclic prefix is utilized to cover the difference in the propagation delays from the multiple cells. MBMS therefore uses an extended cyclic prefix

MBSFN – MBMS Single Frequency Network

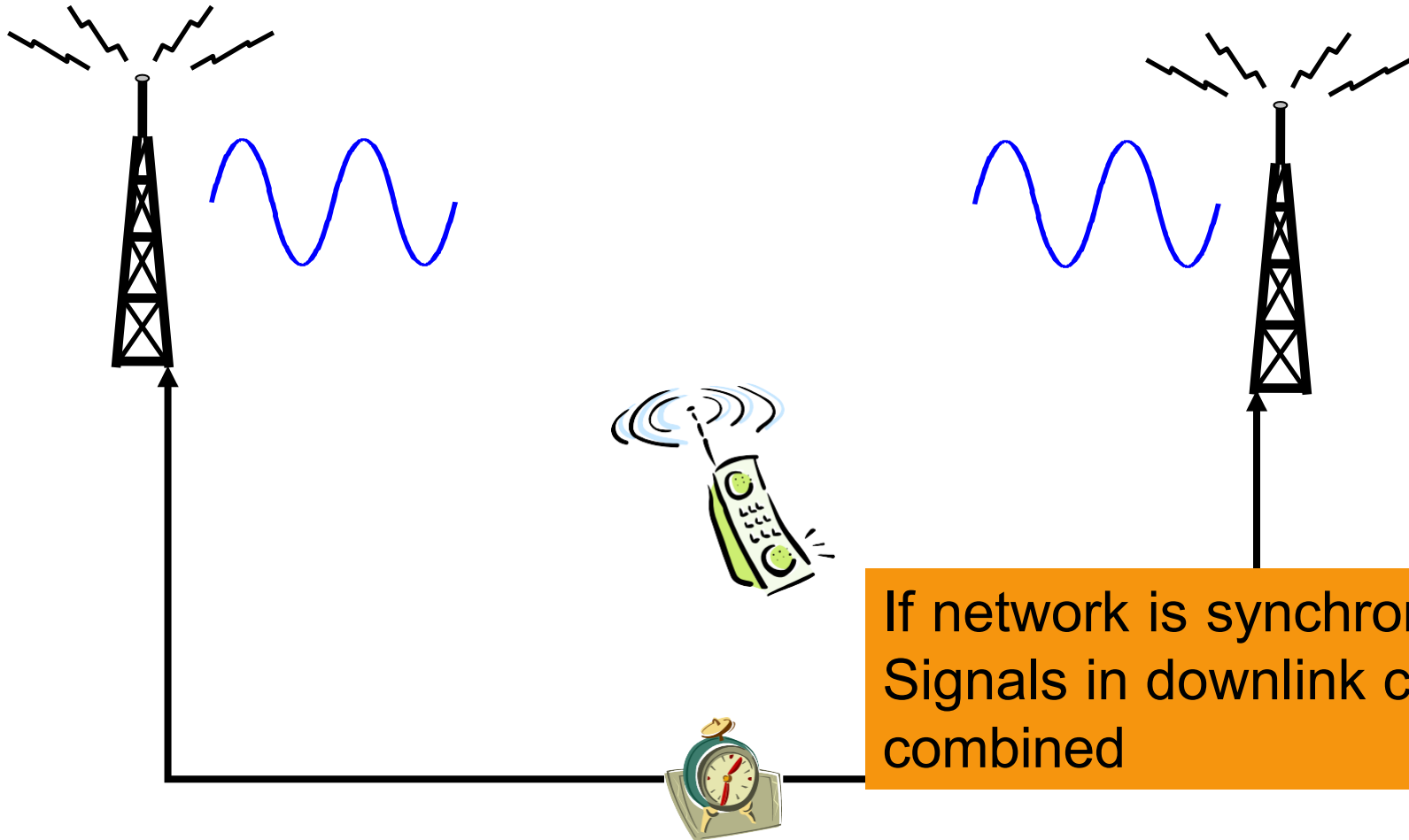


Mobile communication network
each eNode B sends individual
signals



Single Frequency Network
each eNode B sends identical
signals

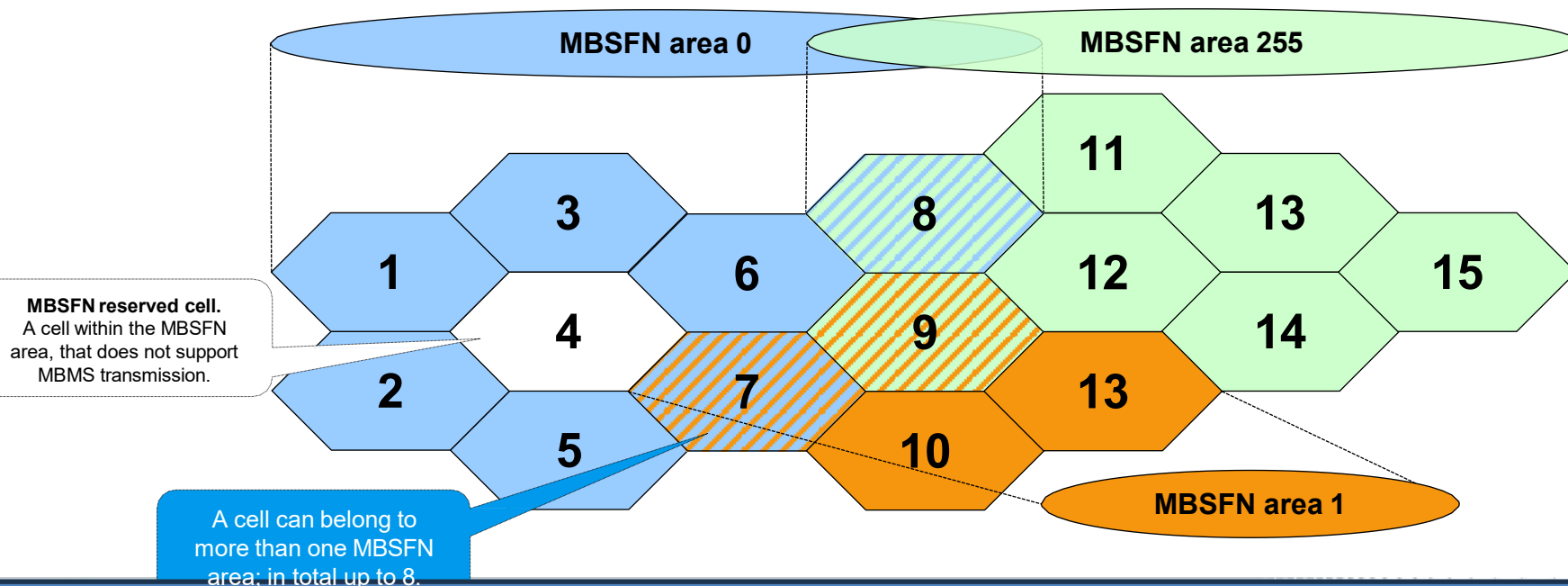
MBSFN



evolved Multimedia Broadcast Multicast Services

Multimedia Broadcast Single Frequency Network (MBSFN) area

- I Useful if a significant number of users want to consume the same data content at the same time in the same area!
- I Same content is transmitted in a specific area is known as MBSFN area.
 - I Each MBSFN area has an own identity (*mbsfn-AreaId* 0...255) and can consists of multiple cells; a cell can belong to more than one MBSFN area.
 - I MBSFN areas do not change dynamically over time.



eMBMS

Downlink Channels

I Downlink channels related to MBMS

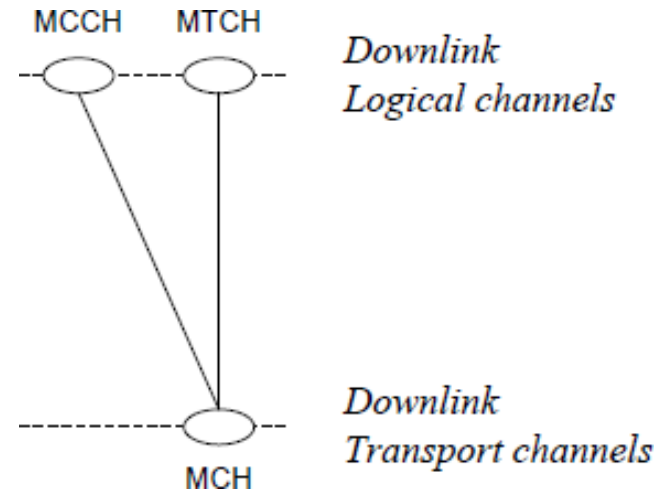
- I MCCH Multicast Control Channel
- I MTCH Multicast Traffic Channel
- I MCH Multicast Channel
- I PMCH Physical Multicast Channel

I MCH is transmitted over MBSFN in specific subframes on physical layer

I MCH is a downlink only channel (no HARQ, no RLC repetitions)

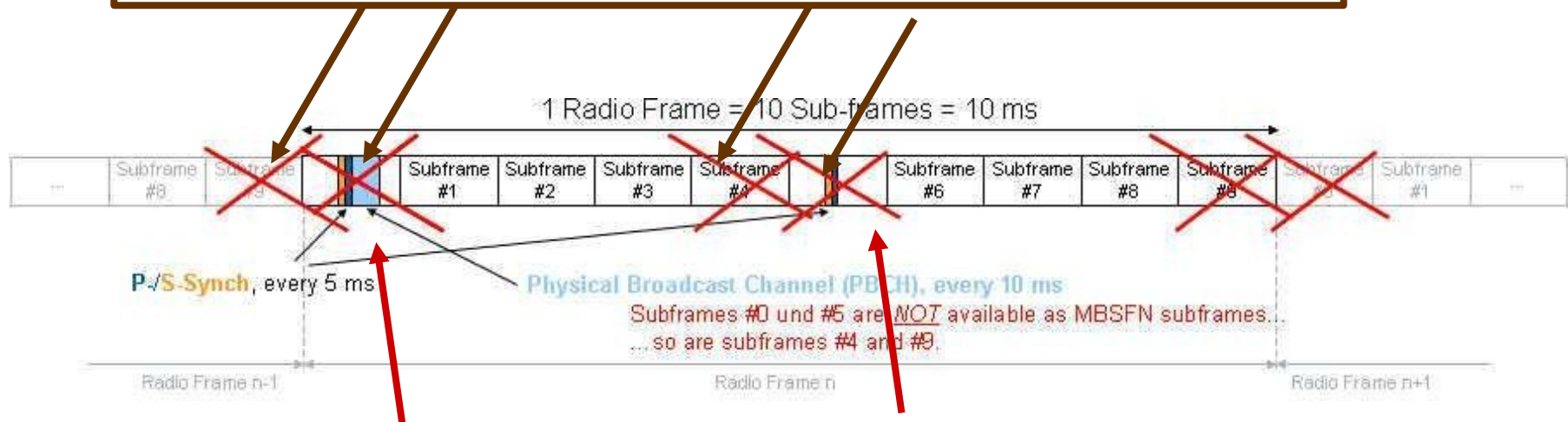
- I Higher Layer Forward Error Correction (see TS26.346)

I Different services (MTCHs and MCCH) can be multiplexed



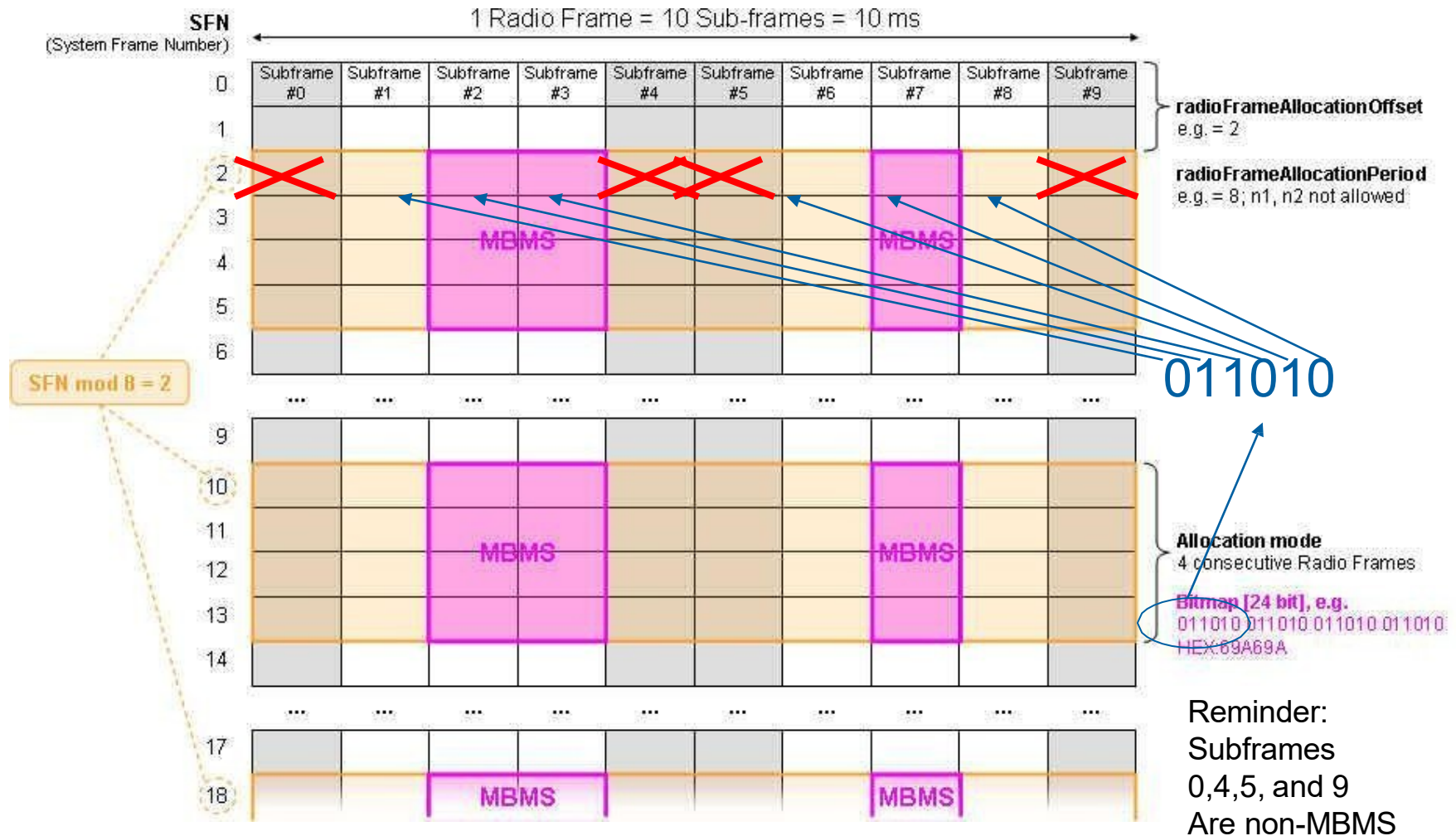
eMBMS channel mapping

Subframes 0,4,5 and 9 are not MBMS, because
Of paging occasion can occur here

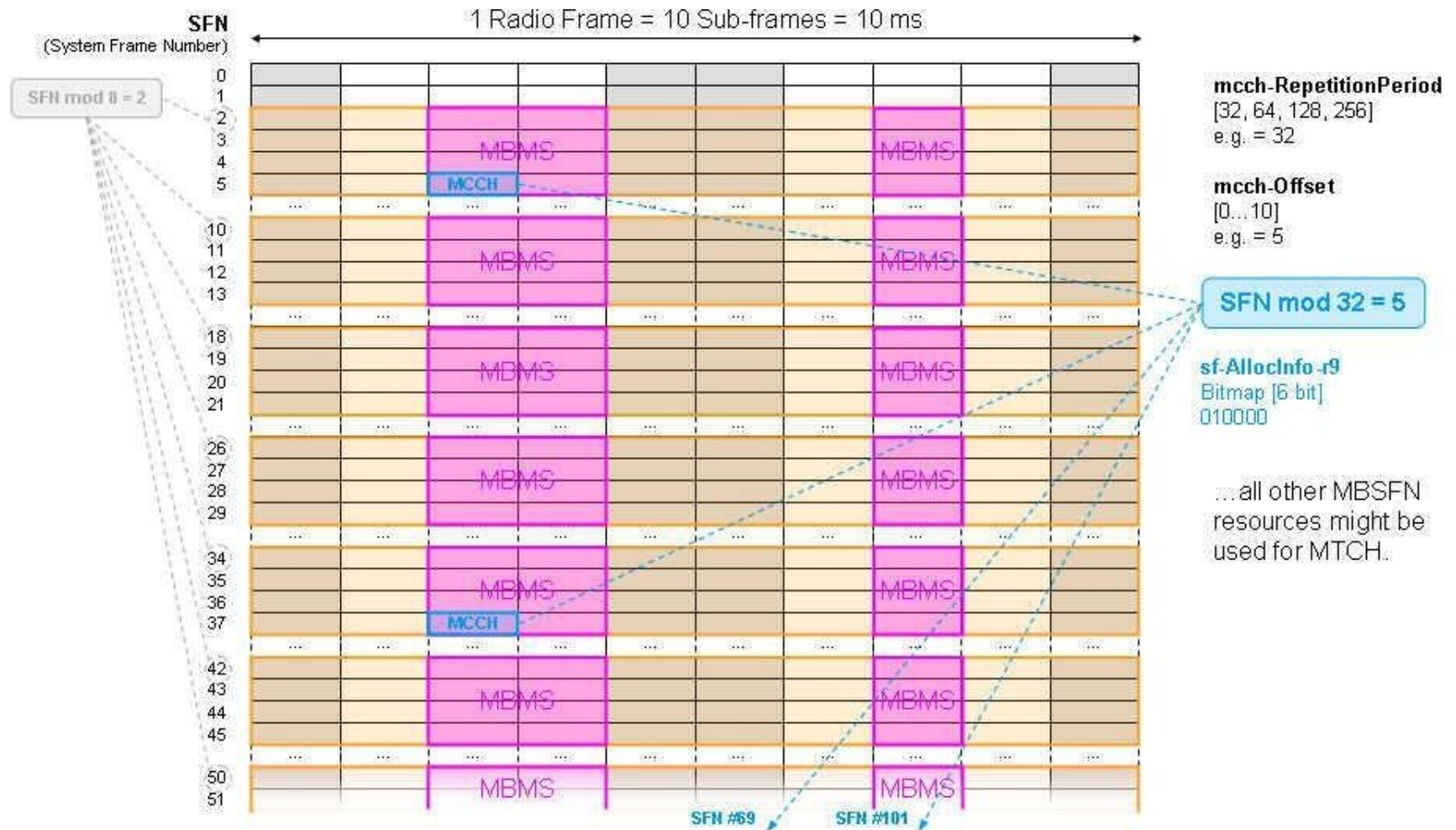


Subframes 0 and 5 are not MBMS, because
of PBCH and Sync Channels

eMBMS allocation based on SIB2 information



eMBMS: MCCH position according to SIB13



LTE Release 9

Dual-layer beamforming

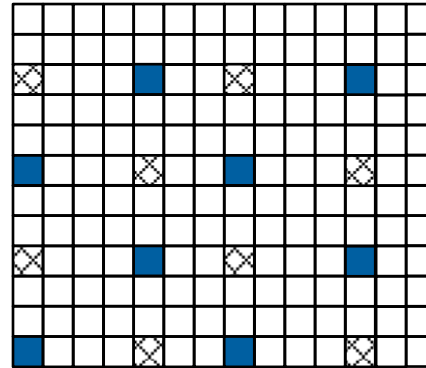
- I **3GPP Rel-8 – Transmission Mode 7 = beamforming without UE feedback, using UE-specific reference signal pattern,**
 - I Estimate the position of the UE (Direction of Arrival, DoA),
 - I Pre-code digital baseband to direct beam at direction of arrival,
 - I **BUT** single-layer beamforming, only one codeword (TB),

- I **3GPP Rel-9 – Transmission Mode 8 = beamforming with or without UE feedback (PMI/RI) using UE-specific reference signal pattern, but dual-layer,**
 - I Mandatory for TDD, optional for FDD,
 - I 2 (new) reference signal pattern for two new antenna ports 7 and 8,
 - I New DCI format 2B to schedule transmission mode 8,
 - I Performance test in 3GPP TS 36.521 Part 1 (Rel-9) are adopted to support testing of transmission mode 8.

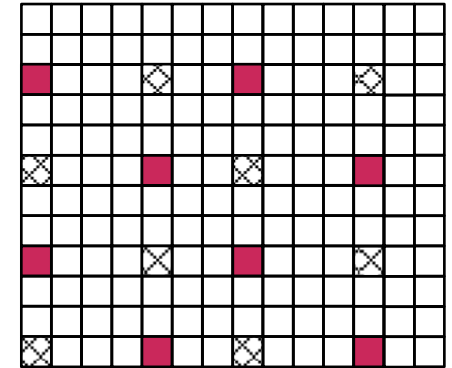
LTE Release 9

Dual-layer beamforming – Reference Symbol Details

- I Cell specific **antenna port 0** and **antenna port 1** reference symbols

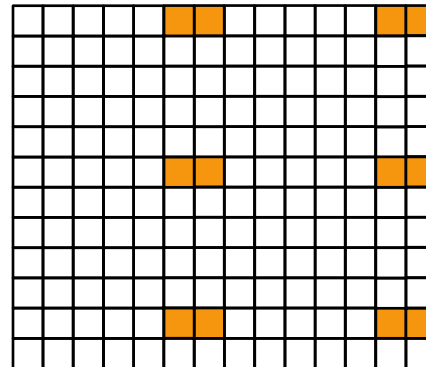


Antenna Port 0

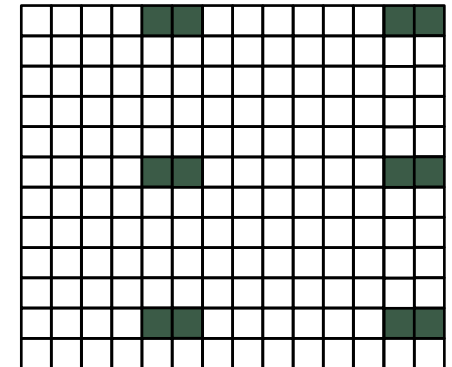


Antenna Port 1

- I UE specific **antenna port 7** and **antenna port 8** reference symbols

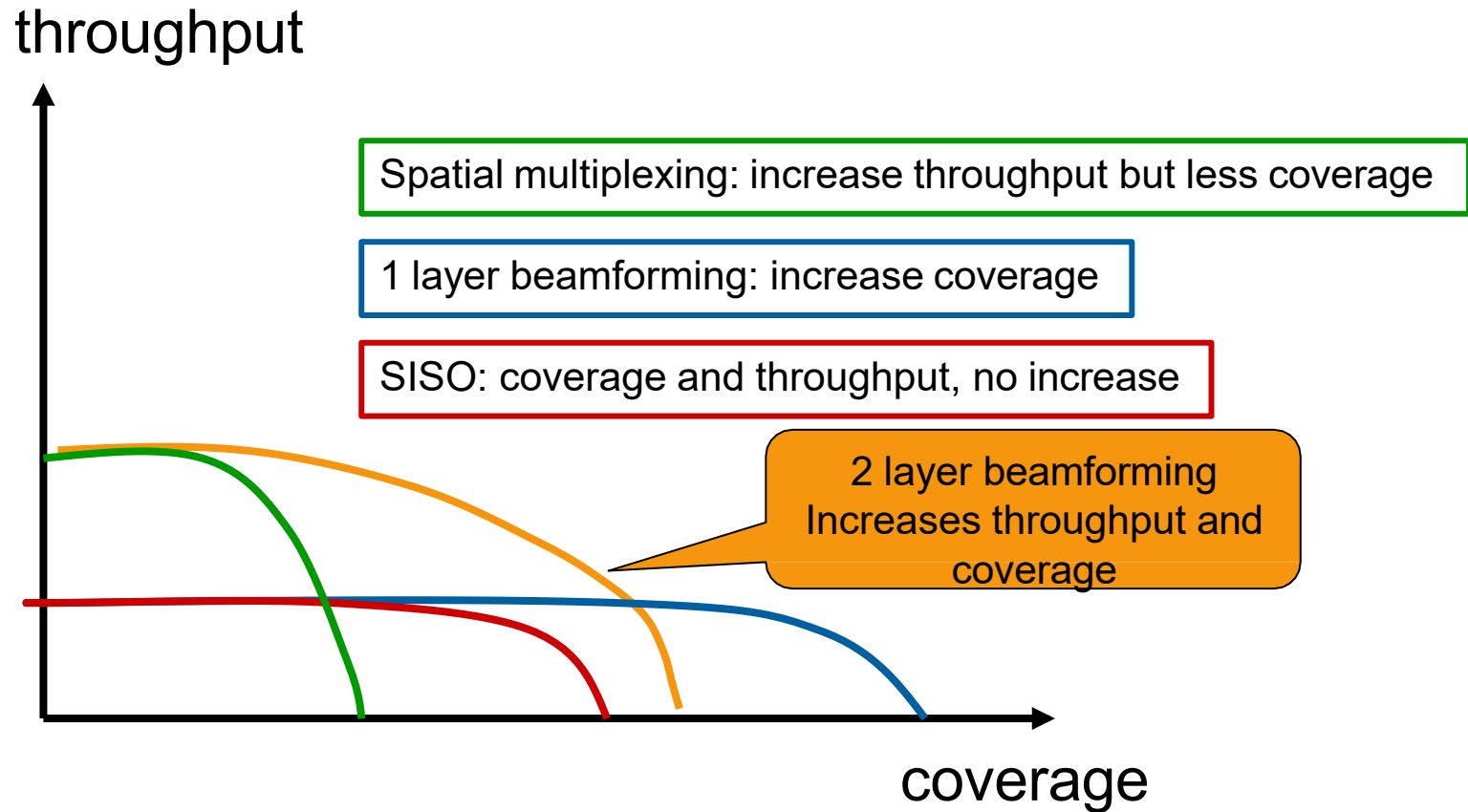


Antenna Port 7



Antenna Port 8

2 layer beamforming



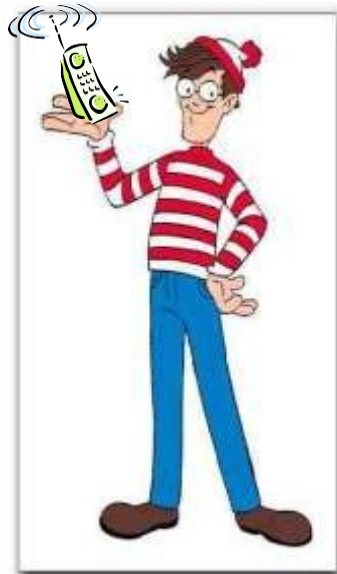
Spatial multiplexing increases throughput, but loses coverage

Location based services

- Location Based Services“
- Products and services which need location information
- Future Trend:
Augmented Reality



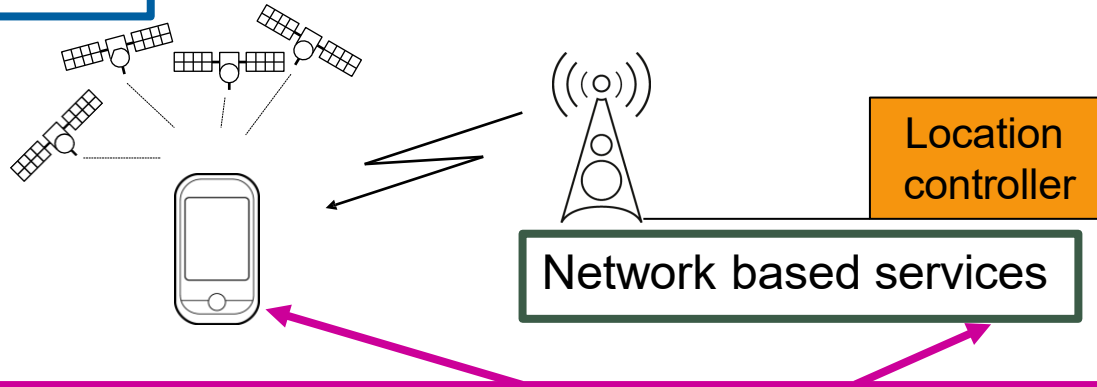
Where is Waldo?



Location based services

The idea is not new, ... so what to discuss?

Satellite based services



Who will do the measurements? The UE or the network? = „assisted“

Who will do the calculation? The UE or the network? = „based“

So what is new?

Several ideas are defined and hybrid mode is possible as well,
Various methods can be combined.

E-UTRA supported positioning methods

Method	UE-based	UE-assisted (LS based)	eNB-assisted (LS based)	Release
A-GNSS	Yes Measurement: UE Estimation: UE	Yes Measurement: UE Estimation: LC	No	Rel-9
Downlink (OTDOA)	No	Yes Measurement: UE Estimation: LS	No	Rel-9
E-CID	No	Yes Measurement: UE Estimation: LS	Yes Measurement: eNB Estimation: LS	Rel-9
Uplink (UTDOA)	No	No	Yes Measurement: eNB Estimation: LS	Rel-11
RF Pattern matching	?	?	?	Rel-11

LTE Release 9

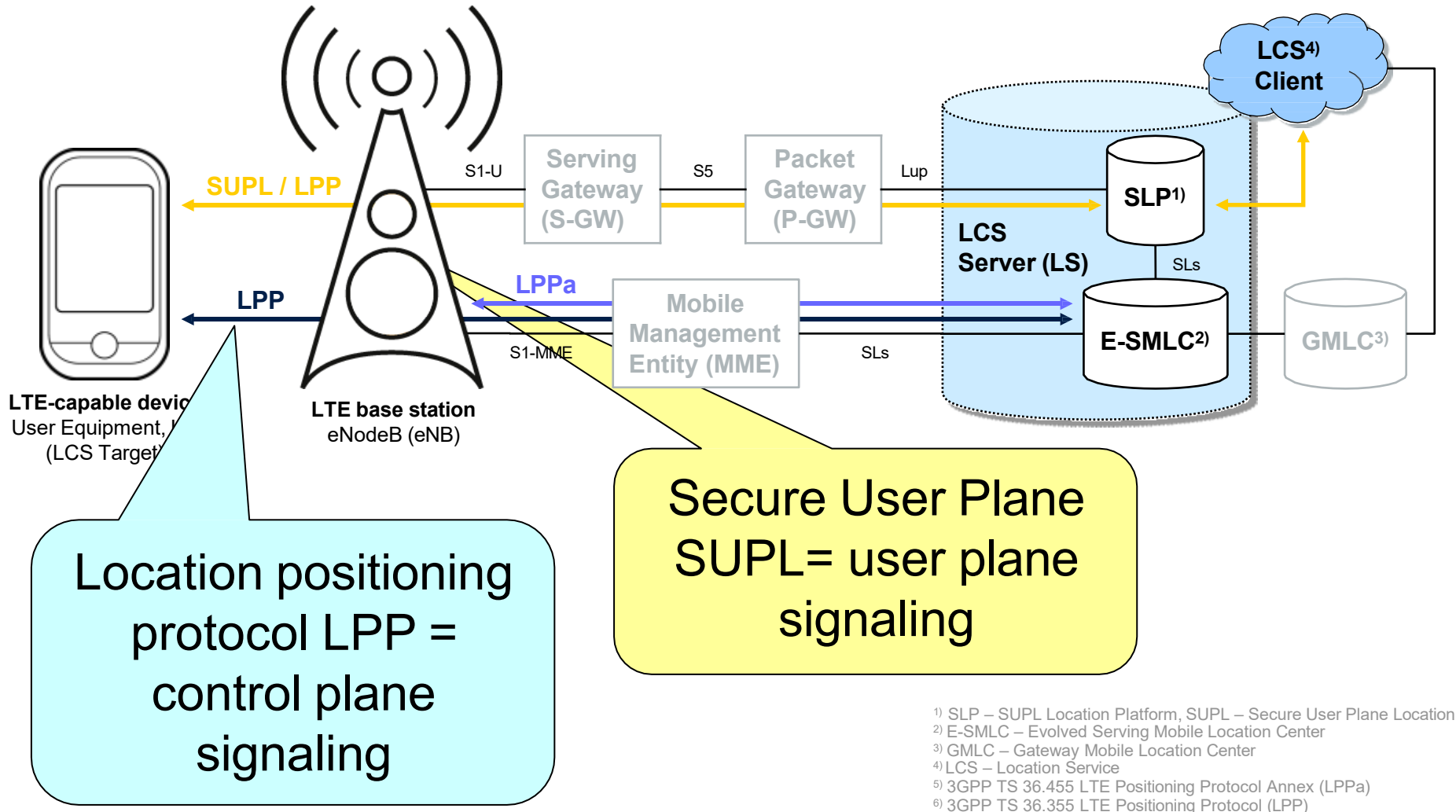
LTE positioning

I **The standard positioning methods supported for E-UTRAN access are:**

- I network-assisted GNSS (Global Navigation Satellite System) methods
 - These methods make use of UEs that are equipped with radio receivers capable of receiving GNSS signals, e.g. GPS.
- I downlink positioning
 - The downlink (OTDOA – Observed Time Difference Of Arrival) positioning method makes use of the measured timing of downlink signals received from multiple eNode Bs at the UE. The UE measures the timing of the received signals using assistance data received from the positioning server, and the resulting measurements are used to locate the UE in relation to the neighbouring eNode Bs.
- I enhanced cell ID method
 - In the Cell ID (CID) positioning method, the position of an UE is estimated with the knowledge of its serving eNode B and cell. The information about the serving eNode B and cell may be obtained by paging, tracking area update, or other methods.

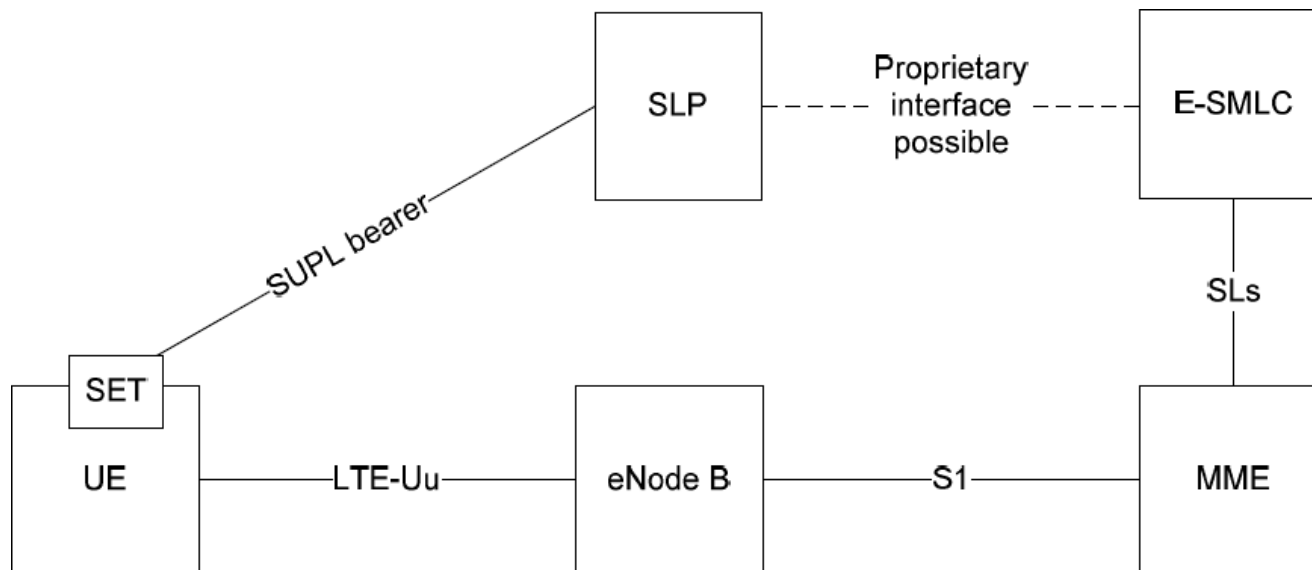
E-UTRA supported positioning network architecture

Control plane and user plane signaling



E-UTRAN UE Positioning Architecture

- I In contrast to GERAN and UTRAN, the E-UTRAN positioning capabilities are intended to be **forward compatible** to **other access types** (e.g. WLAN) and **other positioning methods** (e.g. RAT uplink measurements).
- I Supports user plane solutions, e.g. OMA SUPL 2.0

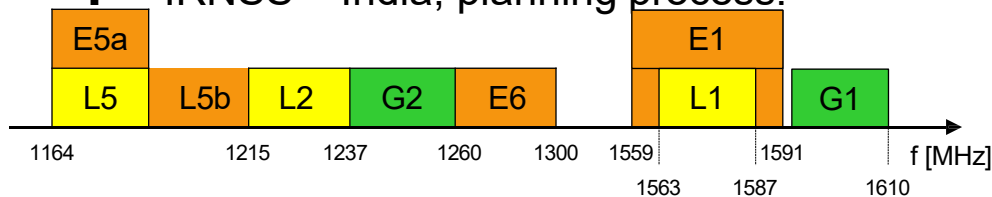


UE = User Equipment
SUPL* = Secure User Plane Location
OMA* = Open Mobile Alliance
SET = SUPL enabled terminal
SLP = SUPL locaiton platform
E-SMLC = Evolved Serving Mobile Location Center
MME = Mobility Management Entity
RAT = Radio Access Technology

Global Navigation Satellite Systems

I GNSS – Global Navigation Satellite Systems; autonomous systems:

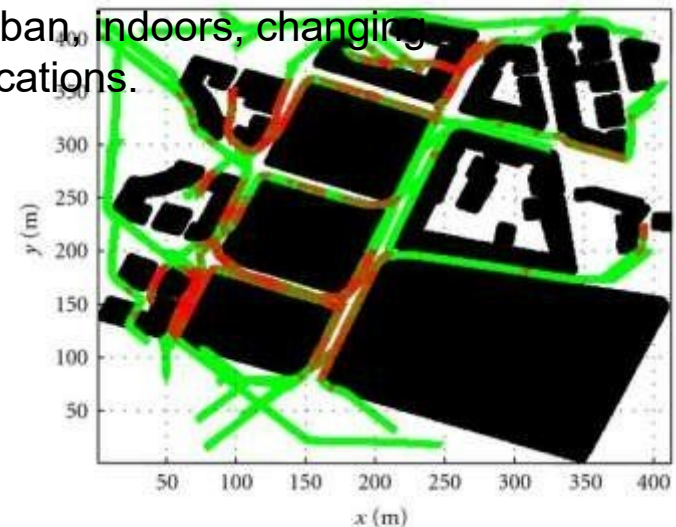
- I GPS – USA 1995.
- I GLONASS – Russia, 2012.
- I Galileo – Europe, target 2017.
- I Compass (Beidou) – China, under development, target 2015.
- I IRNSS – India, planning process.



GALLILEO		GPS		GLONASS	
Signal	f_{Carrier} [MHz]	Signal	f_{Carrier} [MHz]	Signal	f_{Carrier} [MHz]
E1	1575,420	L1C/A	1575,420	G1	$1602 \pm k \cdot 0,5625$
E6	1278,750	L1C	1575,420	G2	$1246 \pm k \cdot 0,5625$
E5	1191,795	L2C	1227,600	$k = -7 \dots 13$	
E5a	1176,450	L5	1176,450		
E5b	1207,140				

I GNSS are designed for continuous reception, outdoors.

- I Challenging environments: urban, indoors, changing locations.

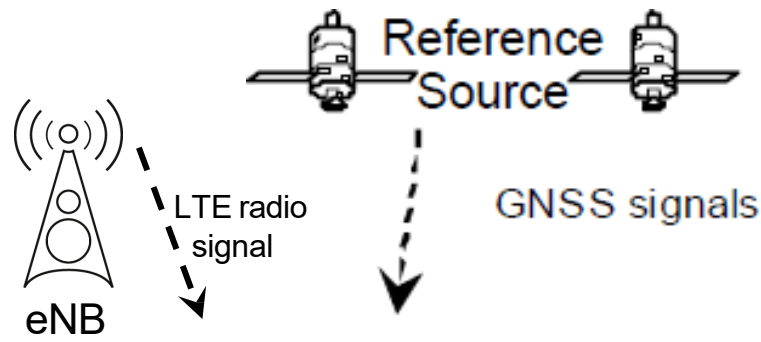


- ▼ No satellite visible
- ▼ 1 satellite visible
- ▼ 2 satellites visible
- ▼ 3 satellites visible
- ▼ => 4 satellites visible
- Building

The image shows the front cover of a 3GPP technical specification document. At the top, the title "3GPP TS 36.355" is prominently displayed in a large, bold, black font, followed by "V9.5.0 (2011-03)" in a smaller font. Below this, the text "Technical Specification Group Radio Access Network" and "Evolved Universal Terrestrial Radio Access (E-UTRAN)" are written in a smaller, black font. The main title "LTE Positioning Protocol (LPP)" is centered in a bold, black font, with "(Release 9)" underneath it. The 3GPP logo is on the right side, and the LTE logo is on the left side. The background is white with a thin black border. There are some orange diagonal lines at the bottom of the page, which appear to be part of the scanning process or a watermark.

Source:
TS 36.355
LTE Positioning
Protocol (LPP)

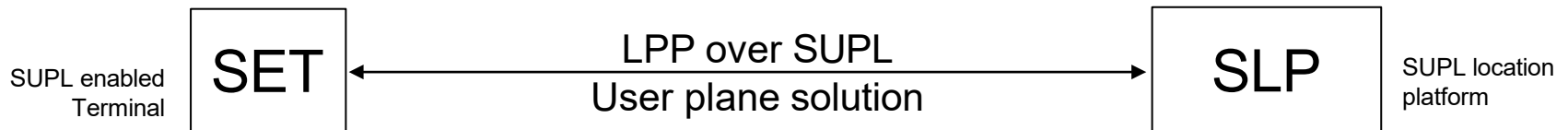
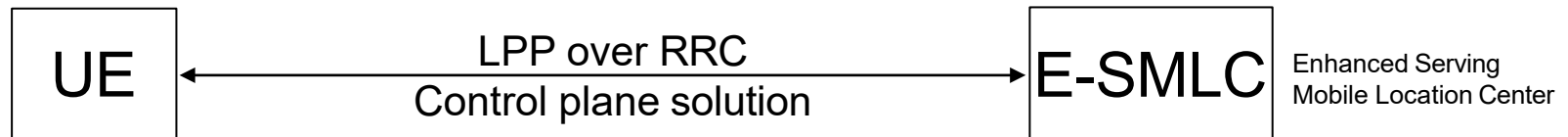
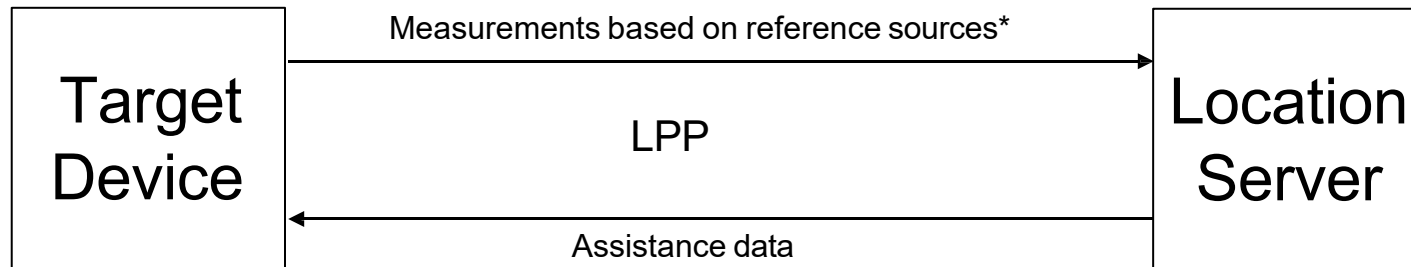
LTE Positioning Protocol (LPP) 3GPP TS 36.355



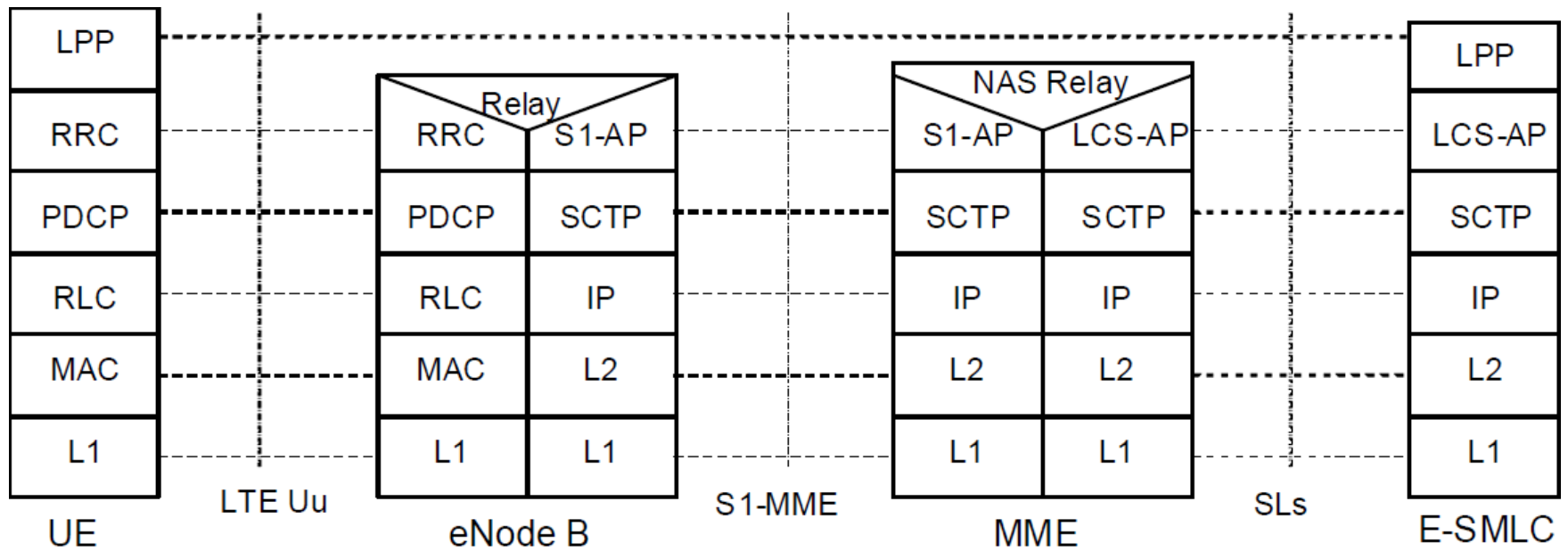
LPP position methods

- A-GNSS Assisted Global Navigation Satellite System
- E-CID Enhanced Cell ID
- OTDOA Observed time difference of arrival

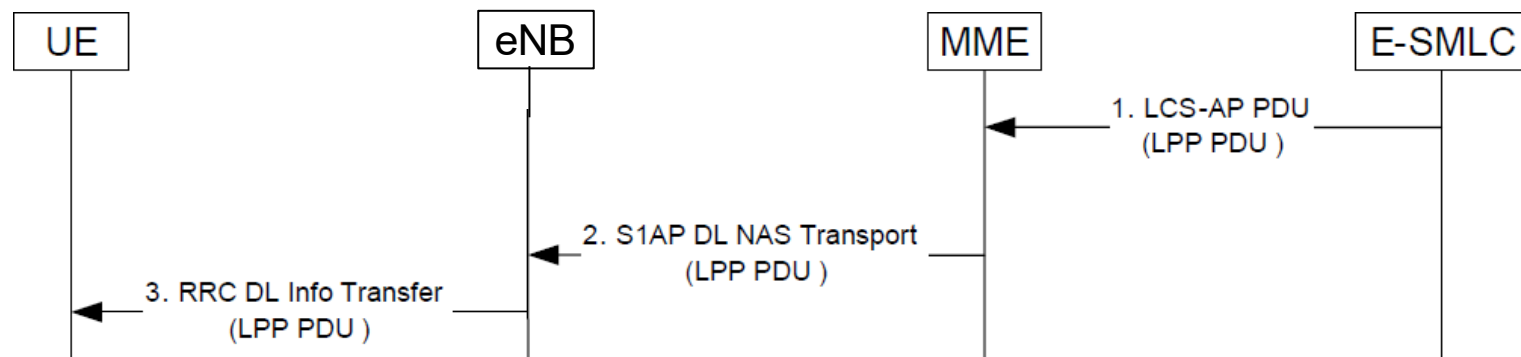
*GNSS and LTE radio signals



LPP and lower layers

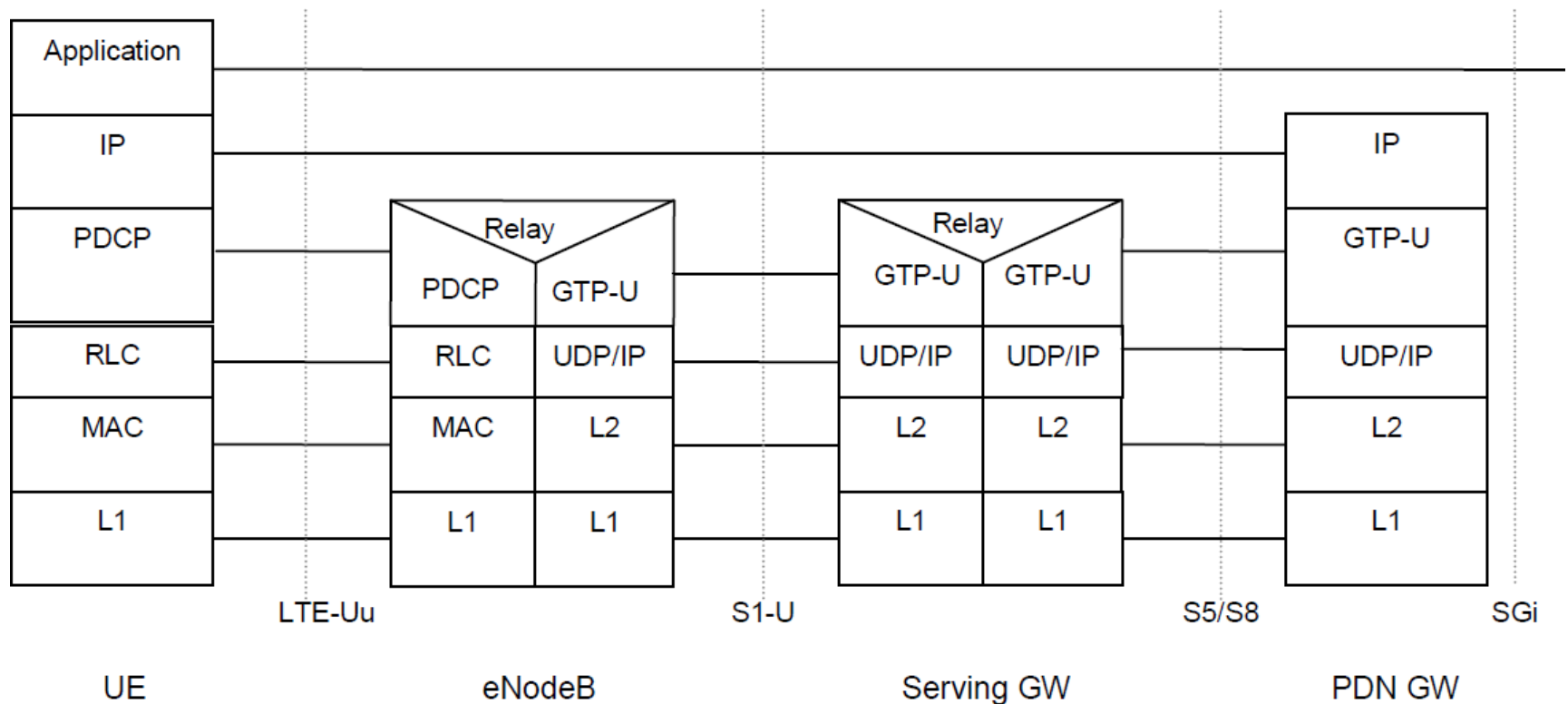


LPP PDU Transfer



User plane stack

SUPL occupies the application layer in the LTE user plane stack, with LPP (or another positioning protocol !) transported as another layer above SUPL.



GNSS positioning methods supported

I Autonomous GNSS

I Assisted GNSS (A-GNSS)

I The network assists the UE GNSS receiver to improve the performance in several aspects:

- Reduce UE GNSS start-up and acquisition times
- Increase UE GNSS sensitivity
- Allow UE to consume less handset power

I UE Assisted

- UE transmits GNSS measurement results to E-SMLC where the position calculation takes place

I UE Based

- UE performs GNSS measurements and position calculation, supported by data ...
 - ... assisting the measurements, e.g. with reference time, visible satellite list etc.
 - ... providing means for position calculation, e.g. reference position, satellite ephemeris, etc.

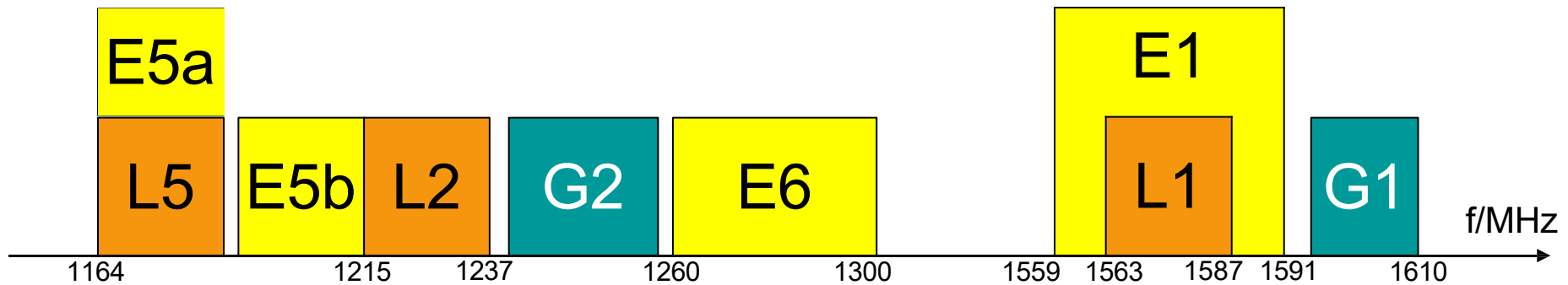
Assistance Data
Reference Time
Reference Location
Ionospheric Models
Earth Orientation Parameters
GNSS-GNSS Time Offsets
Differential GNSS Corrections
Ephemeris and Clock Models
Real-Time Integrity
Data Bit Assistance
Acquisition Assistance
Almanac
UTC Models

GNSS candidates and augmentation systems

- I **GPS/Modernized GPS – Global Positioning System (USA, since 1995)**
- I **Galileo (Europe, under development, target 2013)**
- I **GLONASS (Russia)**
- I **Compass, Beidou-2 (China, under development, target 2015)**
- I **IRNSS (India, in process of planning)**

- I **SBAS – Satellite based augmentation systems**
 - Geostationary satellites supporting error corrections by **overlay signals**
 - WAAS – Wide Area Augmentation System (USA)
 - EGNOS – European Geostationary Navigation Overlay Service (Europe)
 - MSAS – Multi-Functional Satellite Augmentation System (Japan)
 - QZSS – Quasi-Zenith Satellite System (Japan, supports GPS, expected 2013)
 - GAGAN – GPS Aided Geo Augmented Navigation (India)

GNSS band allocations

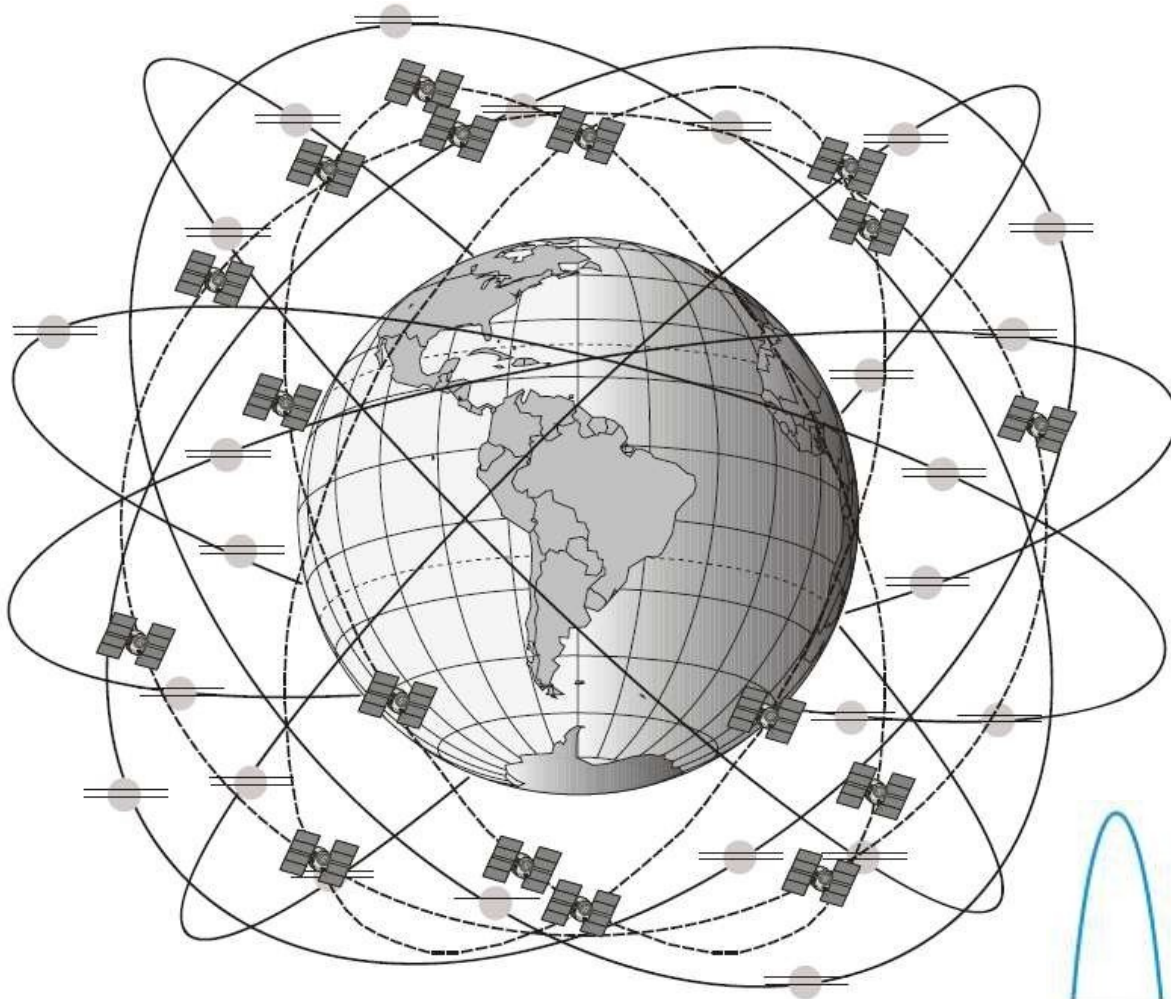


Galileo	
Signal	Carrier Frequency (MHz)
E1	1575,420
E6	1278,750
E5	1191,795
E5a	1176,450
E5b	1207,140

GPS/Modernised GPS	
Signal	Carrier Frequency (MHz)
L1C/A	1575,420
L1C	1575,420
L2C	1227,600
L5	1176,450

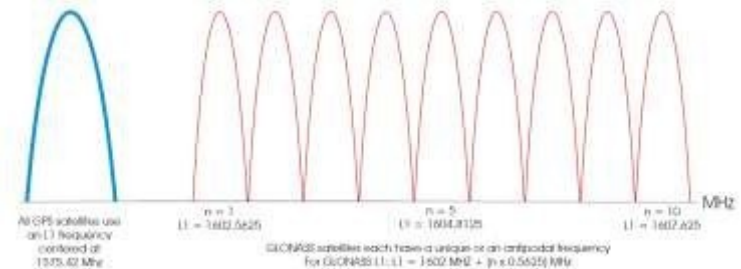
GLONASS	
Signal	Carrier Frequency (MHz)
G1	$1602 \pm k \times 0,5625$
G2	$1246 \pm k \times 0,4375$
	$k = -7 \dots 13$

GPS and GLONASS satellite orbits

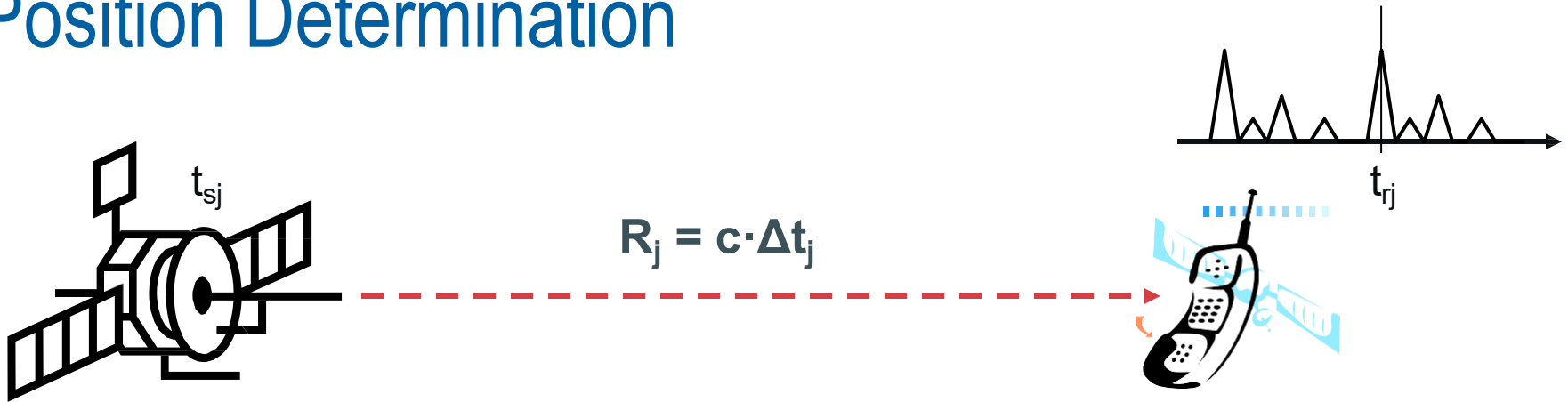


GPS:
26 Satellites
Orbital radius 26560 km

GLONASS:
26 Satellites
Orbital radius 25510 km



Position Determination



I „Pseudo distance“ Satellite – Receiver

I $R_j = c \cdot \Delta t_j = c \cdot (t_{rj} - t_{sj}) = \rho_j + c \cdot \Delta_{\text{clock}} + \Delta_{\text{iono}} + \Delta_{\text{Tropo}} + \Delta_{\text{Mpath}} + \Delta_{\text{Int}} + \Delta_{\text{Noise}}$

- ρ_j = real distance („error-free“)
- $c \cdot \Delta_{\text{clock}}$ = Clock error (4th unknown variable $\rightarrow$ 4th satellite required)
- $\Delta_{\text{iono}}, \Delta_{\text{Tropo}}$ = „speed of light“ error due to ionosphere and troposphere
- $\Delta_{\text{Mpath}}, \Delta_{\text{Int}}, \Delta_{\text{Noise}}$ = t_{rj} uncertainty due to multipath propagation, interference, noise

I Each satellite j broadcasts its current position $\rho_{\text{SAT},j}$ and local time t_{sj} .

I With $\rho_{\text{SAT},j}$ and ρ_j the receiver position can be evaluated

I Additional signal phase measurements to increase accuracy

Backup basic terms

I **Ephemeris**

- I Table of values that gives the positions of objects in the sky at a given time

I **Almanach**

- I Set of data that every satellite transmits. It includes information about the state (health) of the entire satellite constellation and coarse data on every satellite's orbit.

I **Cold start**

- I No ephemeris, almanac or location data available (reset state)

I **Hot start**

- I Location data available

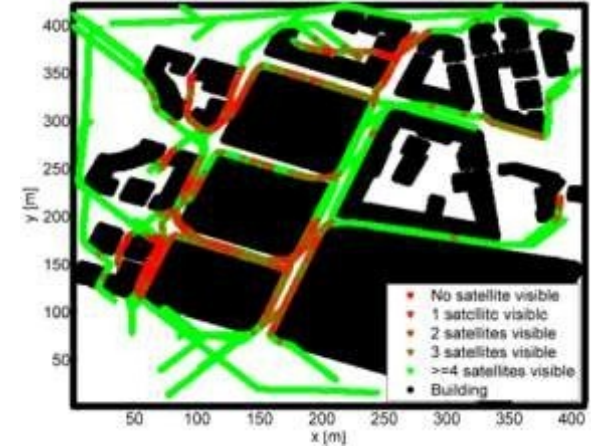
Why is GNSS not sufficient?



Critical scenario



Very critical scenario

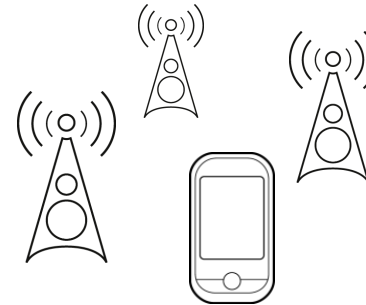
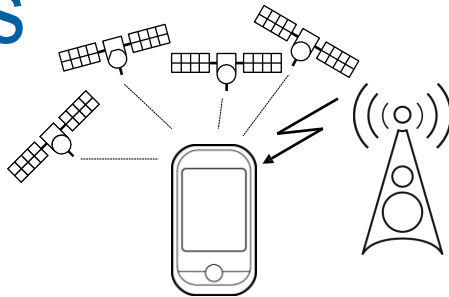


GPS Satellites visibility (Urban)

- Global navigation satellite systems (GNSSs) have restricted performance in certain environments
- Often less than four satellites visible: critical situation for GNSS positioning
 - support required (Assisted GNSS)
 - alternative required (Mobile radio positioning)

Reference [DLR]

(A-)GNSS vs. mobile radio positioning methods

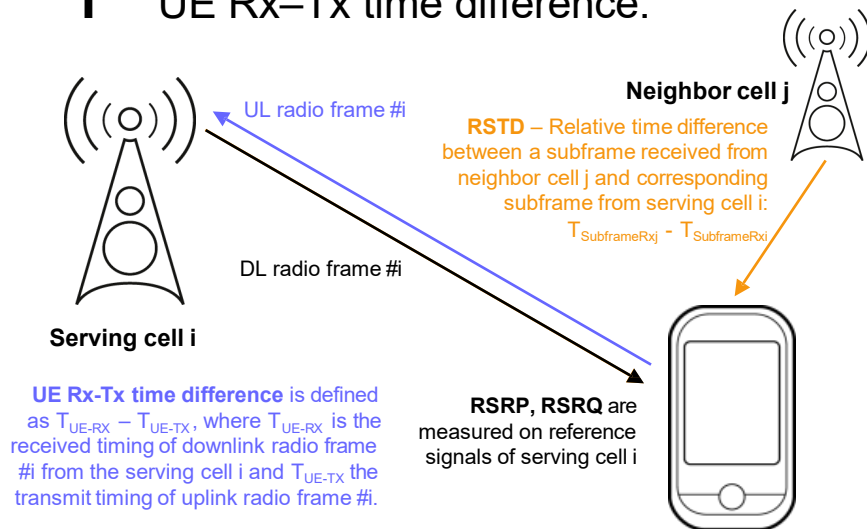


(A-)GNSS	Mobile radio systems
Low bandwidth (1-2 MHz)	High bandwidth (up to 20 MHz for LTE)
Very weak received signals	Comparatively strong received signals
Similar received power levels from all satellites	One strong signal from the serving BS, strong interference situation
Long synchronization sequences	Short synchronization sequences
Signal a-priori known due to low data rates	Complete signal not a-priori known to support high data rates, only certain pilots
Very accurate synchronization of the satellites by atomic clocks	Synchronization of the BSs not a-priori guaranteed
Line of sight (LOS) access as normal case → not suitable for urban / indoor areas	Non line of sight (NLOS) access as normal case → suitable for urban / indoor areas
3-dimensional positioning	2-dimensional positioning

Measurements for positioning

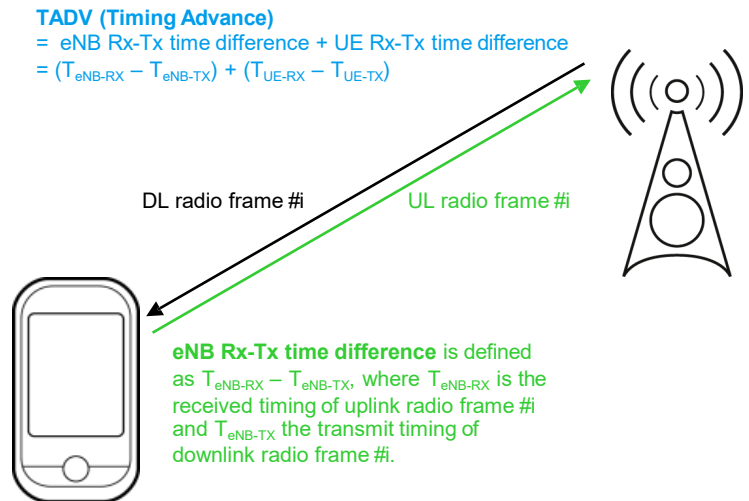
I UE-assisted measurements.

- I Reference Signal Received Power (RSRP) and Reference Signal Received Quality (RSRQ).
- I RSTD – Reference Signal Time Difference.
- I UE Rx–Tx time difference.



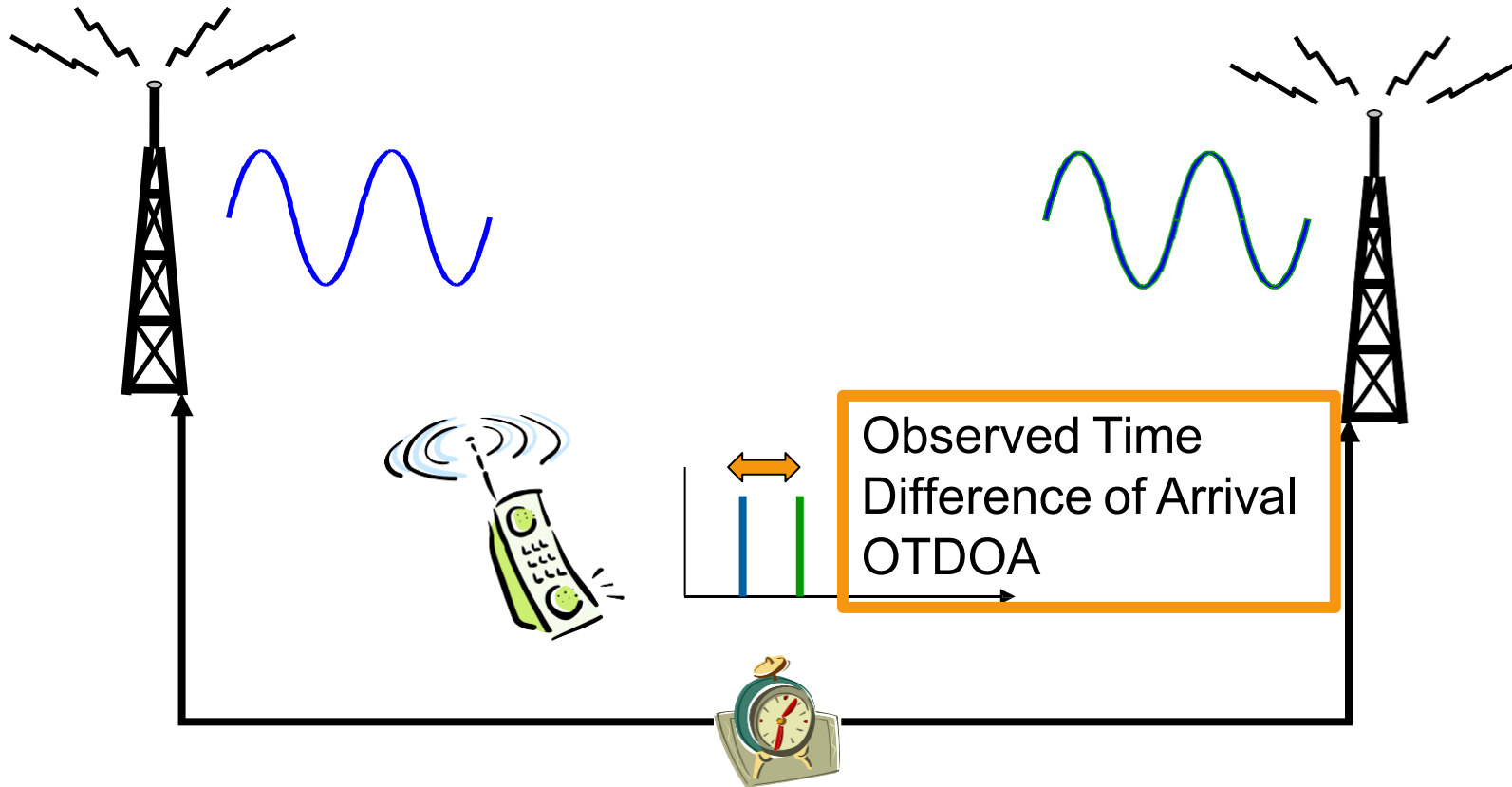
I eNB-assisted measurements.

- I eNB Rx – Tx time difference.
- I TADV – Timing Advance.
 - For positioning Type 1 is of relevance.
- I AoA – Angle of Arrival.
- I UTDOA – Uplink Time Difference of Arrival.



Source: see TS 36.214 Physical Layer measurements for detailed definitions

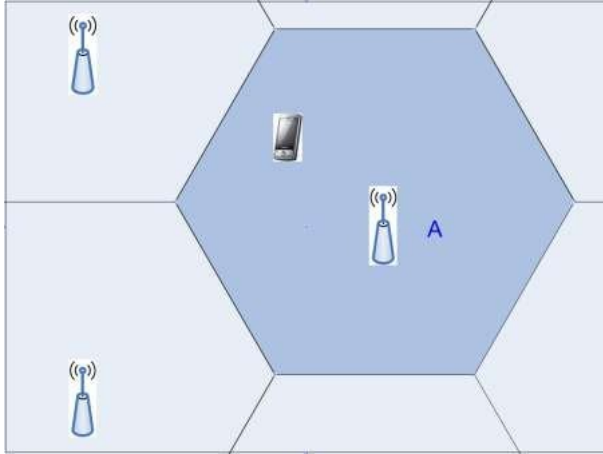
Observed Time difference



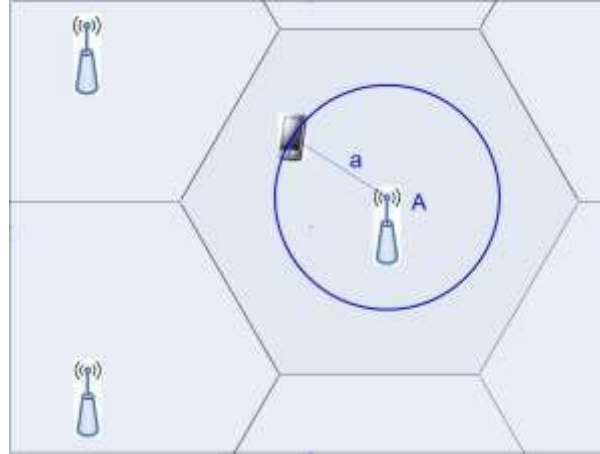
If network is synchronised,
UE can measure time difference

Methods' overview

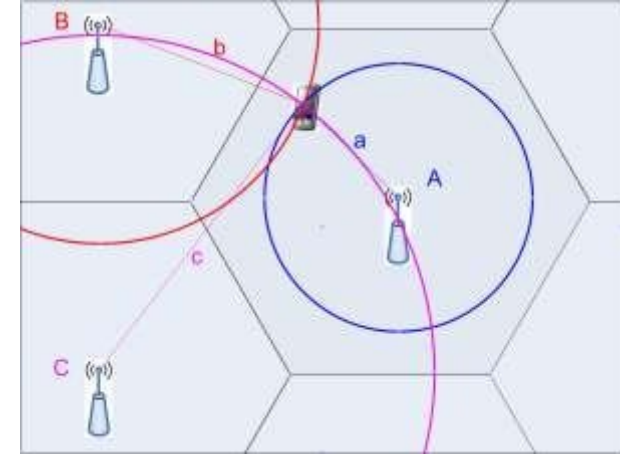
CID



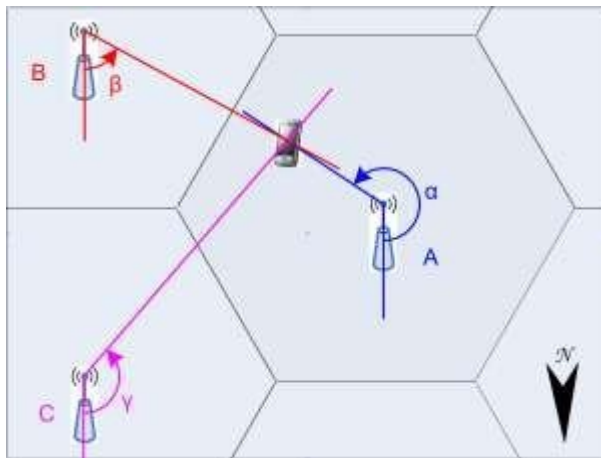
E-CID (RSRP/TOA/TADV)



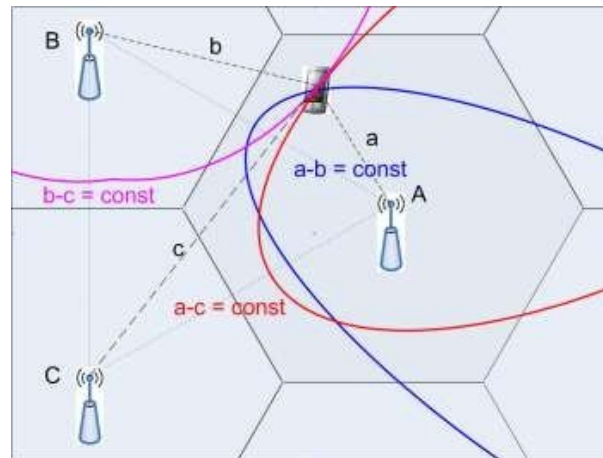
E-CID (RSRP/TOA/TADV) [Trilateration]



E-CID (AOA) [Triangulation]



Downlink / Uplink (O/U-TDOA) [Multilateration]



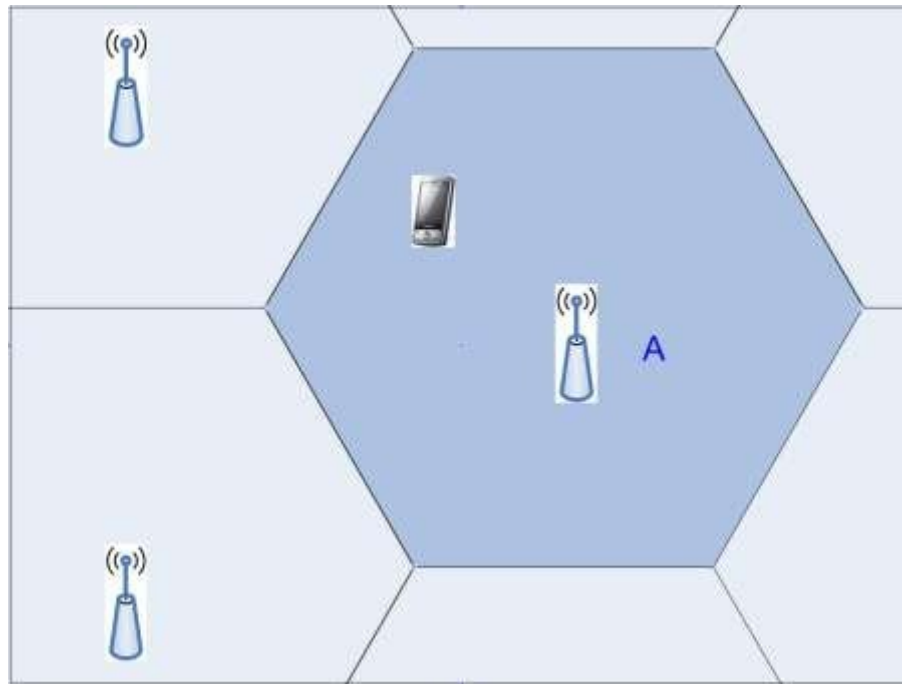
RF Pattern matching

To be updated!!

Cell ID

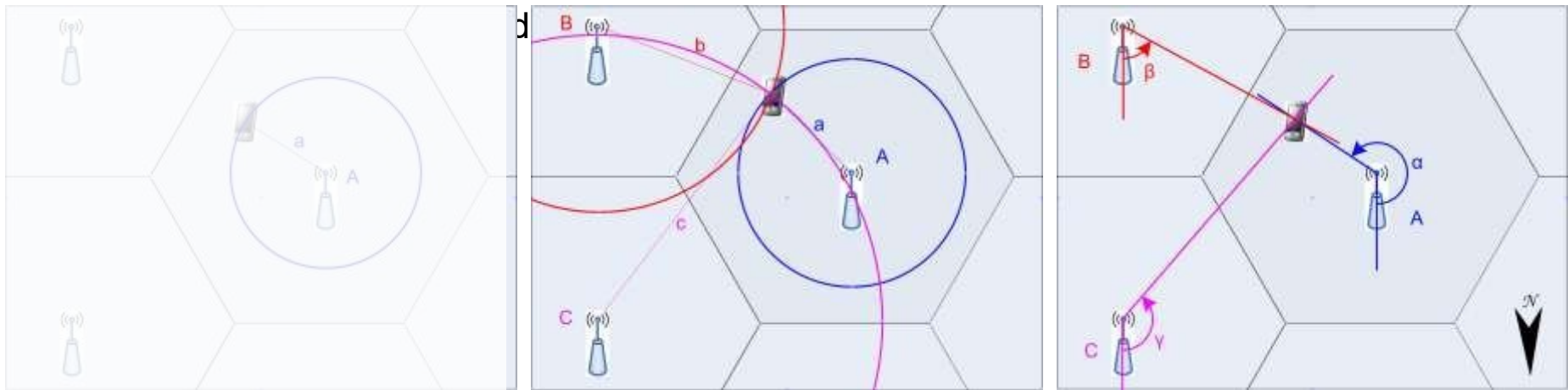
I Not new, other definition: Cell of Origin (COO).

- I UE position is estimated with the knowledge of the geographical coordinates of its serving eNB.
- I Position accuracy = One whole cell .



Enhanced-Cell ID (E-CID)

- I **UE positioning compared to CID is specified more accurately using additional UE and/or E UTRAN radio measurements:**
 - I E-CID with distance from serving eNB → position accuracy: a circle.
 - Distance calculated by measuring RSRP / TOA / TADV (RTT).
 - I E-CID with distances from 3 eNB-s → position accuracy: a point.
 - Distance calculated by measuring RSRP / TOA / TADV (RTT).
 - I E-CID with Angles of Arrival → position accuracy: a point.



RSRP – Reference Signal Received Power
TOA – Time of Arrival
TADV – Timing Advance
RTT – Round Trip Time

TADV – Timing Advance (Round Trip Time, RTT)

I Base station measures:

$$\text{eNB Rx-Tx} = T_{\text{eNB-Rx}} - T_{\text{eNB-Tx}}$$

I eNB orders the device to correct its uplink timing: TA = eNB Rx-Tx.

- I Timing Advance command (MAC).

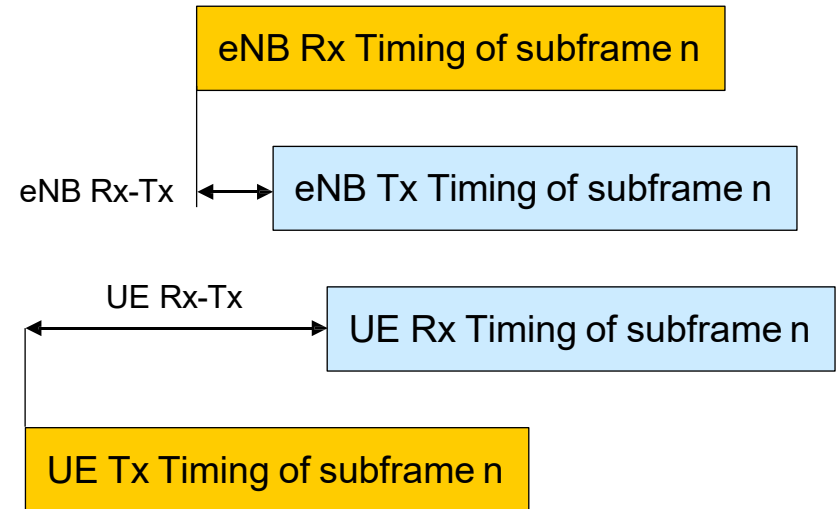
I UE measures and reports:

$$\text{UE Rx-Tx} = T_{\text{UE-Rx}} - T_{\text{UE-Tx}}$$

- I LPP_IE: ue-RxTxTimeDiff.

I eNB calculates and reports to LS: TADV = eNB Rx-Tx + UE Rx-Tx

- I LPPa_IE: timingAdvanceType1/2
- I TADV = Round Trip Time (RTT).



I Distance of UE to eNB is estimated as $d = c \cdot \text{RTT} / 2$.

- I c = speed of light.

I Advantage: No synchronization between eNB's.

Angle of Arrival (AOA)

! **AoA = Estimated angle of a UE with respect to a reference direction (= geographical North), positive in a counter-clockwise direction, as seen from an eNB**

! Determined at eNB antenna based on a received UL signal (SRS).

! **Measurement at eNB:**

! eNB uses antenna array to estimate direction i.e. Angle of Arrival (AOA).

! The larger the array, the more accurate is the estimated AOA.

! eNB reports AOA to LS.

! **Advantage:** No synchronization between eNB's.

! **Drawback:** costly antenna arrays.

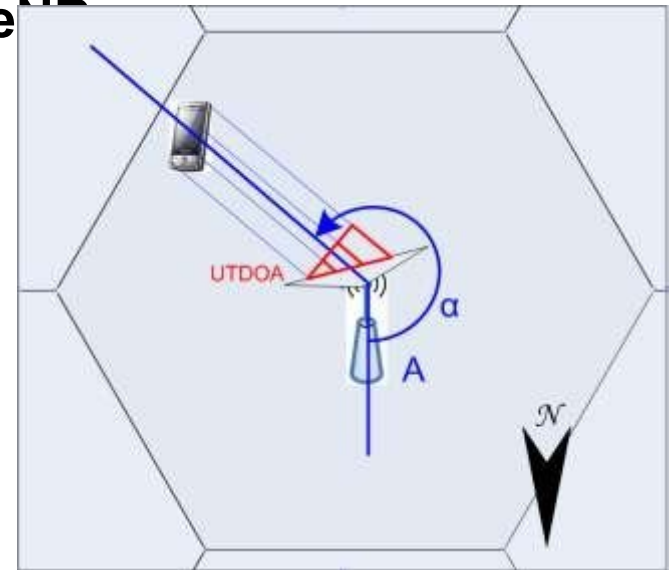
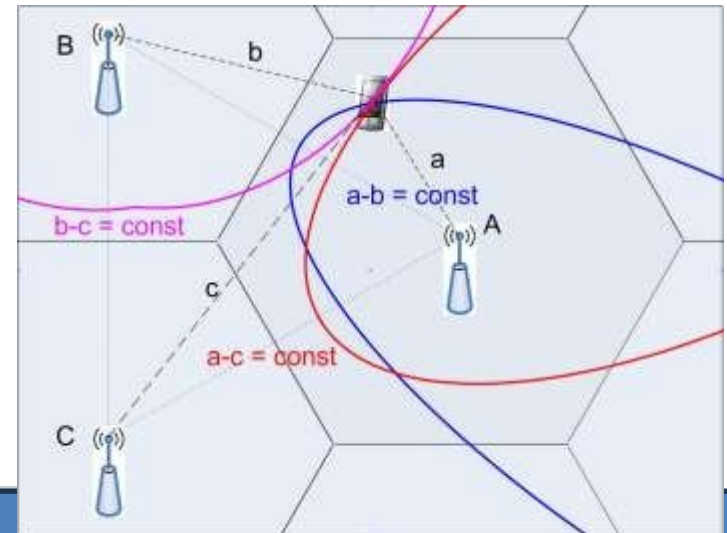
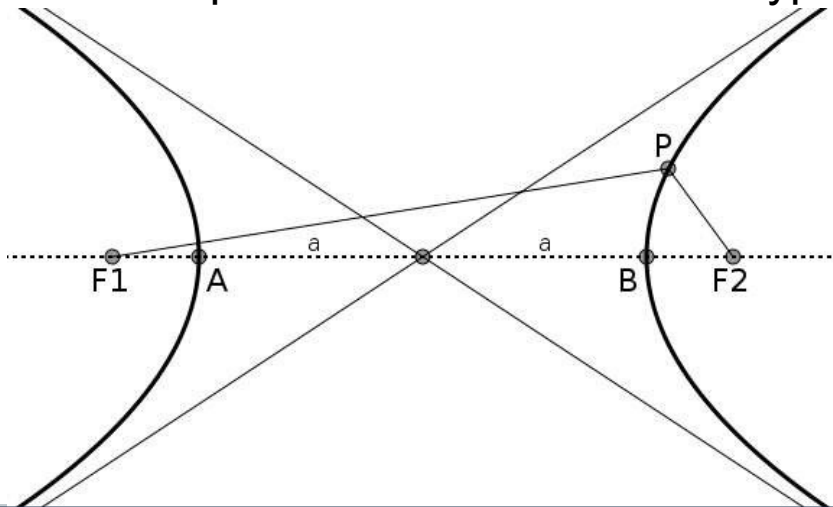


Table 10.2.1-1: AOA measurement report mapping

Reported value	Measured quantity value	Unit
AOA_ANGLE_000	$0 \leq \text{AOA_ANGLE} < 0.5$	degree
AOA_ANGLE_001	$0.5 \leq \text{AOA_ANGLE} < 1$	degree
AOA_ANGLE_002	$1 \leq \text{AOA_ANGLE} < 1.5$	degree
...	...	...
AOA_ANGLE_717	$358.5 \leq \text{AOA_ANGLE} < 359$	degree
AOA_ANGLE_718	$359 \leq \text{AOA_ANGLE} < 359.5$	degree
AOA_ANGLE_719	$359.5 \leq \text{AOA_ANGLE} < 360$	degree

OTDOA – Observed Time Difference of Arrival

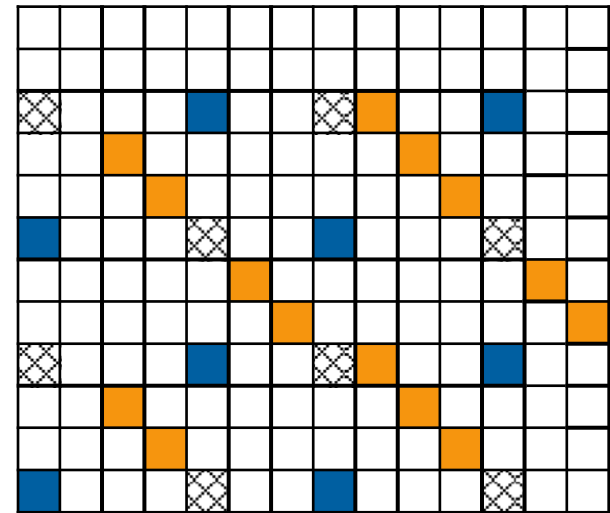
- I **UE position is estimated based on measuring TDOA of Positioning Reference Signals (PRS) embedded into overall DL signal received from different eNB's.**
 - I Each TDOA measurement describes a hyperbola (line of constant difference $2a$), the two focus points of which ($F1$, $F2$) are the two measured eNB-s (PRS sources), and along which the UE may be located.
 - I UE's position = intersection of hyperbolas for at least 3 pairs of



LTE Release 9

UE positioning – Reference Symbol Details

- I PRS is a pseudo-random QPSK sequence similar to CRS
- I **PRS pattern** (baseline for further discussion and CR drafting):
 - I Diagonal pattern with time varying frequency shift,
 - I PRS mapped around CRS (to avoid collisions)



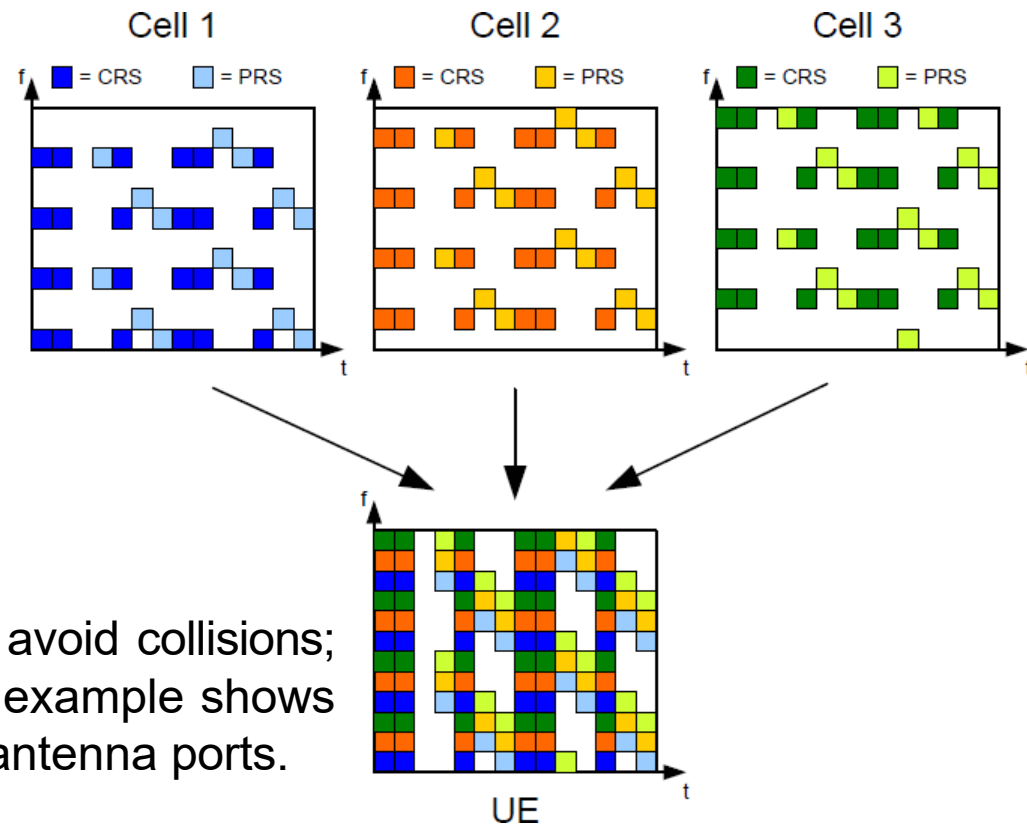
Antenna Port 0

Positioning Reference Signals (PRS) for OTDOA

Definition

- Cell-specific reference signals (CRS) are not sufficient for positioning, introduction of positioning reference signals (PRS) for antenna port 6.

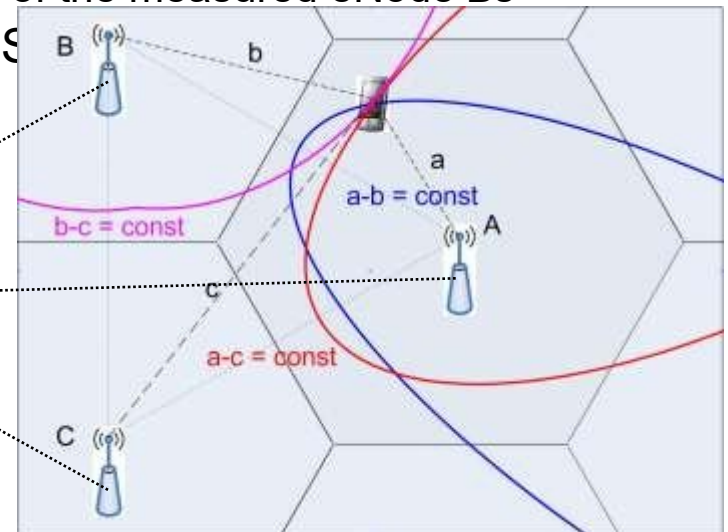
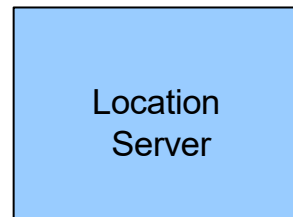
- SINR for synchronization and reference signals of neighboring cells needs to be at least -6 dB.
- PRS is a pseudo-random QPSK sequence similar to CRS; PRS pattern:
 - Diagonal pattern with time varying frequency shift.
 - PRS mapped around CRS to avoid collisions; example shows CRS mapping for usage of 4 antenna ports.



Uplink (UTDOA)

- I UTDOA = Uplink Time Difference of Arrival
- I UE positioning estimated based on:
 - measuring TDOA of UL (SRS) signals received in different eNB-s
 - each TDOA measurement describes a hyperbola (line of constant difference $2a$), the 2 focus points of which (F_1 , F_2) are the two receiving eNB-s (SRS receptors), and along which the UE may be located.
 - UE's position = intersection of hyperbolas for at least three pairs of eNB-s (= 3 eNB-s)
 - knowledge of the geographical coordinates of the measured eNode Bs
- I Method as such not specified for LTE → S

- eNB-s measure and report to eNB_Rx-Tx to LS
-LS calculates UTDOA and estimates the UE position

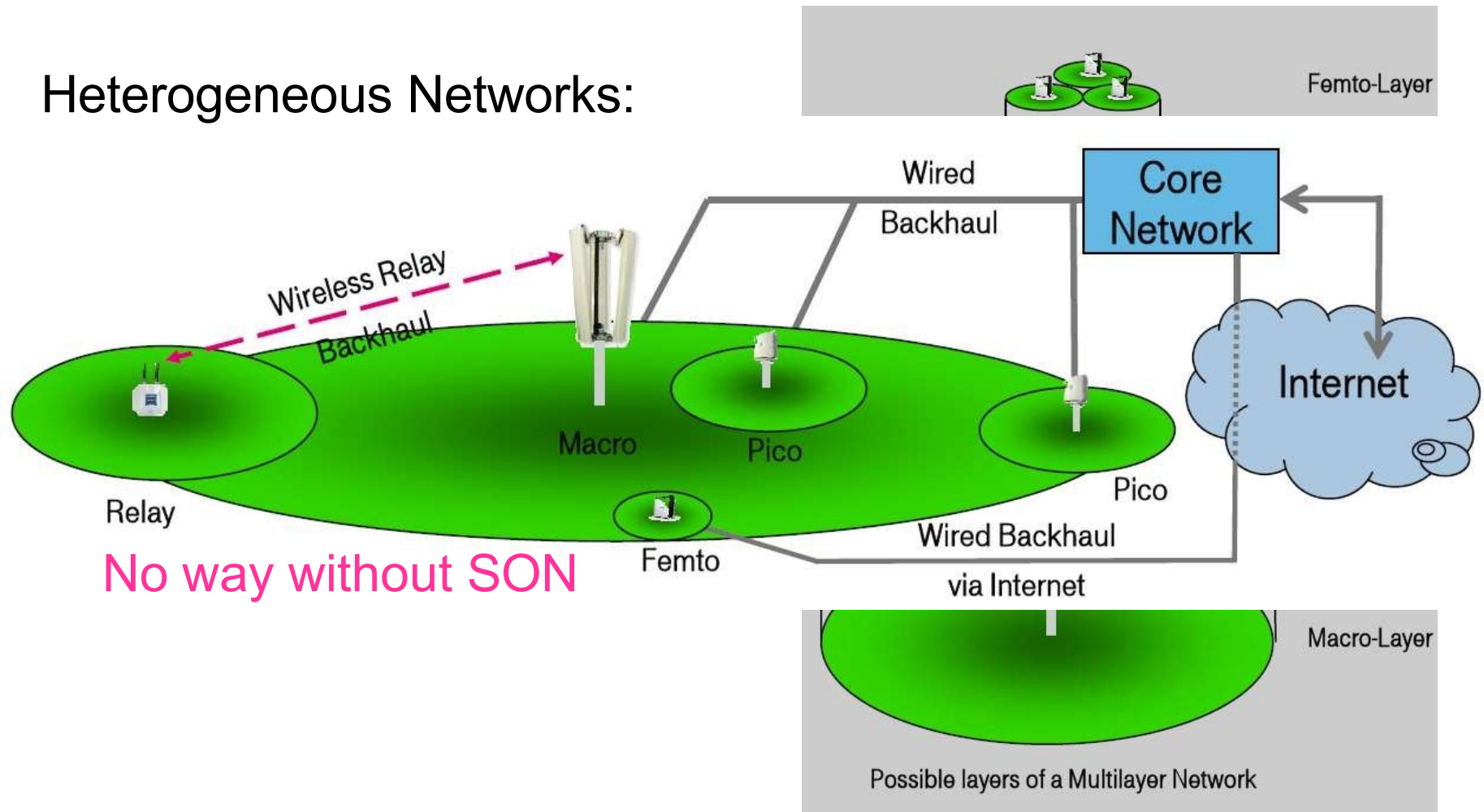


What is Self Organizing Networks, SON?

- SON = Self-Organizing Networks, methods for automatic configuration and optimization of the network
- 3 components:
 - Self-Configuration:
 - basic setup: IP address configuration, association with a Gateway, software download,...
 - Initial radio configuration: neighbour list configuration,...
 - Self-Optimization:
 - Neighbour list optimization, coverage/capacity optimization,...
 - Self-Healing:
 - Failure detection/localization,...

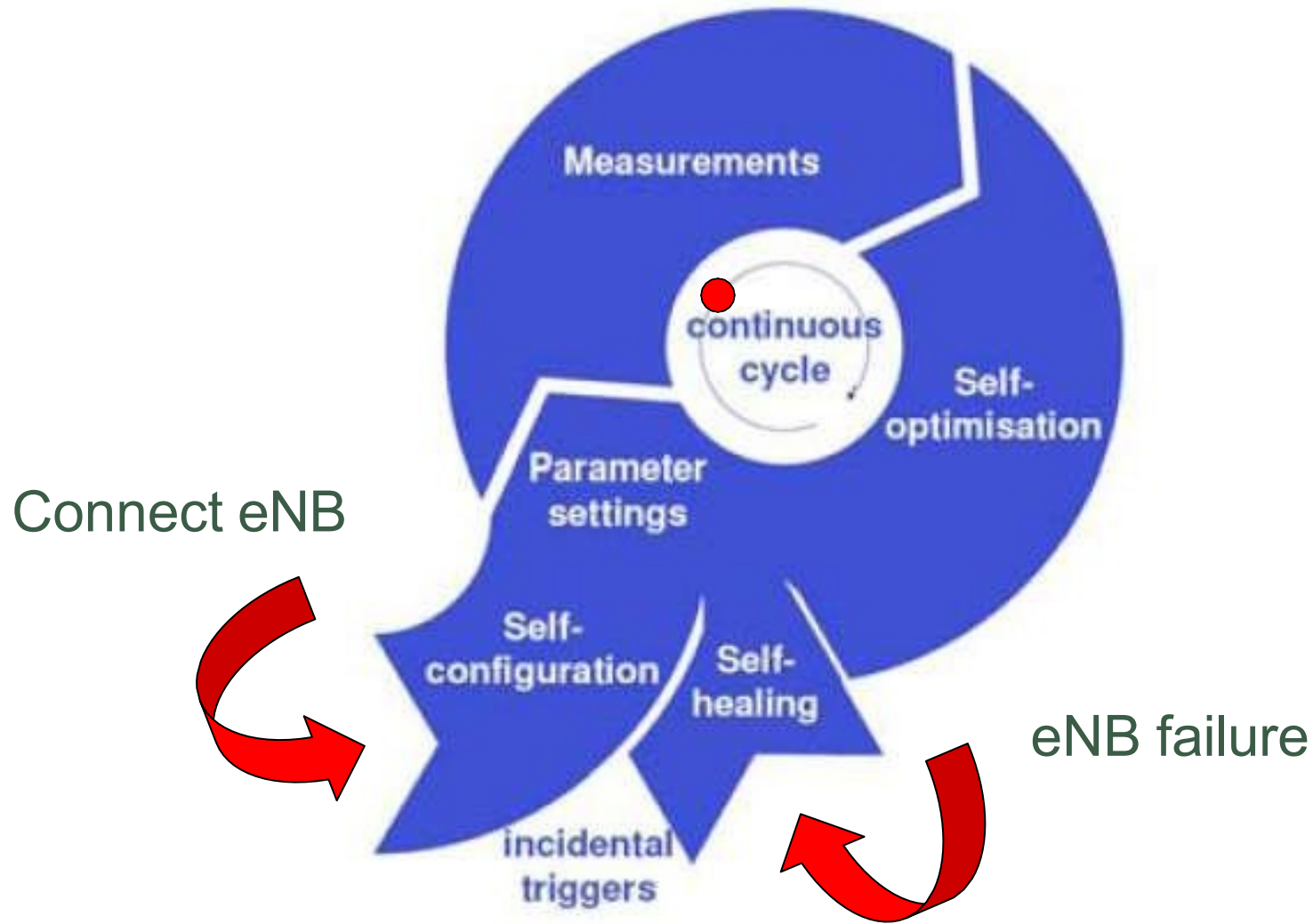
Motivation for SON

Heterogeneous Networks:



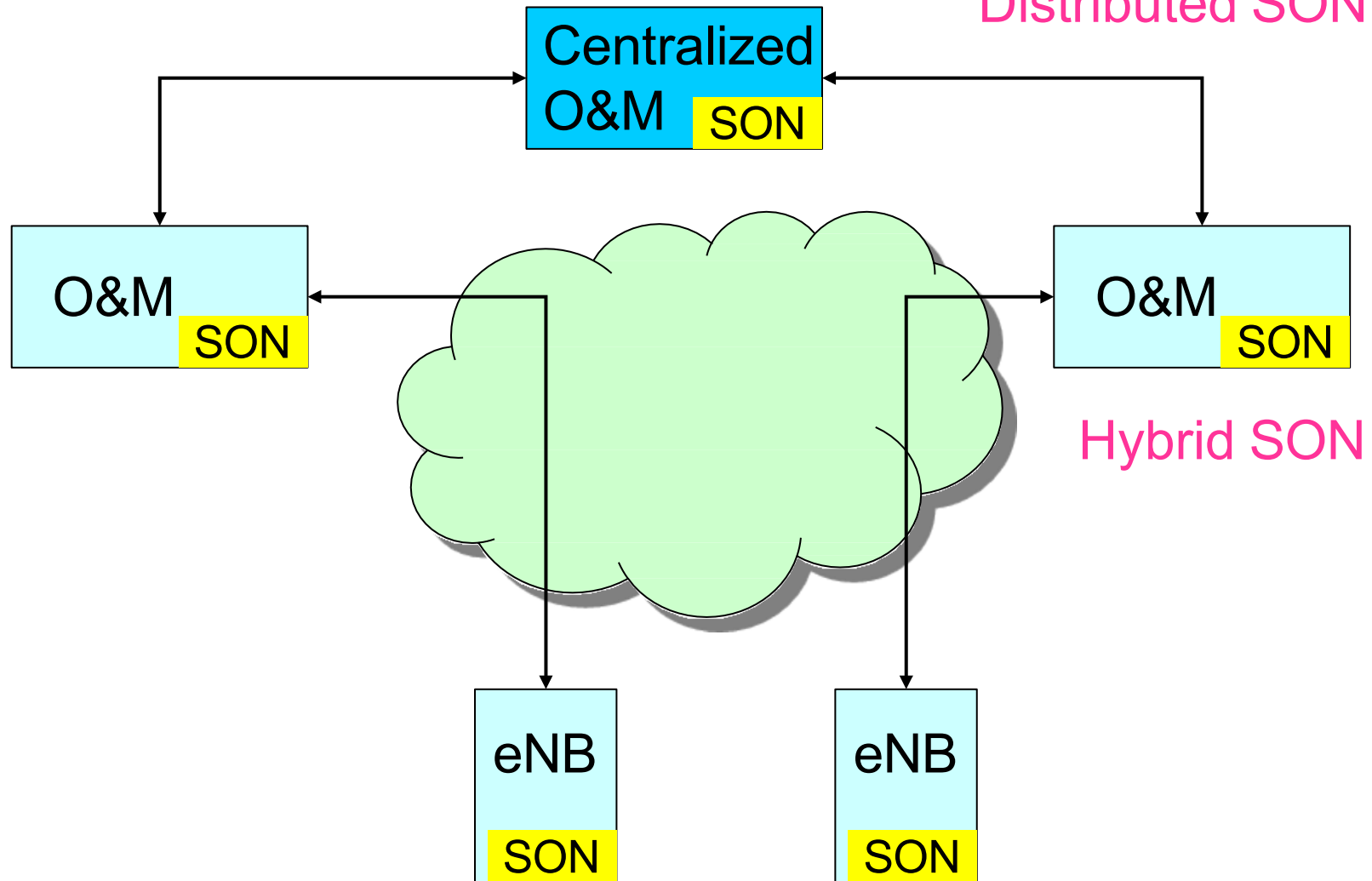
No way without SON

Overview SON



Architecture

Centralized SON
Distributed SON



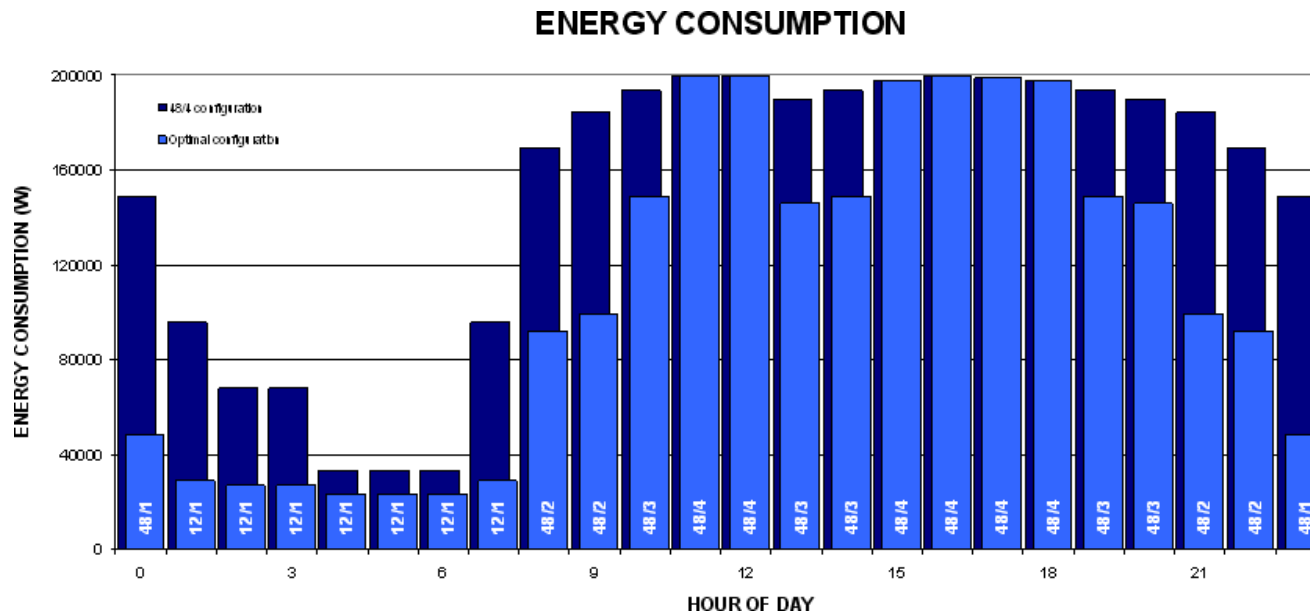
Hybrid SON

Energy Savings

I Match Network Capacity to the required traffic

I Switch on / off cell on demand

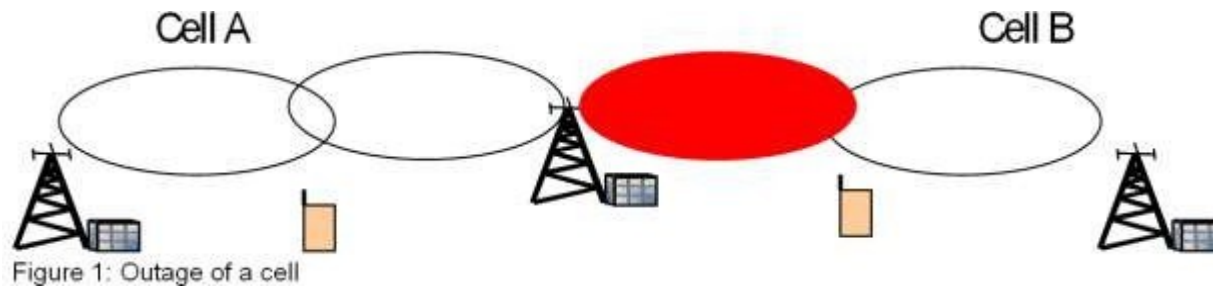
I Vodafone contribution on NGMN:



Self Healing

I Automatic detection of failures (Sleeping Cells)

- I Usually detected by performance statistics
- I Unreliable, because statistics sometimes fluctuate largely



I Cell Outage Compensation

- I Recovery actions
 - Fallback to previous SW
 - Switching to backup units for HW
- I Self optimization of the surrounding NW

Automatic Configuration of Physical Cell ID

Release 8

- I **Automatic configuration of new deployed eNodeBs**

- I 504 Physical Cell IDs are supported

- I **Selection of Physical Cell ID must be**

- I Collision free
 - The ID is unique in the area the cell covers
 - I Confusion free
 - A cell shall not have neighboring cells with identical IDs

- I **Definitely a must for Femto-Cells (HeNBs)**

- I **Centralized or Distributed Architecture**

Mobility Robustness Optimization (MRO)

Release 9

I **Manually setting of HO parameters**

- I Time consuming → Often neglected
- I Optimal settings may depend on momentary radio conditions
 - Difficult to control manually

I **Incorrect HO parameter setting may lead to**

- I Non-optimal use of network resources
 - Unnecessary handovers
 - Prolonged connection to a non-optimal cell
- I HO failures
 - Degradation of the service performance
- I Radio Link failures
 - Combined impact on user experience and network performance
 - **The main objective of MRO is to reduce RLF**

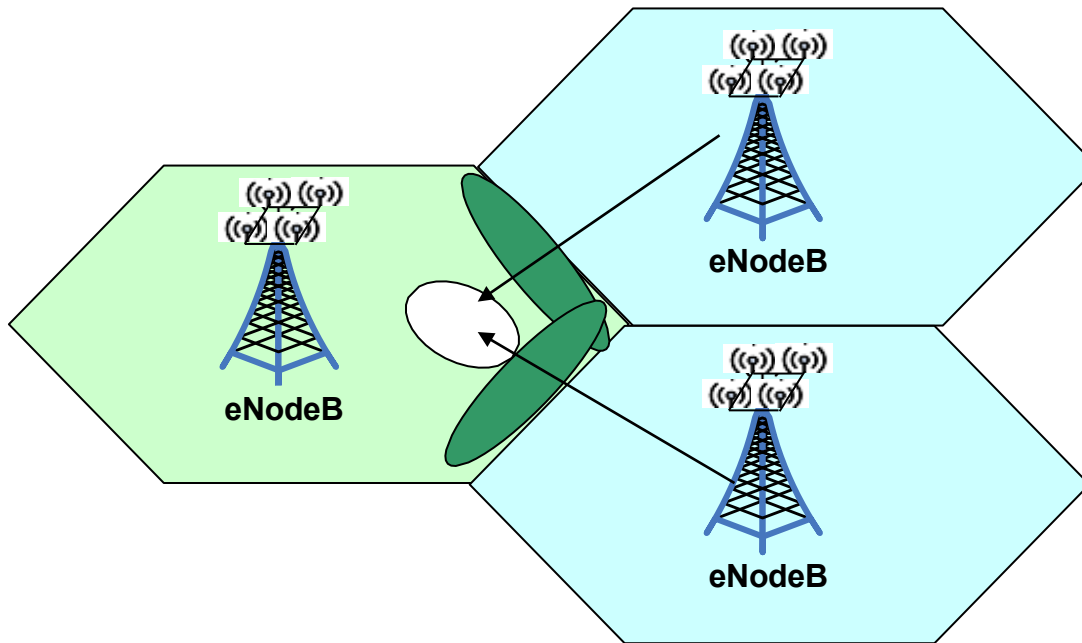
Coverage and Capacity Optimization

Pilot pollution (Interference)

Cell edge performance

Trade-Off between coverage and capacity

→ Coverage has priority



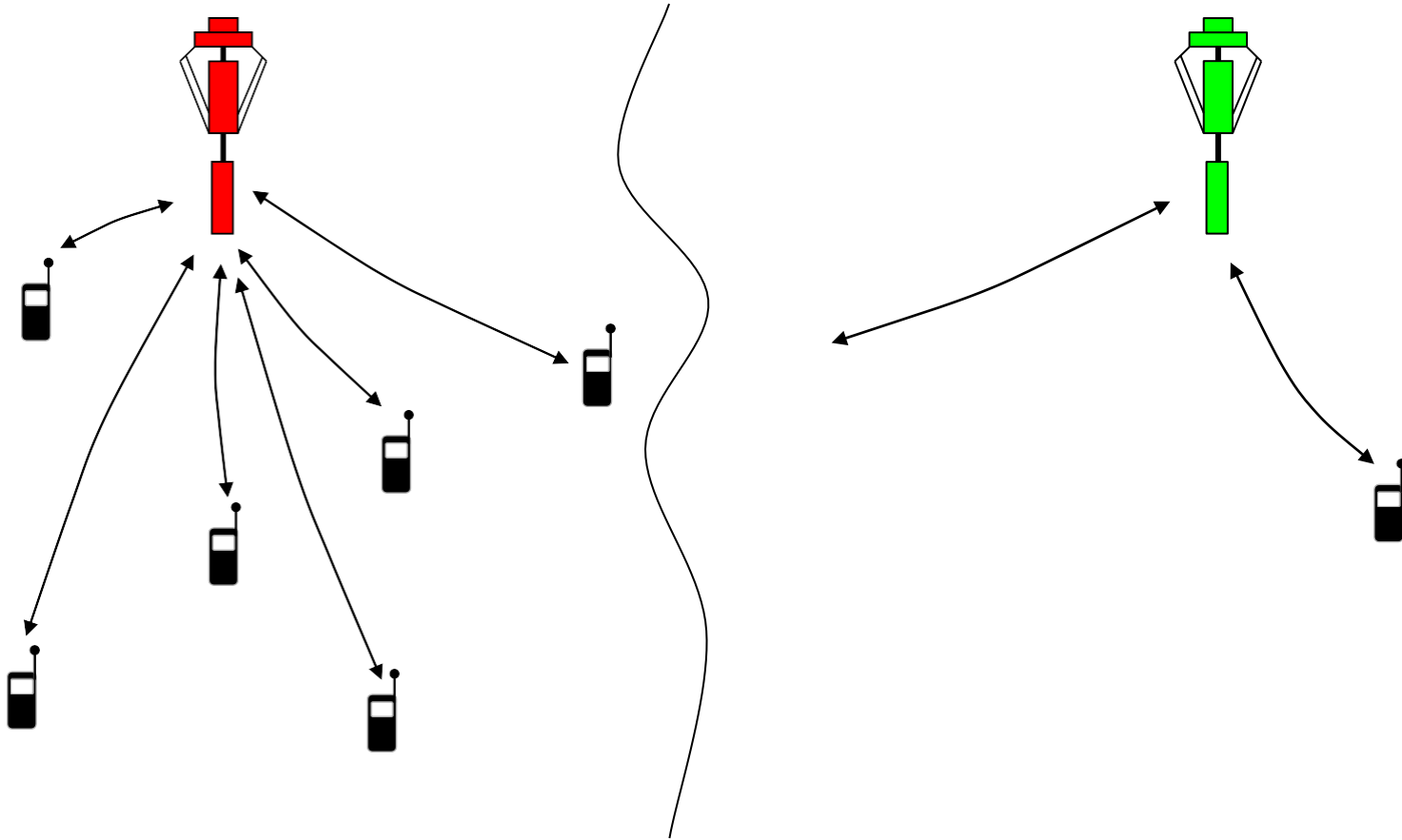
Measurement by Network:

- I Call drop rates**
→ Indication of insufficient coverage
- I Traffic counters**
→ Capacity problems

Detection of unintended coverage holes

Mobility Load Balancing (MLB) Optimization

Release 9



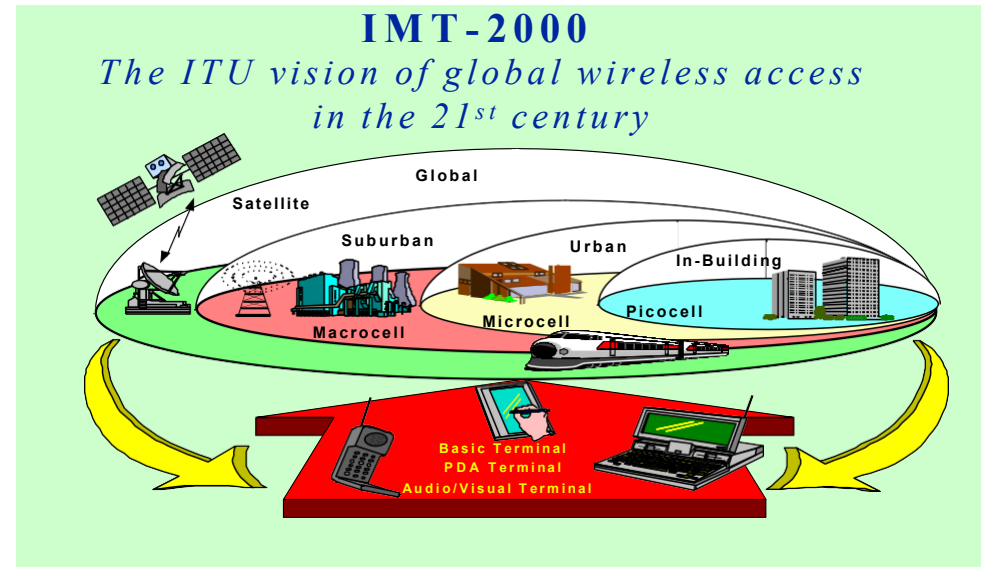
IMT-Advanced Requirements



- I A high degree of commonality of functionality worldwide while retaining the flexibility to support a wide range of services and applications in a cost efficient manner,**
 - I Compatibility of services within IMT and with fixed networks,**
 - I Capability of interworking with other radio access systems,**
 - I High quality mobile services,**
 - I User equipment suitable for worldwide use,**
 - I User-friendly applications, services and equipment,**
 - I Worldwide roaming capability; and**
 - I Enhanced peak data rates to support advanced services and applications,**
 - I 100 Mbit/s for high and**
 - I 1 Gbit/s for low mobility**
- were established as targets for research,**

Do you Remember?

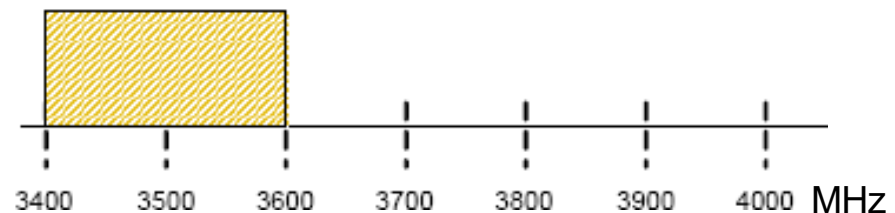
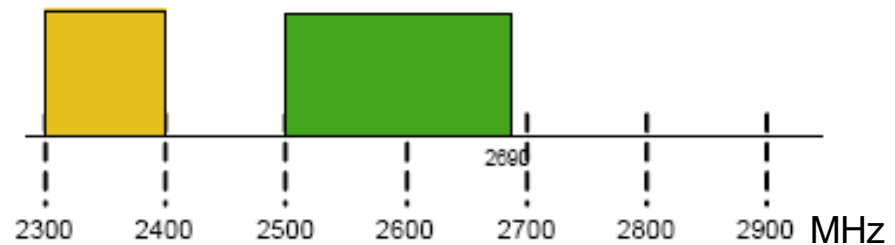
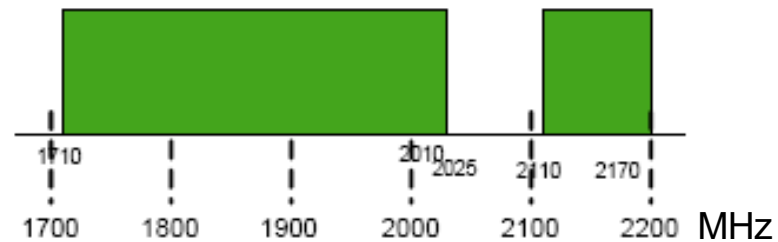
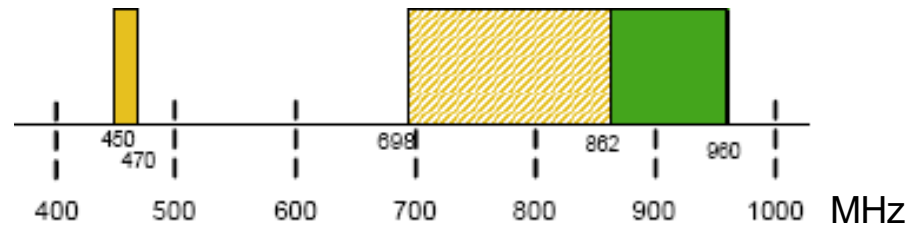
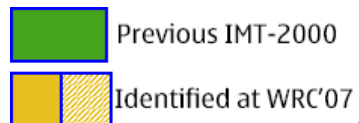
Targets of ITU IMT-2000 Program (1998)



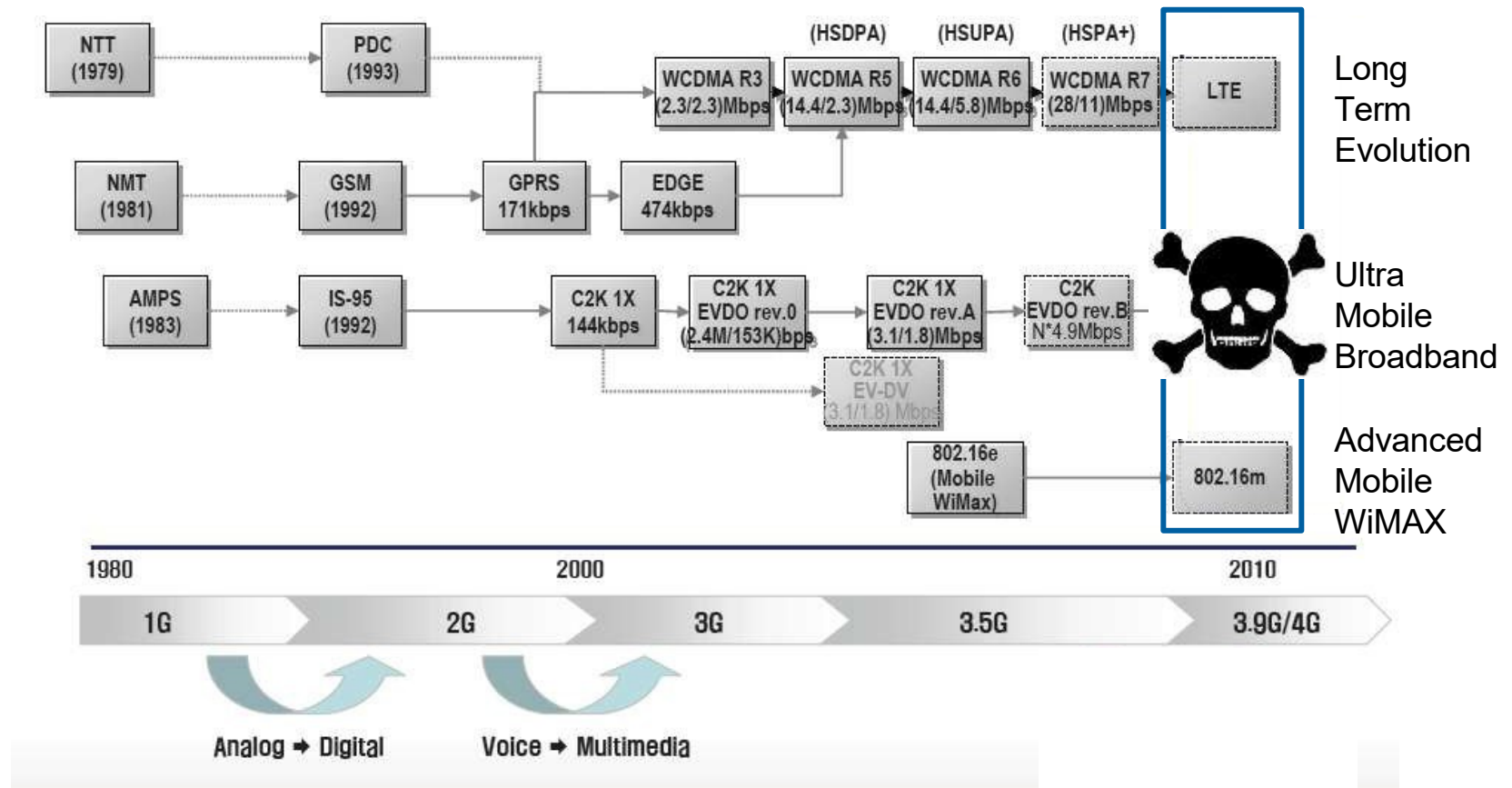
- I Flexible and global
 - Full coverage and mobility at 144 kbps .. 384 kbps
 - Hot spot coverage with limited mobility at 2 Mbps
 - Terrestrial based radio access technologies
- I *The IMT-2000 family of standards now supports four different multiple access technologies:*
 - I *FDMA, TDMA, CDMA (WCDMA) and OFDMA (since 2007)*

IMT Spectrum

Next possible spectrum allocation at WRC 2015!



Expected IMT-Advanced candidates



Source: TTA's workshop for the future of IMT-Advanced technologies, June 2008

IMT – International Mobile Communication

I IMT-2000

- I Was the framework for the third Generation mobile communication systems, i.e. 3GPP-UMTS and 3GPP2-C2K
- I Focus was on high performance transmission schemes:
Link Level Efficiency
- I Originally created to harmonize 3G mobile systems and to increase opportunities for worldwide interoperability, the IMT-2000 family of standards now supports four different access technologies, including OFDMA (WiMAX), FDMA, TDMA and CDMA (WCDMA).

I IMT-Advanced

- I Basis of (really) broadband mobile communication
- I Focus on **System Level Efficiency** (e.g. cognitive network systems)
- I Vision 2010 – 2015

System level efficiency

- I **Todays mobile communication networks use static frequency allocation**
 - I Network planning
 - I Adaptation to traffic load over cell boarder not possible
- I **Dynamic spectrum allocation to increase system efficiency**
 - I Radio resource management to configure the occupied bandwidth of each cell
 - I Dynamic management for inter-cell interference reduction
 - I Cognitive networks (self organizing networks SON) which adjust automatically to traffic load and interference conditions.
- I **Application oriented source and channel coding**
 - I Typically, source and channel coding are separated, i.e. MPEG and convolutional coding. Joint Source & Channel Coding (JSCC) promises better efficiency
- I **Base stations are on 24/7– but why?**
 - I Basisstations/Relaystations operate in „sleep“ or „off“ modes
 - I Enhancements of interference situations and energy consumption
- I **Too many interfaces reduce the throughput**
 - I Reducing the amount of components in the network structure
- I **Heterogenuous networks**
 - I Usage of various radio access technologies of same core network

LTE-Advanced

Possible technology features

**Relaying
technology**

**Wider bandwidth
support**

**Enhanced MIMO
schemes for DL and UL**

**Cooperative
base stations**

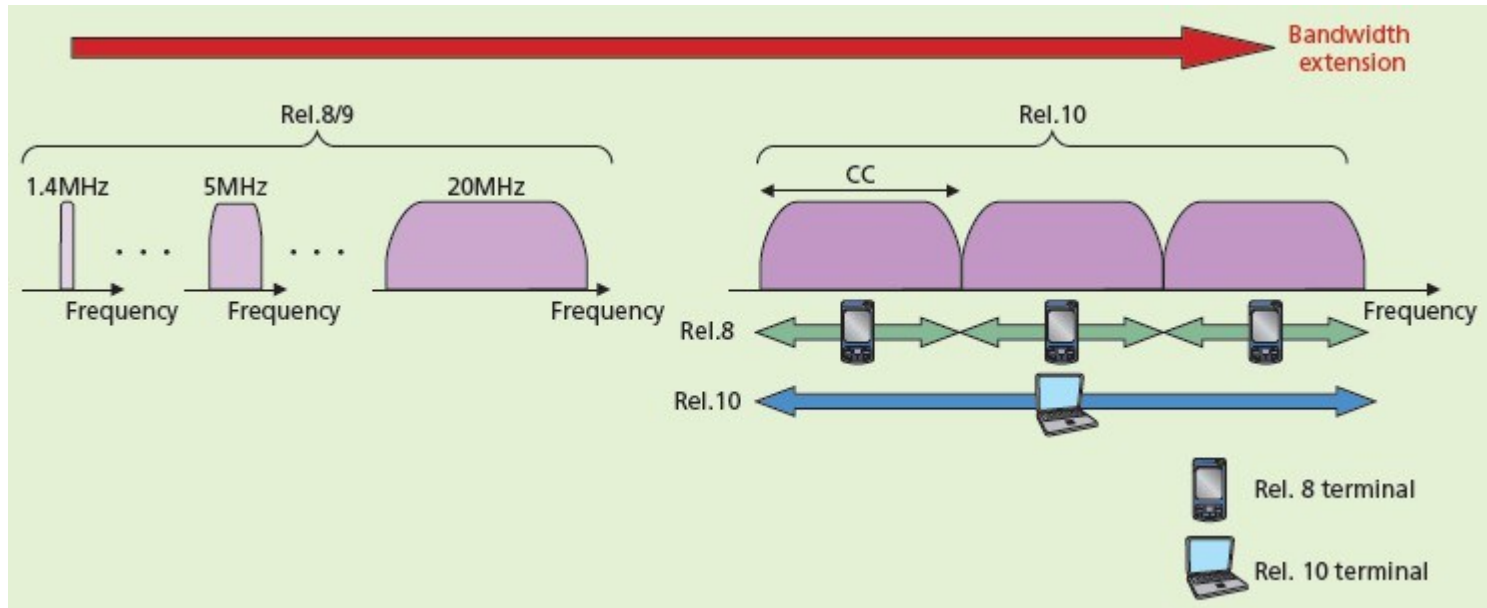
**Interference management
methods**

**Cognitive radio
methods**

Radio network evolution

**Further enhanced
MBMS**

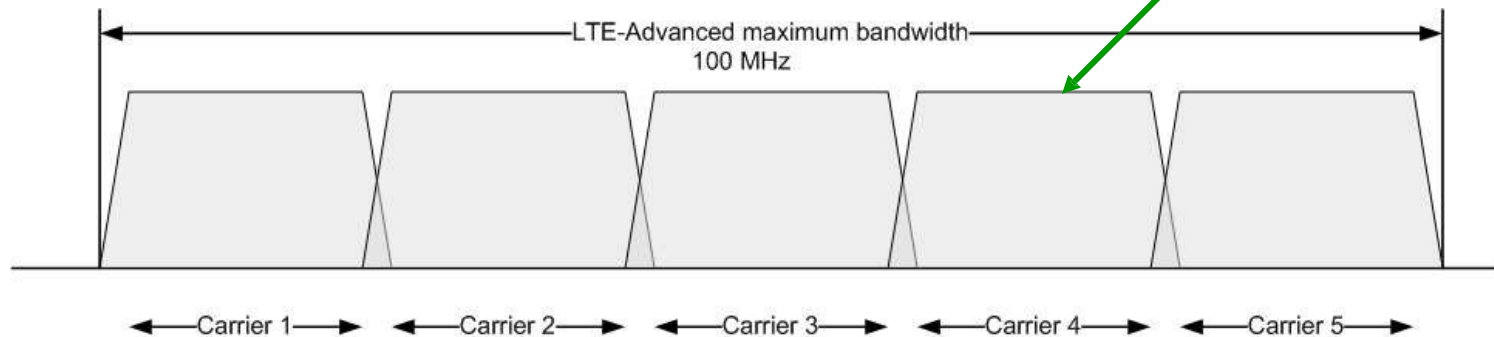
Bandwidth extension with Carrier aggregation



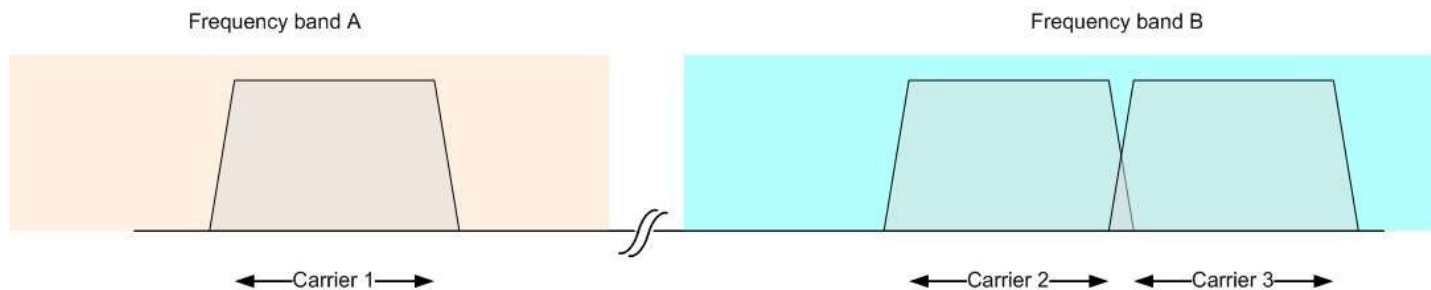
LTE-Advanced Carrier Aggregation



Component carrier CC



Contiguous carrier aggregation



Non-contiguous carrier aggregation

Aggregation

I Contiguous

I Intra-Band



I Non-Contiguous

I Intra (Single) -Band



I Inter (Multi) -Band



I Combination

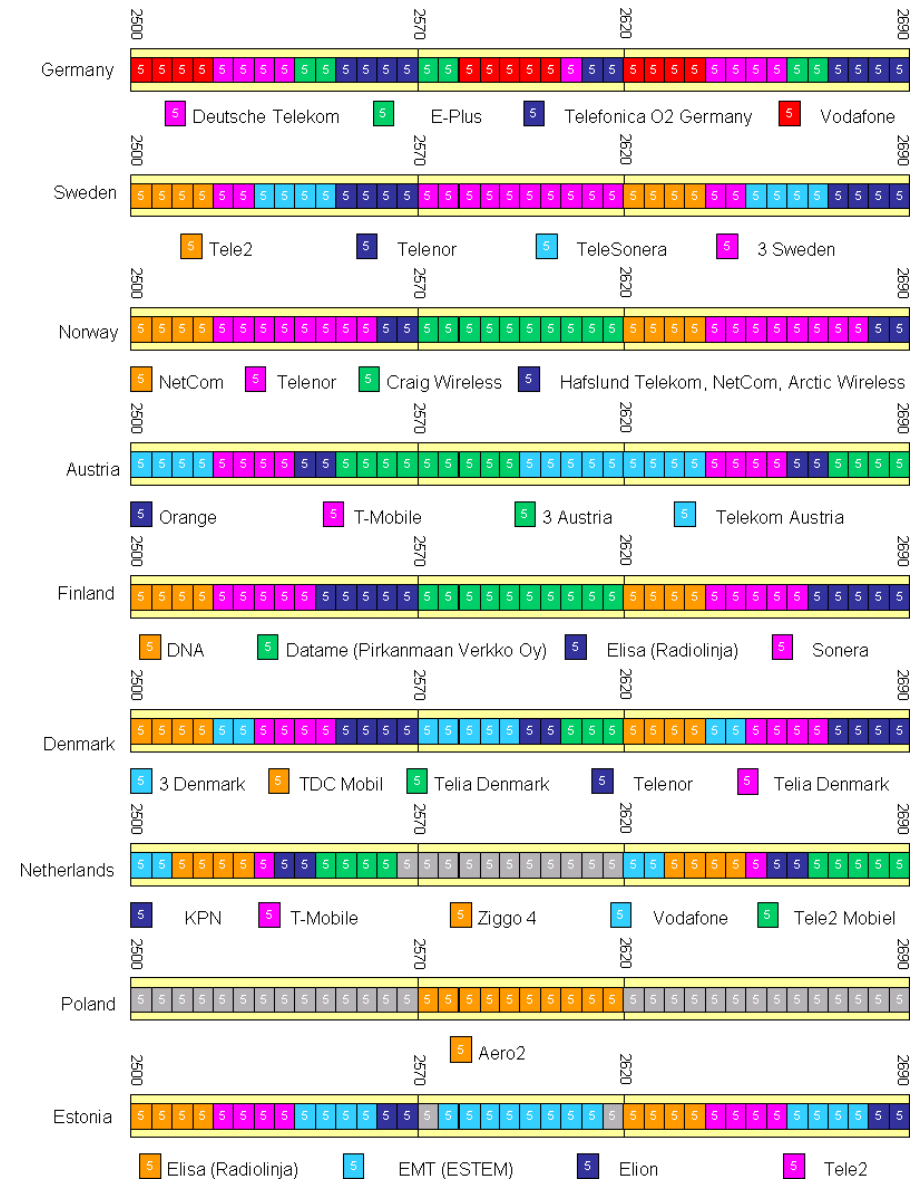
CC1 (20MHz)	CC2 (20MHz)	CC3 (20MHz)	CC4 (20MHz)	CC5 (20MHz)
-------------	-------------	-------------	-------------	-------------

I Up to 5 Rel-8 CC and 100 MHz

I Theoretically all CC-BW combinations possible (e.g. 5+10+20 etc)

Motivation

- 1) Higher data rate through more spectrum allocation
 - to fulfill 4G requirements
 - 1 Gbit/s in downlink (up to 5 component carriers, up to 100 MHz)
 - Other methods (spectral efficiency etc.) already exploited or not relevant
- 2) Exploit the total (combined) BW assigned in form of separated bands
 - scattered spectrum



Carrier aggregation (CA)

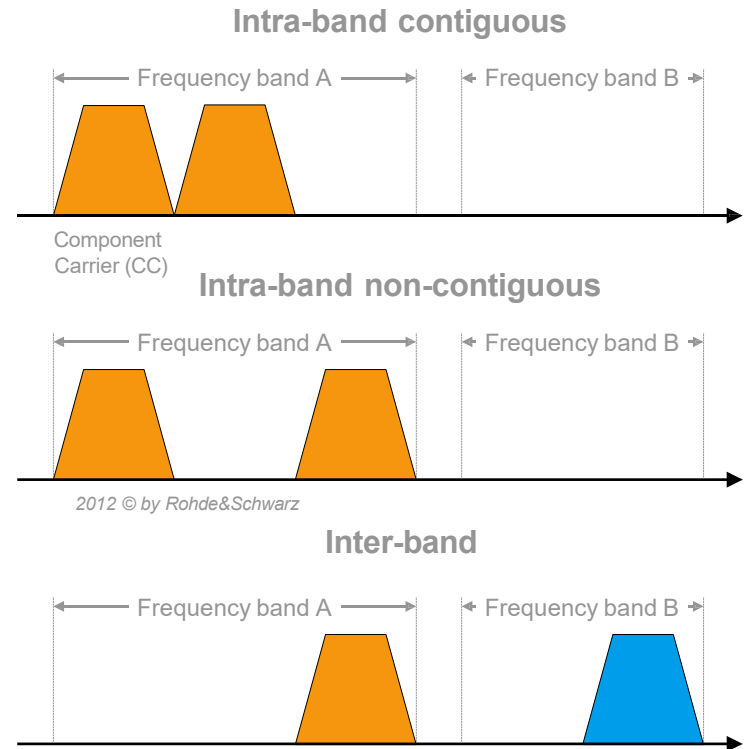
General comments

I Two or more component carriers are aggregated in LTE-Advanced in order to support wider bandwidths up to 100 MHz.

- I Support for contiguous and non-contiguous component carrier aggregation (intra-band) and inter-band carrier aggregation.
- I Different bandwidths per component carrier (CC) are possible.
- I Each CC limited to a max. of 110 RB using the 3GPP Rel-8 numerology (max. 5 carriers, 20 MHz each).

I Motivation.

- I Higher peak data rates to meet IMT-Advanced requirements.
- I NW operators: spectrum aggregation, enabling Heterogonous Networks.

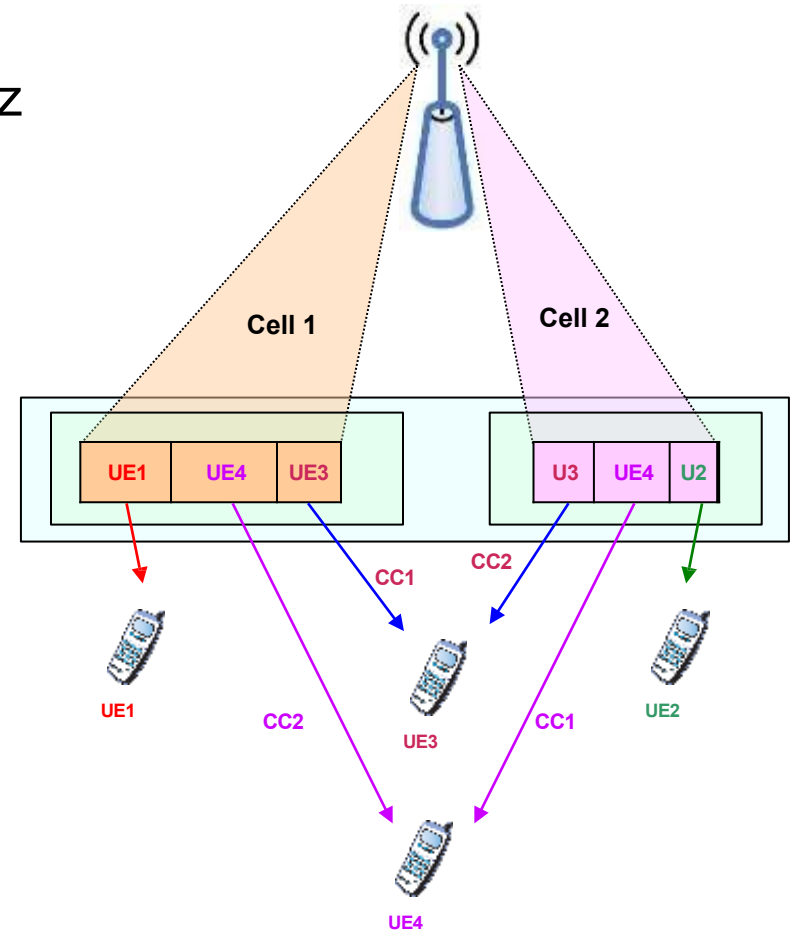


Overview

I Carrier Aggregation (CA)

enables to aggregate up to 5 different cells (component carriers CC), so that a maximum system bandwidth of 100 MHz can be supported (LTE-Advanced requirement).

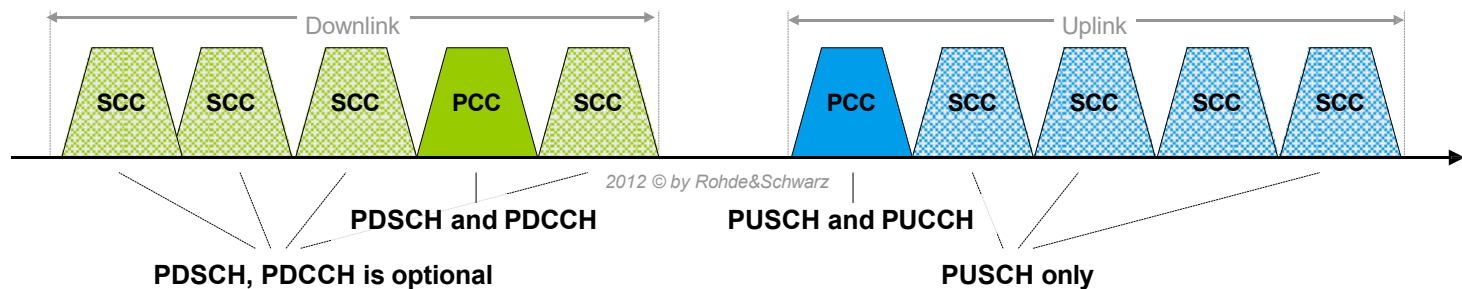
- I Each CC = Rel-8 autonomous cell
 - Backwards compatibility
- I CC-Set is UE specific
 - Registration → Primary (P)CC
 - Additional BW → Secondary (S)CC-s 1-4
- I Network perspective
 - Same single RLC-connection for one UE (independent on the CC-s)
 - Many CC (starting at MAC scheduler) operating the UE
- I For TDD
 - Same UL/DL configuration for all CC-s



Carrier aggregation (CA)

General comments, cont'd.

- I **A device capable of carrier aggregation has 1 DL primary component carrier and 1 associated primary UL component carrier.**
 - I Basic linkage between DL and UL is signaled in SIB Type 2.
 - I Configuration of primary component carrier (PCC) is UE-specific.
 - Downlink: cell search / selection, system information, measurement and mobility.
 - Uplink: access procedure on PCC, control information (PUCCH) on PCC.
 - Network may decide to switch PCC for a device → handover procedure is used.
- I Device may have one or several secondary component carriers. Secondary Component Carriers (SCC) added in RRC_CONNECTED mode only.
 - Symmetric carrier aggregation.
 - Asymmetric carrier aggregation (= Rel-10).



LTE-Advanced

Carrier Aggregation – Initial Deployment



- I Initial LTE-Advanced deployments will likely be limited to the use of two component carrier.
- I The below are focus scenarios identified by 3GPP RAN4.

Intra-band contiguous carrier aggregation	
FDD	UL/DL: 40 MHz in Band 3
TDD	UL/DL: 50 MHz in Band 40
Inter-band non-contiguous carrier aggregation	
Region 1 (Europe)	UL/DL: 40 MHz; 20 MHz CC (Band 7) and 20 MHz CC (Band 20)
	UL/DL: 40 MHz; 20 MHz CC (Band 3) and 20 MHz CC (Band 20)
	UL/DL: 40 MHz; 20 MHz CC (Band 7) and 20 MHz CC (Band 3)
Region 2 (US)	UL/DL: 20 MHz; 10 MHz CC (Band 5) and 10 MHz CC (Band 12)
	UL/DL: 10 MHz; 5 MHz CC (Band 17) and 5 MHz CC (Band 4)
Region 3 (Asia)	UL/DL: 20 MHz; 10 MHz CC (Band 1) and 10 MHz CC (Band 18/19)
	UL/DL: 40 MHz; 20 MHz CC (Band 38) and 20 MHz CC (Band 40)

Bandwidth

I General

- Up to 5 CC
- Up to 100 RB-s pro CC
- Up to 500 RB-s aggregated

I Aggregated transmission bandwidth

- Sum of aggregated channel bandwidths
- Illustration for Intra band contiguous
- Channel raster 300 kHz

I Bandwidth classes

- UE Capability

$$\text{Nominal channel spacing} = \left\lceil \frac{BW_{\text{Channel}(1)} + BW_{\text{Channel}(2)} - 0.1|BW_{\text{Channel}(1)} - BW_{\text{Channel}(2)}|}{0.6} \right\rceil 0.3 \text{ [MHz]}$$

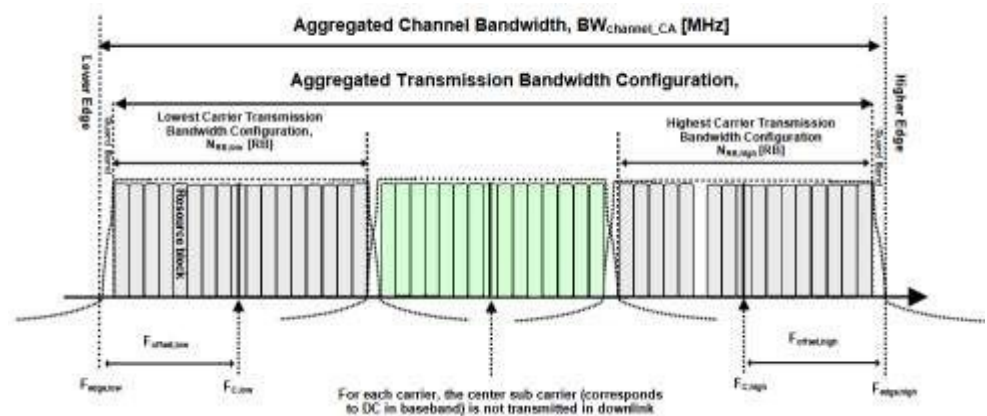
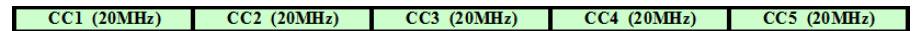


Table 5.6A-1: CA bandwidth classes and corresponding nominal guard bands

CA Bandwidth Class	Aggregated Transmission Bandwidth Configuration	Maximum number of CC	Nominal Guard Band BW_{GB}
A	$N_{RB,agg} \leq 100$	1	$0.05BW_{\text{Channel}(1)}$
B	$N_{RB,agg} \leq 100$	2	FFS
C	$100 < N_{RB,agg} \leq 200$	2	$0.05 \max(BW_{\text{Channel}(1)}, BW_{\text{Channel}(2)})$
D	$200 < N_{RB,agg} \leq [300]$	FFS	FFS
E	$[300] < N_{RB,agg} \leq [400]$	FFS	FFS
F	$[400] < N_{RB,agg} \leq [500]$	FFS	FFS

Note 1: $BW_{\text{Channel}(1)}$ and $BW_{\text{Channel}(2)}$ are channel bandwidths of two E-UTRA component carriers according to Table 5.6-1.

Bands / Band-Combinations (I)

I E-UTRA CA Band

- I Band / Band-Combinations specified in RAN4 for CA scenarios
- I Already 4 CA Bands specified
 - For inter frequency → bands without practical interest to guarantee quick progress of the work

Table 5.5A-1: Intra-band contiguous CA operating bands

E-UTRA CA Band	E-UTRA Band	Uplink (UL) operating band			Downlink (DL) operating band			Duplex Mode
		BS receive / UE transmit			BS transmit / UE receive			
		F _{UL_low} – F _{UL_high}			F _{DL_low} – F _{DL_high}			
CA_1	1	1920 MHz	–	1980 MHz	2110 MHz	–	2170 MHz	FDD
CA_40	40	2300 MHz	–	2400 MHz	2300 MHz	–	2400 MHz	TDD
CA_41	41	2496 MHz		2690 MHz	2496 MHz	–	2690 MHz	TDD

Table 5.5A-2: Inter-band CA operating bands

E-UTRA CA Band	E-UTRA Band	Uplink (UL) operating band			Downlink (DL) operating band			Duplex Mode
		BS receive / UE transmit			BS transmit / UE receive			
		$F_{UL_low} - F_{UL_high}$			$F_{DL_low} - F_{DL_high}$			
CA_1-5	1	1920 MHz	–	1980 MHz	2110 MHz	–	2170 MHz	FDD
	5	824 MHz	–	849 MHz	869 MHz	–	894 MHz	

Bands / Band-Combinations (II)

I Under discussion

- I 25 WI-s in RAN4 with practical interest
 - Inter band (1 UL CC)
 - Intra band cont (1 UL CC)
 - Intra band non cont (1 UL CC)
 - Inter band (2 UL CC)

I Release independency

- I Band performance is release independent
 - Band introduced in Rel-11
 - Performance tested for Rel-10

Intra band contiguous

I	LTE_CA_B7	FDD 2500-2570 / 2620-2690
I	LTE_CA_B25	FDD 1850-1915 / 1930-1995
I	LTE_CA_B38	TDD 2570-2620 / 2570-2620

Intra band non contiguous

I	LTE_CA_NC_B3	FDD 1710-1785 / 1805-1880
I	LTE_CA_NC_B25	FDD 1850-1915 / 1930-1995

Inter band

I	LTE_CA_B1_B7	FDD 1920-1980 / 2110-2170	+ FDD 2500-2570 / 2620-2690
I	LTE_CA_B1_B18	FDD 1920-1980 / 2110-2170	+ FDD 815-830 / 860-875
I	LTE_CA_B1_B19	FDD 1920-1980 / 2110-2170	+ FDD 830-845 / 875-890
I	LTE_CA_B1_B21	FDD 1920-1980 / 2110-2170	+ FDD 1447.9-1462.9 / 1495.9-1510.9
I	LTE_CA_B2_B17	FDD 1850-1910 / 1930-1990	+ FDD 704-716 / 734-746
I	LTE_CA_B3_B5	FDD 1710-1785 / 1805-1880	+ FDD 824-849 / 869-894
I	LTE_CA_B3_B7	FDD 1710-1785 / 1805-1880	+ FDD 2500-2570 / 2620-2690
I	LTE_CA_B3_B8	FDD 1710-1785 / 1805-1880	+ FDD 880-915 / 925-960
I	LTE_CA_B3_B20	FDD 1710-1785 / 1805-1880	+ FDD 832-862 / 791-821
I	LTE_CA_B4_B5	FDD 1710-1755 / 2110-2155	+ FDD 824-849 / 869-894
I	LTE_CA_B4_B7	FDD 1710-1755 / 2110-2155	+ FDD 2500-2570 / 2620-2690
I	LTE_CA_B4_B12	FDD 1710-1755 / 2110-2155	+ FDD 699-716 / 729-746
I	LTE_CA_B4_B13	FDD 1710-1755 / 2110-2155	+ FDD 777-787 / 746-756
I	LTE_CA_B4_B17	FDD 1710-1755 / 2110-2155	+ FDD 704-716 / 734-746
I	LTE_CA_B5_B12	FDD 824-849 / 869-894	+ FDD 699-716 / 729-746
I	LTE_CA_B5_B17	FDD 824-849 / 869-894	+ FDD 704-716 / 734-746
I	LTE_CA_B7_B20	FDD 2500-2570 / 2620-2690	+ FDD 832-862 / 791-821
I	LTE_CA_B8_B20	FDD 880-915 / 925-960	+ FDD 832-862 / 791-821
I	LTE_CA_B11_B18	FDD 1427.9-1447.9 / 1475.9-1495.9	+ FDD 815-830 / 860-875

Inter band with 2UL

I	LTE_CA_B3_B5_2UL	FDD 1710-1785 / 1805-1880	+ FDD 824-849 / 869-894
---	------------------	---------------------------	-------------------------

Carrier aggregation - configurations

I CA Configurations

I E-UTRA CA Band + Allowed BW = CA Configuration

- Intra band contiguous
 - Most requirements
- Inter band
 - Some requirements
 - Main interest of many companies
- Intra band non contiguous
 - No configuration / requirements
 - Feature of later releases?

Table 5.6A.1-1: Supported CC combinations per CA configuration for intra-band contiguous CA

CA Configuration / N _{RB,CA}					
CA Configuration	E-UTRA Band	50RB+100RB (10 MHz + 20 MHz)	75RB+75RB (15 MHz + 15 MHz)	75RB+100RB (15MHz + 20 MHz)	100RB+100RB (20 MHz + 20 MHz)
CA_1C	1		Yes	No	Yes
CA_40C	40	Yes	Yes	No	Yes
CA_41C	41	Yes	Yes	Yes	Yes

Table 5.6A.1-2: Supported E-UTRA bandwidths per CA configuration for inter-band CA

CA operating / Channel bandwidth							
CA Configuration	E-UTRA Bands	1.4 MHz	3 MHz	6 MHz	10 MHz	15 MHz	20 MHz
CA_1A-5A	1				Yes		
	5				Yes		

I CA Requirement applicability

- CL_X → Intra band CA
- CL_X-Y → Inter band
- Non-CA → no CA
(explicitely stated for the Test point which are tested differently for CA and not CA)

Table 8.1.1-1: Applicability of the requirement with respect to the CA capability

CA Capability	CA Capability Description
CL_X	The requirement is applicable to a UE that indicates a CA bandwidth class X on at least one E-UTRA band.
CL_X-Y	The requirement is applicable to a UE that indicates CA bandwidth classes X and Y on at least one E-UTRA band combination.
non-CA	The requirement is applicable to a UE that indicates the CA bandwidth class A (one component carrier) on all supported E-UTRA bands.
Note: The CA bandwidth classes are defined in Table 5.6A-1	

UE categories for Rel-10 *NEW!*

UE categories 6...8 (DL and UL)

UE Category	Maximum number of DL-SCH transport block bits received within a TTI	Maximum number of bits of a DL-SCH transport block received within a TTI	Total number of soft channel bits	Maximum number of supported layers for spatial multiplexing in DL
...	...	...	...	...
Category 6	301504	149776 (4 layers) 75376 (2 layers)	3654144	2 or 4
Category 7	301504	149776 (4 layers) 75376 (2 layers)	3654144	2 or 4
Category 8	2998560	299856	35982720	8

~3 Gbps peak
DL data rate
for 8x8 MIMO

UE Category	Maximum number of UL-SCH transport block bits transmitted within a TTI	Maximum number of bits of an UL-SCH transport block transmitted within a TTI	Support for 64QAM in UL
...	...	...	...
Category 6	51024	51024	No
Category 7	102048	51024	No
Category 8	1497760	149776	Yes

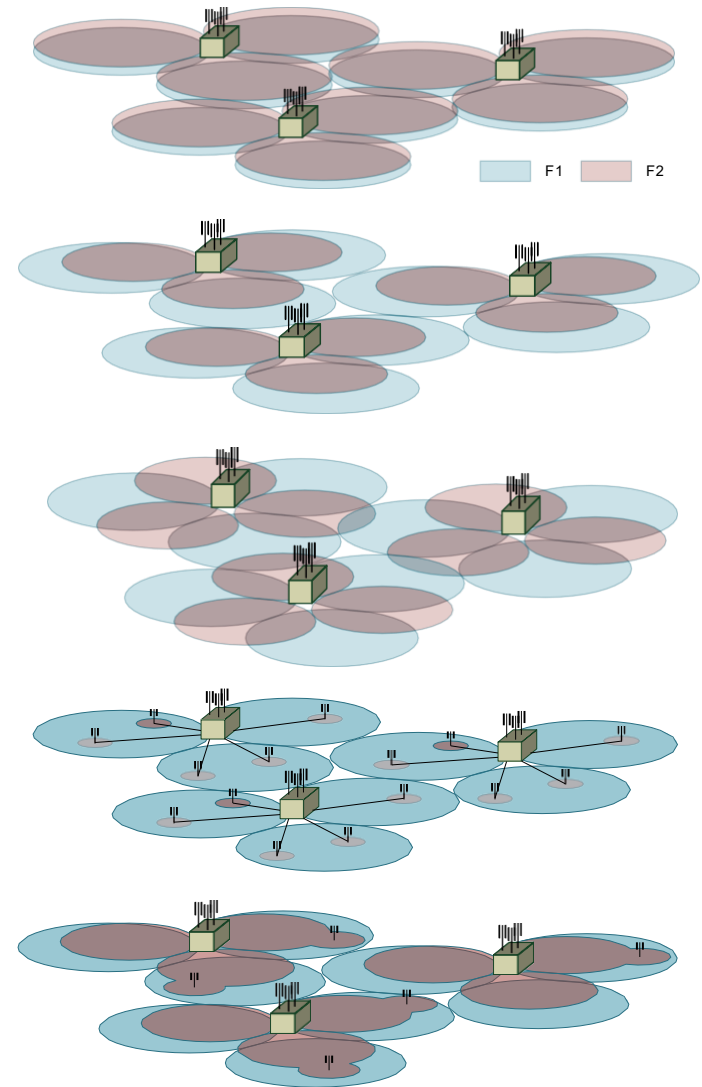
Total layer 2
buffer size
[bytes]

...
3 300 000
3 800 000
42 200 000

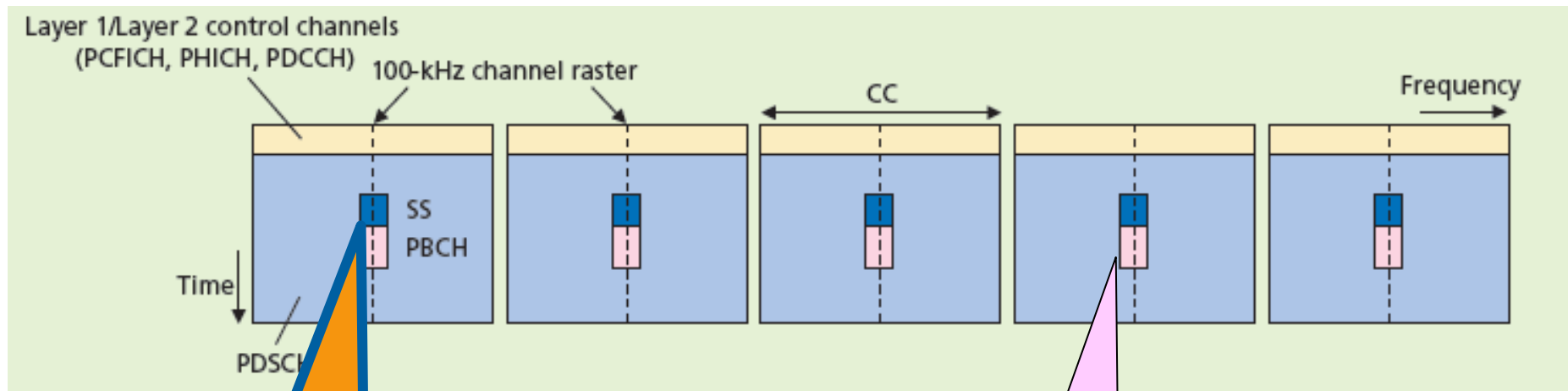
Deployment scenarios

3) Improve coverage

- I #1: Contiguous frequency aggregation
 - Co-located & Same coverage
 - Same f
- I #2: Discontiguous frequency aggregation
 - Co-located & Similar coverage
 - Different f
- I #3: Discontiguous frequency aggregation
 - Co-Located & Different coverage
 - Different f
 - Antenna direction for CC2 to cover blank spots
- I #4: Remote radio heads
 - Not co-located
 - Intelligence in central eNB, radio heads = only transmission antennas
 - Cover spots with more traffic
 - Is the transmission of each radio head within the cell the same?
- I #5: Frequency-selective repeaters
 - Combination #2 & #4
 - Different f
 - Extend the coverage of the 2nd CC with Relays



Physical channel arrangement in downlink



Each component carrier transmits P-SCH and S-SCH, Like Rel.8

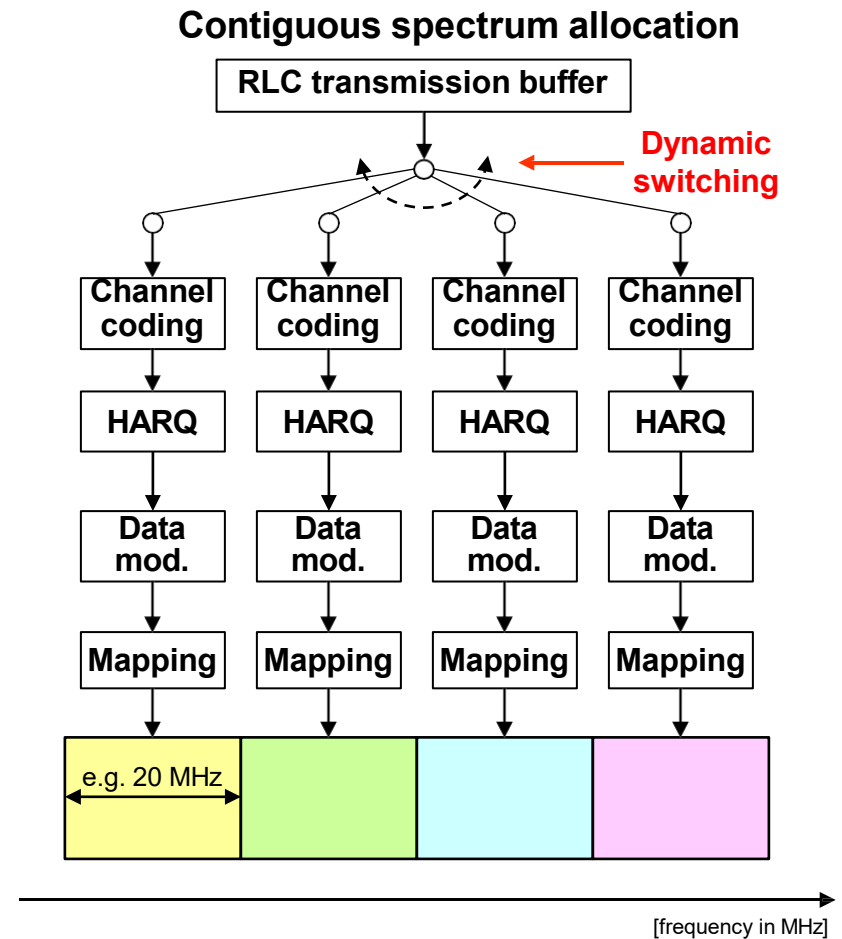
Each component carrier transmits PBCH, Like Rel.8

LTE-Advanced

Carrier Aggregation – Scheduling

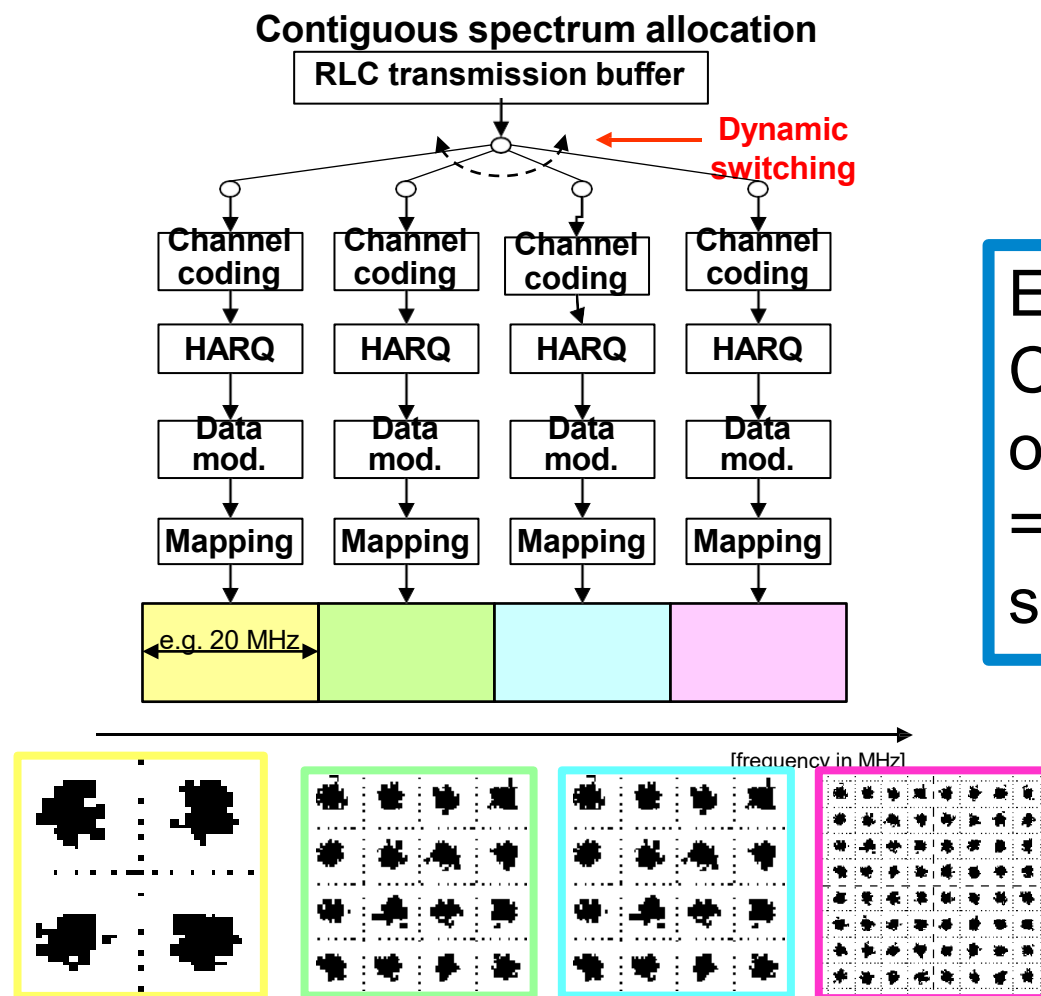


- I There is one transport block (in absence of spatial multiplexing) and one HARQ entity per scheduled component carrier (from the UE perspective),
- I A UE may receive multiple component carriers simultaneously,
- I Two different approaches are discussed how to inform the UE about the scheduling for each band,
 - I Separate PDCCH for each carrier,
 - I Common PDCCH for multiple carrier,



LTE-Advanced

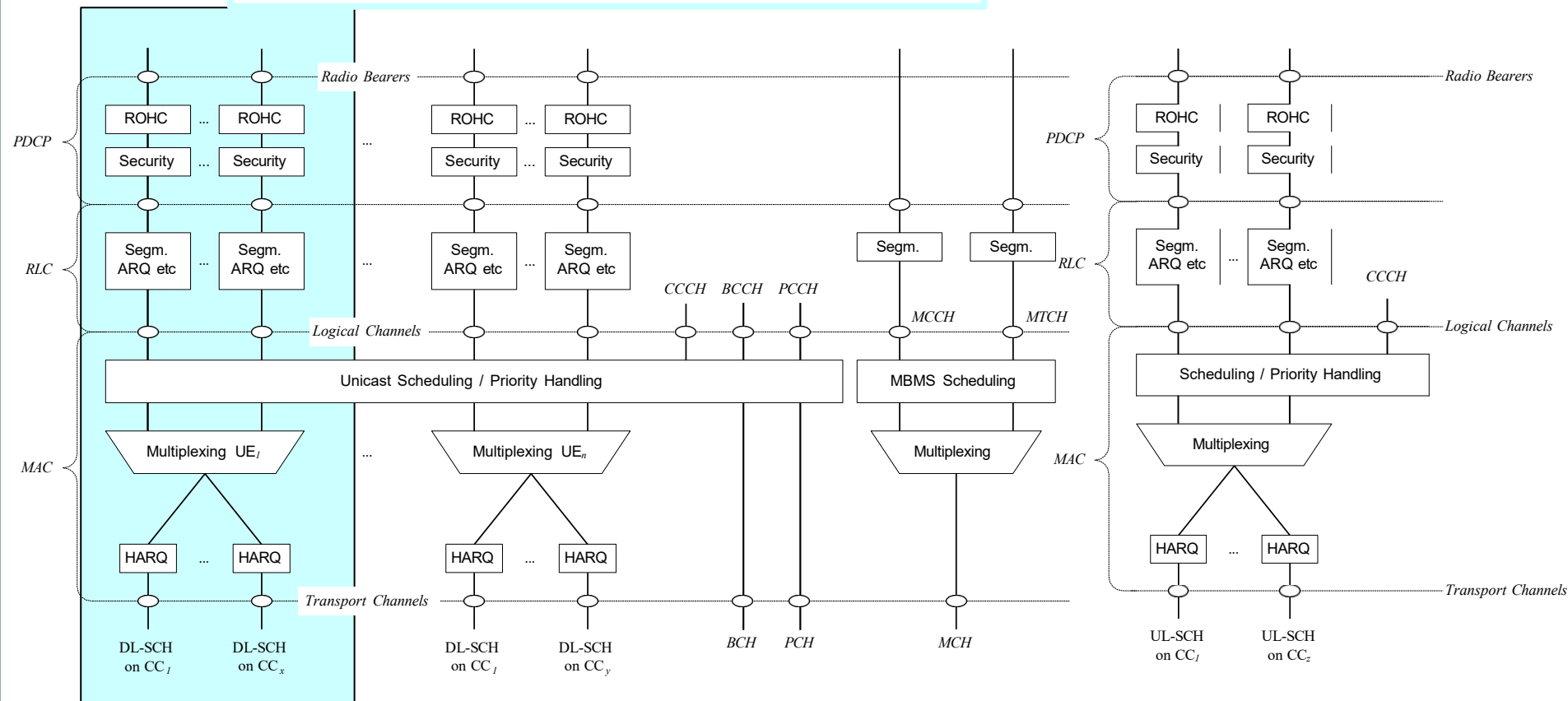
Carrier Aggregation – Scheduling



Each component Carrier may use its own AMC, = modulation + coding scheme

Carrier Aggregation – Architecture downlink

1 UE using carrier aggregation



In case of CA, the multi-carrier nature of the physical layer is only exposed to the MAC layer for which one HARQ entity is required per serving cell

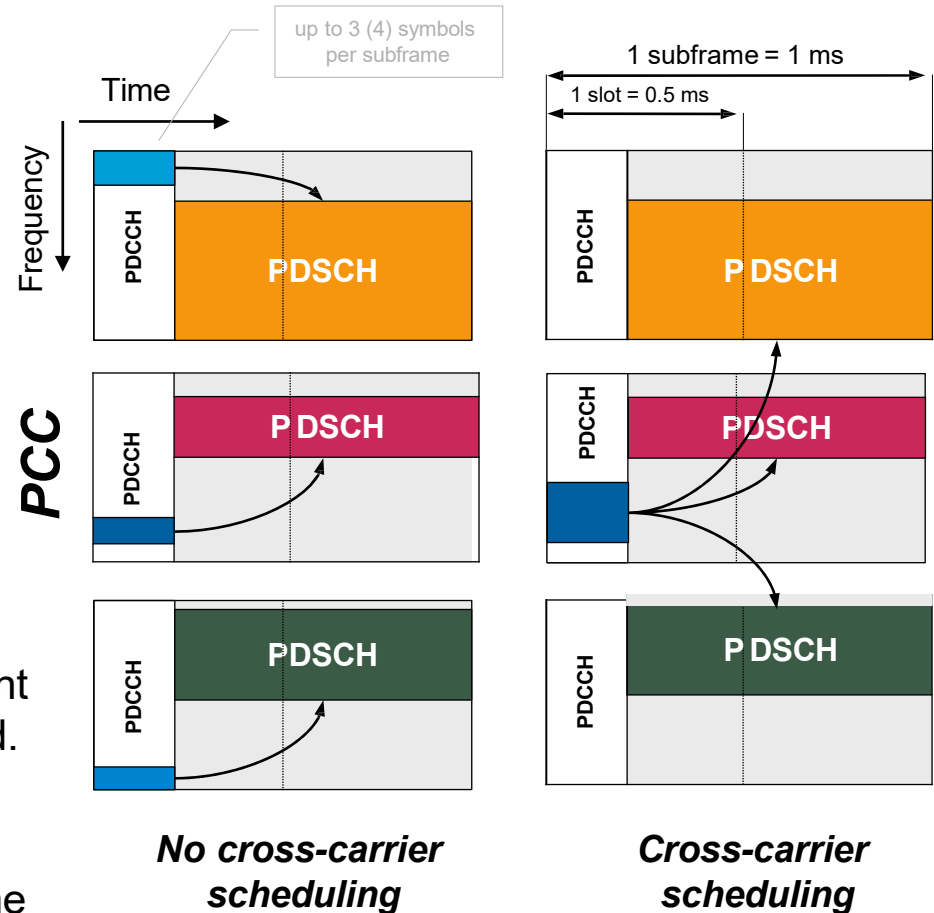
Common or separate PDCCH per Component Carrier?

I No cross-carrier scheduling.

- I PDCCH on a component carrier assigns PDSCH resources on the same component carrier (and PUSCH resources on a single linked UL component carrier).
- I Reuse of Rel-8 PDCCH structure (same coding, same CCE-based resource mapping) and DCI formats.

I Cross-carrier scheduling.

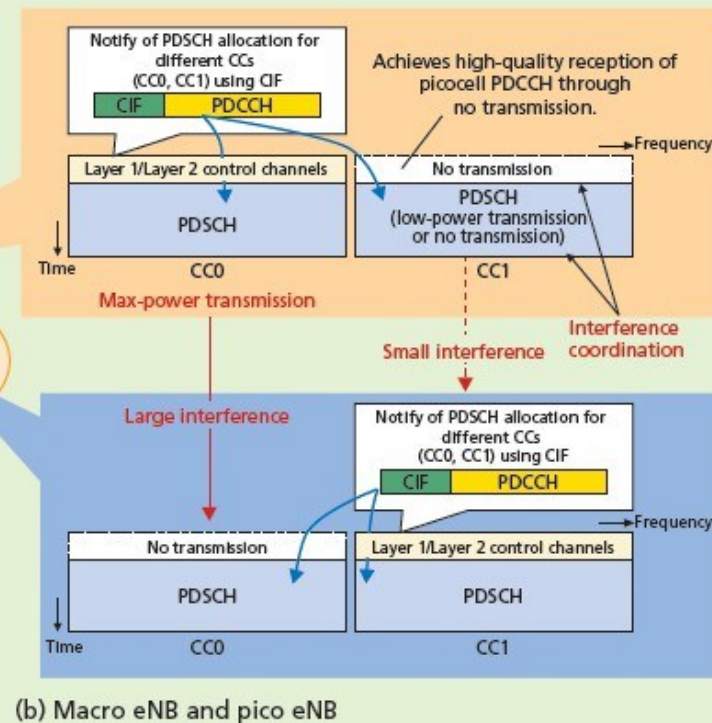
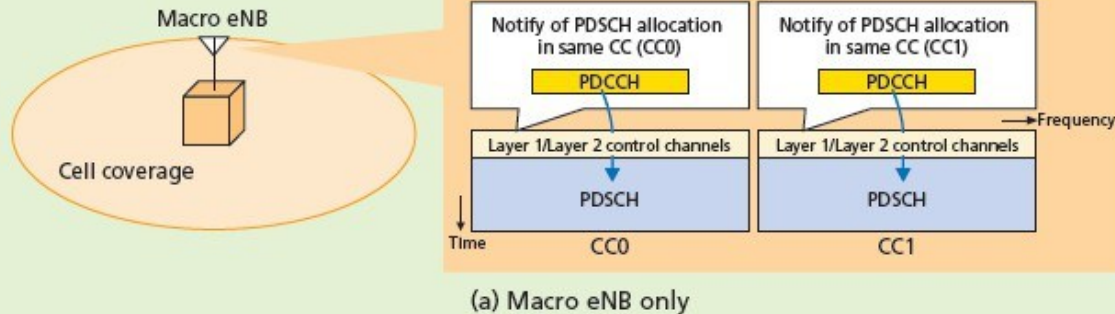
- I PDCCH on a component carrier can assign PDSCH or PUSCH resources in one of multiple component carriers using the carrier indicator field.
- I Rel-8 DCI formats extended with 3 bit carrier indicator field.
- I Reusing Rel-8 PDCCH structure (same coding, same CCE-based resource mapping)



Carrier aggregation: control signals + scheduling

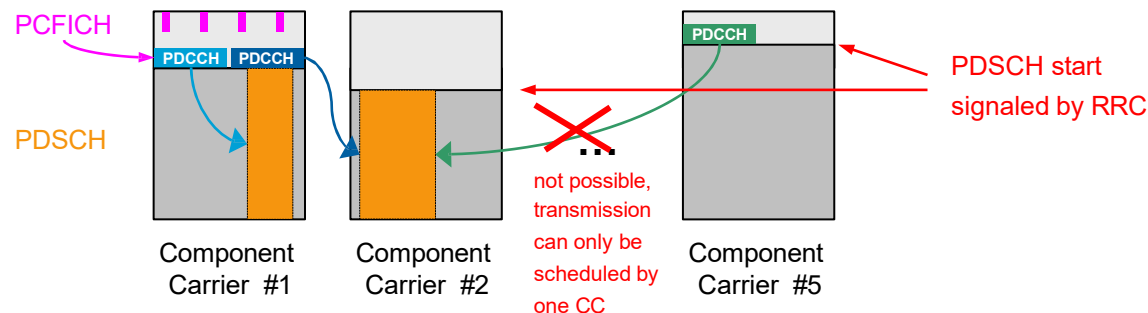
Each CC has its own control channels, like Rel.8

Femto cells:
Risk of interference!
-> main component carrier will send all control information.



Cross-carrier scheduling

- **Main motivation for cross carrier scheduling: Interference management for HetNet (eICIC); load balancing.**
- **Cross carrier scheduling is optional to a UE.**
 - Activated by RRC signaling, if not activated no CFI is present.
 - Component carriers are numbered, Primary Component Carrier (PCC) is always cell index 0.
 - Scheduling on a component carrier is only possible from ONE component, independent if cross-carrier scheduling is ON or OFF:



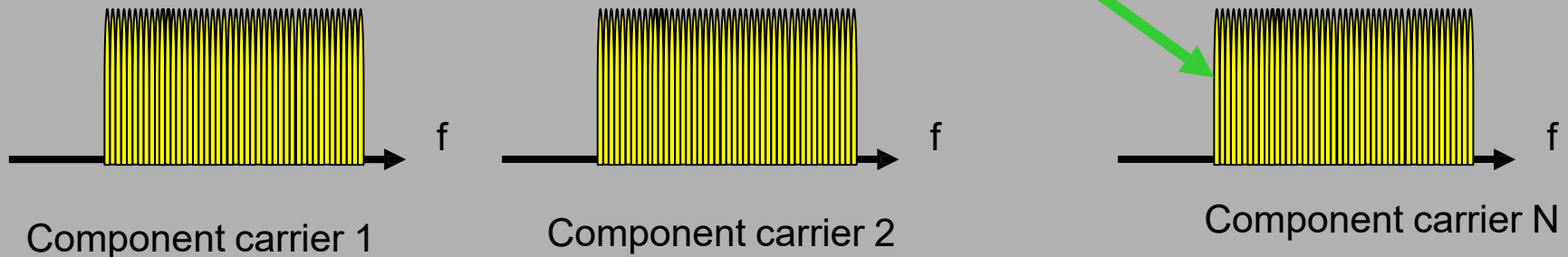
- If cross carrier-scheduling active, UE needs to be informed about PDSCH start on component carrier → RRC signaling.

DCI control elements: CIF field

New field: carrier indicator field gives information, which component carrier is valid.

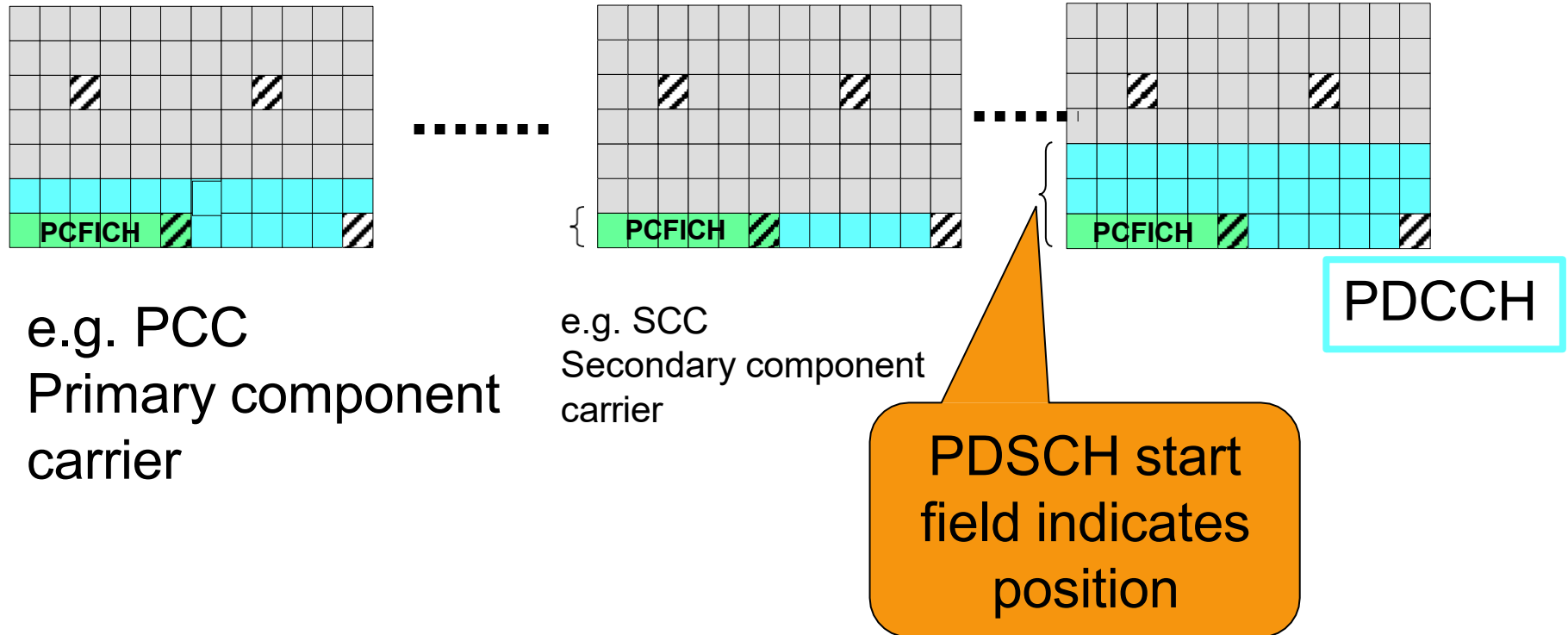
=> reminder: maximum 5 component carriers!

Carrier Indicator Field, CIF, 3bits



PDSCH start field

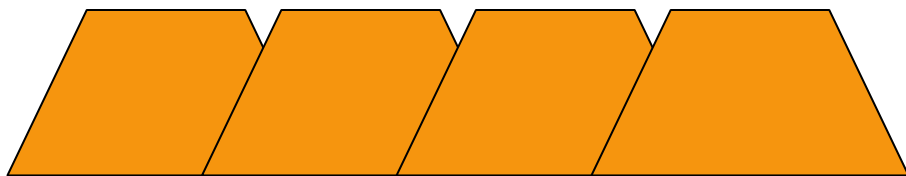
Example: 1 Resource block
Of a component carrier



In cross-carrier scheduling, the UE does not read the PCFICH and PDCCH on SCC, thus it has to know the start of the PDSCH

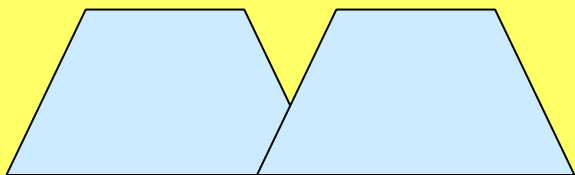
Component Carrier – RACH configuration

Downlink

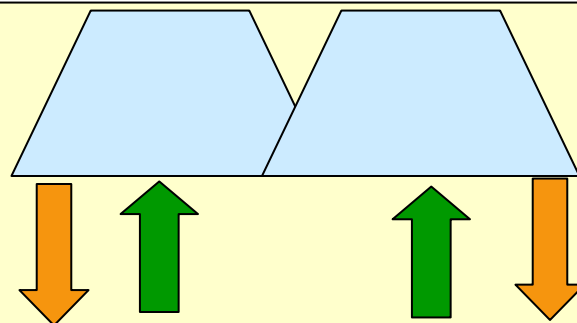


Asymmetric carrier
Aggregation possible,
e.g. DL more CC than UL

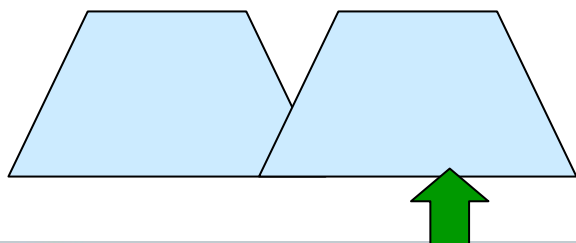
Uplink



Each CC has its own RACH



All CC use same RACH preamble
Network responds on all CCs

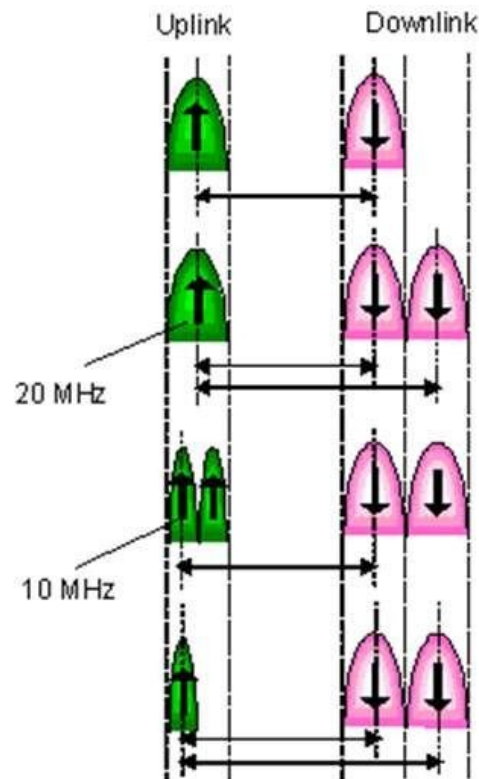


Only 1 CC contains RACH

Symmetry

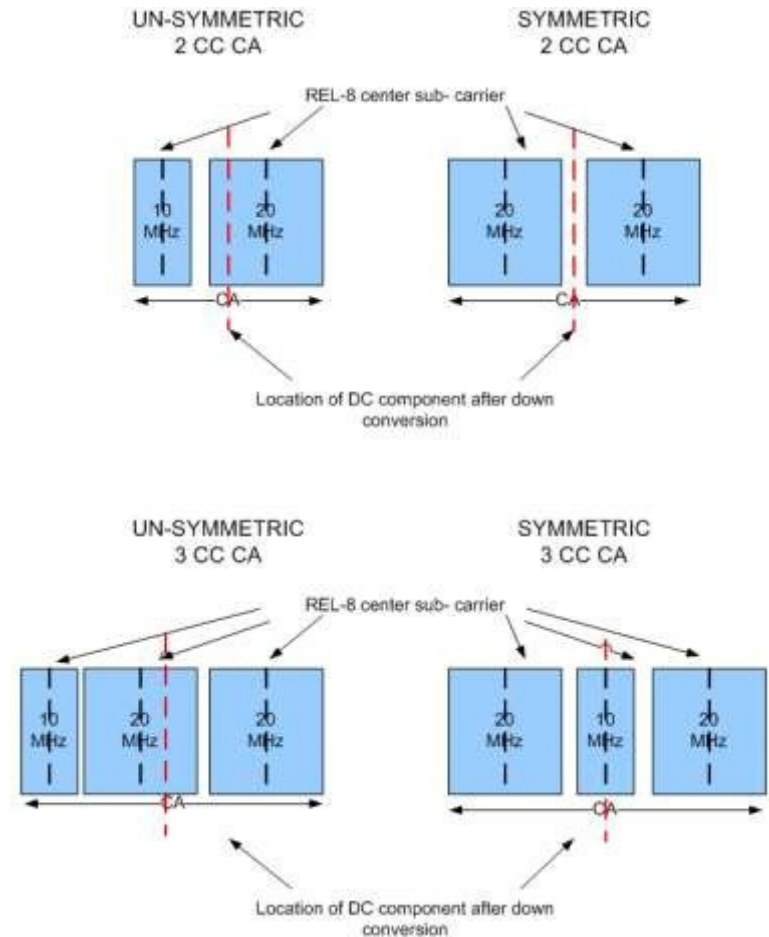
I DL-UL

- I Number of CC
 - TDD: DL = UL
 - FDD: DL $\geq$ UL
- I Bandwidth
- I Both



I Component Carrier

- However position of DC not known to System Simulator



Addition, modification, release of additional CC

I *RRCConnectionReconfiguration* Message contains new Rel-10 information element:

```
...
RRCConnectionReconfiguration-v1020-IEs ::= SEQUENCE {
    sCellToReleaseList-r10          SCellToReleaseList-r10
    sCellToAddModList-r10          SCellToAddModList-r10
    nonCriticalExtension             SEQUENCE {}
}

SCellToAddModList-r10 ::= SEQUENCE (SIZE (1..maxSCell-r10)) OF SCellToAddMod-r10

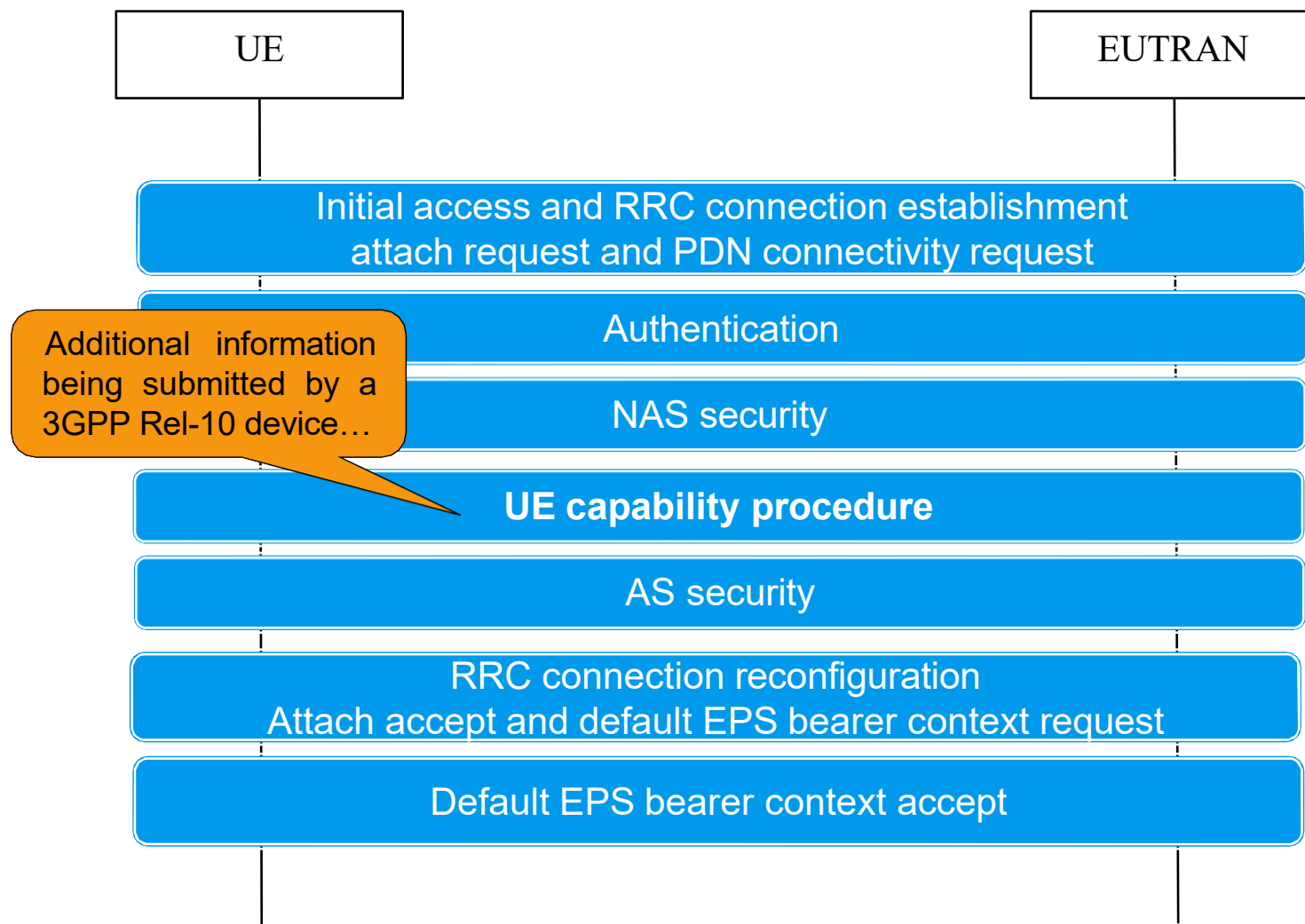
SCellToAddMod-r10 ::= SEQUENCE {
    sCellIndex-r10                SCellIndex-r10,
    cellIdentification-r10        SEQUENCE {
        physCellId-r10            PhysCellId,
        dl-CarrierFreq-r10        ARFCN-ValueEUTRA
    }
    radioResourceConfigCommonSCell-r10 RadioResourceConfigCommonSCell-r10 OPTIONAL, -- Cond SCellAdd
    SCellAdd                       radioResourceConfigDedicatedSCell-r10 RadioResourceConfigDedicatedSCell-r10 OPTIONAL, --
Cond SCellAdd2
    ...
}

SCellToReleaseList-r10 ::= SEQUENCE (SIZE (1..maxSCell-r10)) OF SCellToReleaseList-r10
...
```

DL: bandwidth, antennas, MBSFN subframe configuration, PHICH configuration, PDSCH configuration, TDD config (if TD-LTE SCell)
UL: bandwidth, carrier frequency, additional spectrum emission, P-Max, power control info, uplink channel configuration (PRACH, PUSCH)

UE-specific information; **DL:** cross-carrier scheduling, CSI-RS configuration, PDSCH
UL: PUSCH, uplink power control, CQI, SRS

Default EPS bearer setup (3GPP LTE Rel-8)



UE capability information transfer

3GPP Rel-10 add on's

I New IE's within the for a 3GPP Rel-10 device:

UE-EUTRA-Capability information element

```
-- ASN1START
...
UE-EUTRA-Capability-v1020-IEs ::= SEQUENCE {
    ue-Category-v1020                INTEGER (6..8)
    phyLayerParameters-v1020          PhyLayerParameters-v1020
    rf-Parameters-v1020              RF-Parameters-v1020
    measParameters-v1020             MeasParameters-v1020
    featureGroupIndicators-v1020     BIT STRING (SIZE (32))
    interRAT-ParametersCDMA2000-v1020 IRAT-ParametersCDMA2000-1XRTT-v1020    OPTIONAL,
    ue-BasedNetwPerfMeasParameters-r10 UE-BasedNetwPerfMeasParameters-r10    OPTIONAL,
    interRAT-ParametersUTRA-TDD-v1020 IRAT-ParametersUTRA-TDD-v1020      OPTIONAL,
    nonCriticalExtension              SEQUENCE {}                                OPTIONAL
}
...
PhyLayerParameters-v1020 ::= SEQUENCE {
    twoAntennaPortsForPUCCH-r10      ENUMERATED {supported}                OPTIONAL,
    tm9-With-8Tx-FDD-r10             ENUMERATED {supported}                OPTIONAL,
    pmi-Disabling-r10                ENUMERATED {supported}                OPTIONAL,
    crossCarrierScheduling-r10        ENUMERATED {supported}                OPTIONAL,
    simultaneousPUCCH-PUSCH-r10      ENUMERATED {supported}                OPTIONAL,
    multiClusterPUSCH-WithinCC-r10   ENUMERATED {supported}                OPTIONAL,
    nonContiguousUL-RA-WithinCC-List-r10 NonContiguousUL-RA-WithinCC-List-r10    OPTIONAL
}
...
```

2012 © by Rohde&Schwarz

Not all combinations are allowed!

Carrier aggregation capabilities are signaled separately for DL, UL.

What frequency bands, band combination (CA and MIMO capabilities, inter-band, intra-band contiguous and non-contiguous, bandwidth class does the device support?

!

Carrier Aggregation - Activation

- I **A new MAC control element for Component Carrier Management is defined containing at least the activation respectively deactivation command for the secondary DL component carriers configured for a UE. The new MAC CE is identified by a unique LCID**
- I
- I **For actual deactivation and activation signalling for the DL SCCs, the MAC CE for CC Management includes a 4/5-bit bitmap where each bit is representing one of the DL CCs that can be configured in the UE. A bit set to 1 denotes activation of the corresponding DL CC, a bit set to 0 respectively denotes deactivation**
- I **New timer for implicit CC deactivation**
- I **CC's are "just" additional resources. UL scheduling will assume we do not have different QOS (delay/loss) on different CC's**

Carrier Aggregation

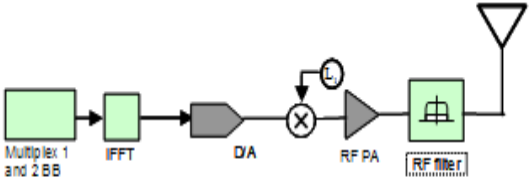
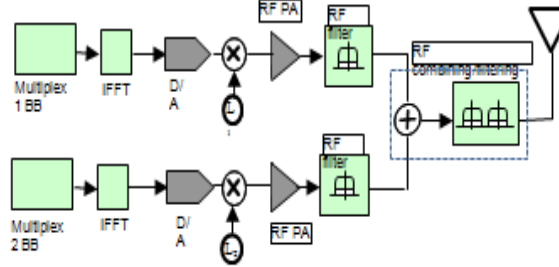
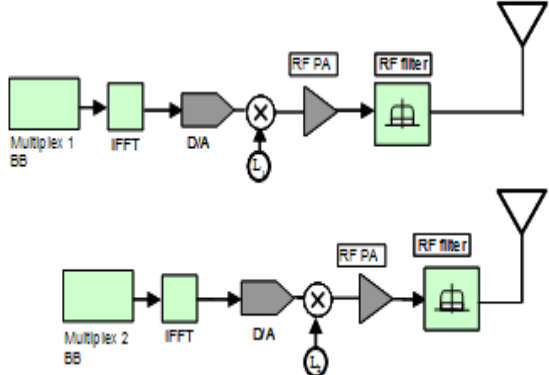
- I The transmission mode is not constrained to be the same on all CCs scheduled for a UE**
- I A single UE-specific UL CC is configured semi-statically for carrying PUCCH A/N, SR, and periodic CSI from a UE**
- I Frequency hopping is not supported simultaneously with non-contiguous PUSCH resource allocation**
- I UCI cannot be carried on more than one PUSCH in a given subframe.**

Carrier Aggregation

- I **Working assumption is confirmed that a single set of PHICH resources is shared by all UEs (Rel-8 to Rel-10)**

- I **If simultaneous PUCCH+PUSCH is configured and there is at least one PUSCH transmission**
 - I UCI can be transmitted on either PUCCH or PUSCH with a dependency on the situation that needs to be further discussed
 - I All UCI mapped onto PUSCH in a given subframe gets mapped onto a single CC irrespective of the number of PUSCH CCs

UE Architectures

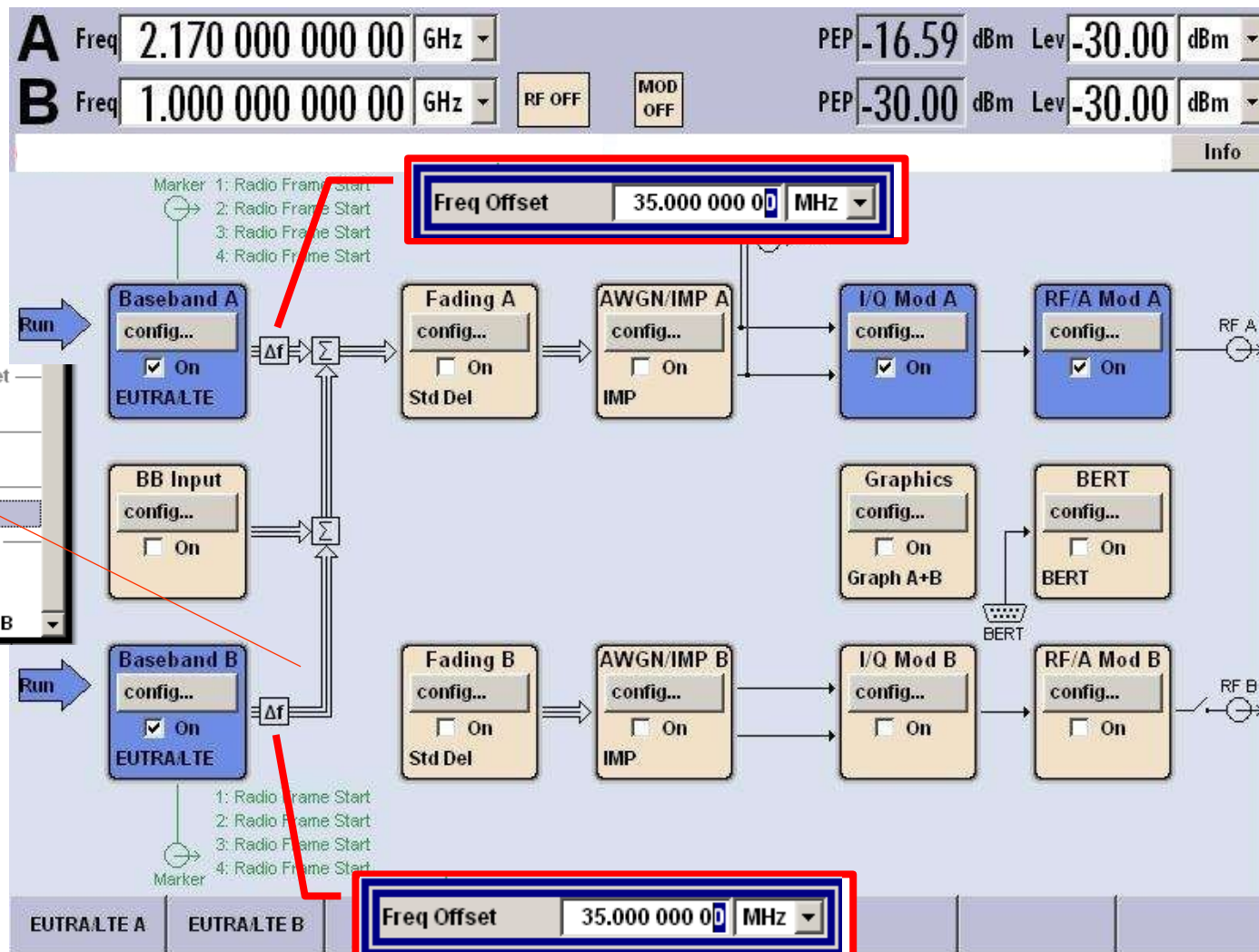
Tx single antenna transmission ~ (Type A)	Tx shared antenna transmission ~ (Type D1)	Tx dual antenna transmission ~ (Type D2)
		
Type A1: Intra-band contiguous: Supports a maximum of two release 8 CC carriers (1.4, 3, 5, 10, 15 and 20MHz) Max CA UL bandwidth is 40 MHz	Type D1-1: Intra-band contiguous: Supports a maximum of two release 8 CC carriers (1.4,3, 5, 10, 15 and 20MHz) Max CA UL bandwidth is 40 MHz.	Type D2-1: Intra-band contiguous: Supports a maximum of two release 8 CC carriers (1.4,3, 5, 10, 15 and 20MHz) Max CA UL bandwidth is 40 MHz. UL MIMO is supported to address ITU-R requirement for peak UL data rate
Type A2: Intra-band non - contiguous: Not supported	Type D1-2: Intra-band non - contiguous: Capability of supporting single band non-contiguous in future releases	Type D2-2: Intra-band non - contiguous: Capability of supporting single band non-contiguous in future releases
Type A3: Inter-band non -contiguous: Supports a maximum of two release 8 CC carriers (1.4,3, 5, 10, 15 and 20MHz) Simultaneous Tx on both bands is not supported	Type D1-3: Inter-band non -contiguous: Supports a maximum of two release 8 CC carriers (1.4, 3, 5, 10, 15 and 20MHz) Max CA UL bandwidth is 20 MHz. Simultaneous Tx on both bands is supported.	Type D2-3: Inter-band non -contiguous: Supports a maximum of two release 8 CC carriers (1.4, 3, 5, 10, 15 and 20MHz) Max CA UL bandwidth is 20 MHz. Simultaneous Tx on both bands is supported.

I Possible TX architectures

- I Same / Different antenna (connectors) for each CC
- I D1/D2 could be switched to support CA or UL MIMO

LTE-Advanced solutions from R&S

R&S® SMU200 Vector Signal Generator

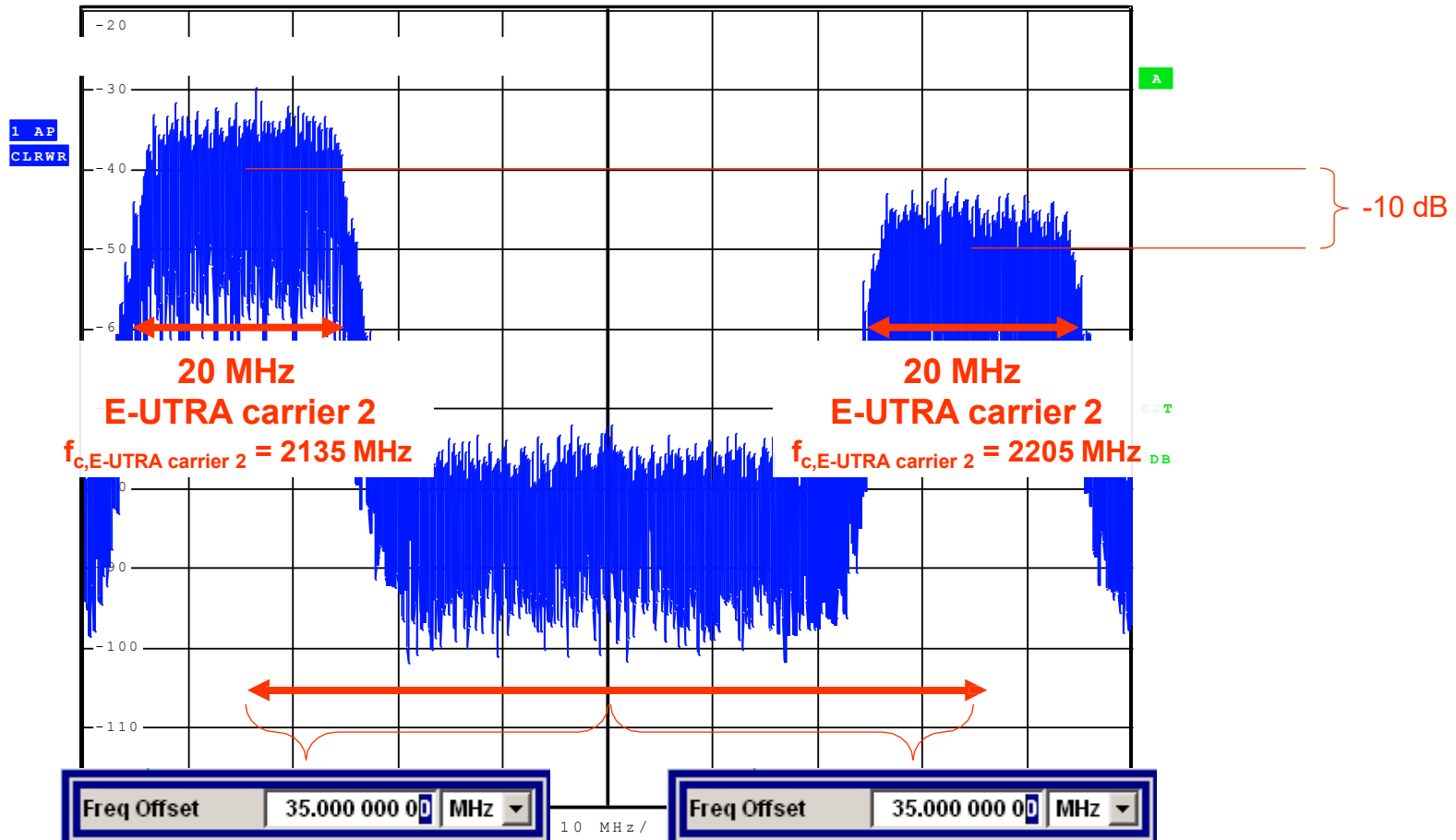


LTE-Advanced solutions from R&S

R&S® FSQ Signal Analyzer



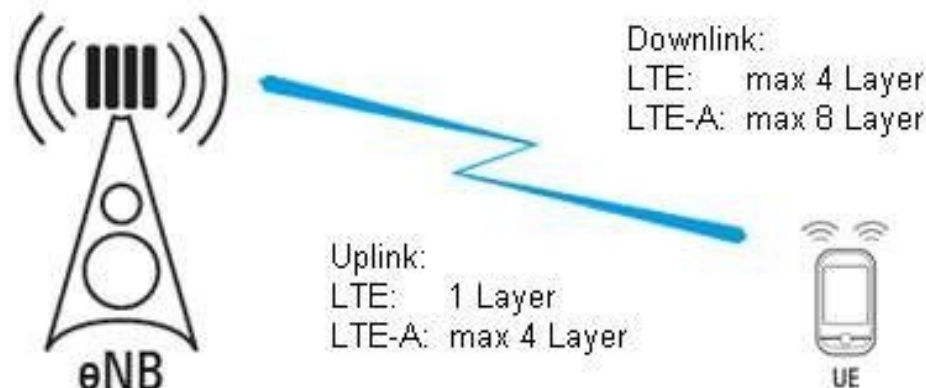
Ref -20 dBm Att 5 dB RBW 2 MHz
VBW 5 MHz SWT 2.5 ms



Enhanced MIMO schemes

I Increased number of layers:

- I Up to 8x8 MIMO in downlink.
- I Up to 4x4 MIMO in uplink.



I In addition the downlink reference signal structure has been enhanced compared with LTE Release 8 by:

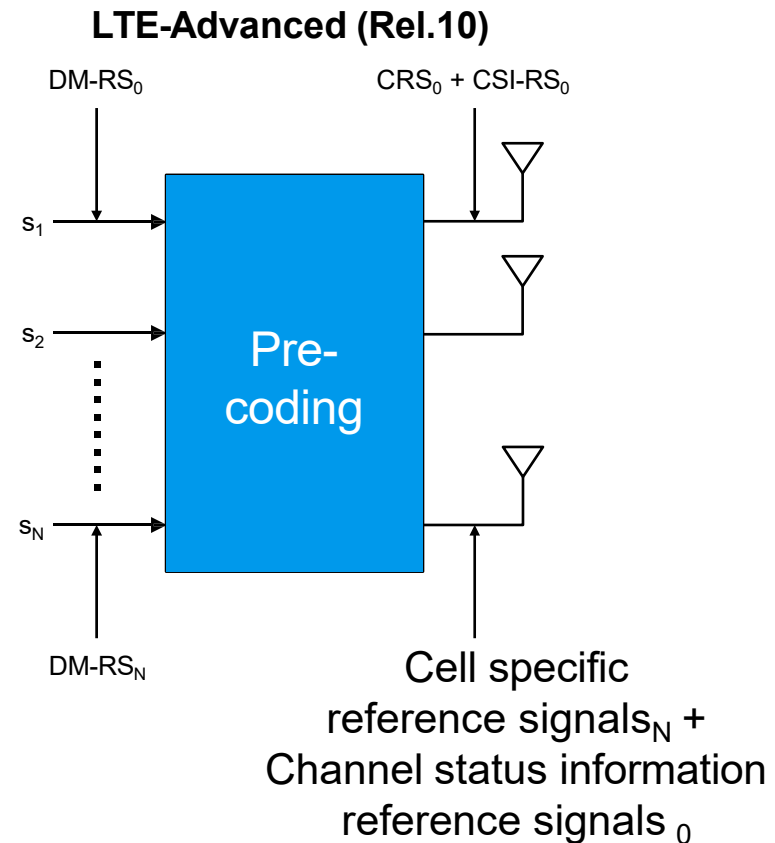
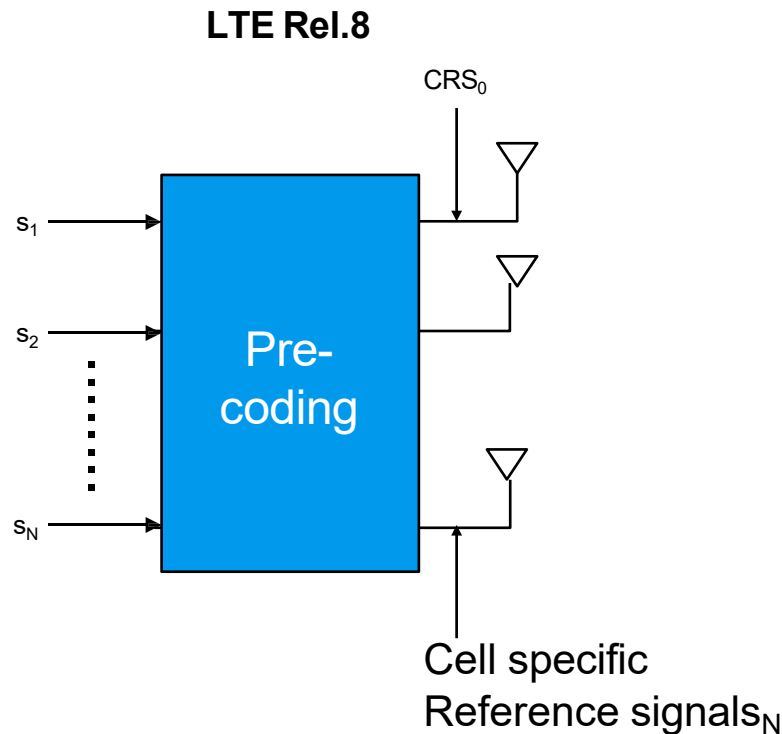
- I Demodulation Reference signals (DM-RS) targeting PDSCH demodulation.
 - UE specific, i.e. an extension to multiple layers of the concept of Release 8 UE-specific reference signals used for beamforming.
- I Reference signals targeting channel state information (CSI-RS) estimation for CQI/PMI/RI/etc reporting when needed.
 - Cell specific, sparse in the frequency and time domain and punctured into the data region of normal subframes.

Downlink reference signals in LTE-Advanced



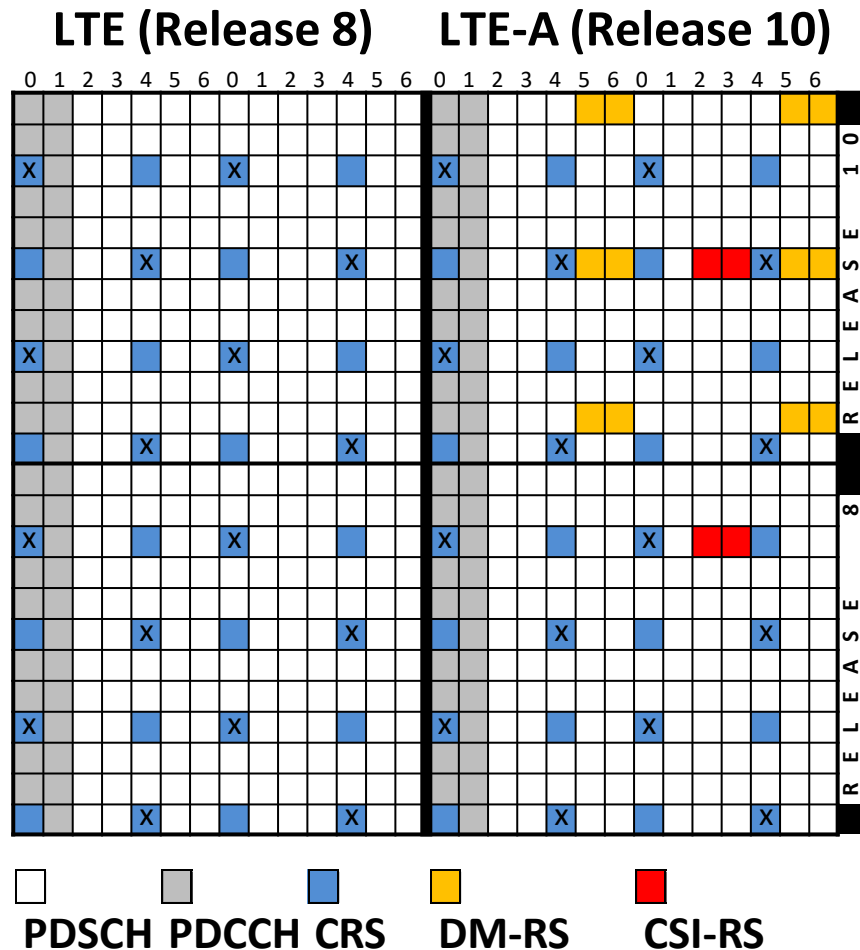
- I **Define two types of RS,**
 - I RS targeting PDSCH demodulation,
 - I RS targeting CSI generation (for CQI/PMI/RI/etc reporting when needed),
- I **RS targeting PDSCH demodulation (for LTE-Advanced operation) are**
 - I UE specific
 - Transmitted only in scheduled RBs and the corresponding layers
 - Different layers can target the same or different UEs
 - Design principle is an extension of the concept of Rel-8 UE-specific RS (used for beamforming) to multiple layersDetails on UE-specific RS pattern, location, etc are FFS
 - I RS on different layers are mutually orthogonal,
- I RS and data are subject to the same pre-coding operation,
- I complementary use of Rel-8 CRS by the UE is not precluded,
- I **RS targeting CSI generation (for LTE-A operation) are**
 - I Cell specific and sparse in frequency and time
- I **Rel-8 transmission schemes using Rel-8 cell-specific and/or UE-specific RS still supported,**

Cell specific Reference Signals vs. DM-RS



- I Demodulation-Reference signals DM-RS and data are precoded the same way, enabling non-codebook based precoding and enhanced multi-user beamforming.**

Ref. signal mapping: Rel.8 vs. LTE-Advanced



I Example:

- I 2 antenna ports, antenna port 0, CSI-RS configuration 8.
- I PDCCH (control) allocated in the first 2 OFDM symbols.
- I **CRS sent on all RBs; DTX sent for the CRS of 2nd antenna port.**
- I **DM-RS sent only for scheduled RBs on all antennas; each set coded differently between the two layers.**
- I **CSI-RS punctures Rel. 8 data; sent periodically over allotted REs (not more than twice per frame)**

DL MIMO

Extension up to 8x8



I Max number of transport blocks: 2

I Number of MCS fields

- I one for each transport block

I ACK/NACK feedback

- I 1 bit per transport block for evaluation as a baseline

I Closed-loop precoding supported

- I Rely on precoded dedicated demodulation RS (decision on DL RS)

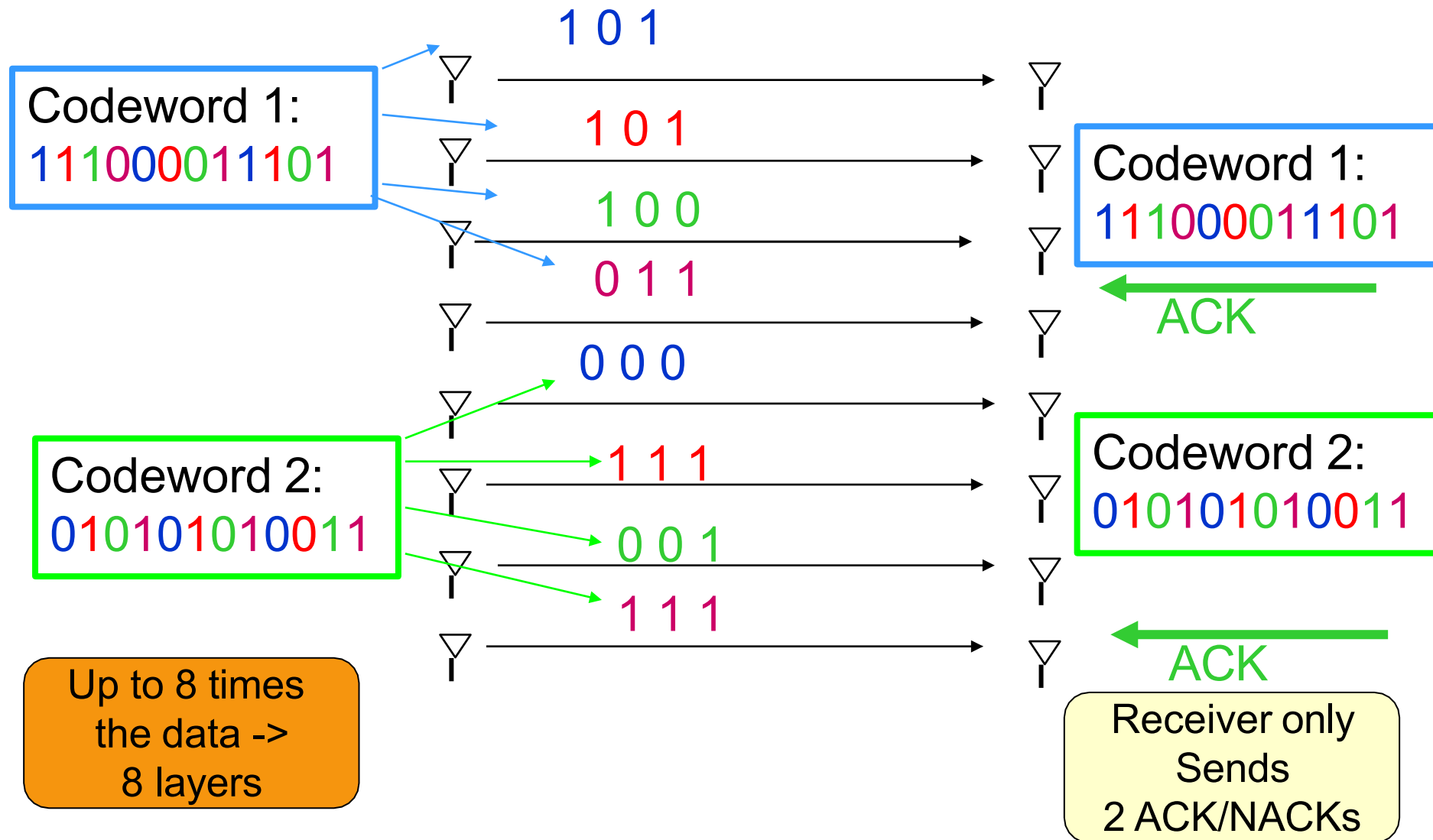
I Conclusion on the codeword-to-layer mapping:

- I DL spatial multiplexing of up to eight layers is considered for LTE-Advanced,
- I Up to 4 layers, reuse LTE codeword-to-layer mapping,
- I Above 4 layers mapping – see table

I Discussion on control signaling details ongoing

Codeword to layer mapping for spatial multiplexing			
Number of layers	Number of code words	Codeword-to-layer mapping $i = 0, 1, \dots, M_{\text{symb}}^{\text{layer}} - 1$	
5	2	$x^{(0)}(i) = d^{(0)}(2i)$ $x^{(1)}(i) = d^{(0)}(2i+1)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} \quad 2 = M_{\text{symb}}^{(1)} \quad 3$
		$x^{(2)}(i) = d^{(1)}(3i)$ $x^{(3)}(i) = d^{(1)}(3i+1)$ $x^{(4)}(i) = d^{(1)}(3i+2)$	
6	2	$x^{(0)}(i) = d^{(0)}(3i)$ $x^{(1)}(i) = d^{(0)}(3i+1)$ $x^{(2)}(i) = d^{(0)}(3i+2)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} \quad 3 = M_{\text{symb}}^{(1)} \quad 3$
		$x^{(3)}(i) = d^{(1)}(3i)$ $x^{(4)}(i) = d^{(1)}(3i+1)$ $x^{(5)}(i) = d^{(1)}(3i+2)$	
7	2	$x^{(0)}(i) = d^{(0)}(3i)$ $x^{(1)}(i) = d^{(0)}(3i+1)$ $x^{(2)}(i) = d^{(0)}(3i+2)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} \quad 3 = M_{\text{symb}}^{(1)} \quad 4$
		$x^{(3)}(i) = d^{(1)}(4i)$ $x^{(4)}(i) = d^{(1)}(4i+1)$ $x^{(5)}(i) = d^{(1)}(4i+2)$ $x^{(6)}(i) = d^{(1)}(4i+3)$	
8	2	$x^{(0)}(i) = d^{(0)}(4i)$ $x^{(1)}(i) = d^{(0)}(4i+1)$ $x^{(2)}(i) = d^{(0)}(4i+2)$ $x^{(3)}(i) = d^{(0)}(4i+3)$	$M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} \quad 4 = M_{\text{symb}}^{(1)} \quad 4$
		$x^{(4)}(i) = d^{(1)}(4i)$ $x^{(5)}(i) = d^{(1)}(4i+1)$ $x^{(6)}(i) = d^{(1)}(4i+2)$ $x^{(7)}(i) = d^{(1)}(4i+3)$	

MIMO – layer and codeword



Scheduling of Transmission Mode 9 (TM9)

I NEW DCI format 2C with 3GPP Rel-10.

- I Used to schedule transmission mode 9 (TM9), which is spatial multiplexing with DM-RS support of up to 8 layers (multi-layer transmission).
 - DM-RS scrambling and number of layers are jointly signaled in a 3-bit field.

I DCI format 2C.

- Carrier indicator [3 bit].
- Resource allocation header [1 bit]
 - Resource Allocation Type 0 and 1.
- TPC command for PUCCH [2 bit].
- Downlink Assignment Index¹⁾ [2 bit].
- HARQ process number [3 bit (FDD), 4 bit (TDD)].
- Antenna ports, scrambling identity and # of layers; see table [3 bit].
- SRS request¹⁾ [0-1 bit].
- MCS, new data indicator, RV for 2 transport block [each 5 bit].

One Codeword: Codeword 0 enabled, Codeword 1 disabled		Two Codewords: Codeword 0 enabled, Codeword 1 enabled	
Value	Message	Value	Message
0	1 layer, port 7, $n_{SCID}=0$	0	2 layers, ports 7-8, $n_{SCID}=0$
1	1 layer, port 7, $n_{SCID}=1$	1	2 layers, ports 7-8, $n_{SCID}=1$
2	1 layer, port 8, $n_{SCID}=0$	2	3 layers, ports 7-9
3	1 layer, port 8, $n_{SCID}=1$	3	4 layers, ports 7-10
4	2 layers, ports 7-8	4	5 layers, ports 7-11
5	3 layers, ports 7-9	5	6 layers, ports 7-12
6	4 layers, ports 7-10	6	7 layers, ports 7-13
7	Reserved	7	8 layers, ports 7-14

¹⁾ TDD only

Uplink MIMO

Extension up to 4x4

4Tx in UL postponed
to Release 11

I Rel-8 LTE.

- I UEs must have 2 antennas for reception.
- I But only 1 amplifier for transmission is available (costs/complexity).
- I UL MIMO only as antenna switching mode (switched diversity).

I 4x4 UL SU-MIMO is needed to fulfill peak data rate requirement of 15 bps/Hz.

I Schemes are very similar to DL MIMO modes.

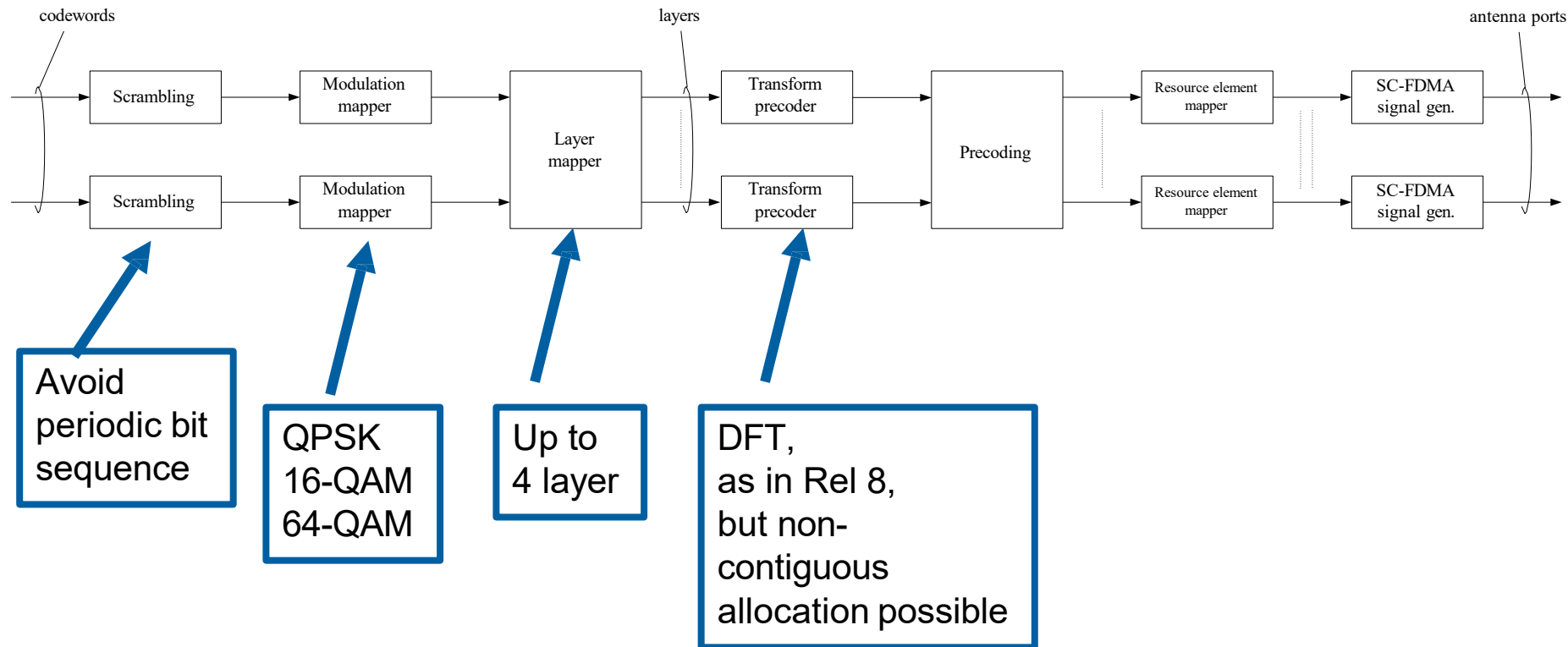
- I UL spatial multiplexing of up to 4 layers is considered for LTE-Advanced.
- I SRS enables link and SU-MIMO adaptation.

I Number of receive antennas are receiver-implementation specific.

- I At least two receive antennas is assumed on the terminal side.

UL MIMO – signal generation in uplink

Similar to Rel.8 Downlink:



UL MIMO – layers and codewords

Number of layers	Number of codewords	Codeword-to-layer mapping $i = 0, 1, \dots, M_{\text{symb}}^{\text{layer}} - 1$
1	1	$x^{(0)}(i) = d^{(0)}(i)$ $M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)}$
2	1	$x^{(0)}(i) = d^{(0)}(2i)$ $x^{(1)}(i) = d^{(0)}(2i+1)$ $M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} / 2$
2	2	$x^{(0)}(i) = d^{(0)}(i)$ $x^{(1)}(i) = d^{(1)}(i)$ $M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} = M_{\text{symb}}^{(1)}$
3	2	$x^{(0)}(i) = d^{(0)}(i)$ $x^{(1)}(i) = d^{(1)}(2i)$ $x^{(2)}(i) = d^{(1)}(2i+1)$ $M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} = M_{\text{symb}}^{(1)} / 2$
4	2	$x^{(0)}(i) = d^{(0)}(2i)$ $x^{(1)}(i) = d^{(0)}(2i+1)$ $x^{(2)}(i) = d^{(1)}(2i)$ $x^{(3)}(i) = d^{(1)}(2i+1)$ $M_{\text{symb}}^{\text{layer}} = M_{\text{symb}}^{(0)} / 2 = M_{\text{symb}}^{(1)} / 2$

Up to 4
Layers
(Rel.11)

2 codewords

UL MIMO scheduling – DCI format 4 **NEW!**

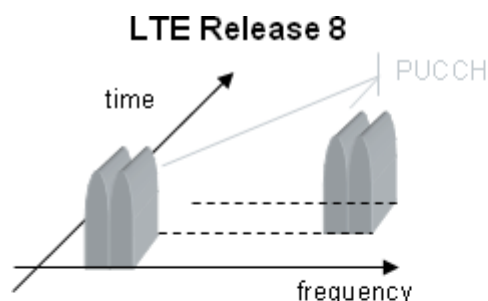
- I **Carrier indicator [0-3 bit].**
- I **Resource Block Assignment:**
 - I [bits] for Resource Allocation Type 0
$$\left\lceil \log_2 (N_{RB}^{UL} (N_{RB}^{UL} + 1) / 2) \right\rceil$$
 - I [bits] for Resource Allocation Type 1.
$$\left\lceil \log_2 \left(\left\lceil \frac{N_{RB}^{UL}}{4} + 1 \right\rceil \right) \right\rceil$$
- I **TPC command for PUSCH [2 bit].**
- I **Cyclic shift for DM RS and OCC index [2 bit].**
- I **UL index [2 bit].**
 - I TDD only for UL-DL configuration 0.
- I **Downlink Assignment Index [2 bit].**
 - I TDD only, UL-DL config. 1-6.
- I **CSI request [1 or 2 bit].**
 - I 2 bit for cells with more than two cells in the DL (carrier aggregation).
- I **SRS request [2 bit].**
- I **Resource Allocation Type [1 bit].**
- I **Transport Block 1.**
 - I MCS, RV [5 bit].
 - I New data indicator [1 bit].
- I **Transport Block 2.**
 - I MCS, RV [5 bit].
 - I New data indicator [1 bit].
- I **Precoding information.**

LTE-Advanced

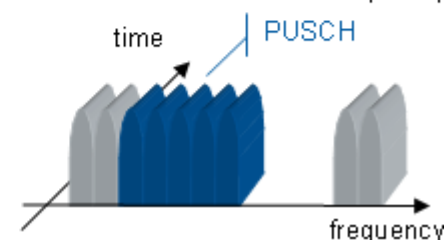
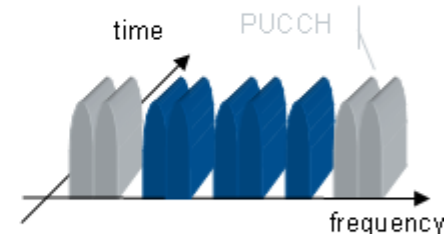
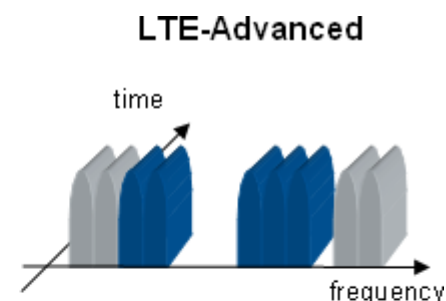
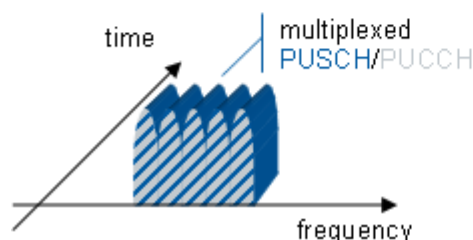
Enhanced uplink SC-FDMA



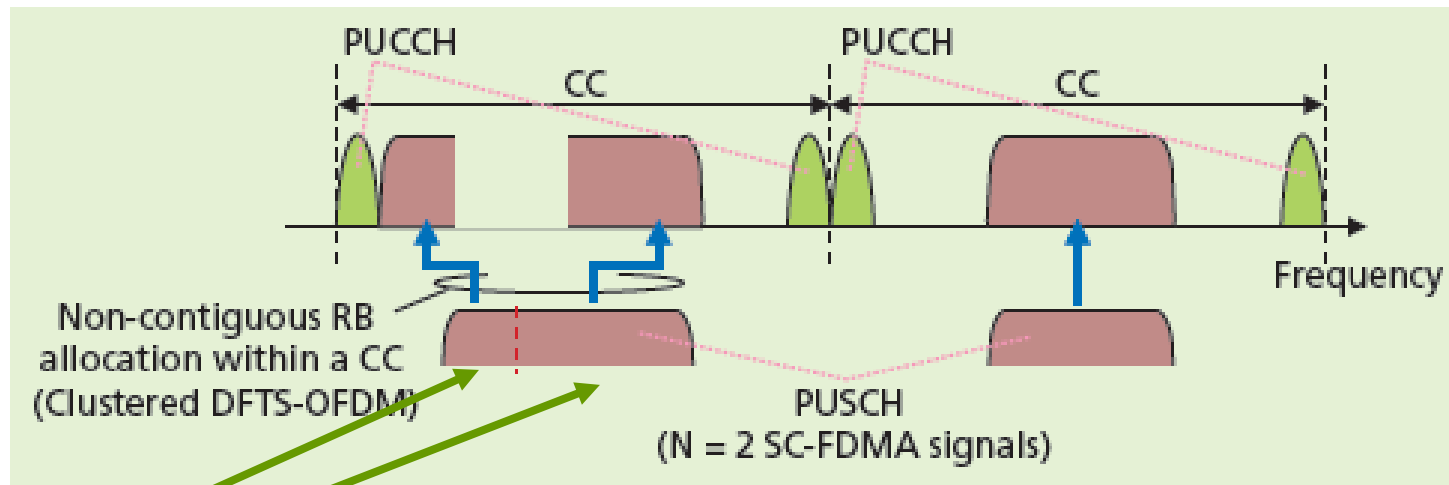
- I The uplink transmission scheme remains SC-FDMA.
- I The transmission of the physical uplink shared channel (PUSCH) uses DFT precoding.
- I Two enhancements:
 - I Control-data decoupling
 - I Non-contiguous data transmission



Either PUCCH only at both edges of UL channel bandwidth or PUCCH multiplexed into PUSCH.



Physical channel arrangement - uplink



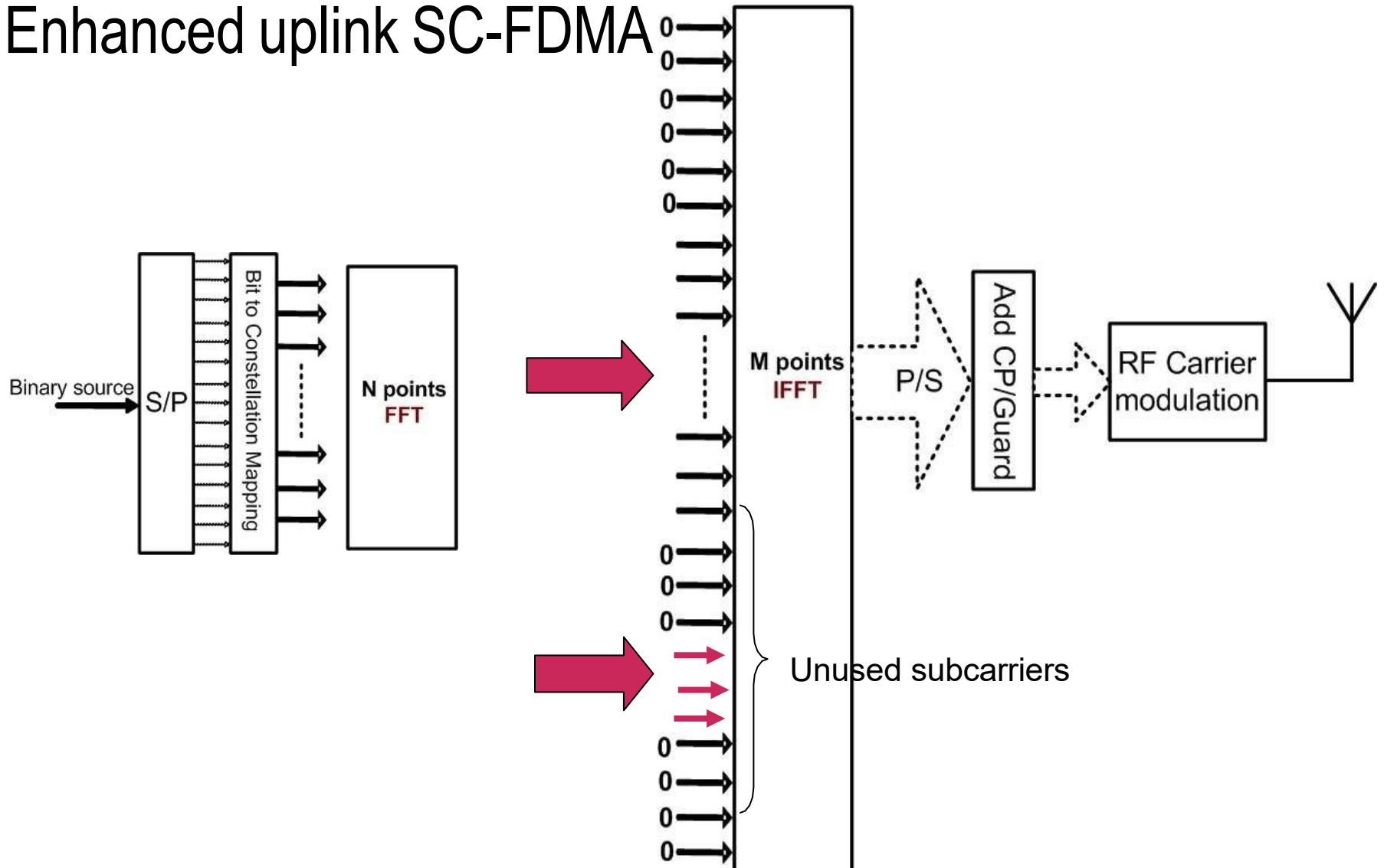
Clustered-DFTS-OFDM
= Clustered DFT spread
OFDM.

Simultaneous transmission of
PUCCH and PUSCH from
the same UE is supported

Non-contiguous resource block allocation => will cause higher
Crest factor at UE side

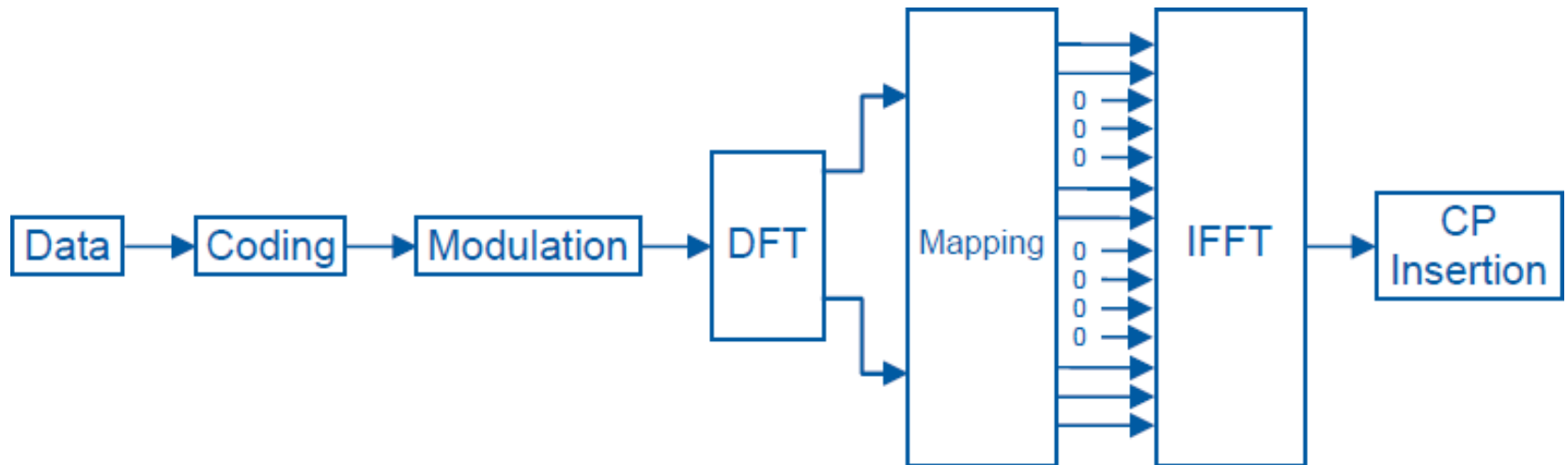
LTE-Advanced

Enhanced uplink SC-FDMA



LTE-Advanced

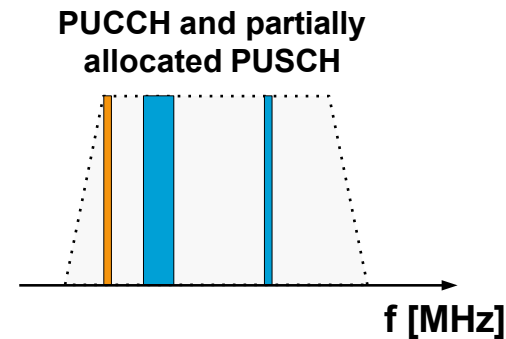
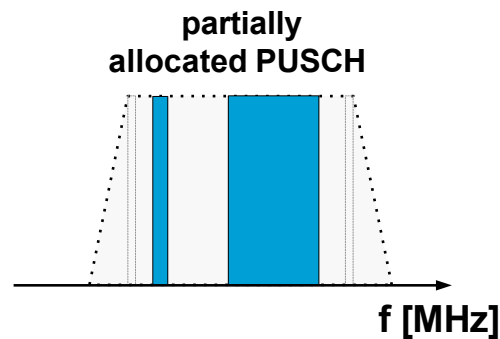
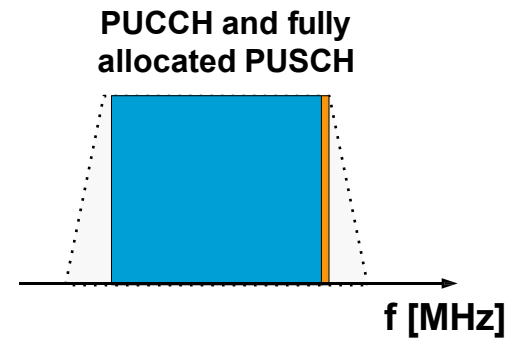
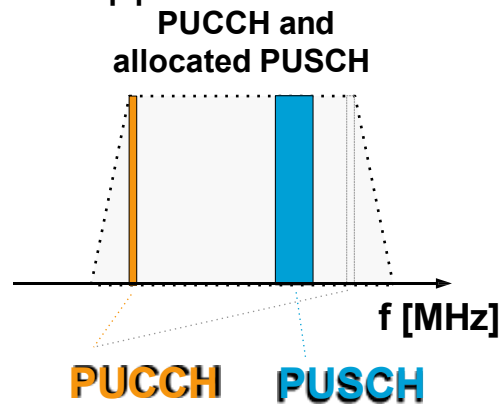
Enhanced uplink SC-FDMA



Due to distribution we get active subcarriers beside non-active subcarriers: worse peak to average ratio, e.g. Crest factor

Simultaneous PUSCH-PUCCH transmission, multi-cluster transmission

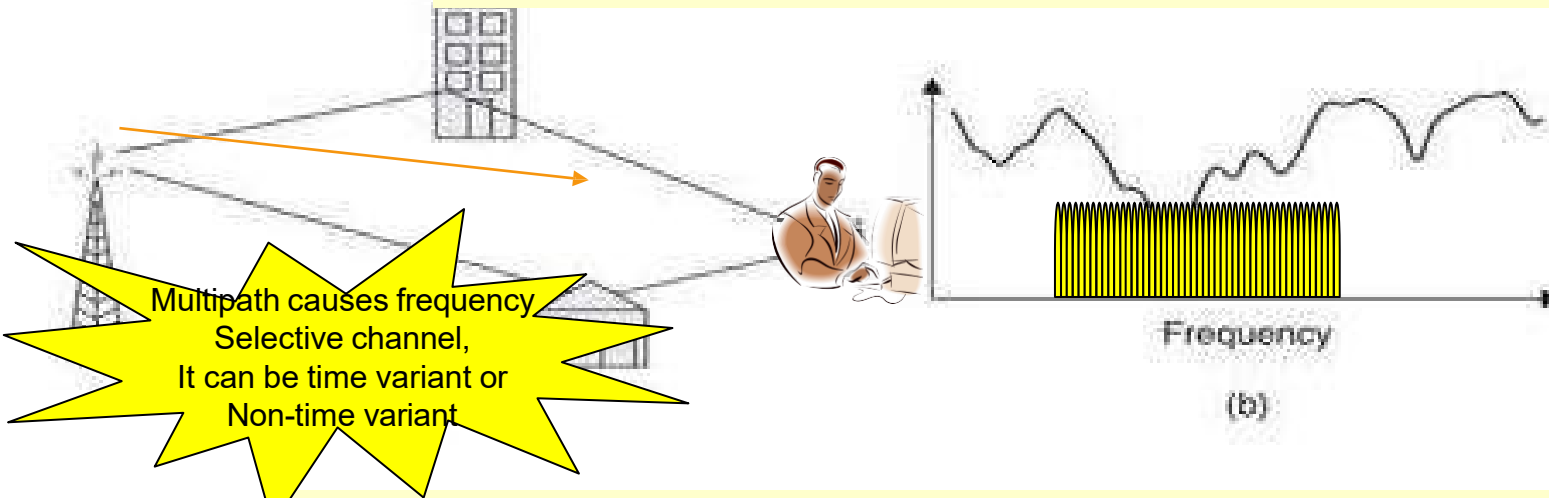
- I Remember, only one UL carrier in 3GPP Release 10;
scenarios:
 - I Feature support is indicated by *PhyLayerParameters-v1020* IE*).



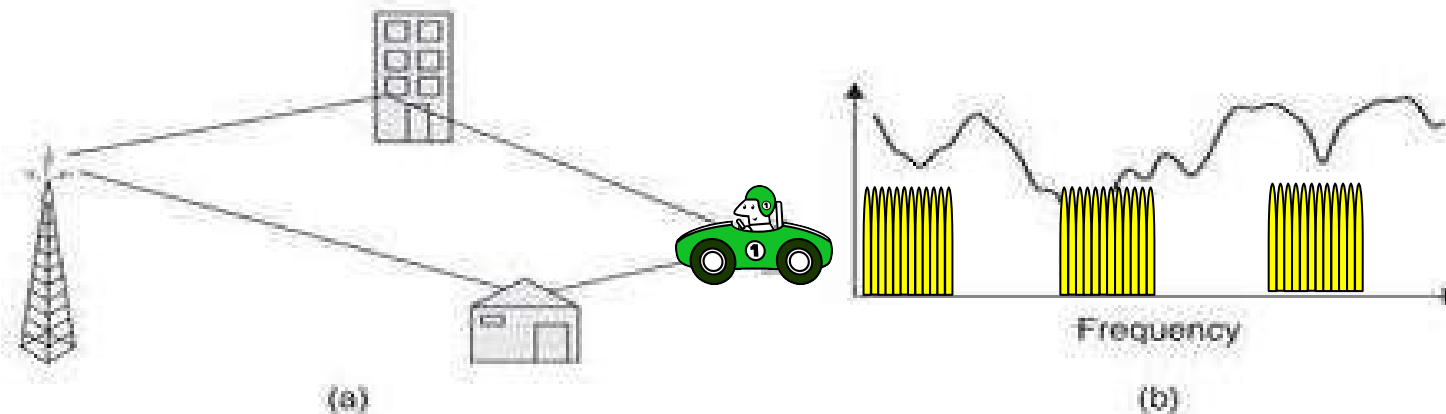
*) see 3GPP TS 36.331 RRC Protocol Specification

Benefit of localized or distributed mode

„static UE“: frequency selectivity is not time variant -> localized allocation



„high velocity UE“: frequency selectivity is time variant -> distributed allocation



Resource Allocation types in the uplink

I Uplink Resource Allocation Type 0.

- I Contiguous allocation as today in 3GPP Release 8.

$$\text{if } (L_{\text{CRBs}} - 1) \leq \lfloor N_{\text{RB}}^{\text{UL}} / 2 \rfloor \text{ then } RIV = N_{\text{RB}}^{\text{UL}} (L_{\text{CRBs}} - 1) + RB_{\text{START}}$$

$$\text{else } RIV = N_{\text{RB}}^{\text{UL}} (N_{\text{RB}}^{\text{UL}} - L_{\text{CRBs}} + 1) + (N_{\text{RB}}^{\text{UL}} - 1 - RB_{\text{START}})$$

I Uplink Resource Allocation Type 1.

- I UL bandwidth is divided into two sets of Resource Blocks (RB).
- I Each set has a number of Resource Block Groups (RBG) of size P.
- I Combinatorial index r, indicates RBG starting and ending index for both set of RB ($s_0, s_1-1 \mid s_2, s_3-1$).

$$r = \sum_{i=0}^{M-1} \binom{N - s_i}{M - i} \{s_i\}_{i=0}^{M-1} \quad 1 \leq s_i \leq N, \quad s_i < s_{i+1}$$

– M = 4, N is bandwidth dependent, $N = \lceil N_{\text{RB}}^{\text{UL}} / P \rceil + 1$

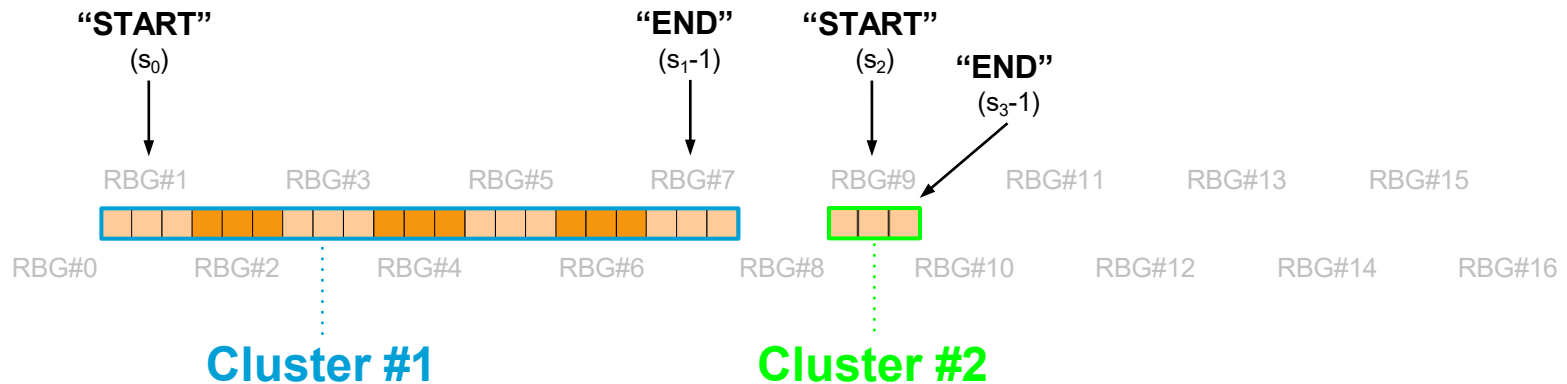
System Bandwidth	RBG Size (P)
≤10	1
11 – 26	2
27 – 63	3
64 – 110	4

Multi-cluster allocation

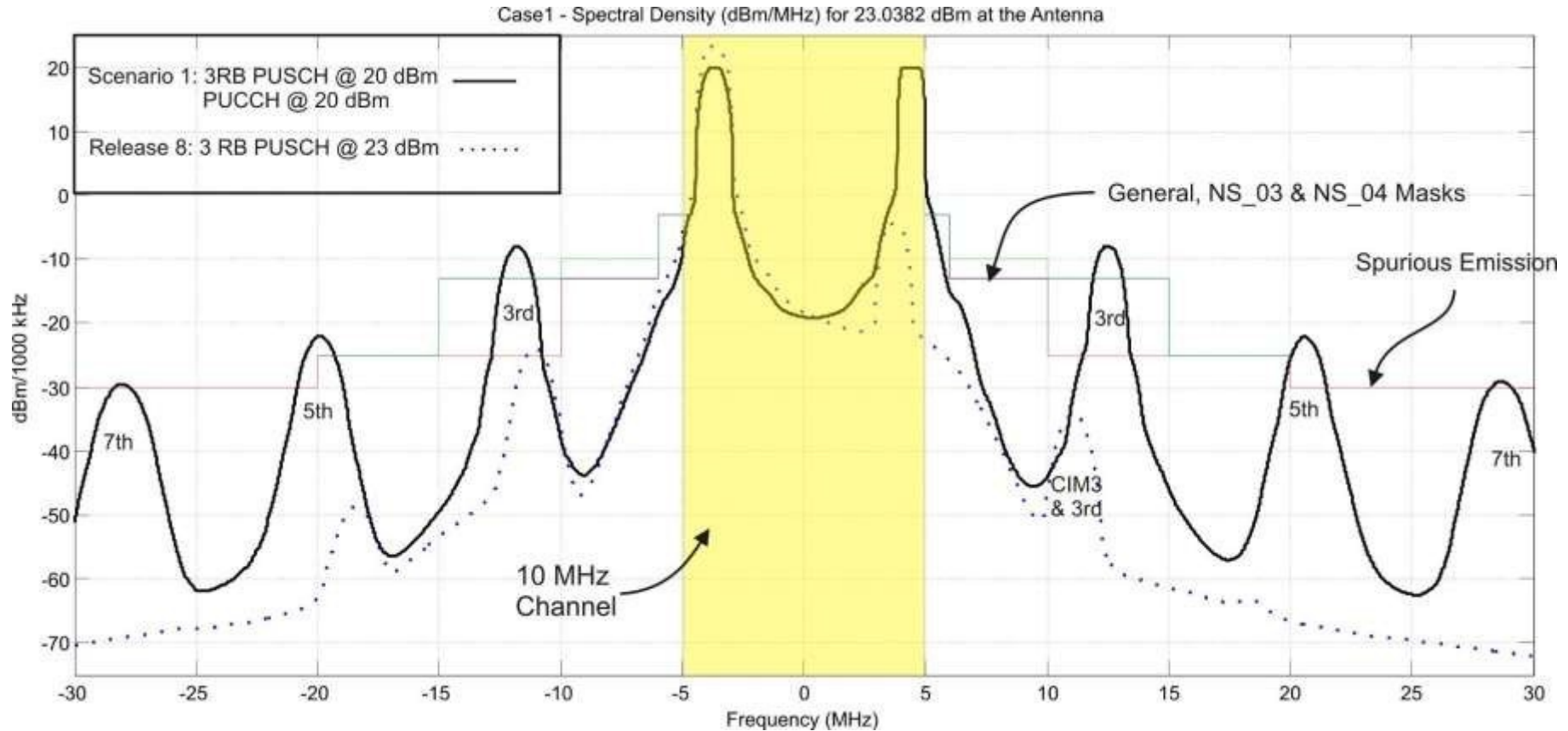
Uplink Resource Allocation Type 1

I **Example: LTE 10 MHz (50 RB), $P = 3$ leads to 17 RBG.**

- I Combinatorial index r indicates RBG starting indices for RB set #1 (s_0, s_1-1 , defining cluster #1) and RB set #2 (s_2, s_3-1 , defining cluster #2).
- I Range for s_0, s_1, s_2, s_3 for 10 MHz (50 RB): 1 to 18 (see previous slide).
- I Applied rule: $s_0 < s_1 < s_2 < s_3$ (see previous slide).
- I Example: $s_0=2, s_1=9, s_2=10, s_3=11$:

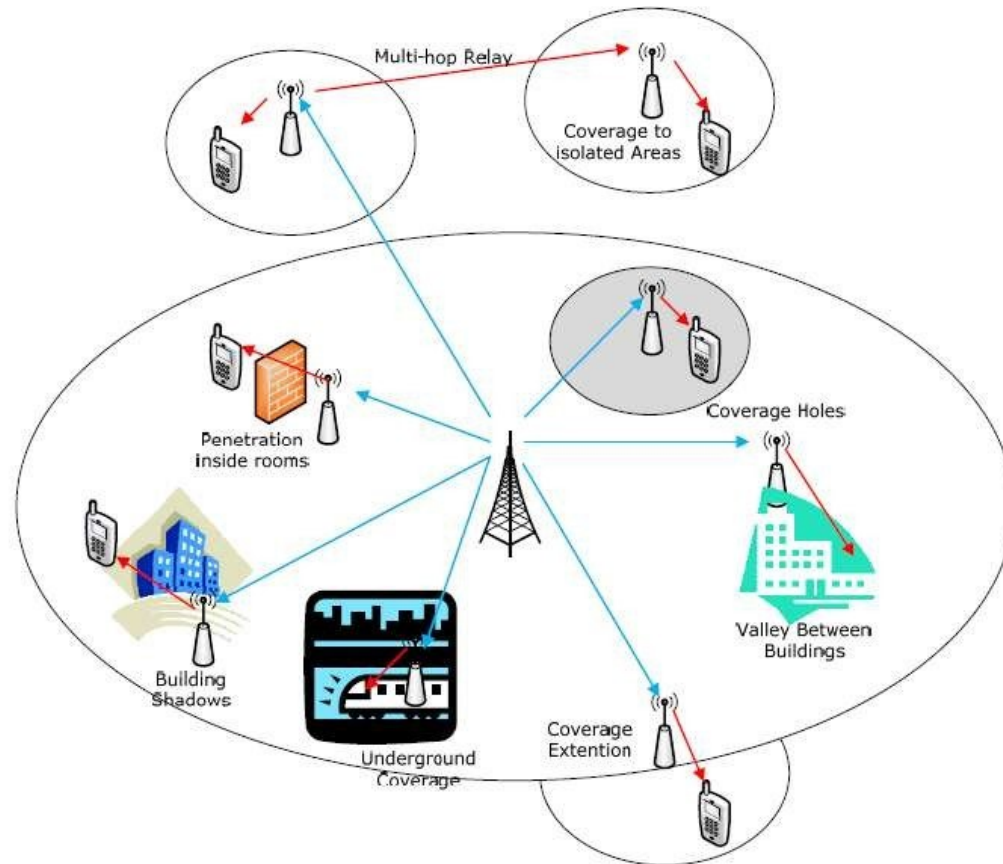


What are the effects of “Enhanced SC-FDMA”?



Source: http://www.3gpp.org/ftp/tsg_ran/WG4_Radio/TSGR4_54/Documents/R4-101056.zip

Significant step towards 4G: Relaying ?



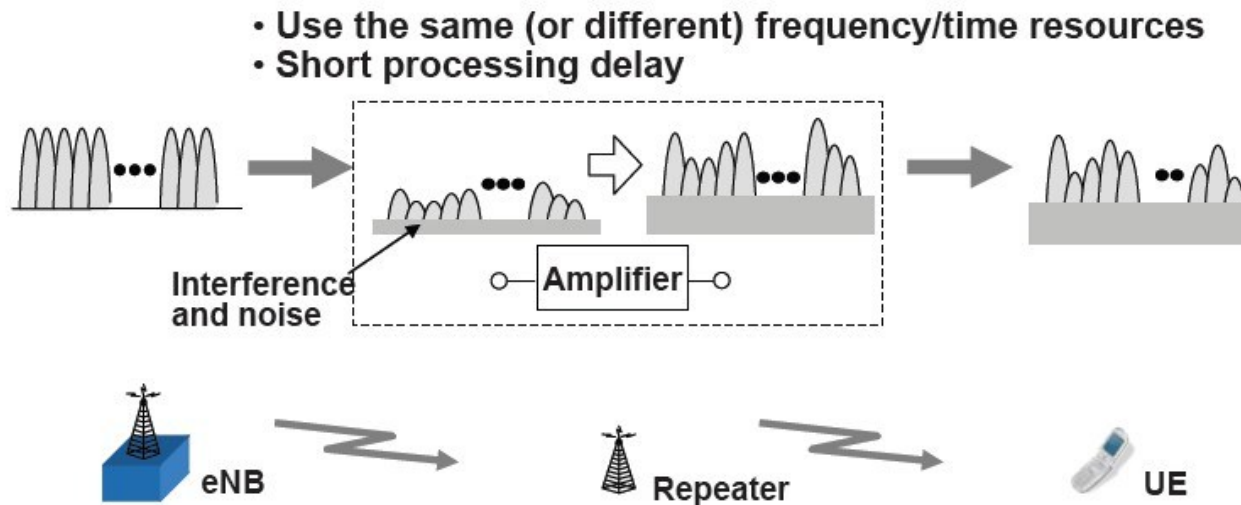
Source: TTA's workshop for the future of IMT-Advanced technologies, June 2008

Radio Relaying approach

■ Relays using radio

• L1 relays with non-regenerative transmission, i.e., repeaters

- ✓ Since delay is shorter than cyclic prefix duration, no additional change to radio interface is necessary
- ✓ Repeaters are effective in improving coverage in existing cells
- ✓ Should be used as well as in 2G/3G networks



No Improvement of SNR resp. CINR

Source: TTA's workshop for the future of IMT-Advanced technologies, June 2008

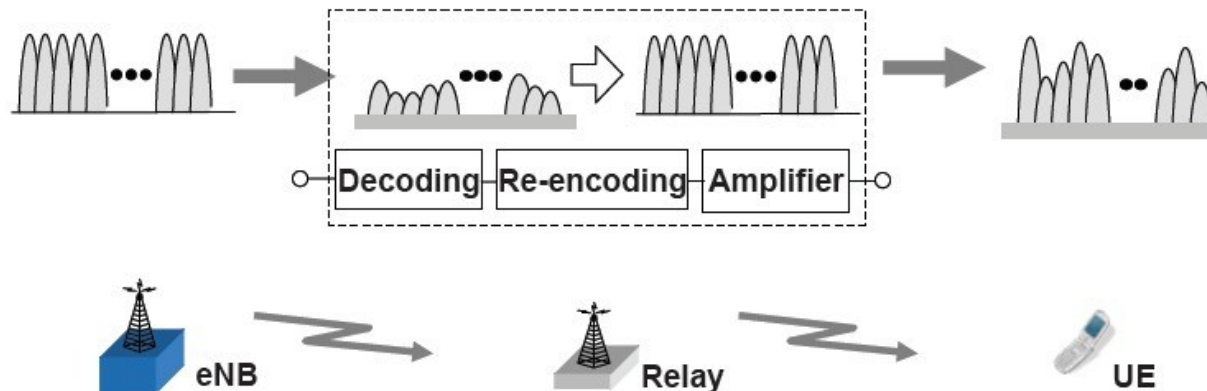
L1/L2 Relaying approach

■ Relays using radio

• L2 and L3 relays

- ✓ L2 and L3 relays can achieve wide coverage extension via increase in SNR
- ✓ Problems to be solved are efficient radio resource assignment to signals to/from relay station, delay due to relay, etc.

- Use different frequency/time resources
- Long processing delay
- Radio resource management at relays (L2 and L3 relays)



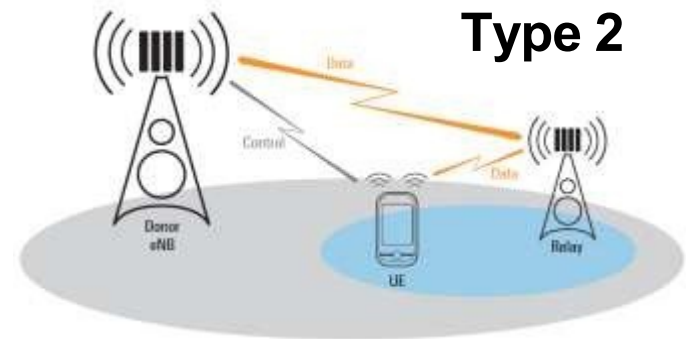
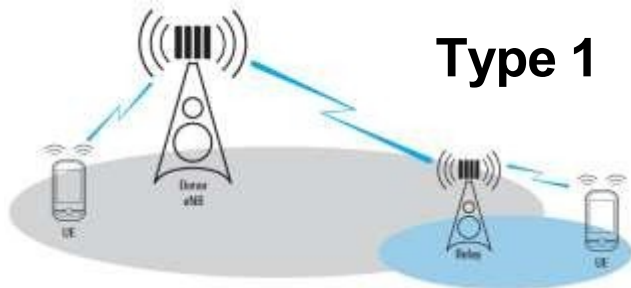
Source: TTA's workshop for the future of IMT-Advanced technologies, June 2008

LTE-Advanced

Relaying

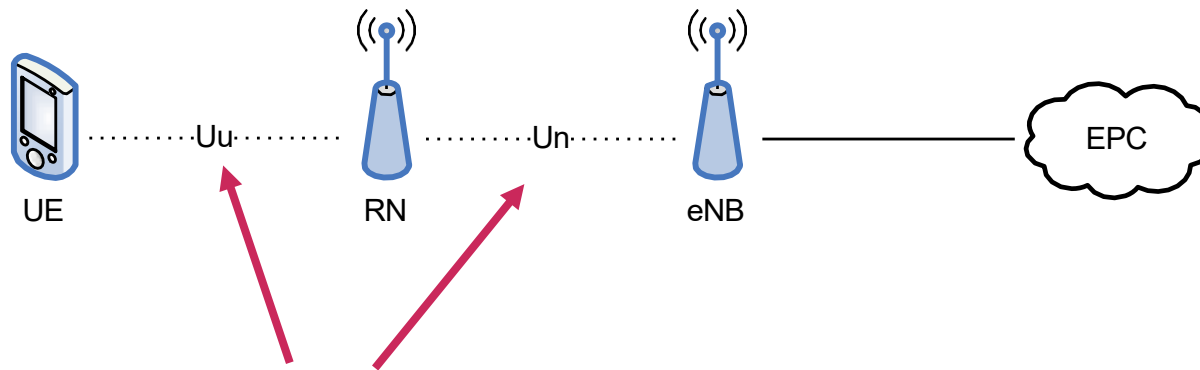


- I **LTE-Advanced extends LTE Release 8 with support for relaying in order to enhance coverage and capacity**
- I **Classification of relays based on**
 - I implemented protocol knowledge...
 - Layer 1 (repeater)
 - Higher Layer (decode and forward or even mobility management, session set-up, handover)
 - I ... and whether the relay has its own cell identity
 - Type 1 relay effectively creates its own cell (own ID and own synchronization and reference channels)
 - Type 2 relay will not have its own Cell_ID



LTE-Advanced: Relaying

The relay node (RN) is wirelessly connected to a *donor cell* of a *donor eNB* via the Un interface, and UEs connect to the RN via the Uu interface



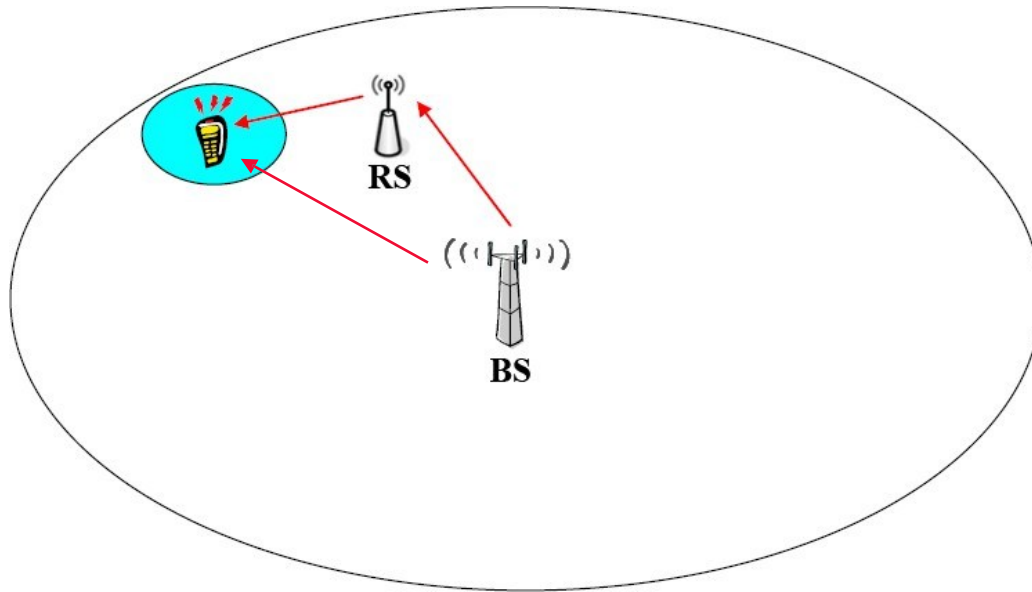
Inband: 2 links operate in the same frequency

Outband: 2 links operate in different frequencies

Co-operative Relaying Approaches

- I **Receiver in MS (or BS) gains from both, the signal origin and the relay station**
- I **Co-operative Relaying creates virtual MIMO**

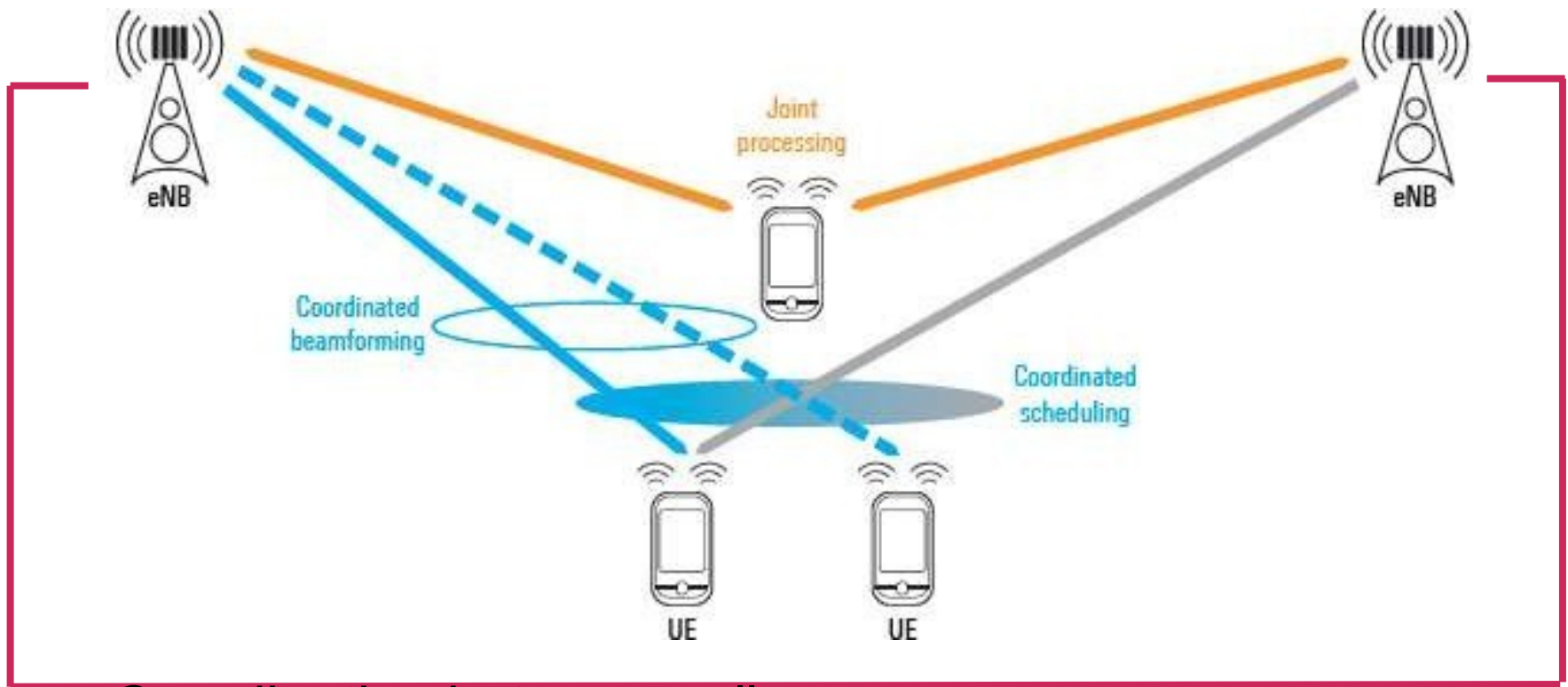
- RSs are „decode-and-forward“ devices



LTE-Advanced

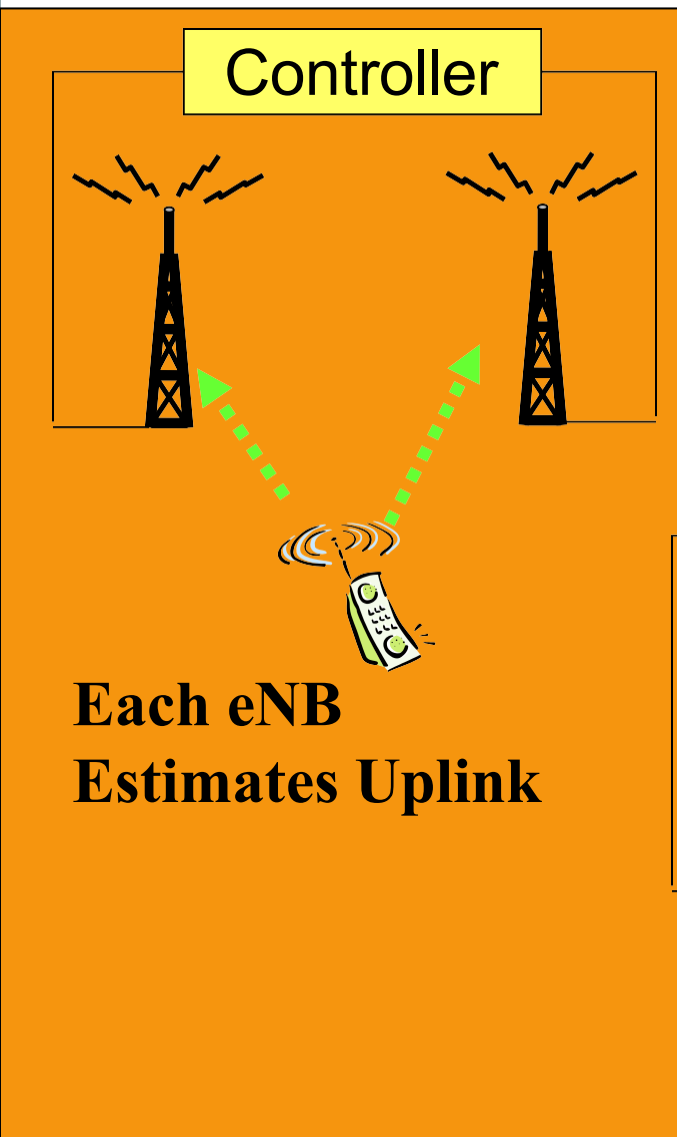
Coordinated Multipoint Tx/Rx (CoMP)

CoMP

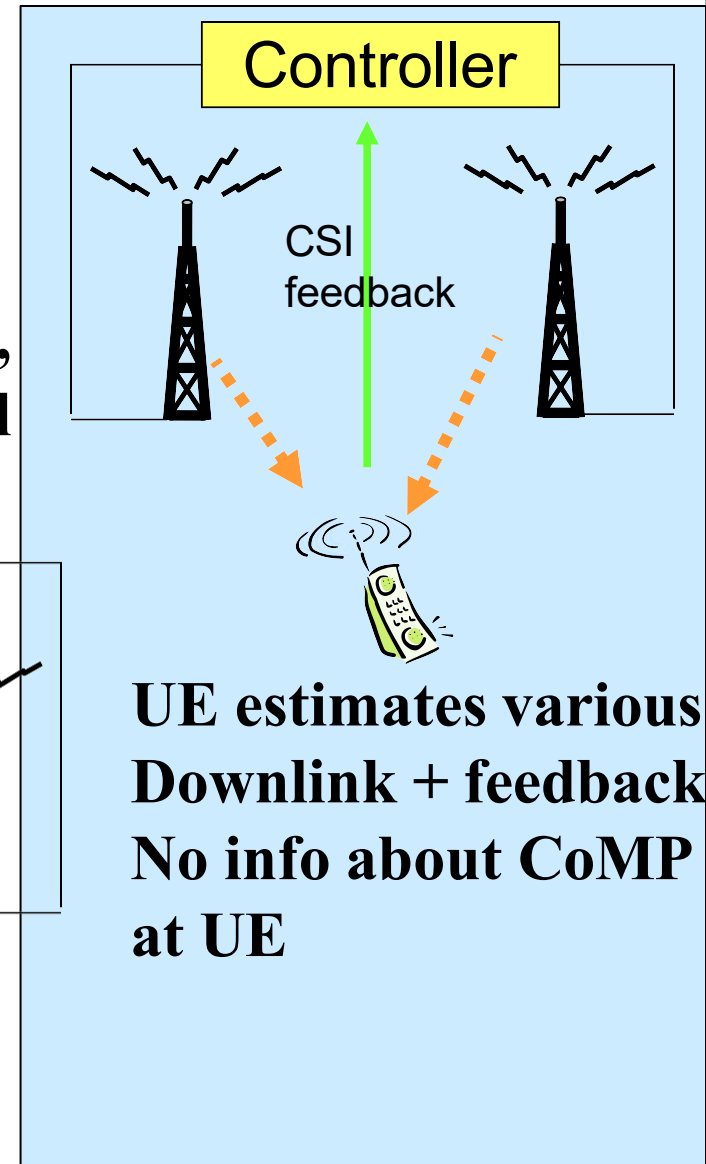
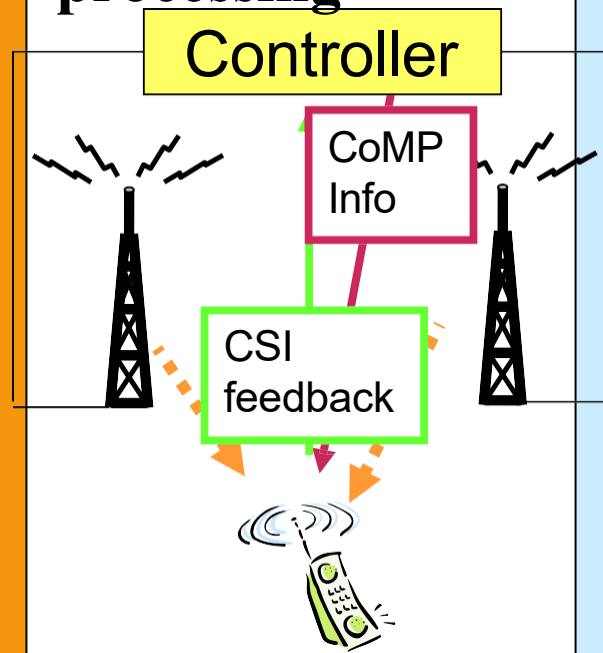


Coordination between cells

Coordinated Multipoint, CoMP



**UE estimates
Various DL
+ feedback
Info about CoMP,
UE perform signal
processing**



LTE-A: PCFICH indicating PDCCH size

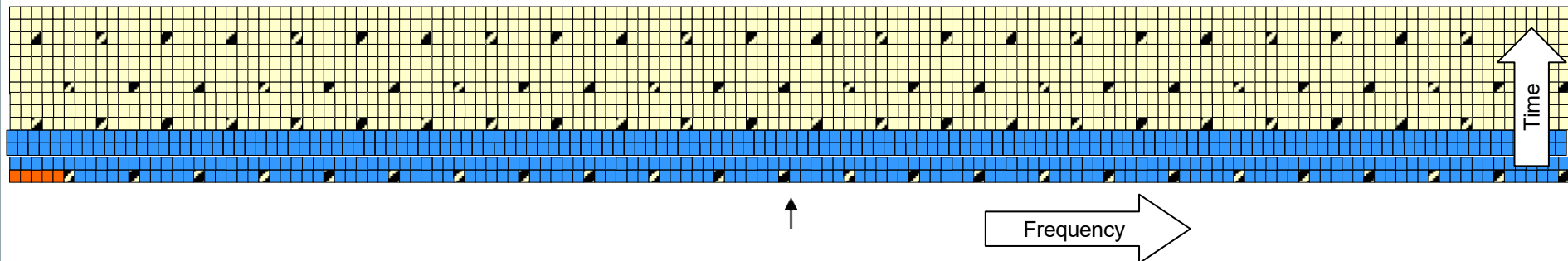
PCFICH content in LTE R-8

Subframe where PCFICH is sent	Number of OFDM symbols for PDCCH when $N_{RB}^{DL} > 10$	Number of OFDM symbols for PDCCH when $N_{RB}^{DL} \leq 10$
	1, 2	2
Subframe 1 and 6 in TDD mode	1, 2	2
Subframe 0 in FDD mode	1, 2, 3	2, 3, 4

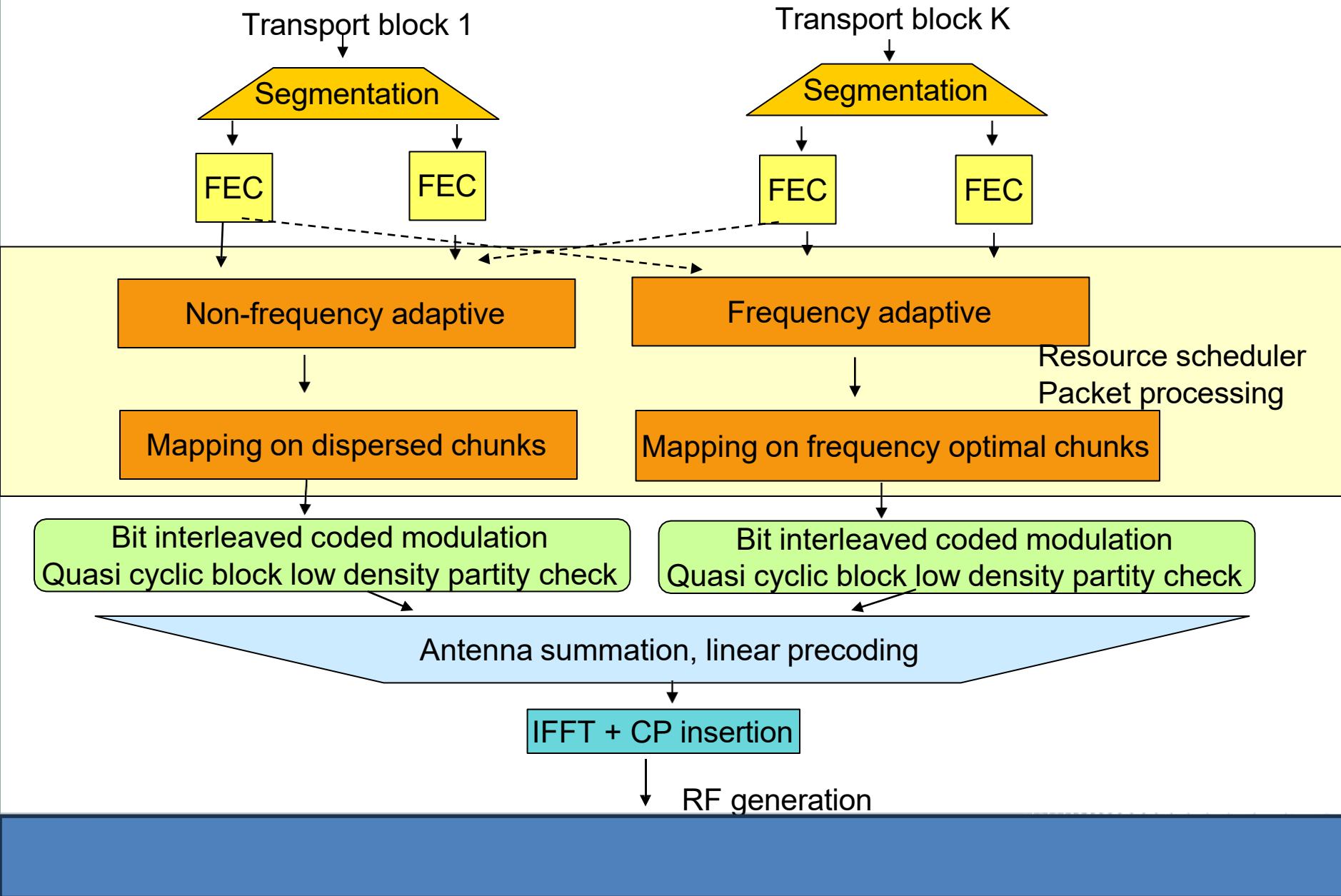
PCFICH content in LTE-A

PCFICH can indicate up to 4 OFDM symbols for used by PDCCH

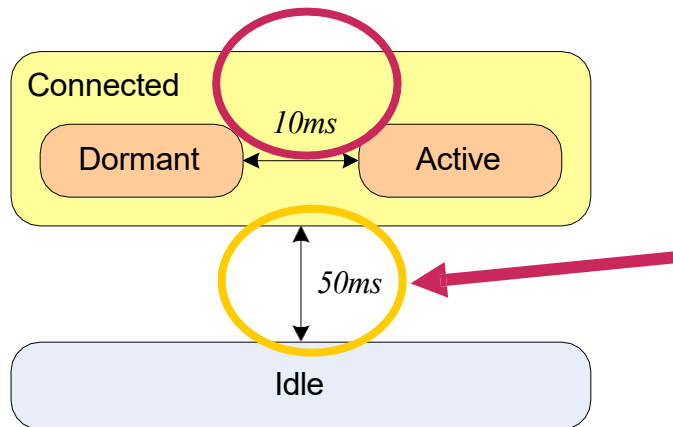
- PDSCH
- PDCCH
- PCFICH
- PBCH
- S-SCH
- P-SCH



LTE-A: Multiple Access MAC layer concept



LTE-Advanced: C-Plane latency



Already fullfills ITU Requirement with Rel.8, but ideas to speed up:

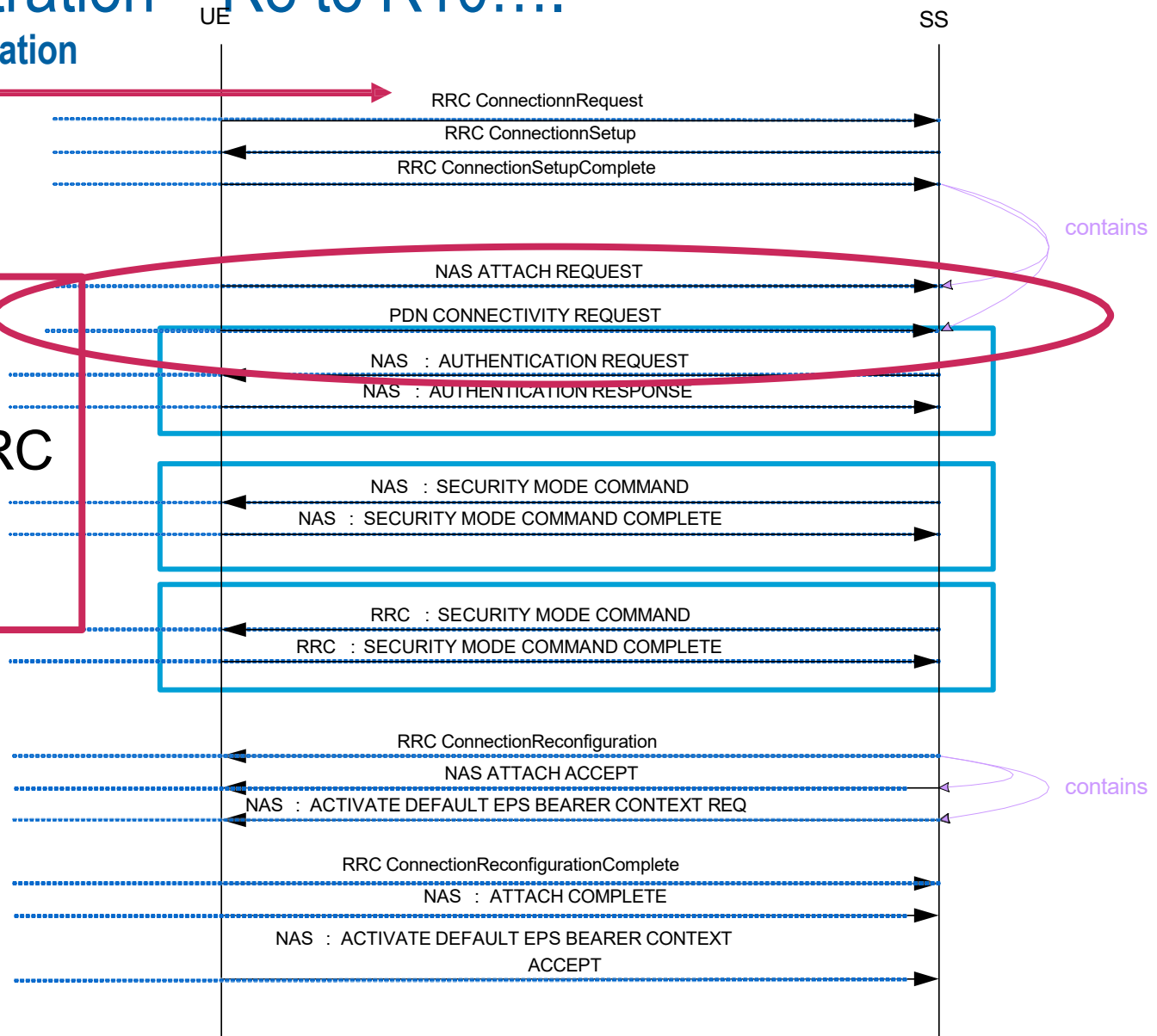
- Combined RRC Connection Request and NAS Service Request
- Reduced processing delays in network nodes
- Reduced RACH scheduling period: 10ms to 5ms

Shorter PUCCH cycle: requests are sent faster
Contention based uplink: UE sends data without previous request

LTE Registration – R8 to R10....

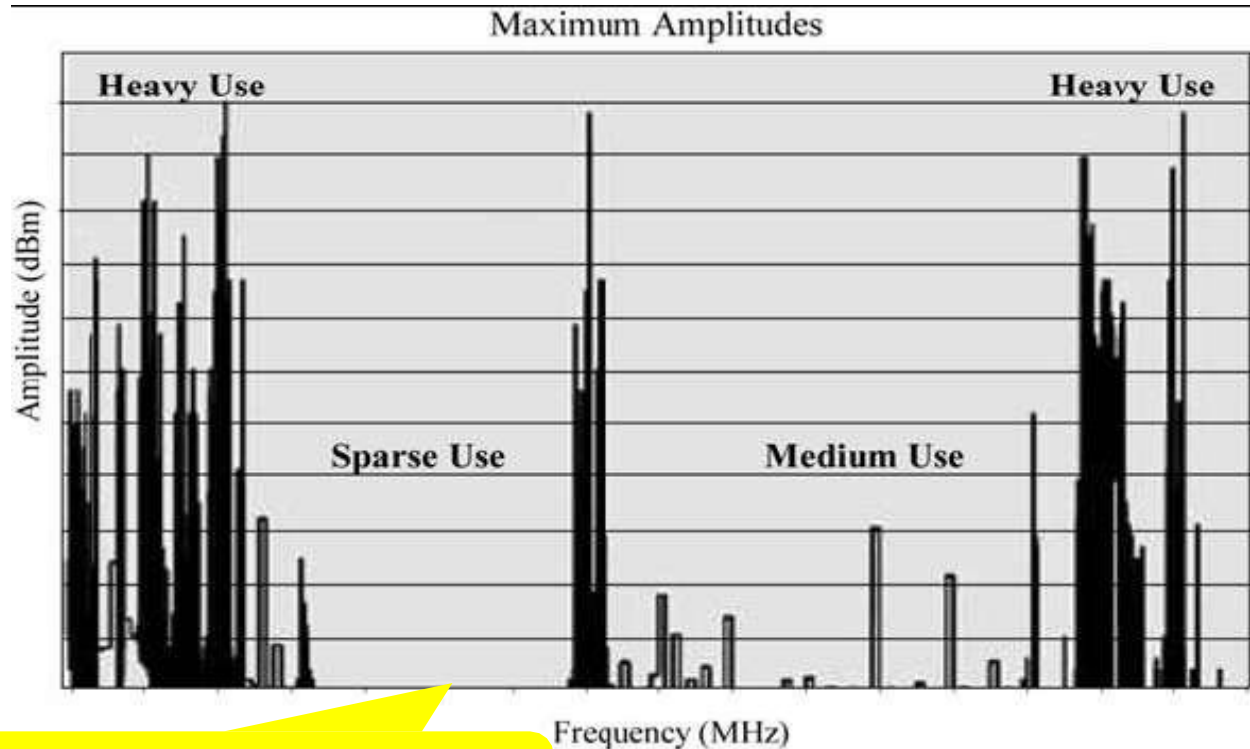
incl. security activation

NAS
Registration
Already in RRC
Connection
Request



Present Thrust- Spectrum Efficiency

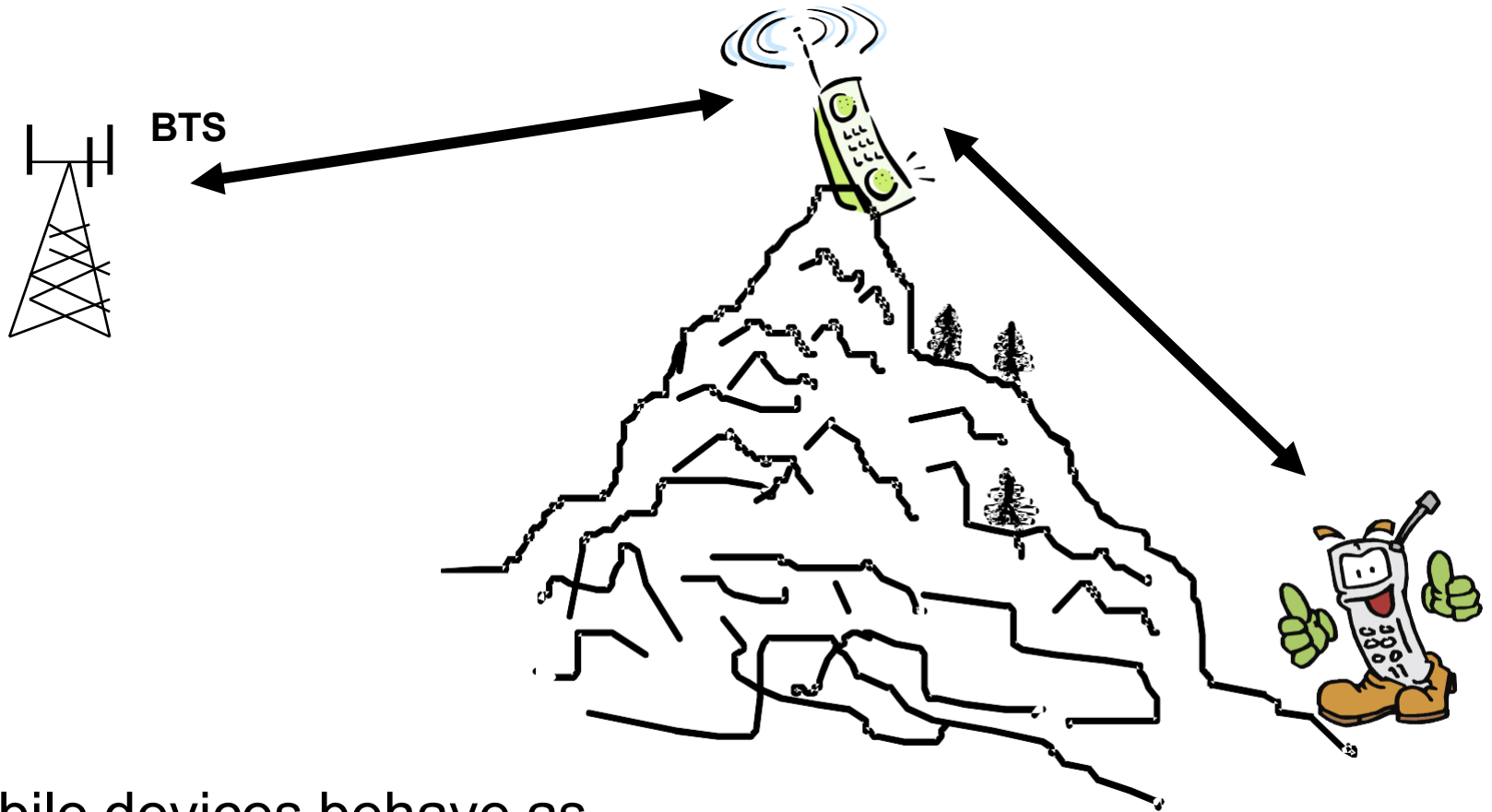
Momentary snapshot of frequency spectrum allocation



Why not use this part of the spectrum?

- I FCC Measurements:- Temporal and geographical variations in the utilization of the assigned spectrum range from 15% to 85%.

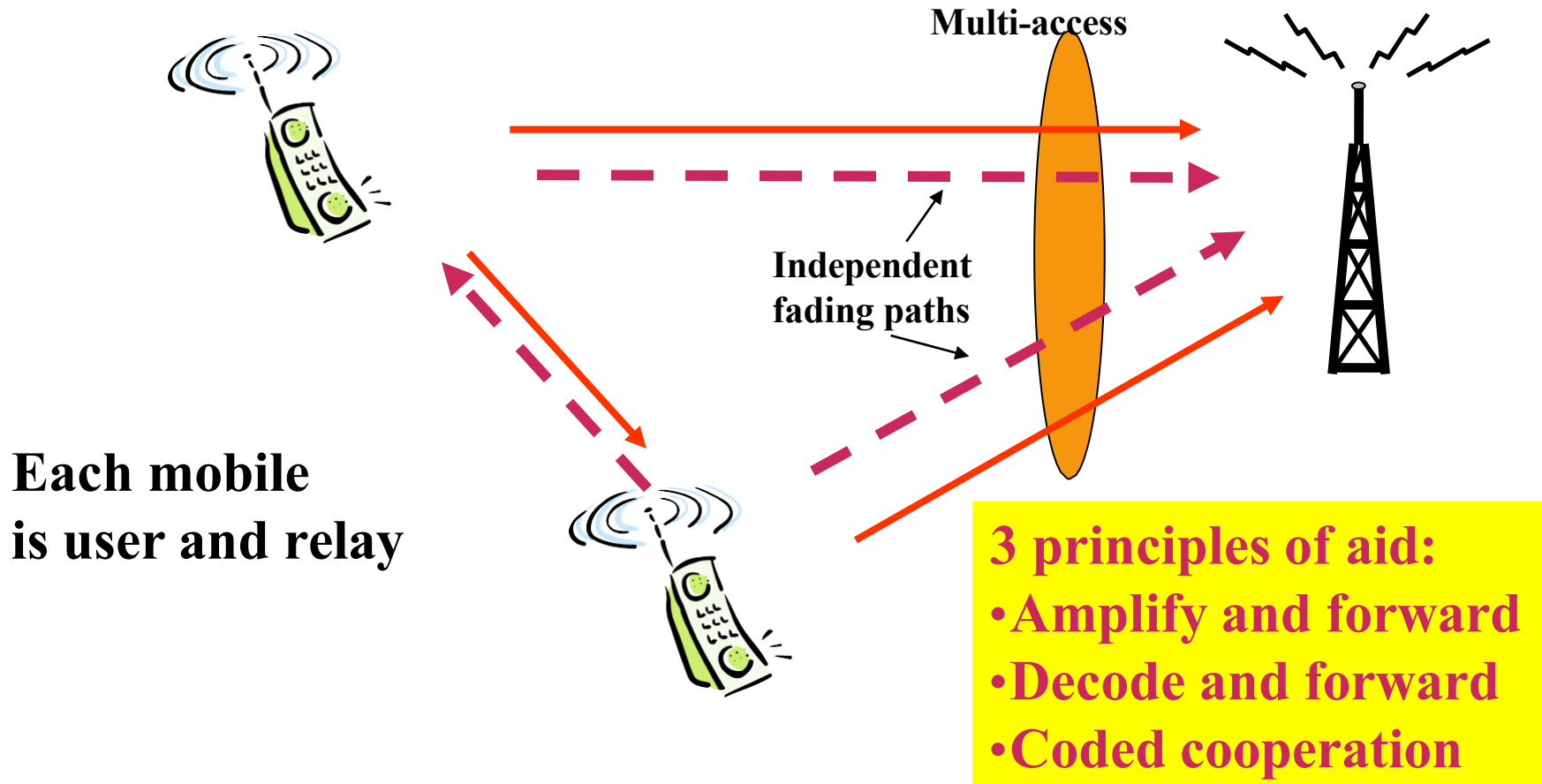
ODMA – some ideas...



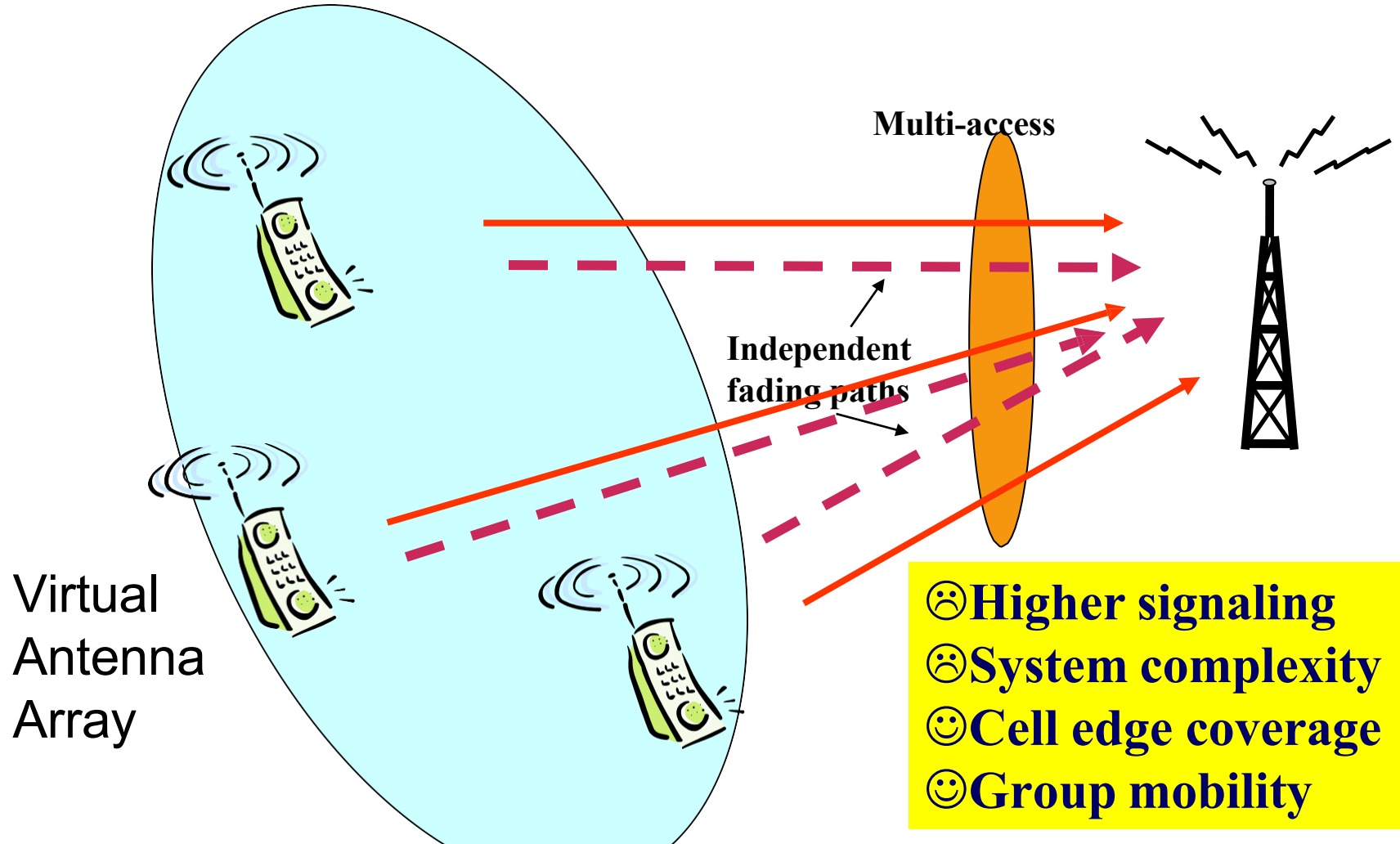
Mobile devices behave as
relay station

Cooperative communication

How to implement antenna arrays in mobile handsets?



Cooperative communication



Mobile going GREEN

